



**GREEN
CLIMATE
FUND**

Meeting of the Board

4 – 7 October 2021

Virtual meeting

Provisional agenda item 15

GCF/B.30/02/Add.06

14 September 2021

Consideration of funding proposals - Addendum VI

Funding proposal package for FP174

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic";
- b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment*;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

** includes the redacted version of the Secretariat's assessment*

Table of Contents

Funding proposal submitted by the accredited entity	3
No-objection letter(s) issued by the national designated authority(ies) or focal point(s)	99
Environmental and social report(s) disclosure	112
Secretariat's assessment	115
Independent Technical Advisory Panel's assessment	133
Response from the accredited entity to the independent Technical Advisory Panel's assessment	148
Gender documentation	151

Funding Proposal

Project/Programme title:	Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic
Country(ies):	Guatemala, Honduras, El Salvador, Nicaragua, Costa Rica, Panama and the Dominican Republic
Accredited Entity:	Central American Bank for Economic Integration (CABEI)
Date of first submission:	2019/11/08
Date of current submission	2021/09/06
Version number	V.10



Section A	PROJECT / PROGRAMME SUMMARY
Section B	PROJECT / PROGRAMME INFORMATION
Section C	FINANCING INFORMATION
Section D	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section E	LOGICAL FRAMEWORK
Section F	RISK ASSESSMENT AND MANAGEMENT
Section G	GCF POLICIES AND STANDARDS
Section H	ANNEXES

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-CABEI-CENTRAL AMERICA-2019/11/08"

PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	Not applicable		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <i>overall</i> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <i>GCF budget</i> devoted to it. The total of the percentages when summed should be 100%.		
	Mitigation: Reduced emissions from: ☐ Energy access and power generation: ☐ Low-emission transport: ☐ Buildings, cities, industries and appliances: ☐ Forestry and land use: Adaptation: Increased resilience of: ☒ Most vulnerable people, communities and regions: ☐ Health and well-being, and food and water security: ☐ Infrastructure and built environment: ☒ Ecosystem and ecosystem services:		GCF contribution: <u>Enter number%</u> <u>Enter number%</u> <u>Enter number%</u> <u>Enter number%</u> 50% <u>Enter number%</u> <u>Enter number%</u> 50%
A.5. Expected mitigation impact	Indicate t CO ₂ eq over lifespan	A.6. Expected adaptation impact	1,084,441 people (direct); 1,364,044 people (indirect)
			N/A
A.7. Total financing (GCF + co-finance)	US\$ 268.4 million	A.9. Project size	Large (Over USD 250 million)
A.8. Total GCF funding requested	US\$ 174.3 million <i>See annex 17.</i>		
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant US\$84.3 million ☒ Loan US\$ 60.0 million ☒ Guarantee US\$ 30 million	☐ Equity <u>Enter number</u> ☐ Results-based payment <u>Enter number</u>	
A.11. Implementation period	7 years	A.12. Total lifespan	12 years
A.13. Expected date of AE internal approval	<i>Click or tap to enter a date.</i>	A.14. ESS category	I-2 ^[OBJ]
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		

A.20. Executing Entity information

The programme will be executed by three institutions with clear roles according to their capacities and role in the region

CABEI will be the accredited entity for the programme, but will also be directly executing the major part of component 2 – output 4 - of the programme (activity 4.1 and activity 4.2) as they will be directly responsible for the set up and management of the loan and guarantee facilities of the programme. FAO will be the Executing entity for the Technical Assistance facility under this component, (activity 4.3). CABEI will also be the executing entity for the grant facility under output 2 (activity 2.3)

The Food and Agriculture Organization (FAO) will play a major role in the execution of the capacity building, technical assistance and implementation of the Ecosystem based Adaptation demonstration sites throughout the region as the Executing Entity of outputs 1 and 2 of the programme.

The Central-American Commission for Environment and Development (CCAD) will play a key role in political coordination among participant countries through the direct execution of output 3 focused of gathering and disseminating key results and analysis derived from the execution of the programme aiming at generating evidence to inform and mainstream adaptation into key national policies and regional strategies.

A.21. Executive summary

1. The Central American Dry Corridor and the Arid Zones of the Dominican Republic are among the world's most vulnerable tropical regions to the impacts of climate change, with Honduras, Nicaragua, Guatemala, the Dominican Republic and El Salvador recognised as being among the sixteen most vulnerable countries in the world to extreme climate events for the period 1998-2017 (Eckstein, Hutfils and Wings 2018). The term "Dry Corridor" emerged in the last three decades to describe the increasing frequency and intensity of droughts in the region. These droughts are linked to El Niño events, which are occurring more frequently and intensely as a result of climate change and have had severe impacts on agriculture and food security in the region. The impacts of droughts during El Niño are contrasted by an increase in the frequency and severity of extreme rainfall events linked to concurrent changes in La Niña. The resulting extreme rainfall leads to reductions in aquifer recharge, as well as increased surface runoff and soil erosion. These impacts are exacerbated by poor land management practices — which result in extensive environmental degradation — and widespread poverty.
2. With more than one million families in the Dry Corridor and Arid Zones reliant on agriculture for their livelihoods, the impacts of increasingly severe droughts over the past five years have resulted in the need for considerable humanitarian assistance. In 2014 alone, intense drought resulted in an estimated US\$465 million worth of agricultural losses across the region, with more than 2.5 million people experiencing food insecurity. From 2015 to 2018, food insecurity has increased annually in the region, with indigenous populations, women and rural communities the most likely to suffer from chronic hunger and malnutrition. By the last quarter of 2018, 2.2 million people had experienced crop losses that year, with 1.4 million urgently needing food assistance. These impacts are expected to continue to worsen under future climate scenarios.
3. Climate projections for the region indicate that, by the end of the century, temperatures will increase by 3–3.5°C under RCP4.5 and by 6–7°C under RCP8.5 (Lyra et al. 2017). At the same time, mean annual rainfall is projected to decrease by 11% on average. These changes will be compounded by further changes in the El Niño-La Niña cycle, which will result in prolonged droughts (Rauscher et al. 2008) as well as more frequent and intense extreme rainfall events across the region. If average global temperatures reach 1.5°C above pre-industrial levels, extreme El Niño events are likely to happen twice as often — approximately every ten years as opposed to every 20 years — with severe consequences for agriculture and food security. The impacts of future climate change will include further decreases in important crop harvests such as maize, beans and coffee, leading to increased poverty, food insecurity and migration from rural areas (Hannah et al. 2017). The land area suitable for growing these crops is also expected to decrease across the region, which will be exacerbated by soil degradation linked to increased rainfall intensity. Furthermore, as a result of climate change-related increases in temperature and drought frequency, the extent of the Dry Corridor and Arid Zones is predicted to expand. Currently, the Dry Corridor encompasses ~64% of the municipalities in Central America and is expected to extend to ~85% by 2050. These changes in temperature all also have an impact on the ranges of certain crop yields. For example, as cooler highland areas warm up, they will become more suitable for specific crops which may be pushed beyond thermal limits in lowland areas. However, the desire to shift crops into these highland areas will likely catalyse forest

clearing in what are critical water recharge zones of river catchments, thereby reducing aquifer recharge. This impact, coupled with an increase in extreme rainfall events, will increase the likelihood and severity of flooding. Climate change will also increase the frequency of wildfires, causing further forest degradation and threatening farming, forest-based livelihoods and fuelwood supply.

4. Without comprehensive adaptation interventions, agricultural losses will continue to increase, along with ecosystem loss, water shortages, poverty and food insecurity. These impacts will ultimately increase the exposure and sensitivity of communities in the Dry Corridor and Arid Zones who are already extremely vulnerable to climate change. However, several barriers currently constrain the adoption of climate-adaptive practices by vulnerable communities. These barriers include limited: i) access to credit for adaptation interventions among vulnerable populations; ii) access to financial resources for the implementation of ecosystem-based adaptation (EbA), water- and resource-efficient technologies among the private sector; iii) knowledge and understanding of climate change impacts as well as appropriate adaptation measures among decision-makers, communities and commercial farmers; iv) limited integration of EbA and other adaptation measures into sectoral policies; v) technical capacity within governments and local communities to implement EbA and other adaptation measures; and vi) development of economic incentives for investment in sustainable natural resource management and EbA.
5. In response to the above-described climate threats, GCF loan, grant and guarantee finance are requested to initiate a paradigm shift in the regional approach to catchment and water demand management in the Dry Corridor and Arid Zones. This shift will see governments of the seven participating countries — Guatemala, Honduras, El Salvador, Nicaragua, Costa Rica, Panama and the Dominican Republic — promoting large-scale EbA. At the same time, smallholder and commercial farmers, as well as entrepreneurs in rural communities, will have increased access to the financial resources and technical skills required to implement impactful adaptation interventions. A combination of GCF loan and grant finance, a guarantee, as well as a senior loan provided as co-finance, will be used to establish three financial facilities at the regional level, namely: i) a grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing (US\$28 million); ii) an EbA lending facility for small- and large-scale EbA investments at farm-, enterprise- and household-level (US\$102.8 million); and iii) a guarantee facility that provides an additional layer of security to financial institutions in order to promote EbA finance (US\$30 million).

The facilities will be managed by CABEL with a blended financing structure, where the relevant funds in the form of senior loans for the EbA lending facility are channelled via two separate streams towards end beneficiaries and are blended at the level of IFIs: (1) one stream for CABEL's co-financing funds, passing from CABEL's balance sheet as senior loans to the IFIs' balance sheet within an existing Global Credit Line (GCL), and (2) a second stream of GCF funds provided by CABEL (as trustee of the GCF funds) via limited recourse loans to the IFIs. The co-financing ratio for the EbA lending facility among the CABEL Co-financing and the GCF funding provided to each selected IFI shall be of 40:60, respectively.. Both streams are hence being blended at IFI level, and from there channelled by the IFIs to the PFIs that directly on-lend to end beneficiaries.

A guarantee facility will be implemented by CABEL with the aim to increase IFIs appetite to take on further financing to channel to PFIs by guaranteeing the PFI EbA finance performance. The grant facility will be channelling grant funds to on-the-ground stakeholders (rural communities, providers of EbA solutions, among others), to be distributed via two competitive bottom-up identification rounds for eligible projects per year, for four years (each round providing USD 3.5 million to an expected number of 70 projects per round). The grant facility aims at strengthening the identification, incubation and even acceleration of promising EbA solution in the localities, to prepare EbA solution markets for commercial and mainstreamed EbA investment and finance favouring the development of a business ecosystem around EbA. These mechanisms will make US\$130.8 million available for investment in EbA and water- and resource-efficient technologies in the region (US\$102.8 million from loans and US\$28 million from grants). The total grant request of US\$84.3 million will be used in the following ways; i) US\$10.1 million will be used to strengthen the technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures; ii) US\$ 28.5 million will be used to implement large-scale adaptation interventions — including EbA, water- and resource-efficient technologies — in rural communities across the region; iii) US\$28 million will form the grant facility through which vulnerable communities can access on-grants channelled through financial institutions; iv) US\$6.3 million will be used to generate and share information on climate change adaptation and finance across the region, and mainstream it into local and national policies; v) US\$6.5 million will be used to set up the financial mechanisms and strengthen the technical capacity of CABEL accredited and non-regulated financial institutions to access and channel funds through them; vi) US\$4.2 million will be used for overall management and monitoring of programme activities (Independent evaluations and PMC costs) ; and vii) US\$0.7M will be requested as contingency. Additionally, national

governments have committed US\$51.3 million in co-finance which will further scale and mainstream EbA, water and resource-efficiency in their respective countries.

6. Together, these interventions will catalyse the autonomous upscaling of EbA and water- and resource-efficient technologies by combining technical capacity-building, policy instruments and accessible financial mechanisms. This approach will transform the financial landscape in the region to enable vulnerable groups to access climate finance and implement climate-resilient measures to strengthen their adaptive capacity. The proposed programme will directly benefit ~1,084,441 people across the target catchments through on-the-ground interventions, with ~134,694 of these people also able to access loans for the implementation of adaptation measures through financial mechanisms established under the programme (lending and guarantee facilities). Furthermore, the combined impact of programme interventions will facilitate the uptake of adaptation measures beyond the target municipalities, ultimately indirectly benefitting ~1,364,044 people across the region during the programme lifetime, and thousands more into the future.
7. The programme will be executed by three institutions with clear roles according to their capacities and role in the region. CABI will be the accredited entity for the programme, but will also be directly executing the major part of output 2 and 4 of the programme as they will be directly responsible for the set up and management of the three financial facilities of the programme CABI will therefore be the executing entity for activities 2.3 (grant facility), 4.1 and 4.2 (lending and guarantee facilities). FAO will be the Executing entity for the Technical Assistance facility under output 4 (activity 4.3), supported by UNEP as technical advisor. The responsibility of output 4 is thus shared between CABI and FAO. FAO will also play a major role in the execution of the capacity building, technical assistance and implementation of the Ecosystem based Adaptation demonstration sites throughout the region as the Executing Entity of outputs 1 and 2 of the programme. CCAD will play a key role in political coordination among participant countries through the direct execution of output 3 focused of gathering and disseminating key results and analysis derived from the execution of the programme aiming at generating evidence to inform and mainstream adaptation into key national policies and regional strategies. UNEP will provide technical advice to CCAD for the development of technical knowledge products on financial mechanisms for EbA, methodologies for ecosystem services valuation and technical guidelines for EbA implementation. UNEP will play a key role also as technical advisor mainly on the financial component and the regional hub component contributing with all the technical experience already developed in the region on the design and deployment of financial mechanisms for EbA and on EbA mainstreaming into national policies.

PROJECT/PROGRAMME INFORMATION

B.1. Climate context

Context

8. The Dry Corridor of Central America consists of semi-arid and sub-humid ecosystems that cover most of the Pacific slope of Guatemala, Honduras, El Salvador and Nicaragua, as well as smaller areas of Costa Rica and Panama (Figure 1). Similar climatic conditions are found in the arid and semi-arid areas (hereafter 'Arid Zones') in the west of the Dominican Republic (Figure 1). The Dry Corridor is the most densely populated region of Central America¹, and the Arid Zones are home to a significant proportion of the Dominican Republic's population — 12% of the total², amounting to ~1.2 million people. In addition, Guatemala, Honduras, El Salvador and Nicaragua have comparatively small Gross Domestic Product (GDP) per capita, while Costa Rica, Panama and the Dominican Republic are comparatively wealthy and have relatively low poverty levels³. However, all these countries have high income inequality, with instances of extreme poverty — particularly in rural areas.

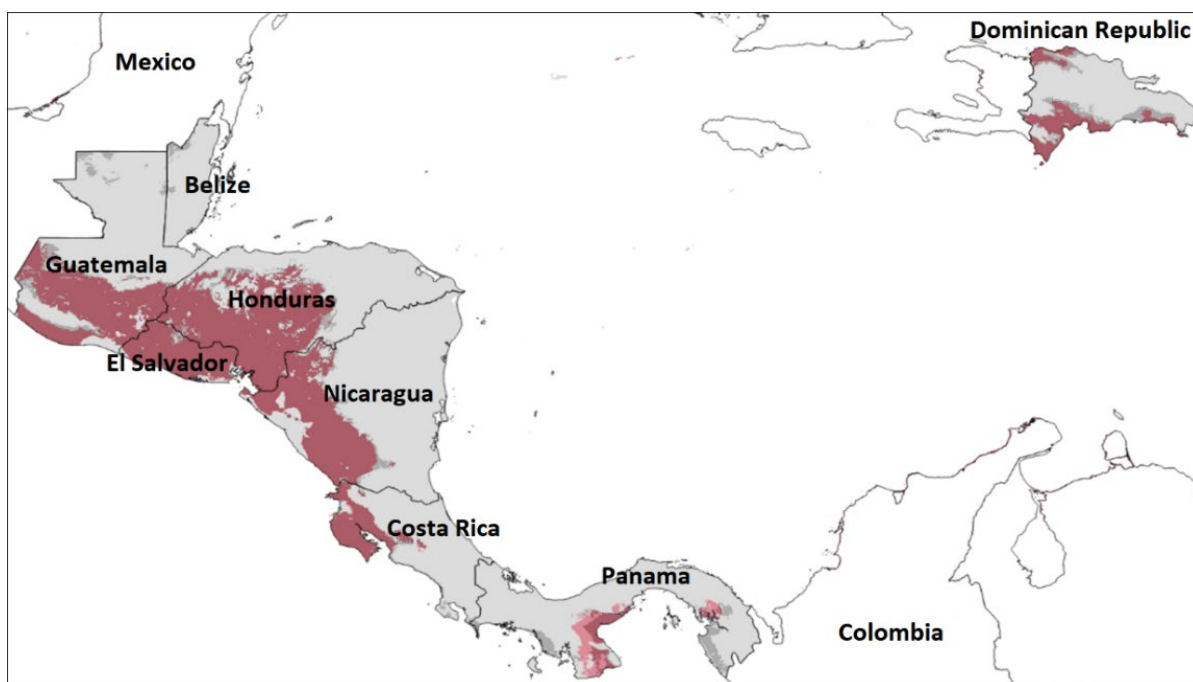


Figure 1. Current (red) and projected extent in 2050 (red and dark grey) of the Central American Dry Corridor and the Dominican Republic's Arid Zones⁴. The map is based on calculations by the International Centre for Tropical Agriculture (CIAT), using the current climate distribution⁵, future scenarios⁶ and evapotranspiration estimations^{7,8}.

9. Across the region, poverty in rural areas is significantly higher than in urban areas. Nearly 60% of people in the Dry Corridor live in poverty⁹, while the Arid Zones of the Dominican Republic have the highest poverty rate in the country¹⁰. A further example is that in Costa Rica double the number of people in rural areas are living in extreme poverty compared with in urban areas¹¹. Similarly, in Panama, prevalence of extreme poverty is 27%.

¹ with a population of 10.5 million, representing ~22% of the total population of the six countries across which it extends

² Brinkhoff, T. 2017. *City Population*. Available at: <http://www.citypopulation.de>.

³ Respective countries' economic situations are described in Section B.5, with additional detail provided in Section 2.3 of Annex 2: Feasibility Study.

⁴ Projected 2050 extent based on Representative Concentration Trajectory (RCP) 8.5, since according to the latest Emissions Gap Report published by UN Environment (2017), national commitments will only reach one third of the emissions reduction required by 2030 to meet current mitigation objectives.

⁵ Hijmans et al., 2005. Very high resolution interpolated climate surfaces for global land areas. *International Journal of Climatology*, 25: 1965-1978.

⁶ Ramirez-Villegas & Jarvis, 2010. Downscaling Global Circulation Model outputs: The Delta Method. Decision and Policy Analysis Working Paper No. 1, CIAT. Available at: <http://ccafs-climate.org/downloads/docs/Downscaling-WP-01.pdf>

⁷ Allen et al., 2006. Evapotranspiration del cultivo: guías para la determinación de los requerimientos de agua de los cultivos. FAO, Rome; Droogers & Allen, 2002. Estimating reference evapotranspiration under inaccurate data conditions. *Irrigation and Drainage Systems* 16: 33-45.

⁸ The calculations consider that a month is dry when the relationship between precipitation and evapotranspiration is less than 0.5.

⁹ living on under US\$ 3.20 per day

¹⁰ As estimated in 2017. Thomas Brinkhoff: City Population, <http://www.citypopulation.de>

¹¹ Borgen Project, 2017. Costa Rica's Poverty Rate. Available at: <https://borgenproject.org/about-costa-ricas-poverty-rate/>.

in rural areas compared with only 4% in urban areas¹². Most rural populations in the Dry Corridor and Arid Zones depend on agriculture for their livelihoods, practising both subsistence and commercial farming, and farming both crops and livestock. In addition to agriculture, plantation forestry supports the livelihoods of some local communities.

Climate rationale

10. The Dry Corridor and Arid Zones are among the world's most vulnerable tropical regions to the impacts of climate change¹³, with Honduras, Nicaragua, Guatemala, the Dominican Republic and El Salvador recognised as being among the fifteen most vulnerable countries in the world to extreme climate events¹⁴. The term "Dry Corridor" emerged in the last three decades to describe the increasing frequency and intensity of droughts in the region. These droughts are linked to El Niño¹⁵ events, which are occurring more frequently and intensely as a result of climate change¹⁶, and have had severe impacts on agriculture and food security in the region. Since agriculture and forestry are greatly affected by climatic conditions, the livelihoods of rural people in the Dry Corridor and Arid Zones are extremely sensitive to the impacts of climate change. For example, during the 2015 drought, over three million people experienced acute food insecurity¹⁷. These impacts are exacerbated by poor land management practices — which result in extensive environmental degradation — and widespread poverty. Furthermore, these communities have limited access to financial resources for the implementation of adaptation interventions. Without urgent adaptation measures, the vulnerability of the region's population to climate change impacts will increase. Further details of climate change risks and impacts for the Dry Corridor and Arid Zones are described below.

Climate change risks

11. Past climate trends and future projections show that the region's climate is changing in several ways, including:
 - i) temperatures increasing across the region; and ii) shifting rainfall patterns. In most parts of the Dry Corridor and Arid Zones, rain seasons are shortening and the intensity of midsummer droughts¹⁸ is increasing¹⁹. Droughts that extend over a year or more are also becoming increasingly frequent and severe, mainly because of the increasing frequency and intensity of El Niño events. Concurrently, extreme rainfall events are increasing in frequency and severity because of changes in La Niña. Country-specific observed and predicted changes in temperature and rainfall are presented in Section 3 of Annex 2: Feasibility Study.
12. Overall, temperatures across Central America are anticipated to continue to increase in future. By the end of the century, temperatures are projected to increase by 3–3.5°C under a medium emissions scenario (RCP4.5) and by as much as 6–7°C under a high emissions scenario (RCP8.5)²⁰ relative to the baseline of 1961–1990. While the trend of increasing temperatures is consistent across Central America and the Caribbean, the magnitude of change is expected to vary among countries and between geographical areas within countries (Figure 2). Mean annual rainfall is projected to decrease across much of Central America by 2070 for both medium (RCP4.5) and high (RCP8.5) emissions scenarios (Figure 3). In addition to the predicted future trends in mean annual temperature and rainfall, extreme temperatures are predicted to increase in the coming decades (Figure 4). From 2021–2050, daytime high temperatures (TXx)²¹ and night-time high temperatures (TNx)²² are projected to increase everywhere in the region. Rates of warming are expected to be strongest over the northern portion of the Dry Corridor, particularly over the eastern mountain slopes in Guatemala, Honduras,

¹² Borgen Project, 2017. Panama's Poverty Rate. Available at: <https://borgenproject.org/panama-poverty-rate/>.

¹³ IPCC (2014). Fifth Assessment Report of the Intergovernmental Panel on Climate Change.

¹⁴ Kreft et al., 2016. Global Climate Risk Index 2017. Who suffers most from extreme weather events? Weather-related loss events in 2015 and 1996 to 2015. Bonn: Germanwatch e.V.

¹⁵ These form part of the El Niño Southern Oscillation (ENSO) cycle, which is an irregularly periodic variation in winds and sea surface temperatures over the tropical eastern Pacific Ocean. The warming phase is known as El Niño and the cooling phase as La Niña. The ENSO cycle strongly influences the region's climate.

¹⁶ Cai et al. 2015. ENSO and greenhouse warming. *Nature Climate Change*, 5: 849.

¹⁷ FAO. 2017. Chronology of the Dry Corridor: The impetus for resilience in Central America. Available at: <http://www.fao.org/in-action/agronoticias/detail/en/c/1024539>

¹⁸ The rain season in the Dry Corridor lasts from May to October, interrupted in August by a period of lower precipitation known as the mid-summer drought, *canícula* or *veranillo*.

¹⁹ Rauscher et al., 2008. Extension and intensification of the Meso-American mid-summer drought in the twenty-first century. *Climate Dynamics* 31: 551-571.

²⁰ Lyra, A., Imbach, P., Rodriguez, D., Chou, S.C., Georgiou, S. and Garofolo, L., 2017. Projections of climate change impacts on central America tropical rainforest. *Climatic Change*, 141, pp.93-105.

²¹ Monthly maximum value of daily maximum temperature

²² Monthly maximum value of daily minimum temperature

and Nicaragua. Similarly, temperatures of cold days²³ are predicted to increase across the region. The rates of these increases are projected to be strongest over the Pacific coasts of Nicaragua, El Salvador, Honduras and Guatemala²⁴. In addition — and although rainfall in the Dry Corridor and Arid Zones is expected to decrease by 11% on average by the end of the century — extreme rainfall is predicted to increase significantly across parts of the region by mid-century. The number of intense rainfall events²⁵ and extremely intense rainfall events²⁶ occurring each year is predicted to increase (Figure 5). The R50mm index is used to represent precipitation exceeding 50 mm/day which represents heavy rains, and projections show an increase in R50mm in the Atlantic coast of Costa Rica and northern parts of Panama²⁷. A similar trend is observed in projections for the 90th percentile of precipitation index (R90p), which represents extreme precipitation, with rains heavier than the R50mm, that are less frequent. The R90p indicator has been projected to have a strong increase in the Atlantic coast of Panama, Costa Rica, and Nicaragua. The Pacific coast of Panama and Costa Rica also shows some increase, but at a smaller rate. In the Pacific coast of El Salvador, Honduras, and Guatemala the R90p projections show a decreasing trend, which is attributed to the decrease of winds or hurricane activity²⁸.

13. Recent evidence²⁹ and model simulations predict that the risk of drought will increase with climate change, as more intense and prolonged El Niño events occur across the region³⁰. Indeed, if average global temperatures reach 1.5°C above pre-industrial levels, extreme El Niño events are likely to happen twice as often — approximately every 10 years as opposed to every 20 years. Both droughts and dry spells will increase in the Dry Corridor and Arid Zones as a result of climate change. Droughts are years or seasons when rainfall is below average, while dry spells are defined as large number of consecutive days when minimal rain falls. From 2021–2050, it is projected that parts of the Dry Corridor and Arid Zone will experience an increase in dry spells^{31,32}. While the number of dry spells is projected to decrease in parts of Panama, Costa Rica, Nicaragua and Honduras, it is expected to increase over the Pacific basin in Nicaragua and El Salvador, as well as in Guatemala (Figure 6)³³. In addition, annual mean water excess³⁴ is projected to decrease across most parts of the Dry Corridor (Figure 6). Projections for drought events in the region show an increase in drought length, intensity and frequency under both a medium emissions scenario (RCP4.5) high emissions scenario (RCP8.5), for both short-term and long-term droughts³⁵. These increases are projected to be moderate in early decades of the century (2011–2040), then become more pronounced at the mid-century (2041–2070) before becoming extreme at the end of the century (2071–2100). During dry spells and droughts, there is a pronounced risk of the spread of aridity in parts of the Dry Corridor and Arid Zone as the availability of water is reduced, which could have lasting impacts on the local environment³⁶. On an annual scale, the region currently experiences a rainy season with two peaks separated by a dry period of weeks to over a month in duration, termed the mid-summer drought (MSD). Projections for MSD trends under a high emissions scenario (RCP8.5) show an increase in duration for the MSD period, which is projected to increase by an average of over a week. Additionally, the MSD minimum precipitation is projected to decrease by an average of over 26%, with

²³ Monthly minimum value of daily maximum temperature

²⁴ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

²⁵ Heavy rainfall events are defined as those in which rainfall reaches 50 mm per day.

²⁶ Extremely heavy rainfall events are defined as those in which rainfall exceeds 50 mm per day.

²⁷ Imbach resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

²⁸ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

²⁹ Guojian, W., 2017. Continued increase of extreme El Niño frequency long after 1.5 °C warming stabilization. *Nature Climate Change* 7, 568–572.

Available at: <https://www.nature.com/articles/nclimate3351>.

³⁰ Fasullo, J. T., Otto-Bliesner, B. L., & Stevenson, S., 2018. ENSO's changing influence on temperature, precipitation, and wildfire in a warming climate. *Geophysical Research Letters*, 45, 9216–9225. <https://doi.org/10.1029/2018GL079022>.

³¹ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

³² Dry spells affecting vegetation growth can be assessed through the relationship between actual (ET) and potential (ETP) evapotranspiration, which combines the effects of precipitation, temperature, soil moisture, and vegetation cover, among other atmospheric conditions. Dry spells are defined as the annual largest number of consecutive days when $ET \leq 0.5 \cdot ETP$.

³³ Imbach P, Molina L, Locatelli B, Roupsard O, Mahe' G, Neilson R, et al. Modeling potential equilibrium states of vegetation and terrestrial water cycle of Mesoamerica under climate change scenarios. *J Hydrometeorol*. 2012; 13:665–680.

³⁴ Annual mean water excess, defined as P-ETP, is a proxy of water availability from runoff. When P-ETP is positive, there is a surplus of water availability, while when P-ET is negative, there is a deficit of water availability.

³⁵ Depsky, N. and Pons, D., 2020. Meteorological droughts are projected to worsen in Central America's Dry Corridor throughout the 21st century. *Environmental Research Letters*, 16(1), p.014001.

³⁶ Quesada-Hernandez, L.E., Calvo-Solano, O.D., Hidalgo, H.G., Perez-Briceno, P.M. and Alfaro, E.J., 2019. Dynamical delimitation of the Central American Dry Corridor (CADC) using drought indices and aridity values. *Progress in Physical Geography: Earth and Environment*, 43(5), pp.627-642.

statistically significant changes for the mountains and Pacific side in most of Nicaragua, Honduras, El Salvador, and Guatemala³⁷. Other studies assessing climate change impacts on water in the region have modelled decreases in water availability by 2100^{38, 39}. As a result of these climate change-related increases in temperature and drought frequency, the Dry Corridor and Arid Zones are predicted to expand. Currently, the Dry Corridor encompasses ~64% of the municipalities in Central America; it is expected to extend to ~85% by 2050 (Figure 1).

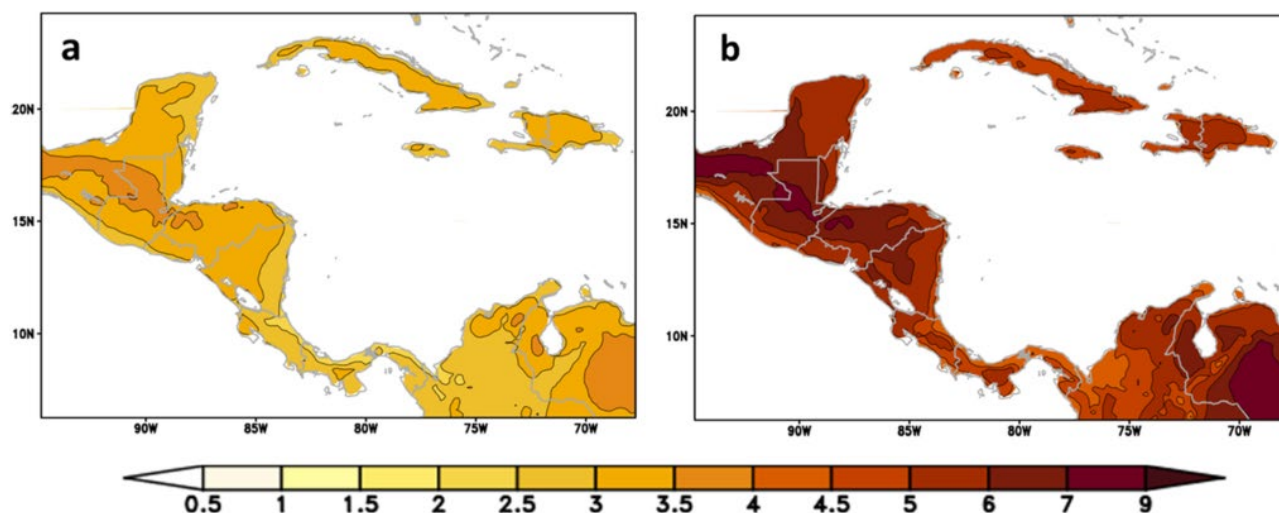


Figure 2. Difference between the baseline (1961–1990) and the future (2071–2100) annual mean temperature (°C) simulated by the Eta-HadGEM2 model for: a) RCP4.5 (medium emissions); and b) RCP8.5 (high emissions) scenarios⁴⁰.

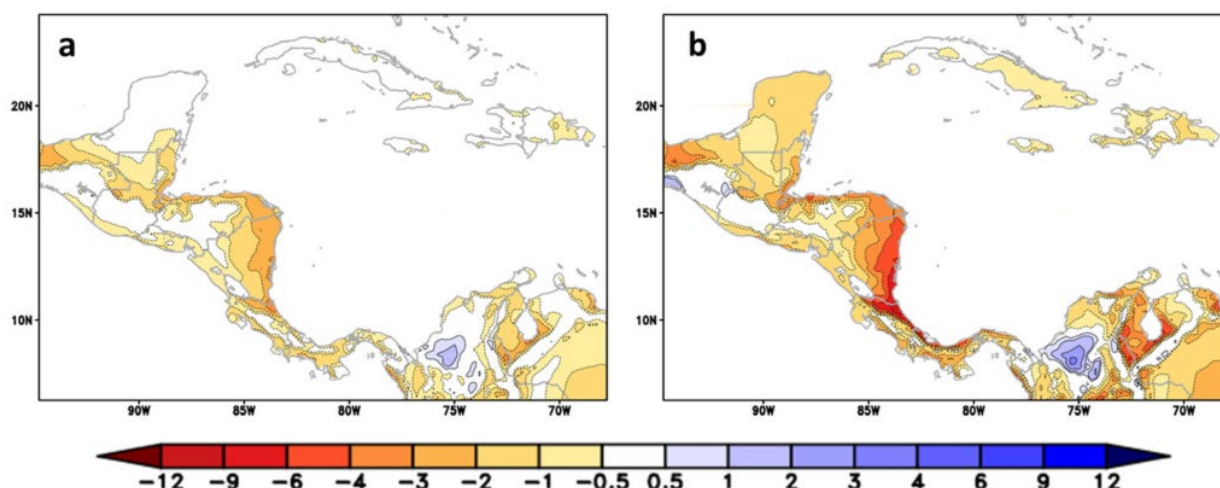


Figure 3. Difference between the baseline (1961–1990) and the future (2070–2100) mean annual rainfall (mm/day) simulated by the Eta-HadGEM2 model for a) RCP4.5 (medium emissions) and b) RCP8.5 (high emissions) scenarios⁴¹.

³⁷ Maurer, E.P., Roby, N., Stewart-Frey, I.T. and Bacon, C.M., 2017. Projected twenty-first-century changes in the Central American mid-summer drought using statistically downscaled climate projections. *Regional Environmental Change*, 17(8), pp.2421-2432.

³⁸ Imbach P, Molina L, Locatelli B, Roupsard O, Mahe' G, Neilson R, et al. Modeling potential equilibrium states of vegetation and terrestrial water cycle of Mesoamerica under climate change scenarios. *J Hydrometeor*. 2012; 13:665–680.

³⁹ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

⁴⁰ Lyra, A., Imbach, P., Rodriguez, D., Chou, S.C., Georgiou, S. and Garofolo, L., 2017. Projections of climate change impacts on central America tropical rainforest. *Climatic Change*, 141, pp.93-105.

⁴¹ Lyra, A., Imbach, P., Rodriguez, D., Chou, S.C., Georgiou, S. and Garofolo, L., 2017. Projections of climate change impacts on central America tropical rainforest. *Climatic Change*, 141(1), pp.93-105.

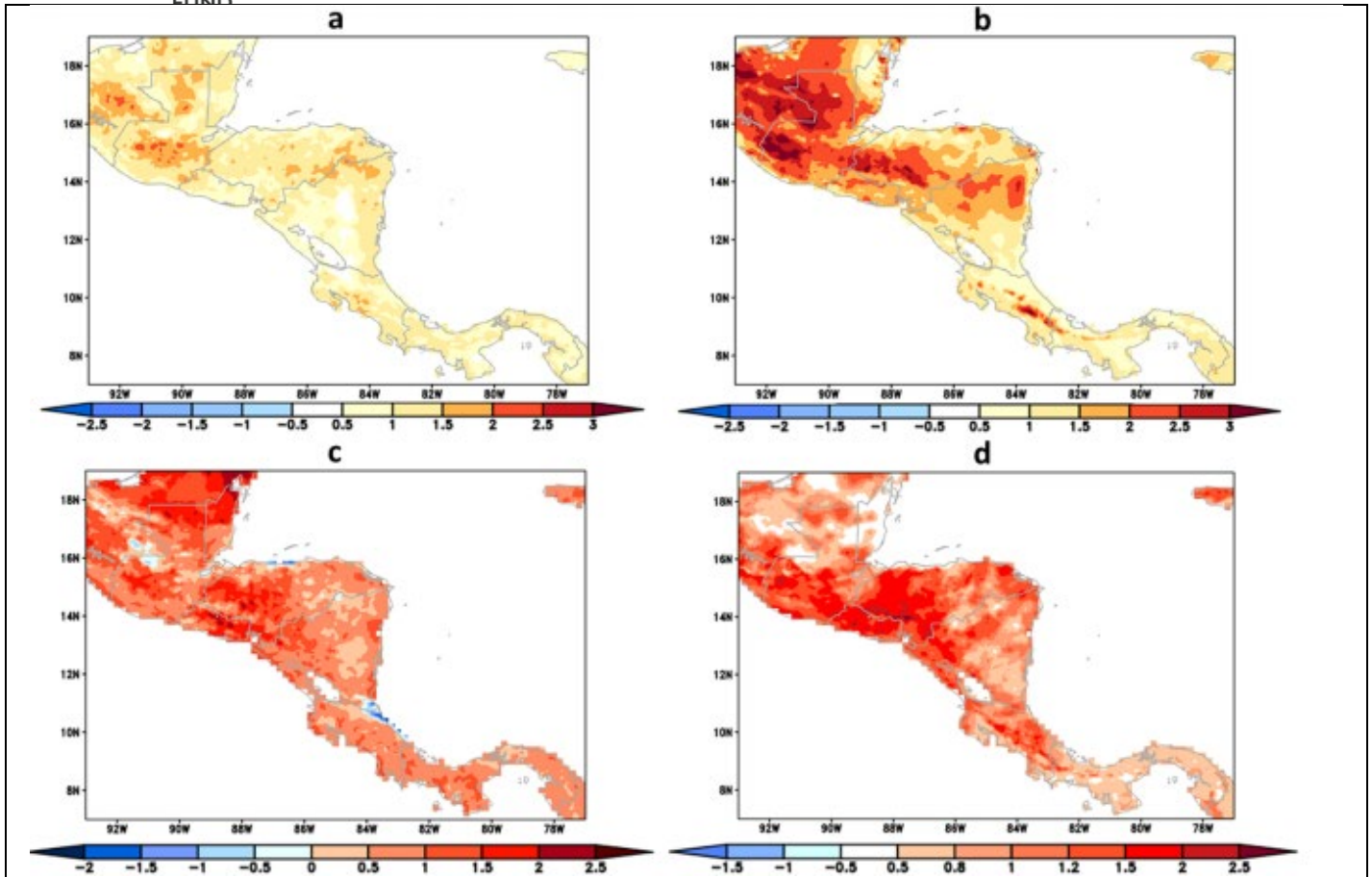


Figure 4. Projected trends of extreme temperature indicators for Central America over the period 2021–2050, namely: a) TNx – temperature of warm nights ($^{\circ}\text{C}$); b) TXx – temperature of warm days ($^{\circ}\text{C}$); c) TNn – temperature of cold nights⁴² ($^{\circ}\text{C}$); d) TXn – temperature of cold days⁴³ ($^{\circ}\text{C}$).⁴⁴

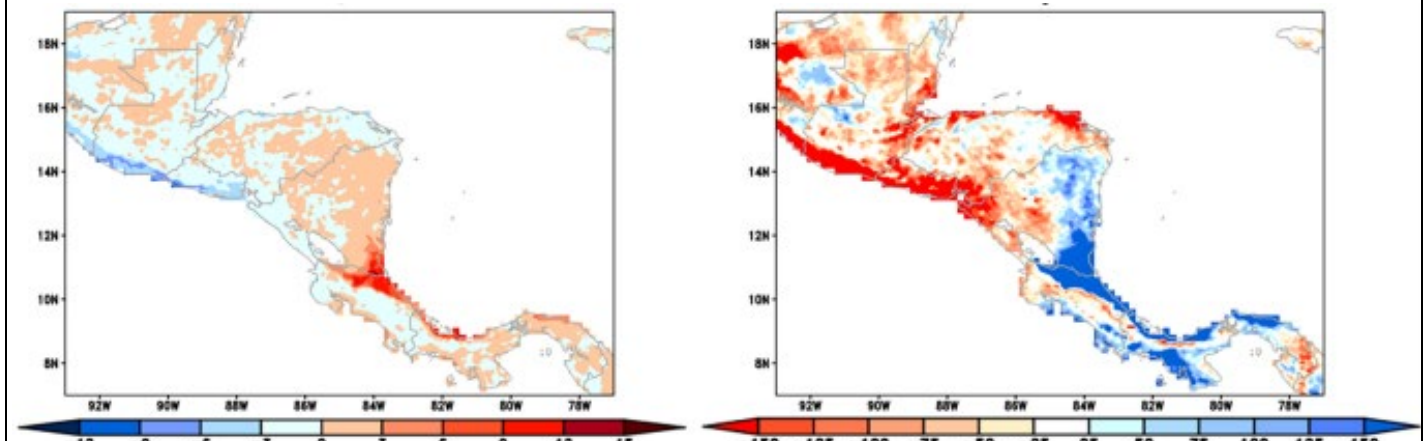


Figure 5. Projected trends of extreme rainfall indicators for Central America over the period 2021–2050, namely: i) R50mm (days) – heavy rainfall events, where rainfall exceeds 50 mm/day (left); and ii) R90p (mm) – extreme (90th percentile of) rainfall (right)⁴⁵.

⁴² Monthly minimum value of daily minimum temperature

⁴³ Monthly minimum value of daily maximum temperature

⁴⁴ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

⁴⁵ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

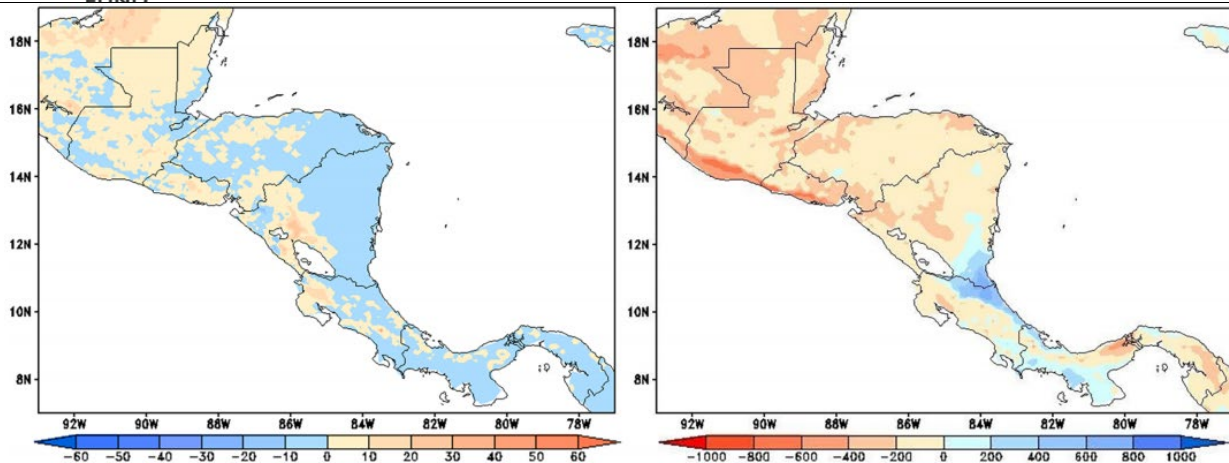


Figure 6. Projected trends of drought indicators for Central America over the period 2021–2050, namely: i) dry spells, defined as annual largest number of consecutive days when $ET \leq 0.5 \cdot ETP$ (left); and ii) annual P-ETP in (mm per year, right)⁴⁶.

Climate change impacts

14. Climate change is already having severe impacts on communities in the Dry Corridor and Arid Zones. Indeed, its impact on agriculture is commonly cited as a major driver of migration out of the region to North America, especially from Honduras, Guatemala and El Salvador⁴⁷. In 2014, for example, severe drought across the region resulted in an estimated US\$465 million worth of agricultural losses, including basic grains, coffee and livestock. In Honduras, Guatemala and El Salvador alone, a total ~2.5 million people experienced food insecurity as a result of approximately 75% loss of maize and bean crops⁴⁸. In 2016, 50–90% of crop loss resulted in 3.5 million people requiring humanitarian assistance across the region⁴⁹. This trend is continuing, with subsistence farmers in the Dry Corridor of Guatemala, El Salvador, Honduras and Nicaragua having lost more than half of their maize and bean crops in 2018, with 2.2 million people affected⁵⁰. These recent droughts have been unprecedented in their severity and the magnitude of their impacts. Such intense droughts are expected to increase in frequency under future climate conditions and, in the absence of adaptation interventions, people living in the Dry Corridor and Arid Zones will continue to experience adverse impacts of climate change⁵¹.
15. Future predicted impacts of climate change include decreased harvests of maize, beans, coffee and other important crops, leading to increased poverty, food insecurity and migration from rural areas⁵². Reductions in coffee yields have been predicted to be between 7% in 2020 and 39% in 2100 across all Central American countries under climate change conditions⁵³, while a similar study predicted reductions in maize yields of between 4% in 2020 and 18% in 2100⁵⁴. These reductions will result from direct climate impacts (e.g. droughts and floods) as well as indirect impacts on crops (e.g. declines in crop pollinators)⁵⁵. Climate change will also impact the land area suitable for growing specific crops, with declining areas expected for coffee and beans across the region⁵⁶. For example, El Salvador and Costa Rica are expected to experience decreases in land suitability for growing coffee of 40% or more⁵⁷. Similarly, modelling of maize indicates that land suitability is

⁴⁶ Imbach, P., Chou, S.C., Lyra, A., Rodrigues, D., Rodriguez, D., Latinovic, D. et al., 2018. Future climate change scenarios in Central America at high spatial resolution. *PLoS ONE* 13(4): e0193570. <https://doi.org/10.1371/journal.pone.0193570>.

⁴⁷ World Food Programme et al., 2017. Food security and emigration: Why people flee and the impact on family members left behind in El Salvador, Guatemala and Honduras. Available at: <https://www.wfp.org/content/2017-food-security-emigration-why-people-flee-salvador-guatemala-honduras>

⁴⁸ United Nations Office for the Coordination of Humanitarian Affairs (OCHA), Regional Office for Latin America and the Caribbean, 2014. Drought in Central America Situation Report No. 01 (as of December 10, 2014).

⁴⁹ Food and Agriculture Organisation of the United Nations (FAO), 2016. Dry Corridor, Central America, Situation Report — June 2016.

⁵⁰ FAO, 2018. Central America, Drought causes crop losses in “Dry Corridor” in Central America. GIEWS update 27 August 2018.

⁵¹ Further details regarding climate change impacts are presented in Section 7 of Annex 2: Feasibility Study.

⁵² Hannah et al., 2017. Regional modeling of climate change impacts on smallholder agriculture and ecosystems in Central America. *Climatic Change* 141: 29–45.

⁵³ CEPAL y CAC/SICA, 2014a. Impactos potenciales del cambio climático sobre el café en Centroamérica, LC/MEX/L.1169, México, DF.

⁵⁴ CEPAL y CAC/SICA, 2014b. Impactos potenciales del cambio climático sobre los granos básicos en Centroamérica. LC/MEX/L.1123, México, DF.

⁵⁵ Imbach et al., 2017. Coupling of pollination services and coffee suitability under climate change. *Proceedings of the National Academy of Sciences*.

⁵⁶ Details of expected impacts are presented in Tables 7.1 and 7.2; Section 7 of Annex 2: Feasibility Study.

⁵⁷ Hannah et al., 2017. Regional modeling of climate change impacts on smallholder agriculture and ecosystems in Central America. *Climatic Change* 141: 29–45.

likely to decrease by up to 34% in El Salvador, Guatemala, Honduras and Nicaragua⁵⁸. Although some cooler highland areas that are currently unsuitable for these crops may become more suitable as temperature increases, desire to shift crops into these areas will likely catalyse forest clearing in critical water recharge zones of river catchments thereby reducing aquifer recharge⁵⁹. This impact, coupled with an increase in extreme rainfall events, will increase the likelihood and severity of flooding. Climate change will likely also increase the frequency of wildfires, causing further forest degradation and threatening farming, forest-based livelihoods and fuelwood supply⁶⁰.

16. The climate change impacts described above are exacerbated by the baseline problems of forest loss and land degradation, which result in decreased water supply and food security⁶¹. Without comprehensive adaptation interventions, agricultural losses will continue to increase, along with ecosystem loss, water shortages, poverty and food insecurity. This will ultimately increase the exposure and sensitivity of communities in the Dry Corridor and Arid Zones who are already extremely vulnerable to climate change.
17. The programme will target seven catchments for interventions — one in each of the participating countries. In these catchments, the application of the Soil and Water Assessment Tool (SWAT) allows for the analysis of potential impacts of climate change in the programme's target areas. The availability of historical observed meteorological data and site-specific climate change projections are very limited. As observed in the following figure, there are only 24 meteorological stations within the limits of the Dry Corridor. Only for three of the countries, there were stations within the prioritized pilot sites and just one of them (Honduras) had complete observed rainfall and temperature data for a period of over 30 years with less than 15% of discontinuity.

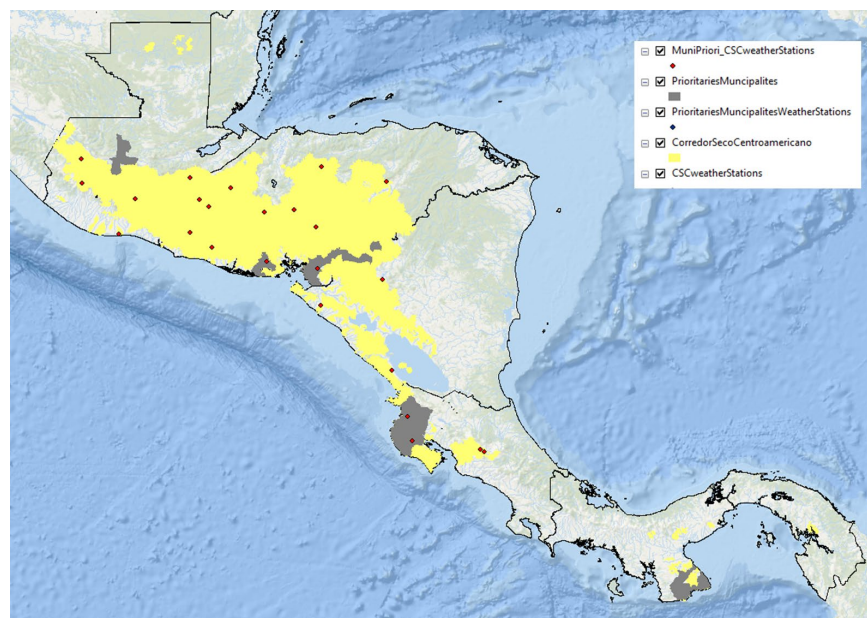


Figure 7a. Map of the meteorological stations in the Dry Corridor
Source: Authors, from <http://climate.onebuilding.org/default.html>

18. Input data: Data from Choluteca-baja are measurement stations that were verified locally and statistically. The regional data set for the rainfall and temperature that was used in the Dominican Republic and Costa Rica, was complemented and used as input was the CFSR obtained by the SWAT modelling online data system. The Global Weather Data (SWAT, <https://globalweather.tamu.edu>), is a product from the National Center for Environmental Prediction (NCEP, <https://climatedataguide.ucar.edu/climate-data/climate-forecast-system-reanalysis-cfsr>) Climate Forecast System Reanalysis (CFSR). These have been continuously run over the 36-

⁵⁸ Hannah et al., 2017. Regional modeling of climate change impacts on smallholder agriculture and ecosystems in Central America. *Climatic Change* 141: 29-45.

⁵⁹ Bouroncle et al., 2017. Mapping climate change adaptive capacity and vulnerability of smallholder agricultural livelihoods in Central America. *Climatic Change*, 141: 123-137.

⁶⁰ Imbach et al., 2012. Modeling potential equilibrium states of vegetation and terrestrial water cycle of Mesoamerica under climate change scenarios. *Journal of Hydrometeorology*, 13: 665-680.

⁶¹ Details are provided in Section 5 of Annex 2: Feasibility Study.

year period of 1979 through 2014, and continuously updated in real time. The CFSR was designed and executed as a global, high resolution, coupled atmosphere-ocean-land surface-sea ice system to provide the best estimate of the state of these coupled domains. These results have been used in several publications and evaluated with good validation results in different climates.

19. The results from the analysis undertaken in three basins in the Dry Corridor indicate that the foreseen impact will be severe in the programme areas, with a total predicted decrease in average rainfall of 11%, which is equivalent to a decrease of 655 mm/year under RCP 4.5 for 2050 and reaching up to 15% under RCP 8.5 for 2050. This decrease will also likely result in a significant delay in the timing of peak annual rainfall, with a shift from the month of May to the month of June, with important implications for agricultural activities. Additionally, climate change projections also predict substantial changes in the main hydrological variables in the programme areas, with a significant decrease between 4.1% and 5.7% in soil humidity and 18% to 25% in percolation for the Choluteca basin. These variables are critical for ecosystem maintenance and also for the sustainability of the agricultural activity and soil stability. Results from the hydrological model are included in the Feasibility Study in Annex 2 and Annex 31 of this proposal. Figure 7b below summarizes the changes in main hydrological variables under RCP 4.5 and RCP 8.5 for 2050 for the main hydrological variables in two selected basins in the Dry Corridor. These analysis is also further specified and discussed in the draft EbA site-specific plans developed for the programme sites in Honduras and Costa Rica (Annex 31).

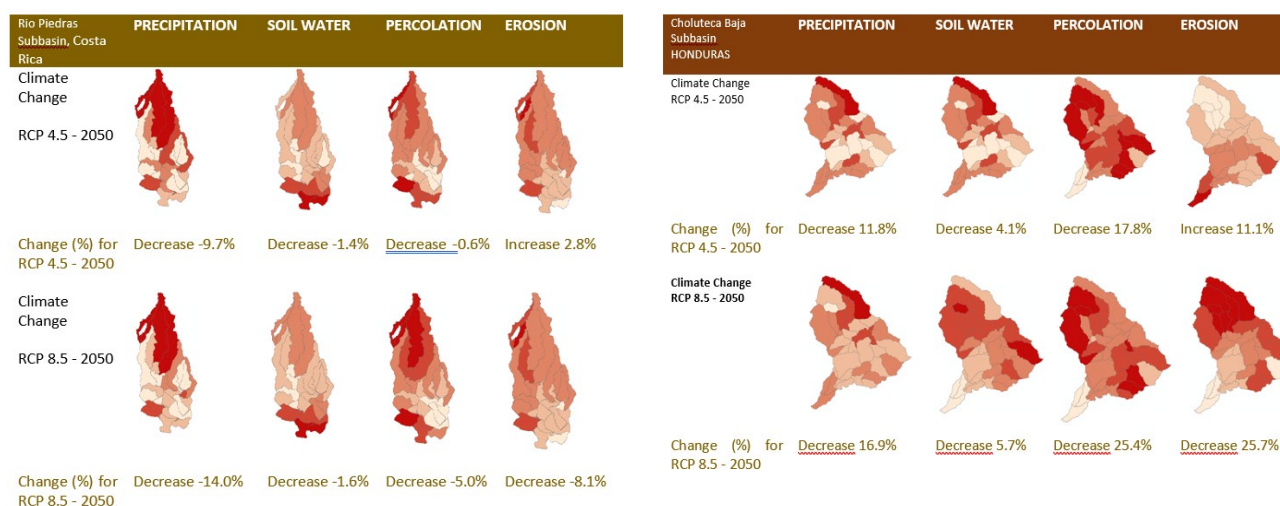


Figure 7b. Modelled changes in Soil humidity, percolation and erosion under RCP 4.5 and RCP 8.5 for 2050 for the Choluteca (right), Rio Piedras (left) basins through SWAT. Complete analysis included in Annex 2 Feasibility Study and Annex 31 Site specific plans

Countries	Time period	Emission scenario	Future climate change in top indicators				
			Temperature (annual mean)	Precipitation (annual mean)	Aridity actual (annual mean)	Soils moisture (annual mean)	Water runoff (annual mean)
Guatemala (average from 5 selected municipalities)	2041-2070	RCP 4.5	+2°C	-2%	29%	-12%	1%
		RCP 8.5	+3°C	-14%	2%	-26%	-19%
	2071-2100	RCP 4.5	+2°C	-7%	2%	-18%	-22%
		RCP 8.5	+4°C	-25%	-16%	-44%	-45%
		RCP 8.5	+5.67°C	-21%	30%	-27%	-11%
Nicaragua (average from 5 selected municipalities)	2041-2070	RCP 4.5	+2°C	-4%	17%	-10%	50%
		RCP 8.5	+3°C	4%	33%	-2%	14%
	2071-2100	RCP 4.5	+2.4°C	-6%	25%	-14%	9%
		RCP 8.5	+5°C	-14%	27%	-25%	3%
El Salvador	2041-2070	RCP 4.5	+1.2°C	-3%	8%	-8%	-6%
		RCP 8.5	+2°C	-8%	2%	-13%	-24%
		RCP 4.5	+2°C	-10%	2%	-15%	-16%

(average from 5 selected municipalities)	2071-2100	RCP 8.5	+4°C	-14%	4%	-27%	-42%
		RCP 8.5	+4.8°C	19%	18%	-4%	71%
Panama (average from 5 selected municipalities)	2041-2070	RCP 4.5	+2°C	26%	-19%	N/A	N/A
		RCP 8.5	+3°C	44%	-13%	N/A	N/A
	2071-2100	RCP 4.5	+2°C	38%	-3%	N/A	N/A
		RCP 8.5	+4°C	100%	-8%	N/A	N/A
The Dominican Republic (average from 5 selected municipalities)	2041-2070	RCP 4.5	+1.8°C	-10%	6%	-13%	-21%
		RCP 8.5	+2°C	-14%	12%	-22%	-25%
	2071-2100	RCP 4.5	+2°C	-15%	10%	-18%	-33%
		RCP 8.5	+4°C	-19%	26%	-29%	-41%

Figure 7c. Modelled changes in Temperature, Precipitation, Aridity, Soils moisture, and Water runoff under RCP 4.5 and RCP 8.5 for 2041-2070 and 2071-2100. Average calculated based on data from 5 and 6 selected municipalities in each of the 7 countries.

Source: Authors, from SMHI, Climate Information, <https://climateinformation.org/>, last accessed: June 15th 2021.

20. In addition to the 2 detailed hydrological modelings produced, Figure 7c provides a summary of the climate projections for the other 5 countries based on the Coupled Model Intercomparison project Phase 5 (CMIP5 from the Global Climate Models - GCM) and the Regional Climate Models from the Coordinated Regional Downscaling experiment (CORDEX)⁶². The projected changes in soil humidity, aridity and water runoff are consistent with the trends identified by the hydrological modelling: the annual precipitation is projected to decrease significantly, from a range of -14% to -21% for RCP 8.5 in 2071-2100 in four of the five countries. A similar trend is observed for soil moisture and water runoff's variation, that are generally projected to decrease throughout all four scenarios, while the annual mean erosion is generally increasing. Panamá's modelling does not seem to follow these general trends, but this is believed to be linked to the limited consistence among models (no projections are provided on water runoff and soil moisture) which could explain this inconsistency with the projections in the other countries.
21. Further results of the above hydrological modelling show a general positive impact of the interventions proposed for the programme. The modelling has allowed for the comparison between different alternatives of interventions, and demonstrates how increasing benefits are realised when the adaptation solutions are introduced in combination rather than in isolation.
22. The study shows that adaptation measures have the potential to generate hydrological benefits at the level of intervened use and at the basin scale; Increased water available for plants through improved soil moisture, contribute in water recharge and reduced water erosion, are some main benefits; positive changes depend on the type and magnitude of intervention. Implementation of adaptation measures holistically, meaning simultaneous implementation of a group of these, shows the best results in hydrological processes and avoided erosion. According to the results of this study, the best results occurred under the implementation of combined adaptation measures at the medium and high levels.
23. Figure 8 summarises the changes in the main hydrological variables after the implementation of EbA interventions in comparison to the change expected under climate change (in red). The introduced EbA options include: forest conservation (at the top), reforestation, agroforestry systems, irrigation systems, instalment of vegetative barriers and the combination of all these EbA options (at the bottom). As stated above, the aggregated analysis of all hydrological variables concludes that the combination of EbA options show the best results. This combination of interventions results in substantial improvements for all key hydrological variables in the catchment areas. Soil humidity increases from large losses under the climate change base scenario to positive values under EbA combinations; this results in benefits for water availability both for agriculture and forests that draw water from the top soil layers. The changes in water production (surface water) and percolation (water recharge) show even greater benefits from the implementation of this combination of EbA measures. These values are key for ensuring water availability throughout the year from a scenario of significant water

⁶² Read more on data production: <https://climateinformation.org/data-production-and-tailoring/production-of-climate-indicators/>

deficit under climate change conditions. Lastly, the implementation of this combination of EbA measures also shows very positive impacts on soil erosion rates.

SCENARIOS				Scenarios name	Watershed level				COMMENTS
Adaptation measures			Level		SW	PERC	WYLD	EROSION	
NO				Climate Change	-1,38	-10,61	-14,05	1,55	CC (% change)
Adaptation measure 1: Conservation and restoration forest	1a	ForestConservation	low	ForestConservationlow	1,35	10,66	14,02	-3,68	Change compared to Climate change scenario
		ForestConservation	middle	ForestConservationmiddle	1,23	11,42	13,93	-13,84	
		ForestConservation	high	ForestConservationhigh	1,08	12,23	13,80	-28,03	
	1b	ForesRestoration	low	ForesRestorationlow	1,33	10,68	14,01	-4,01	
		ForesRestoration	middle	ForesRestorationmiddle	2,63	22,73	30,26	-11,02	
		ForesRestoration	high	ForesRestorationhigh	1,09	11,02	13,78	-16,96	
Adaptation measure 2: Establishment of agroforestry systems	SAF_Increase		low	SAF_Increase_low	1,37	10,75	14,04	-2,99	
	SAF_Increase		middle	SAF_Increase_middle	1,32	10,69	14,00	-4,52	
	SAF_Increase		high	SAF_Increase_high	0,57	11,78	13,30	-44,37	
Adaptation measure 3: Implementation of efficient irrigation systems	Agriculture with irrigation		low	Agriculture with irrigation_low	1,37	10,62	14,05	-1,71	
	Agriculture with irrigation		middle	Agriculture with irrigation_middle	1,36	10,71	14,04	-2,51	
	Agriculture with irrigation		high	Agriculture with irrigation_high	1,34	10,88	14,02	-3,97	
Adaptation measure 4: Vegetative barriers	LivingBarriers		low	LivingBarriers_low	1,38	10,67	14,05	-2,21	
	LivingBarriers		middle	LivingBarriers_middle	1,38	10,70	14,05	-3,50	
	LivingBarriers		high	LivingBarriers_high	1,39	11,66	14,05	-20,23	
Combined scenarios	Mix	low	Mix_low	1,27	10,83	13,95	-7,54		
	Mix	middle	Mix_middle	1,02	11,27	13,72	-21,70		
	Mix	high	Mix_high	0,61	12,00	13,35	-43,72		

Figure 8. Changes in hydrological variables (water production, groundwater, percolation and erosion) analyzed at land use level and at watershed level under climate change scenario (RCP 4.5 2050) without EbA interventions and with EbA interventions for the Choluteca basin. For complete analysis of the three basins please refer to Annex 2 and Annex 31 of the proposal.

24. The above results demonstrate the benefits and positive impacts that the proposed programme interventions will have in the target areas, substantially reducing the expected negative impacts of climate change. . In addition to the results from hydrological modelling undertaken for the proposed programme, other studies have demonstrated links between the proposed interventions — EbA, water- and resource-efficient technologies — and improved climate resilience among vulnerable communities in the region.
25. Economic analysis of the benefits of land restoration in Latin American provides evidence to support that the adoption of policy measures and removal of barriers to land restoration efforts by national and local governments result in substantial benefits, including gains in agricultural production, alleviation of food insecurity and carbon sequestration⁶³. The economic benefits will contribute to the climate resilience of communities in combination with the direct benefits of EbA and water- and resource-efficient technologies on land and ecosystems. The use of soil conservation methods and EbA interventions, such as vegetative barriers, is recognised in the region as increasing climate adaptation by increasing soil water infiltration, retaining soil moisture and reducing soil erosion⁶⁴. Land restoration also decreases the demand for agricultural expansion by enabling improvements in production on degraded lands through agroforestry, silvopastoral systems⁶⁵, and other types of EbA and water- and resource-efficient technologies. The proposed landscape-scale adaptation interventions implemented in target catchments will restore degraded lands and prevent subsequent forest clearing in critical recharge zones and other ecosystems. This is expected to protect communities from risks of flooding and will decrease the need for migration from rural areas.
26. The proposed EbA interventions and water- and resource-efficient technologies will aim to reduce the vulnerability of targeted beneficiaries by increasing resilience to drought and extreme rainfall impacts on agricultural production through improved hydrological flow and restoration of catchments. During the initial stage of implementation, sound stakeholder consultations will be undertaken in each of the pilot sites to build on the information developed during the proposal design and the EbA interventions will be validated from a wide range of identified EbA interventions that are adjusted to the specific climate risks and socio-environmental

⁶³ World Resources Institute. 2018. The Economic Case for Landscape Restoration in Latin America. Available at: https://wriorg.s3.amazonaws.com/s3fs-public/The_Economic_Case_for_Landscape_Restoration_in_Latin_America.pdf?_ga=2.223405598.480839473.1559569688-232533270.1535385279

⁶⁴ Harvey et al., 2017. The use of Ecosystem-based adaptation practices by smallholder farmers in Central America. Agriculture, Ecosystems and Environment, 246:279-290. Available at: <https://www.sciencedirect.com/science/article/pii/S0167880917301810>

⁶⁵ World Resources Institute. 2018. The Economic Case for Landscape Restoration in Latin America. Available at: https://wriorg.s3.amazonaws.com/s3fs-public/The_Economic_Case_for_Landscape_Restoration_in_Latin_America.pdf?_ga=2.223405598.480839473.1559569688-232533270.1535385279

context of the catchment. Further specific criteria that will be applied for the selection and validation of these EbA interventions are described in paragraph 50. Improved ecosystem health through EbA interventions — such as reforestation, vegetative barriers and agroforestry systems — will create enhanced groundwater resources for use during droughts and decrease the risk of flooding and erosion during extreme rainfall events. Reducing demand for scarce water resources through water-efficient technologies in farms and households will further reduce the stress on ecosystems' hydrological cycles and will improve ecosystems' resilience to climate change impacts. Further to this, the economic benefits and increased agricultural productivity gained from land restoration in the region will help improve livelihood resilience and food security in communities.

Complementarity with related projects, programmes and initiatives

27. Considerable investments have been and are being made through public expenditure and donor-funded initiatives to increase the resilience of vulnerable communities to climate change throughout the Dry Corridor and Arid Zones. Lessons learned and best practices from these investments have been incorporated into the design of the proposed programme to replicate successful techniques and ensure that there is complementarity between the programme and existing initiatives. The most relevant investments that will be complemented by the proposed programme are presented below⁶⁶.
28. CABEL is implementing the GCF-funded 'Productive Investment Initiative for Adaptation to Climate Change' (CAMBio II) programme in the same seven countries the proposed programme will work in but in areas outside of the Dry Corridor and Arid Zones. Specifically, CAMBio II is working to: i) provide loans to producers through CABEL's accredited intermediary financial institutions (IFIs) for the implementation of adaptation measures; ii) build capacity of micro, small and medium-sized enterprises (MSMEs) and IFIs through grants to improve technical and business skills; and iii) establish an incentive scheme to promote adaptation measures implemented by MSMEs.
29. The proposed programme will be complementary to CAMBio II in several regards. First, the two operate in complementary geographic regions. While CAMBio II works in an expanded range across the region — focusing on areas where there is demand from MSMEs — the proposed programme will only work within the Dry Corridor. A detailed vulnerability assessment of the region at the municipal level has been undertaken, resulting in the selection of the 36 most vulnerable municipalities⁶⁷ in which this programme will be focusing. Second, the two will target different groups. CAMBio II is explicitly focused on MSMEs, while the proposed programme will work with individual producers and their associated groups, focusing specifically on the most vulnerable smallholder farmers. Third, the two will operate through different financial mechanisms. CAMBio II is working through financial intermediation with IFIs, channelling funds through credit.

In contrast, the proposed programme will operate through a trust structure for GCF funds, and CABEL, as direct trustee for GCF, will then provide senior limited recourse loans to the IFIs, combined with an on-lending structure via senior loans for CABEL co-financing, both to existing Intermediary Financial Institutions (IFIs) that will blend the funds on their balance sheet to channel to Partner Financial Institutions (PFIs) in the programme areas. PFIs in that context are smaller institutions such as small banks, cooperatives, NGOs, or rural community banks, that serve the last mile for the end beneficiaries. Fourth, the proposed programme will expand on the range of activities offered — focusing on water security as a whole, rather than only agriculture. Fifth, by providing a guarantee mechanism to be funded by GCF to exclusively cover non-GCF/non-CABEL co-financing and leveraged funds, the programme will enable the scale-up and mainstreaming of successful EbA finance by further leveraging additional private and possibly public funds for the on-lending structure, giving the GCF fund's trust structure the catalytic role it is intended for. This expanded focus will include on-the-ground ecosystem-based adaptation (EbA)⁶⁸ and restoration.
30. In addition to the above, the technical assistance provided by the two programmes will be complementary. CAMBio II is providing a variety of technical assistance to IFIs and MSMEs, where the beneficiaries receive initial training on the programme operations and thereafter can request assistance on specific matters. The proposed programme otherwise offers technical assistance — both practical and theoretical trainings — for local government, farmers and rural communities, as well as IFI's and smaller financial institutions, that focus on: i) the technical areas in climate change; ii) accessing and channeling funds through established financial

⁶⁶ A comprehensive list of past and ongoing projects, programmes and initiatives, as well as details of their focus and alignment with the proposed programme, is presented in Section 9 of Annex 2: Feasibility Study.

⁶⁷ A list of these municipalities is provided in the first paragraph of Section B.3 Project/Programme Description.

⁶⁸ Ecosystem-based adaptation refers to "the use of biodiversity and ecosystem services to help people adapt to the adverse effects of climate change" (Convention on Biological Diversity, 2009)

mechanisms; and iii) implementing EbA and other adaptation measures. Furthermore, the proposed programme will expand on the outputs of CAMBio II, drawing on experiences during implementation to inform higher-level outcomes. For example, CAMBio II offers incentives (Adapt-Award) for IFIs and MSMEs which use loans to implement adaptation measures. However, these incentives will not scale well beyond programme implementation. The proposed programme will improve on this approach by promoting the mainstreaming of adaptation incentives into local and, ultimately, national policy through capacity building activities for decision-makers and mainstreaming policy initiatives. Overall complementarity and sharing of best practices will be ensured as both will be executed by CABEL, with lessons learned being directly exchanged between management units within CABEL. However, it is important to highlight that, although complementary, these initiatives have a fundamental difference in their overall goals. In the case of the Dry Corridor, the approach is to ensure water security for communities at the landscape/basin level, while for CAMBio II, the goal is to remove funding barriers for MSMEs.

31. In addition to CAMBio II, the proposed programme will align with the Microfinance for EbA (MEbA) project that is being implemented in eight countries across Latin America and Africa, including three of the targeted Central American countries, namely Costa Rica, Dominican Republic and El Salvador. MEbA is promoting private sector engagement in adaptation finance by facilitating microfinance products aimed at small-scale farmers to invest in EbA options to improve their resilience to climate change. The proposed programme will draw on lessons learned and best practices established through MEbA to guide the establishment of financial mechanisms for the Dry Corridor and Arid Zones.
32. Several national projects are being implemented in Central America with a focus on water-efficient technologies, EbA and land restoration to improve ecosystem services and livelihoods, increase food security, and build climate resilience of vulnerable communities. Examples of these include 'Upscaling climate resilience measures in the Dry Corridor agro-ecosystems of El Salvador (RECLIMA)' and 'Building livelihood resilience to climate change in the upper basins of Guatemala's highlands', both funded by GCF. These projects are implementing several adaptation measures that will be replicated and scaled up regionally by the proposed programme. These include: i) riparian forest restoration; ii) establishing agroforestry and silvopastoral systems; iii) rainwater collection; and iv) water-efficient agricultural technologies such as drip irrigation. The proposed programme will use best practices from these projects to scale up interventions across the region. In addition, increased access to financial products and services under Component 2 of the proposed programme will ensure that a larger number of individuals and enterprises are able to implement these kinds of interventions. The proposed programme will also contribute directly to restoration commitments made by several countries under Initiative 20x20. This initiative is a country-led effort⁶⁹ whose primary objective is to restore 20 million hectares of degraded land across Central America and the Caribbean by 2020. The proposed programme will contribute directly to these goals by restoring ~19,200 hectares of forested area across the seven catchments targeted by the programme.
33. Finally, a project funded by the German Federal Ministry for the Environment, Nature Conservation and Nuclear Safety (BMU) has been implemented from 2012–2019 by the International Union for Conservation of Nature (IUCN) in several countries in Central America, including Costa Rica, El Salvador, Guatemala, Honduras, Mexico and Panama. This project — 'Transforming evidence into change: a holistic approach to governance for EbA' (Go4EbA) — focused on improving existing governance frameworks for implementing EbA through developing an evidence base for its implementation. One catchment per country was selected to demonstrate EbA measures and ~400 people received capacity-building. Go4EbA also established a database including information about over 100 EbA-related cases in the region, and a virtual platform known as 'EbA Solutions'⁷⁰ to share knowledge and provide technical assistance. The platform includes information on: i) adaptation projects by type of activity; ii) availability and management of water resources; and iii) ecosystem protection and recovery. The proposed programme will draw lessons learned from Go4EbA with regards to best practices when implementing a bottom-up approach focused on promoting water and food security in the region. It will also upscale measures implemented under Go4EbA in several additional catchments within the Dry Corridor and Arid Zones.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

⁶⁹ In the Dry Corridor, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua and Panama have committed to restoring 6.4 million hectares in total, while the Dominican Republic has committed to restoring 0.09 million hectares. El Salvador, Honduras, Nicaragua and Panama: 1 million hectares per country; Guatemala: 1.2 million hectares; Costa Rica: 2.7 million hectares. Available at: <http://www.wri.org/our-work/project/initiative-20x20>.

⁷⁰ www.solucionesabe.org

Description of the problem

34. Climate change is causing increasingly frequent and intense droughts, as well as a higher occurrence of extreme rainfall events, in the Dry Corridor region of Central America and the Arid Zones of the Dominican Republic. These areas are densely populated and have high levels of poverty — concentrated in rural areas — when compared with the rest of the region. In addition, most rural communities in the Dry Corridor and Arid Zones are dependent on agriculture and plantation forestry for their livelihoods, both of which are extremely sensitive to climate impacts — specifically those impacts that result in changes in water availability. These communities also have limited knowledge of climate change adaptation measures, coupled with an inability to access funds to finance adaptation. As a result, rural communities in this region are extremely vulnerable to the impacts of climate change. This vulnerability is exacerbated by widespread environmental degradation and an associated loss of ecosystem services such as water production. Future climate projections indicate that the Dry Corridor and Arid Zones will continue to become warmer and drier, with increasingly variable rainfall and an associated increased risk of severe drought⁷¹. Past and ongoing adaptation efforts in the region have not been implemented at a sufficient scale to prevent rural communities experiencing severe negative impacts of climate change, which will likely only increase in the future.

Preferred adaptation solution

35. The preferred adaptation solution to the above-described problem is to improve the climate resilience of rural communities in the Dry Corridor and Arid Zones by: i) implementing integrated catchment management and restoring catchments; ii) improving hydrological flow and infiltration of rainwater into groundwater reserves through forest and ecosystem restoration; and iii) reducing demand for scarce water resources by implementing water-efficient technologies at farm- and household-level. The selection/validation of the specific EbA interventions will be based on scientific climate risk assessments and sound stakeholder consultations as described in paragraphs 51 and 99. Specifically, under the preferred adaptation solution, commercial farmers and entrepreneurs in rural communities would have access to the financial resources and technical skills required for the widespread implementation of adaptation measures — including EbA, and water- and resource-efficient techniques. This access to financial and technical resources would strengthen the adaptive capacity of rural communities to withstand increasingly frequent and intense droughts, which result in water shortages, and subsequent reductions in economic activity, productivity and food security. In addition, the mainstreaming of adaptation considerations into policy and planning by government, and the generation of knowledge and awareness of adaptation options should take place across the region. Mainstreaming in this way would enable the uptake of large-scale EbA and other adaptation measures. The overall approach under the preferred adaptation solution would ensure access to adaptation measures for all impacted communities in the Dry Corridor and Arid Zones, building their climate resilience in the long-term.

Barriers to the preferred adaptation solution

36. The proposed programme will address the following eight barriers to the preferred adaptation solution in the Dry Corridor and Arid Zones. These barriers are described in further detail in Section 11 of Annex 2: Feasibility Study.

- **Barrier 1: Limited knowledge and understanding of climate change impacts among decision-makers.** Decision-makers in Central America and the Dominican Republic often have knowledge gaps with regards to climate change impacts, and how these impacts affect farmers. These gaps include knowledge on: i) water availability; ii) areas susceptible to extreme weather events; iii) locations of smallholder farmers; iv) projected changes in temperature and rainfall; v) expected impacts on crop yields; and vi) adaptation strategies. These knowledge gaps impede the mainstreaming of climate change adaptation into the policies and plans of the different sectors. In this regard and in particular for the selected pilot sites, the availability of historical observed meteorological data is very limited as well as the availability of site-specific climate change projections. There are only 24 meteorological stations within the limits of the Dry Corridor. Only for three of the countries, there were stations within the prioritized pilot sites and just one of them (Honduras) had complete observed rainfall and temperature data for a period of over 30 years with less than 15% of discontinuity.
- **Barrier 2: Limited knowledge and understanding of EbA and other adaptation measures among decision-makers and communities.** Decision-makers and communities in the Dry Corridor and Arid Zones frequently have limited knowledge and understanding of EbA, and of adaptation measures such as water- and resource-efficient technologies. They have limited access to information and limited understanding of the value of ecosystem services

⁷¹ Details are provided in Section B.1 and Section 6 of Annex 2: Feasibility Study.

and the costs and benefits of ecosystem restoration, in particular because of the large scales at which many ecosystem services are generated. As a result, they have limited capacity to implement these interventions and reap their benefits.

- *Barrier 3: Insufficient implementation of existing policies on climate change adaptation and limited integration of EbA and other adaptation measures into sectoral policies.* Climate change adaptation is a priority in the NDCs of all seven participating countries. National policies also indirectly support EbA; for example, conservation policies, and the need to restore, conserve and sustainably manage the Dry Corridor and Arid Zone forests, wetlands and agroforestry systems is widely recognised. However, institutions at both national and local level have insufficient knowledge and technical capacity to implement existing policies, preventing policies from achieving the necessary impacts.
- *Barrier 4: Limited technical capacity within governments and local communities to implement EbA and other adaptation measures.* Governments have gradually reduced public budgets for the provision of extension services that could promote EbA. As a result, communities receive limited support, training and technical assistance to implement climate-resilient practices at landscape-, community- and household-level.
- *Barrier 5: Limited knowledge and technical capacity to adopt technologies and approaches for efficient water use.* Technologies that support efficient water use have been implemented successfully in the region, but these cases have not been well documented. As a result, there is limited information on water management and improved technologies, both for policy-makers at the local-level, and for water users. In the agricultural sector, specifically, successful water-efficient technologies are known for large commercial farming (e.g. drip irrigation of sugarcane and oil palms), but knowledge of water-efficient technologies is less widespread among smallholder farmers.
- *Barrier 6: Limited access for financial institutions to specific refinancing options for the implementation of EbA, water- and resource-efficient technologies in the private sector.* Private sector investors, credit agencies and financial institutions have limited evidence of the benefits of EbA and other adaptation measures and limited access to specific EbA refinancing options. In addition, valuations by these institutions do not take into account important long-term benefits of EbA. As a result, these institutions have limited opportunity to assess the commercial viability of EbA approaches and natural resource-based businesses, and to develop related financial services and products.
- *Barrier 7: Limited access to credit for adaptation interventions among vulnerable populations.* Smallholder farmers and rural households require credit to cover the initial costs of implementing EbA and water- and resource-efficient technologies. However, the high risk of such credit to financing institutions limits the availability of credit to vulnerable communities. This is because banks often lack guarantees from central banks/multi-national development banks to reduce their risk in providing credit to individuals and small businesses with little collateral and small farm sizes.
- *Barrier 8: Absence or limited development of economic incentives for investment in sustainable natural resource management and EbA.* Although some water funds and schemes for payment for environmental services (PES) exist in the participating countries, the majority of these funds and schemes operate at limited and local scales. Barriers to the implementation of such economic incentives include lack of enabling strategic, legal and institutional frameworks and lack of transparent monitoring arrangements.

Programme objective

37. The proposed programme's interventions have been designed to overcome the above-mentioned barriers and contribute to the programme objective of strengthening the adaptive capacity of vulnerable people, including smallholder and commercial farmers, as well as entrepreneurs in rural communities in the Dry Corridor and Arid Zones. They will also contribute to the conservation and restoration of degraded, climate-vulnerable landscapes through improved agricultural and agroforestry practices, as well as reforestation and improved forest management practices. This will be achieved through the facilitation of a shift in the approach to catchment and water demand management, seeing governments promoting large-scale implementation of EbA and other adaptation measures, while simultaneously unlocking finance for the adoption of such sustainable practices that will build resilience against extreme climate events such as severe drought and extreme rainfall. This shift will take place through the implementation of four interconnected Outputs, described in detail in Section B.3. A description of how the programme's activities will overcome the identified barriers is provided below.
38. **Barriers 1 and 2** will be overcome predominantly through activities under Output 3, including through the dissemination of knowledge products on climate change impacts, EbA and other adaptation measures through

a regional knowledge hub linked to the existing Environmental Observatory based at the Central American Commission for Environment and Development (CCAD). In addition, policy- and decision-makers will be trained on relevant aspects of climate change — including the disproportionate impact of climate change on women and other vulnerable groups — and the role of EbA and other adaptation measures in mitigating these impacts. To ensure the sustainability of programme interventions in overcoming these barriers, local community organisations — including environmental and women's groups — will be trained to continue the development and dissemination of climate information and knowledge products. Policy- and decision-makers will also be trained on how to integrate climate change adaptation into local, national and regional policies and strategies; and their experiences in doing so will be disseminated throughout the Dry Corridor and Arid Zones to promote this process across different sectors throughout the region. Component 3 will also contribute to overcoming barrier 1 through the collaboration with regional research institutions as one of the main areas of research will be focused on strengthening the availability of downscaled climate projections for the Dry Corridor and the analysis of impacts over hydrological variables. This training will overcome **Barrier 3**. The above activities will combine to result in the dissemination of information on climate change adaptation and its financing across the Dry Corridor and Arid Zones, as well as the mainstreaming of this information into local and national policies. This result will contribute to the outcome of strengthened awareness of climate threats and risk-reduction processes in the seven target countries.

39. The technical capacity of local government, farmers and rural communities will be strengthened through activities under Output 1, enabling them to implement EbA and other adaptation measures, and overcoming **Barriers 4 and 5**. To achieve this, country-specific protocols will be developed for the: i) implementation of sustainable EbA practices; ii) adoption of water-efficient technologies by households; and iii) adoption of resource-efficient technologies that reduce fuelwood demand. These protocols will be developed by the local experts in the Project Execution Units with support from the regional experts and the monitoring and evaluation committees under the supervision of the Executing Entity (FAO) prior to the implementation of on-the-ground interventions in each country, and will be validated through stakeholder engagements in each municipality. Training in the use of these protocols will be provided to local government representatives using a training-of-trainers approach. In addition, local communities will be trained on EbA and other adaptation interventions, including visits to demonstration sites to facilitate farmer-to-farmer knowledge transfer. Through activities under Output 2, on-the-ground EbA interventions and water- and resource-efficient technologies will be introduced across seven catchments, the implementation of which will result in lessons learned and best practices to contribute to the knowledge base regarding the adoption of these interventions. The evidence base generated through the implementation and monitoring of these interventions will be shared with local decision-makers, the private sector (including commercial farmers) and local communities through the regional knowledge hub linked to the CCAD's Environmental Observatory. This evidence base will facilitate the uptake by communities of financial products and services to finance EbA and other adaptation measures as described further below. In addition, the activities that have been designed to overcome Barriers 4 and 5 will result in: i) strengthened technical capacity of beneficiaries to implement EbA and other adaptation measures; and ii) the implementation of large-scale adaptation interventions in rural communities across seven target catchments. These results will both contribute to the outcome of strengthened adaptive capacity and reduced exposure to climate risks among vulnerable communities in the Dry Corridor and Arid Zones.
40. **Barriers 6 and 7** will be overcome mainly through activities under Output 4. These include the establishment of financial products and services to finance EbA investments, including via a blended EbA finance structure with two main facilities, namely a; i) a blended EbA lending facility for small- and large-scale EbA investments at farm-, enterprise- and household-level via two different funding streams provided by CABI to the IFIs; and ii) a guarantee facility to provide additional security for participating intermediary financial institutions for on-lending to partner financial institutions to engage in the riskier target beneficiary segment. The technical capacity of financial institutions will be increased to enable the operationalisation and delivery of these new financial products and services, whilst awareness will be raised among the private sector and vulnerable communities on how to access these mechanisms for the implementation of adaptation measures. These activities will result in financial products and services to finance EbA investments being offered by last mile partner financial institutions to the communities that need them the most. To increase economic incentives for investing in EbA and other sustainable land management practices, an evidence base will be constructed based on the introduction of adequate and where possible digital monitoring, reporting and verification systems for all activities financed. This will help to demonstrate the value and effectiveness of proposed adaptation activities to policy-makers and financial institutions so that these considerations will be integrated into development plans. A methodology will also be developed to value ecosystem services. It will be used to guide local governments in making policy changes and implementing economic incentives — such as water funds — for

sustainable land management practices. This will overcome **Barrier 8**. The activities described above will combine to contribute towards the outcome of strengthened institutional and regulatory systems for climate-responsive planning and development in the seven target countries.

41. Together, the above-described interventions will overcome the barriers to improving climate resilience of rural communities in the Dry Corridor and Arid Zones. They will contribute to the achievement of three outcomes (as detailed in the Theory of Change diagram below) and, ultimately, to the goal of increased climate resilience of vulnerable communities across the region. The sustainability of these interventions will be ensured through a detailed exit strategy centred around the principles of: i) long-term community ownership; ii) mainstreaming initiatives into catchment management and planning frameworks; iii) incentivising private sector investments into EbA; and iv) sustainable financial instruments.

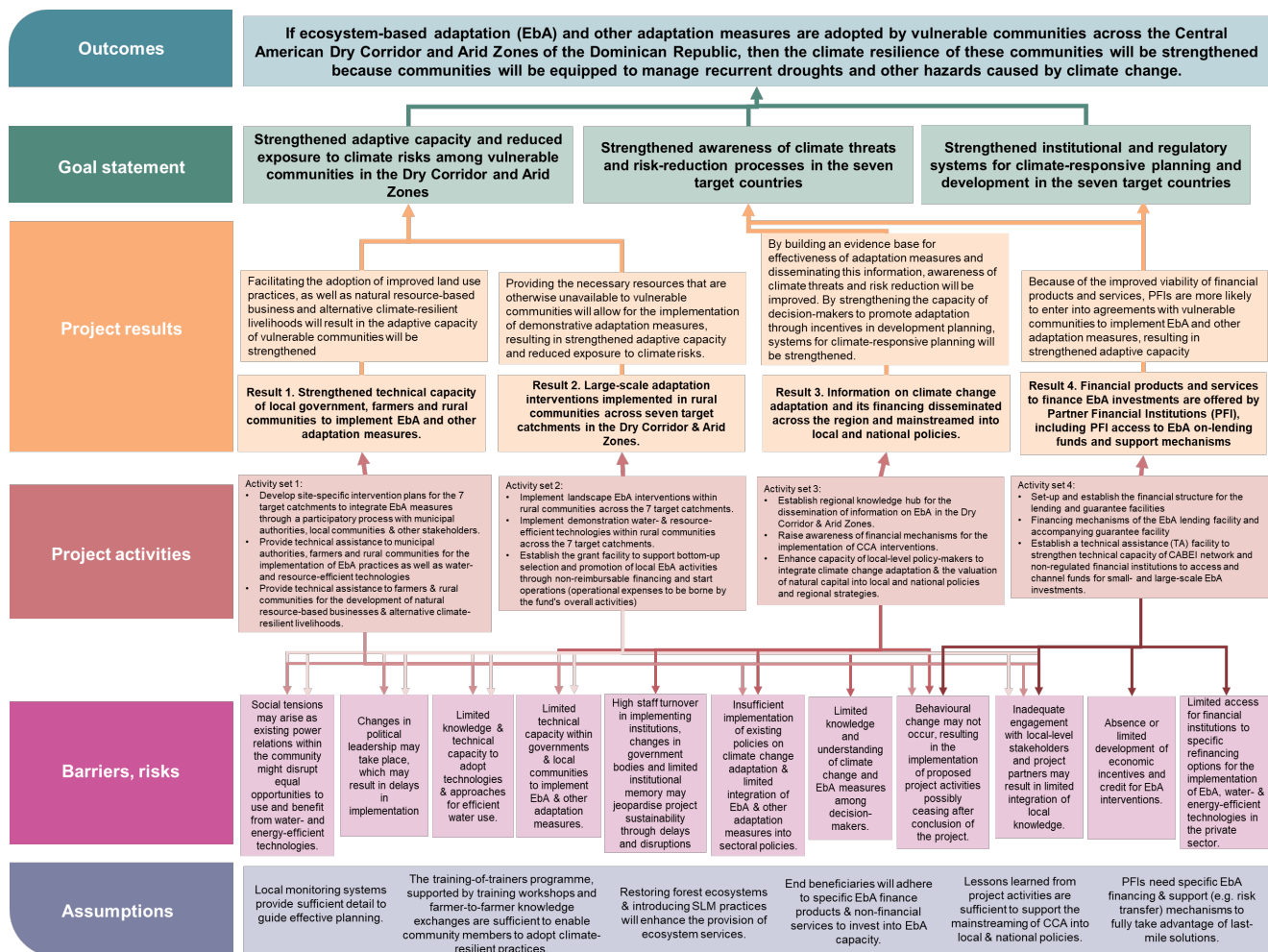


Figure 9. Theory of Change diagram.

B.3. Project/programme description

42. Programme interventions will be implemented in all six countries of the Dry Corridor — Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua and Panama — as well as the Arid Zones of the Dominican Republic. A single catchment has been selected for each country based on a detailed climate change vulnerability assessment, with five to six municipalities targeted within each catchment (see Section 12 of Annex 2: Feasibility Study). The target municipalities for each country are:

- Guatemala — Canillá, Chicamán, San Andrés Sajcabaja, Uspantán and Zacualpa in the Río Salinas watershed;
- Honduras — Apacilagua, Choluteca, Duyure, Marcovia, Morolica, and Orocuina in the Choluteca watershed;

- El Salvador — Concepción Batres, El Carmen, El Tránsito, Jucuarán and San Miguel in the San Miguel watershed;
- Nicaragua — El Jícaro, Palacagüina, Somoto, Telpaneca and Yalagüina in the Rio Coco watershed;
- Costa Rica — Bagaces, Carrillo, Liberia, Nicoya and Santa Cruz in the Tempisque watershed;
- Panama — Guarar, Macaracas, Pocrí, Pedasí and Tonosí in the Los Santos watershed; and
- Dominican Republic — El Pino, Los Almácigos, Monción, Partido and San Ignacio de Sabaneta in the Guayubín and Mao watersheds.

The programme's regional approach will create an enabling environment for scalable and sustainable catchment and water demand management, facilitating large-scale investment in, and implementation of, EbA and water-efficient technologies. The total grant request of US\$84.3 million will be used in the following ways: i) US\$10.1 million will be used to strengthen the technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures through the activities comprised in output 1; ii) US\$ 28.5 million will be used to implement large-scale adaptation interventions — including EbA, water- and resource-efficient technologies — in rural communities across the region throughout output 2; iii) US\$28 million will form the grant facility through which vulnerable communities can access on-grants to foster the creation of a business ecosystem around EbA available technologies as per activity 1.2.; iv) US\$6.3 million will be used to generate and share information on climate change adaptation and finance across the region, and mainstream it into local and national policies within the output 3; v) US\$6.5 million will be used to set up the financial mechanisms and strengthen the technical capacity of CABEL accredited and non-regulated financial institutions to access and channel funds through them throughout output 4; vi) US\$4.2 million will be used for overall management and monitoring of programme activities (Independent evaluations and PMC costs); and vii) US\$0.7 million is requested for contingency.

Component 1. Mainstreaming of EbA, water-efficient technologies and natural resource-based businesses into selected catchments

43. Under Component 1, three interrelated outputs will facilitate the widespread implementation of adaptation measures across the region. This will include: i) providing technical assistance to local government, farmers and rural communities to implement adaptation measures; ii) mainstreaming adaptation measures and incentives into policy and planning as well as disseminating information on climate change and adaptation across the Dry Corridor and Arid Zones; and iii) implementing on-the-ground EbA and water/resource-efficient technologies. These outputs will complement Output 4 by providing the technical assistance needed by local government, farmers and rural communities to implement EbA activities and water/resource-efficient technologies financed by the mechanisms established under Output 4. In addition, increased knowledge disseminated under Component 1 regarding EbA and water-efficient technologies and newly-available financial mechanisms will facilitate access for those previously unable to fund these adaptation measures, as well as creating an enabling environment for the establishment of economic incentives for investment in EbA.

Output 1. Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures.

44. Output 1 will be fully executed by FAO who will be responsible for the establishment of the National Project Execution Units in each country as well as hosting the Regional Project Management Unit in their Regional Office in Panama. CABEL will sign a Subsidiary Agreement with FAO for the execution of this output. This output will be executed directly through the national structure of FAO as well as through agreements with national partners when necessary for the undertaking of capacity building activities, trainings and community engagement as well as for the promotion of opportunities through the financial mechanism of the programme and the monitoring of the impact of the investments.
45. Activities under Output 1 will contribute to GCF Outcome A7.0 — Strengthened adaptive capacity and reduced exposure to climate risks — by facilitating the adoption of improved land use practices, as well as natural resource-based businesses (NRBs) and alternative climate-resilient livelihoods. A community-based approach will be adopted to identify site-specific interventions in each of the 36 municipalities targeted by the programme during the first year of the programme as described in paragraphs 51 and 99, which will be implemented under Output 2. FAO as the Executing Entity for these activities will ensure that the site-specific intervention plans meet all eligibility criteria and ultimately retain the final decision to approve implementation of all interventions under this output. Technical assistance will be provided to local governments to adopt and implement these intervention plans during the first year of the programme and with follow up activities along the programme.

46. Technical assistance will also be provided to farmers and rural communities in 36 municipalities, on the implementation of on-the-ground EbA practices, including silvopasture, agroforestry and sustainable land management⁷². The technical assistance under this component will also support the integration of EbA in policy and regulatory frameworks at the local level focusing mainly on water management regulations and water management community structures as well as local land use planning instruments. Target municipalities were selected following a rigorous vulnerability assessment and extensive consultations with key stakeholders in each country. Using IPCC guidelines, vulnerability was determined as a function of: i) the profile, magnitude and rate of climate changes to which a system is exposed; ii) the sensitivity of the system to those changes; and iii) the adaptive capacity of that system. Based on the result of the vulnerability assessment, a single watershed was identified in the most vulnerable area for each country (except for Dominican Republic, where two adjacent watersheds were selected), and then 5-6 municipalities were selected within the target watersheds. Details of the vulnerability assessment and site selection process are provided in Section 12 of Annex 2: Feasibility Study.
47. Country-specific implementation protocols will be developed for: i) the implementation of sustainable EbA practices; ii) the adoption of water-efficient technologies; and iii) the adoption of resource-efficient technologies that reduce fuelwood demand. These protocols will be developed by the local experts in the Project Execution Units with support from the regional experts and the monitoring and evaluation committees under the supervision of the Executing Entity (FAO) prior to the implementation of on-the-ground interventions in each country, and will be validated through stakeholder engagements in each municipality. To operationalise these protocols, a training-of-trainers programme will be introduced for representatives of local departments of environment/water, community-based organisations, water committees, local cooperatives, environmental and women's organisations within each country. Particular attention will be paid to the inclusion of women in this training, especially considering their active role in water and fuelwood use. Site visits will be arranged to demonstration sites that will be established in each of the target municipalities (Output 2), as well as showcasing mature plots where EbA interventions are already providing demonstrable benefits regarding water security. Commercial farmers, smallholder subsistence farmers, local communities and individual households will also be provided with technical assistance to adopt water-efficient technologies, including drip irrigation and rainwater harvesting, as well as for resource-efficient practices, including fuelwood drying and efficient charcoal production which will reduce the pressure of fuelwood collection on remnant forests.
48. In addition to the technical assistance provided for on-the-ground implementation of EbA practices and water/resource-efficient technologies, training will be provided to communities on the development of small, locally appropriate livelihoods. This will include training for natural resource-based businesses (NRBs) that: i) will support the implementation and maintenance of EbA interventions — such as tree nurseries and the manufacturing, distribution or repair of water-efficient technologies; and ii) rely on non-destructive use of ecosystem goods and services provided through EbA — such as businesses developed around the ecotourism industry, or non-timber forest products (NTFPs). The establishment of NRBs not only contributes to the upscaling of EbA interventions, but also enhances the resilience of vulnerable communities by providing climate-resilient income-generating opportunities. The promotion of NRBs — in conjunction with increased access to financing under Output 4 and increased knowledge and understanding of adaptation options under Output 3 — will strengthen the adaptive capacity of smallholder farming communities under future climate conditions. Programme activities that will be implemented under Output 1 are listed below.
- Activity 1.1. Develop site-specific intervention plans for the seven target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities and other stakeholders.
 - Activity 1.2. Provide technical assistance to municipal authorities, farmers and rural communities for the implementation of EbA practices as well as water- and resource-efficient technologies.
 - Activity 1.3. Provide technical assistance to farmers and rural communities for the development of natural resource-based businesses and alternative climate-resilient livelihoods.

Output 2. Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.

⁷² Technical assistance will not be limited to farmers accessing the financial mechanisms from Output 1, but to any farmers in the target municipalities that are open to adopting improved practices but lack the technical knowledge to do so.

49. Activities under Output 2 will contribute to GCF Outcome A7.0 — Strengthened adaptive capacity and reduced exposure to climate risks — by providing the necessary resources for the on-the-ground implementation of EbA interventions as well as water- and resource-efficient technologies in rural communities across the seven target catchments. Support will be targeted at the most vulnerable communities in 36 municipalities⁷³, that lack the capacity to access financing and equipment to implement adaptation interventions without external assistance. The hydrological modeling undertaken for the programme areas (described in Section B.1 of the proposal and in further detail in Section 13 of Annex 2: Feasibility Study) has demonstrated that a combination of adaptation interventions, such as forest conservation, reforestation, transition to agroforestry systems, introduction of irrigation systems and vegetative barriers results in substantial improvements for all key hydrological variables in the catchment areas. As a result, a combination of adaptation measures will be implemented in each of the target catchments to maximise the benefits conveyed and the reduction of climate change impacts, specifically on hydrological parameters. FAO will be the EE for the implementation of these measures under activities 2.1 and 2.2. The grant-supported interventions all also serve as demonstration sites that will provide practical examples to surrounding communities on how to implement sustainable practices. CABI will be the EE for the implementation of the grant-facility under activity 2.3.
50. The programme will work with local organisations in each municipality to implement on-the-ground EbA interventions that provide multiple ecosystem goods and services to vulnerable communities. The relevant organisations will be identified during the development of site-specific intervention plans (Activity 1.1). Relevant interventions have been identified for each municipality based on community consultations, sectoral expert opinion obtained for the Feasibility Study and the results of the hydrological models for three of the pilot catchments. The selected interventions remain as indicative and are subject to validation during the inception phase and in consultation with the specific communities. These indicative interventions, pre-identified with the available and developed information during the formulation process include: i) establishing one tree nursery per municipality focusing on native species and fuelwood trees (36 total); ii) establishing 4,160 ha of forest protection zones across the seven catchment areas; iii) protecting 1,820 ha of natural forest in major recharge areas and riparian zones, including assisted regeneration; iv) restoring 5,170 ha of natural forest areas across seven catchments; v) restoring 820 ha of native pine forests in Guatemala, Honduras and Nicaragua; vi) establishing 1,293 km of agroforestry systems using diversified living fence arrangements in basic grains crops; vii) establishing 612 ha of agroforestry systems for natural shade in coffee plantations; viii) establishing 1,280 km of silvopasture systems using diversified living fence arrangements; ix) establishing 1,280 ha of silvopasture systems across the 36 target municipalities; x) establishing 5,305 ha of sustainable fuelwood and timber plantations across the 36 target municipalities; xi) establishing 129 km of firebreaks for forests and plantations; xii) constructing 118 km of living barriers for soil conservation around agricultural plots; and xiii) constructing 118 km of superficial drainage for soil conservation. In addition to these EbA practices, activities under Output 2 will also address the projected increase in water scarcity under future climate conditions by introducing water-efficient technologies to reduce demand for scarce water resources in rural communities. This will include installing: i) 109 rainwater harvesting systems at the community level targeting clinics and schools; ii) 156 community-level rainwater reservoirs (500 m³ each); and iii) 38 community-level Water pumping systems by solar panels. Finally, resource-efficient technologies will be introduced in each of the target municipalities, focussing on efficient charcoal production methods using half-orange kilns (43 systems). This will maximise the energy output per unit of biomass, thereby reducing the pressure of fuelwood collection on natural ecosystems.
51. The validation of these indicative pre-selected interventions will be based on the development of the site-specific plans (activity 1.1) that will include both a scientific analysis of specific climate projections through the hydrological models, as well as sound stakeholder consultations that will allow to identify the most appropriate interventions. The validation of these indicative interventions will therefore be based on the following criteria:
- - Relevance to site-specific climate risks
 - - Positive impact potential over climate risks
 - - Familiarity and acceptance by stakeholders
 - - Positive impact potential over most vulnerable sectors of the community
 - - Positive impact potential on gender balance
 - - Technical feasibility of the intervention
 - - Compliance with social and environmental safeguards
 - - Potential co-benefits of the intervention
 - - Minimal level of risks and constraints for implementation
 - - Potential for self-sustainability

⁷³ A description of the target municipalities and catchments is provided in Section 12 of Annex 2: Feasibility Study.

- - Potential for sustained monitoring capacity

52. To foster grant support for identification, incubation or even acceleration of site-specific solutions and distribution models, a US\$28 million grant facility will be established to support site-specific EbA and water-/resource-efficient technologies for the most vulnerable communities, who are unable to access loan finance. A bottom-up approach will be used to identify eligible projects, following stringent eligibility checks for all grant applications. The Regional Project Steering Committee (RPSC) — comprising representatives from FAO, CCAD, CABI, UNEP, national-level CCAD liaisons and the Regional Project Manager — will define the eligibility criteria for each country at programme inception, with final sign off provided by CABI as the the Executing Entity. This process will include identifying specific companies and organisations that have a strong commitment to delivering agreed results of EbA activities and water technologies. The grant eligibility criteria and selection process is foreseen to focus on individuals or legal entities:

- with a proven track record in implementing or providing EbA solutions or water technologies;
- situated or attending the programme areas of operations in the defined catchment areas;
- with a clear vision or plan on how to scale EbA solution or water technology provision;
- able to provide clear and measurable result metrics that can be verified and audited; and
- that adhere to the established grant management principles.

53. Legal entities can refer to business companies, unions or cooperatives, associations, cooperatives, sector organisations, public research and training centres and NGOs that directly cater to targeted end beneficiaries.

54. The qualifying stakeholders will be invited to apply for funding following the application process of the grant manual. Grant finance will be available for the following purposes – in four innovation areas of business linkage innovation, service and input innovation and sector organizations and institutions strengthening:

- New or additional working capital of existing or new projects;
- New or additional investment capital of existing or new projects;
- Seed capital for new projects; and
- New or additional technical assistance for new or existing projects.

55. National Project Execution Units (NPEU) will guide communities on how to develop proposals to access the funds. The NPEUs will be directly hired by FAO as the Executing Entity for output 2 (for activities 2.1 and 2.2) and will be based in the selected municipalities for the execution of the EbA demonstration sites. NPEUs will identify projects that can be supported with the above-mentioned grant purposes to promote innovation in three categories: 1) Business linkage innovation, focusing on strengthening commercial cooperation agreements between different service providers, such as between PFIs (financing EbA solutions packages, rather than stand-alone solutions), a set of EbA solution providers (e.g. drip irrigation system, water reservoir installation, organic fertilizer) and agricultural extension service providers to drive capacity building on how to access the EbA solutions and their finance, bundled in a comprehensive offer to end beneficiaries; 2) Service and input innovation, focusing on identifying locally relevant EbA solutions that are not yet available in the market and can be “imported” to be made available; and 3) the strengthening of sector organization and institutions, such as multi-stakeholder water funds, producer associations or municipal and community coordination entities.

56. On-the-ground EbA-oriented grant support will benefit community-based institutions including *mancomunidades*⁷⁴, municipal associations, farmers cooperatives, solution providers or civil society organisations (CSOs). Eligible projects will include community-level investments, the establishment of water funds, seed funding for payment for ecosystem services (PES) and support for the adoption of selected EbA practices and/or water- and resource-efficient technologies. The grant facility will channel investment to selected projects in each target municipality. The selection of eligible projects will be managed within the proposed governance structure in two competitive selection rounds per year during year 2, 3, 4 and 5 of the programme by i) identifying suitable EbA projects at Project Execution Unit level, ii) apply first screening at National Advisory Committees level, iii) pre-selecting proposed EbA projects from the countries at National Advisory Committees level and iii) deciding upon EbA projects in the Regional Steering Committee and iv) final approval by CABI as the Executing Entity. Further details on the selection process are described in the Operational Guidelines (Annex 23). In some less developed areas in terms of solutions available, successful

⁷⁴ *Mancomunidades* are associations of municipalities with legal personality that provide a practical framework for resource management across municipalities.

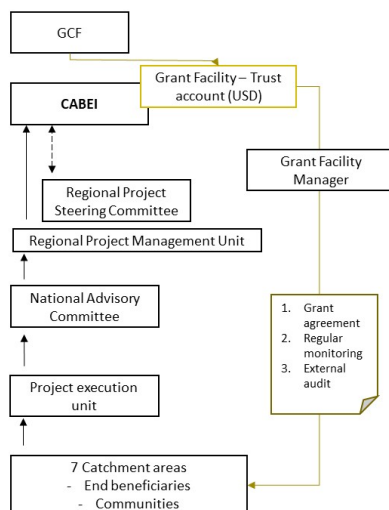
implementation will catalyse the uptake of EbA practices as well as water- and resource-efficient technologies to be financed through a credit facility (see below), in other more developed areas, the grant facility will support enhanced mainstreaming of EbA investment and finance. Training and support activities will also be provided to communities in the target municipalities (Output 1) to enhance the technical capacity of communities to access the Grant facility.

The following figure describes the proposed operation of the grant facility

Grant Facility

Flow of funds

- GCF disburses grant funds of USD 28 million to CABEI trust account
- CABEI disburses funds to grantees as per final decision on recommended grant project portfolio



Contractual arrangement

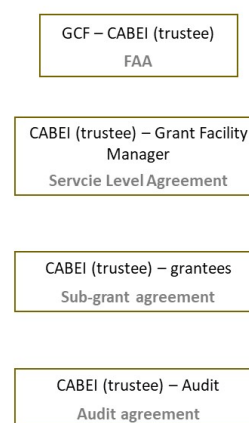


Figure 10. Grant Facility.

57. In addition to directly strengthening the adaptive capacity of vulnerable communities, on-the-ground implementation of EbA and water-efficient technologies will contribute to the existing evidence base of successful adaptation interventions in the region. Specific programme activities that will be implemented under Output 2 are listed below.

- Activity 2.1. Implement landscape EbA interventions within rural communities across the seven target catchments.
- Activity 2.2. Implement demonstration water- and resource-efficient technologies within rural communities across the seven target catchments.
- Activity 2.3. Establish the grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing and start operations.

Output 3. Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies.

58. This component will be executed by CCAD as the regional entity that leads the formulation of the regional strategy for Environmental Management and the regional strategy for Climate Change. The Regional Environmental Observatory gathers information from all countries in the region and provides the opportunity to build scientific information into the regional decision-making process. It is expected that this programme will feed a significant amount of information into this platform that will also be strengthened through communities of practice on EbA to foster science-based decision making at the regional level. CCAD has collaborated in the past with regional technical organizations such as IUCN, FAO, UNDP and UNEP in the production of technical studies for the region and it provides a key political platform for the dissemination of this information to influence decision making at the national level. As such, their leadership in this output of the programme is essential to ensure that results from the programme are being disseminated among countries and used to inform both key regional strategies and national policies. UN Environment will also provide technical advice through this component developing knowledge products including sustainable landscape management standards and

procedures, opportunities for access to financing and women economic empowerment as well as technical guidelines for the assessment of ecosystem services.

59. Activities under Output 3 will contribute to GCF Outcome A8.0 — Strengthened awareness of climate threats and risk-reduction processes — by building an evidence base for the effectiveness of EbA and other adaptation measures in reducing the impacts of climate change and by disseminating this information to local decision-makers, the private sector (including commercial farmers) and local communities. The dissemination of information will be managed through the establishment of a regional knowledge hub linked to the CCAD's Environmental Observatory. The knowledge hub will serve two primary functions. First, it will act as a repository for EbA information, including, but not limited to, all protocols developed by the programme as well as the best practices and lessons learned collected through the programme's monitoring and evaluation processes, previous and ongoing complementary projects. This will include guidance on how to access opportunities for accessing finance provided through the programme, with specific focus on women's economic empowerment. Second, the hub will provide users with specific knowledge products that will guide policy- and decision-makers to effectively implement sustainable landscape management practices. The knowledge products will also include technical assistance for creating business opportunities as well as technical guidelines for the assessment of ecosystem services. This will support decision-making for climate change adaptation, as well as drought and flood risk management. Training will also be provided to policy- and decision-makers at the municipal level on how to use the knowledge hub and associated tools. Furthermore, community organisations – including women's groups – and traditional leaders will be trained to act as champions in developing, revising and disseminating information products to support the adoption of EbA and other resilient practices. This approach will facilitate the sharing of ideas and sustainable practices between communities from the grass roots level, using existing structures and traditional methods which will not only extend the reach of the dissemination campaign, but also the uptake by local communities. Awareness will also be raised regarding the financial mechanisms established under Output 4, thereby facilitating access to climate financing for local governments, farmers and local communities. The information products made available through the knowledge hub will be demand-driven and will draw on best practices and lessons learned from governments, municipalities, community-based organisations, water managers and private sector stakeholders. By centralising the knowledge hub, information will be effectively disseminated across the region and decision-makers will be able to learn from the experiences of other countries within the region.
60. The hub will be led by a Regional Knowledge Coordinator, who will be hired by CCAD for this output. The coordinator will be responsible for the strategic positioning of the hub as the region's foremost online platform for evidence-based products to facilitate the mainstreaming of an ecosystem-based adaptation approach for long term catchment and municipal level planning. Additionally, an administrative officer will support the coordinator in all aspects related to the daily management of hub activities and the implementation of its annual work plans. Part-time senior and junior researchers will be hired in every country to assist with the synthesis of findings from catchments, programme advances, impact and experiences learned for dissemination across the region. The researchers will develop regional knowledge products informed by the compilation and analysis of data and information from the seven catchments where the programme is implemented. Additionally, the researchers will work closely with the Project Execution Units to help with the preparation of evidence-based communication products at country and regional levels.
61. Output 3 will also contribute to GCF Outcome A5.0 — Strengthened institutional and regulatory systems for climate-responsive planning and development — by building the technical capacity of national- and local-level decision-makers to promote climate change adaptation in the Dry Corridor and Arid Zones. This will be supported by the development or adjustment of a methodology for the valuation of ecosystem services to develop accounts profiles of natural capital for integration into development plans in the Dry Corridor and Arid Zones. The methodology will draw on best practices and lessons learned from ongoing initiatives, including the: i) "incentivo forestal" in Guatemala; ii) payment for environmental services in Panama and Costa Rica; iii) Water Fund of the city of Quito in Ecuador ; and iv) water funds established in cities in Guatemala, Costa Rica and the Dominican Republic . Ultimately, the valuation of ecosystem services will be used to promote the establishment of economic incentives for EbA investments — for example, water funds and payments for environmental services. The capacity of local governments to make policy changes that include these kinds of incentives will also be strengthened, including the use of protocols and criteria for the adoption of EbA and implementation of economic incentives for sustainable land management. This output will also support the integration of EbA into national sectoral policies by providing inputs into the currently under development National Adaptation Plans (NAPs). At least 5 out of the 7 countries are currently developing their NAPs with a sectoral or local focus and have identified water security, ecosystem restoration and resilient agriculture as main priorities. The technical assistance provided through this output to policy makers will also focus on the

integration of EbA into these policy frameworks at the national level. Additionally, the technical assistance under this component will support the integration of EbA into the four relevant regional strategies convened by the Central American Integration System (SICA) including the environmental, climate change, water resources and agricultural regional strategies. Programme activities that will be implemented under Output 3 are listed below.

62. Output 3 will also be a key component of the programme in the generation and strengthening of regional climate information and climate impact analysis. The collaboration with regional researchers (sub-activity 3.3.1 under Activity 3.3) will provide the opportunity to strengthen the availability of site-specific climate projections and climate risk analysis, as well as the generation of evidence on the benefits of Nature based Solutions as an effective strategy to build climate resilience. The participation of these research institutions will be key during the first year of implementation generating the scientific inputs for the development of the site specific plans.

- Activity 3.1. Establish regional knowledge hub for the dissemination of information on EbA in the Dry Corridor and Arid Zones.
- Activity 3.2. Raise awareness of financial mechanisms for the implementation of CCA interventions.
- Activity 3.3. Enhance capacity of national and local-level policy-makers, in partnership with knowledge network of universities, research centers and private sector, to integrate climate change adaptation and the valuation of natural capital into local policies.

Component 2. Financing and implementation of EbA as well as water- and resource-efficient technologies across selected catchments

63. The execution arrangements for this component will be distributed among CABEI and FAO. CABEI will be responsible for those activities related to the set up and management of the EbA financing mechanism including the lending, guarantee and grant facilities and the set up of the Regional Project Steering Committee guidelines in relation to the financial mechanism, (Activities 4.1 and 4.2). FAO will be responsible for the execution of the awareness raising activities for the financial component of the programme as well as for the monitoring of the investments undertaken at local level by the local financial institutions through the Project Execution Units (Activity 4.3). UN Environment will deliver technical support in this component through the development of training materials for financial institutions, support in the development of MRV systems and annual support in the monitoring of results from the investments undertaken by financial institutions (under activity 4.3). UNEP will be engaged by FAO through a UN to UN agreement.

64. The second component under the proposed programme will establish and operationalise financial mechanisms to facilitate investment in EbA practices, as well as the adoption of water- and resource-efficient technologies (Activity 4.2). This component will build on the experience of CABEI's regional CAMBio II project as well as the regional MEbA project to increase access to climate financing for local organisations and communities — including non-regulated financial institutions, commercial and subsistence farmers, as well as vulnerable households — through an existing network of intermediary financial institutions (IFIs). Using the increasing body of knowledge on EbA financing generated through CAMBio II and other similar initiatives, this component will focus on providing grants and blended finance for EbA investments as well as water- and resource-efficient technologies. This will promote an approach of bottom-up identification of adaptation interventions and de-risked top-down financing for adaptation. The types of EbA and water- and resource-efficient technologies eligible for financing through the established financial mechanisms include (among others):

- agroforestry systems;
- silvopasture systems;
- agrosilvopastoral systems;
- windbreaks;
- firebreaks;
- mixed plant nurseries;
- agricultural terraces;
- soil conditioning;
- efficient biomass stoves;
- firewood drying and charcoal production;
- rainwater collection in small reservoirs;
- rooftop rainwater harvest systems;
- solar water pumping systems;
- drip irrigation;

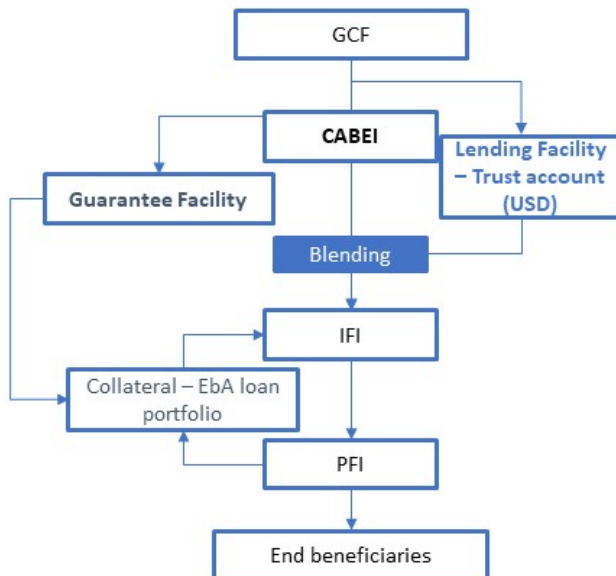
- ceramic filters to purify water
- fog catchers; and
- infiltration pits for water retention;

Further details on these measures, including descriptions, and direct and co-benefits are presented in Annex 22.

Output 4. Financial products and services to finance EbA investments are offered by Partner Financial Institutions (PFI), including PFI access to EbA on-lending funds and support mechanisms.

65. Activities under Output 4 will contribute to GCF Outcome A7.0 — Strengthened adaptive capacity and reduced exposure to climate risks — by establishing financial mechanisms to support the adoption of EbA practices as well as water- and resource-efficient technologies. To that end two financing facilities will be established, concessional lending for EbA solutions relevant to reduce climate risks in the Dry Corridor as well as credit guarantee to further streamline and disseminate EbA lending across the entire Dry Corridor.
66. The EbA lending facility (EbA credit line) will be established and operationalised with the GCF senior loan provision of US\$60 million and CABI US\$42.8 million co-finance together being offered to PFIs as blended finance mechanism at below-market conditions for small- and large-scale EbA investments at the farm-, enterprise- and household-level. The objective of the EbA lending facility is to establish and operationalize financial mechanisms to facilitate investment in EbA practices, as well as the adoption of water- and resource-efficient technologies. The EbA Lending facility deployed under sub-activity 4.2.1 will have a revolving character at the PFI level: while it will be established for a period of 10 years, loans to be disbursed to PFIs are expected to have a tenor of 2 - 5 years maximum (as a function of the expected EbA finance portfolio to be created with end beneficiaries, i.e. focusing more on working capital or investment capital, and in accordance with market standards). Additionally, loan agreements with PFIs will foresee capital (and interest) payments over the tenor of the PFI loans. Returning funds will be reinvested to available PFIs, ensuring a closing of laid out loans before the facility's closure.
67. The guarantee mechanism can be used to co-finance, to support the initial introduction of the lending facility and provide further de-risking. This is needed to support the introduction of EbA finance to the most vulnerable, an end beneficiary segment that is usually without access to any finance due to its elevated risk profile. The remaining guarantee mechanisms will be used to leverage additional finance.
68. The following figure describes the lending and guarantee facilities deployed under sub-activity 4.2.1 and 4.2.2. respectively.

Financing Scheme - Lending and Guarantee facilities



Loan through the Global Credit Line (GCL)

- Channel the resources of the GCF through the **CABEI Global Credit Line (GCL)**.
- These resources will have a liability limitation in case of non-payment by the PFIs, the liability limitation only applies to GCF funds.

Important points:

- The current LGCs will be used.
- There is a tax shield on credit operations.
- There is extensive knowledge on the part of the IFIs about the GCL.
- A new program must be developed.
- The funds are received on balance by the IFIs, which can make the resources expensive to the final beneficiary.
- Operational complexity by dividing disbursements into two tranches.

An **Umbrella Approval** will not require subsequent approval by the CABEI Board of Directors each time an IFI requests a disbursement within the Global Credit Line.

1. The funds from the GCF should not affect the quota available to the IFI.
2. Each IFI that decides to intermediate funds under the GCF program must sign a complementary contract to the GCL contract in which the terms and conditions applicable to the GCF program are defined.
3. Disbursements will be divided into tranches, one from the GCF and the other from CABEI funds.

CONFIDENCIAL EXTERNO

Figure 11. Lending and guarantee facilities' Financing Scheme

69. By providing a blended financing mechanism composed of concessional finance from GCF in combination with a market rate senior debt as co-financing from CABEI (and an associated guarantee facility – see above), incentives are created for financial institutions on different levels to engage in EbA finance, for which there exist no or only limited offerings as of today. Intermediary Financial Institutions in CABEI's established network will manage the EbA lending facility to local partner financial institutions such as cooperatives, small banks, NGOs, rural community banks, among others - Partner Financial Institutions. In parallel these IFI will receive GCF funds and CABEI Co-financing funds via senior loans to on-lend to the same PFIs. PFIs in turn will on-lend to end beneficiaries. Investment will be targeted towards community-level investment, rural MSMEs or other institutions that can foster EbA, as well as water- and resource-efficient technologies, for end beneficiaries across different sectors.
70. Partner Financial institutions will be able to access the EbA Lending facility at concessional below-market rates for direct on-lending to final customers for EbA investments. Investment will be targeted to finance small- and large-scale EbA investments at farm-, enterprise- and household-level to initiate and support a paradigm shift in the regional approach to catchment and water demand management in the Dry Corridor and Arid Zones. As such, the facility will aim to invest in initiatives that improve the climate resilience of rural communities in the Dry Corridor and Arid Zones by: i) implementing integrated catchment management and restoring catchments; ii) improving hydrological flow and infiltration of rainwater into groundwater reserves through forest and ecosystem restoration; and iii) reducing demand for scarce water resources by implementing water-efficient technologies at the farm- and household-level.
71. Eligibility criteria for PFIs to access the EbA lending facility are outlined in Section 6.7.1 of Annex 23 and in section B4, including legal and capacity requirements. Where PFIs are not compliant with the minimum requirements set out in the eligibility criteria, the TA facility will be available to strengthen respective institutional capacities. Technical Assistance in EbA finance methodologies will be a condition precedent to gain access to the EbA Lending facility and Guarantee facility.

72. End beneficiaries, as well as EbA investment eligibility criteria will be defined by the Regional Project Steering Committee at programme inception and finally approved by CABEL as the Executing Entity. Sub-credits⁷⁵ will be made available in the local currency to end-beneficiaries in the eligible geographic regions for both working capital and investment purposes. The sub-credits will be provided under a sub-credit agreement (SCA) between the local PFIs and the beneficiary using the PFI's loan agreement forms and including the standard provision to be developed in the credit manual. Further details on the eligibility criteria are provided in Annex 23. Eligibility criteria will also be guided by vulnerability assessments and learnings from the lending mechanism itself as the execution advances.
73. Additionally, a guarantee facility⁷⁶ will be linked to the EbA lending facility to — in the case of documented success EbA, i.e. increasing productivity and lowering production cost vis-à-vis non-programme scenarios — secure PFI's on-lending and mobilise additional lending from IFIs' and PFIs own and additional public and private investors' resources by mitigating against credit risks associated with commercial EbA Finance to end beneficiaries. CABEL will manage the guarantee facility as a window in its recently established Guarantee Fund (managed in-house), guaranteeing PFIs' sub-credit portfolio to the end borrowers (which in turn will be pledged by the PFIs to the IFIs) Such loans are usually – especially for subsistence farmers – not fully covered by real guarantees such as real estate or other moveable assets pledges, hence making such constructs prohibitively expensive. As digital EbA lending facility MRV systems will be integrated, CABEL will be in position to automate its extension of guarantee certificates over end beneficiary loans in line with current guarantee fund stipulations – IFIs will indirectly have partial coverage by CABEL's guarantee fund, which will lower collateral requirements by PFIs and enhance access to the lending facility and leveraged finance for end beneficiaries.. The guarantee facility will issue a capped partial credit risk guarantees to IFIs for their on lending to PFI portfolio. If successful, the guarantee facility will hence be able to provide guarantee coverage to additional funds. The EbA Guarantee will be established as a liquid facility managed directly by CABEL with a US\$30.0 million guarantee funded by GCF. A recent guarantee facility established by CABEL to attend Covid-19 impacts for eligible end borrowers will provide the mechanism for the Guarantee facility in the proposed programme. Once operational and with increased EbA Finance experience at hand in the programme areas, its coverage will be expanded, following established risk management principles and depending on the guarantee's performance (in terms of claims and outreach). In the same way, in case of claims exceeding expected levels, the emission of additional guarantee obligations will be limited. The guarantee is foreseen to cover 25–75% of end beneficiary sub-loan portfolios to eligible PFIs and hence targeted to leverage additional EbA finance with a ratio of 1:4.33 in the range of US\$150.8 million (see 6-1, section 5 in Annex 3). To that end, the Guarantee facility will de-risk operations in these new end beneficiary segments and therefor increase the acceptance of the Lending facility, as well as to gather data that will be reviewed annually to establish the baseline of actual expected default of resulting sub-loan portfolios. As function of this baseline analysis, the coverage of the Guarantee facility will be expanded up to the above expected leverage.
74. IFIs will have the flexibility to use the guarantee facility and the EbA lending facility as stand-alone solutions or in combination. Potential options for IFIs include:
1. EbA lending facility only for EbA investment to be financed where the IFI is comfortable taking the full credit risk of PFIs and for which it needs liquidity from the Lending facility;
 2. EbA guarantee facility only for EbA investment to be financed for which the IFI would be willing to pay for credit enhancement but for which it does not need external liquidity. This mechanism is being targeted to leverage additional finance from IFIs for on-lending to eligible PFIs directly; and
 3. Blended guarantee/lending facility mechanisms for EbA investment to be financed for which the IFIs seek both risk mitigation and funding support from the two facilities. In that way, IFI can be incentivised to finance parts of eligible portfolios through the EbA lending facility and finance remaining balances from resources making use of the de-risking EbA guarantee facility offer.
75. The financial facilities will be accompanied by a technical assistance (TA) facility (see below) that will develop capacity within municipalities and PFIs to monitor and assess the implementation of EbA and water/resource-efficient investment projects as well as to monitor and document the respective economic and ecosystem benefits. Where possible, Information and Communication Technologies will be introduced to provide data

⁷⁵ Sub-credits are defined as working capital and investment loans for the purpose of financing all or a portion of the expenditures incurred by the beneficiary for goods and works under a sub-project.

⁷⁶ The Guarantee Facility will be managed as a specific window under a recently established Guarantee fund managed by CABEL in-house.

analytics and satellite data insights into the EbA lending portfolio and climate impacts on the portfolio. FAO will be the Executing Entity for this TA Facility (under Activity 4.3) and will receive technical support from UNEP.

76. To facilitate access to and operation of the facilities, Activity 4.3 will provide TA to selected PFIs. Technical assistance will be a mandatory condition for PFIs to access the EbA lending facility. This will ensure that capacities, especially for adequate monitoring, reporting and verification (MRV) systems, are in place to manage the EbA finance. Such technical assistance will focus on developing capacity within municipalities and PFIs to monitor and assess the implementation of EbA and water/resource-efficient investment projects as well as to monitor and document the respective economic and ecosystem benefits. Where possible, Information and Communication Technologies will be introduced to provide data analytics and satellite data insights into the EbA lending portfolio and climate impacts on the portfolio.
77. The TA facility is geared to focus primarily on direct technical assistance and a comprehensive Training-of-Trainers or online course mechanism to strengthen: i) the awareness of climate change, ecosystem degradation and water scarcity; ii) the identification and promotion of solutions to overcome these threats; iii) the application of methodologies, technologies and tools to foster EbA finance and manage climate and ecosystem risks via modern information management solutions, iv) the provision of specific EbA oriented tailor-made financial products and services to end beneficiaries, as well as of non-financial services such as information services or training; and v) the monitoring, reporting and verification systems in place via technology transfer to support learning on successful EbA and water/resource-efficient technologies. The TA will comprise capacity-building for financial intermediaries on the concepts, financial methodology, and tools to provide products and services oriented to EbA and water-efficient technologies. Expert pools will be established in each country to provide country-specific technical assistance to PFIs based on a standardised methodology.
78. The deployment of the loan facility will follow a phased approach along the programme to ensure cohesive and substantive impact of the financed activities as well as proper vulnerability assessments and identification of priorities for other vulnerable regions within the Dry Corridor where the intervention could be scaled up. This phased approach will also allow for more time to strengthen capacities of the financial institutions and to promote the financial mechanism with communities and producers.
79. The three implementation phases of the loan facility are described below:

Launching the loan facility

The loan facility will be initially deployed through local PFIs to attend priorities and requests from producers and communities within the 5 prioritized municipalities in the Tempisque basin. The specific eligibility criteria for the type of interventions to be included in the loan facility will be detailed during the first year of the programme through consultations with communities, experts and financial institutions – subject to final approval by CABEL. An MRV system will also be structured to gather information about the deployment of the loan and guarantee facilities and the impacts of the EbA interventions financed. Additionally, the technical team (RPMU in collaboration with PEUs and technical experts engaged through output 3) will initiate a climate risk assessment and prioritization of other municipalities within the Dry Corridor.

Replicating the loan facility

As a result of the initial phase, the technical team (as above) and PFIs will count on information about the deployment rate of the loan facility and initial results from the impacts of the financed interventions. The larger climate risk assessments will have also provided information about climate risks and adaptation potential for additional municipalities within the Dry Corridor. In this second phase, loans will be disbursed within the 5 initially prioritized municipalities and other surrounding municipalities that have been prioritized and characterized through complete climate risk assessments.

Scale up of the loan facility

Finally, and after gathering more detailed information about the results of the deployment of the loan facility, the beneficial impacts of EbA interventions and undertaking substantive capacity building both with community members and PFIs in the 5 prioritized municipalities and other vulnerable ones within the Dry Corridor, the loan facility will be expended geographically to cover for additional municipalities and always within the limits of the Dry Corridor to ensure substantive action to build resilience.

80. Demand analysis for the loan facility in the Dry Corridor

The potential of the lending facility was modelled at the regional and national levels to understand the absorption capacity on the ground, and is further described in annex 3 (financial model) and 30 (demand analysis) of this proposal.

An initial review of existing and publicly available literature, from academic research to practitioner report and projects' references, shows not only a growing interest into the topic of adaptation and EbA Finance of the global and Central American financial sector, but also a diversifying set of methodologies available for the capacity building of financial institutions willing and interested to engage in the field.

Furthermore, data from publicly available databases is accessible to define solid assumptions on potential demand for the implementation of EbA solutions and in consequence the demand for EbA Finance.

81. REDCAMIF (Microfinance Network of Central America and the Caribbean) is the most important network of microfinance institutions in the region. The organization provides a centralized database with data on its members' performance.

The network is organized as a network of networks, organizing 7 countries' national networks into a regional one, representing 130 microfinance institutions that reach nearly 1.5 million clients via 1,211 branches. Of the aggregated outstanding loan portfolios of over US\$ 2.1 billion, almost 49% is deployed to rural clients.

While REDCAMIF is not representing the total of institutions active in all 7 countries, the network clearly provides the most aggregated reference data sets regarding especially inclusive rural finance which is one of the focus of this programme

In order to understand financial institutions' installed capacity, two dimensions were reviewed as preparation for the analysis:

1. The traction of inclusive agricultural finance in the seven countries (see also section 4.2.4 in annex 30):
2. The presence of financial institutions in the programme pilot areas in order to understand the level of potential offerings to disseminate EbA finance in pilot areas, to generate the required scaling of EbA finance as part of the programme.

82. According to latest statistics from the REDCAMIF's MIFIndex for June 2020, the networks PFIs manage a total inclusive agricultural portfolio of US\$324 million across all 7 countries for 221,477 clients. This represents more than 3 times the total size of the targeted lending facility's volume of US\$102.8 million. The microfinance sector's involvement in inclusive agricultural finance is a good proxy for its ability to serve EbA finance, which will primarily be targeted at Small-scale farmers.

Usually, the microfinance sector refrains of serving these populations, due to key challenges, such as:

1. Rural and agricultural finance are more costly due to greater distances to cover for loan origination as well as recovery.
2. Especially agricultural producers are considered more risky, as the low levels of technification of Small-scale producers makes them more prone to adverse climate (change) impacts.
3. Little incentives are being offered from investors to incentivize greater activities and outreach in agricultural microfinance, especially due to the lack of concessional finance.

Further, especially for EbA and other green finance, markets are not developed and documentation requirements are regularly higher than for traditional finance, which increases cost due to

1. The need for more client awareness raising especially to not well known alternatives to traditional agricultural practices,
2. More time to be spend with client assessment, due diligence and loan evaluation,
3. Increased personnel training and re-training activities as consequence of higher complexity of EBA and climate finance interdynamics.

83. It can hence be expected that the proposed programme will address all of these barriers by raising awareness and ultimately demand of target populations for EbA alternatives, providing digital process and data management solutions for better loan management, build capacities to better identify and mitigate (physical) climate-related financial risks and provide concessional finance to confront perceived risks.

84. As the distribution of financial flows from the Lending facility are expected to be distributed among countries based on prioritization of the Regional Project Steering Committee, under the chair of CABEI, the following considerations are based on the assumption of an equal distribution – and put the total amount to be disbursed

over the initial 4 years in relation to the existing agricultural microfinance portfolio. Hereby it is assumed that additional concessional finance from the Lending facility will

1. Replace existing investments and finance through enhanced documentation of eligible EbA investment as well as
2. Increase the appetite of partner financial institutions to engage with Small-scale farmers that are deemed more risky.

85. The results of the analysis of microfinance portfolios at the national level are summarized below:

86. Costa Rica

In Costa Rica, at June 2020, a total agricultural loan portfolio of US\$ 16 million to 10,181 clients has been reported – representing the highest share of a total microfinance portfolio of US\$38 million with 42%. All loans reported are being managed following individual lending methodologies (as opposed to group lending or community bank lending).

The existing US\$ 16 million loan portfolio to 10,181 clients is almost equal to the total amount disbursed via the proposed Lending facility of US\$15 million (assuming equal distribution of loan across countries for discussion purposes). According to the assumption in the financial modelling of the Lending facility, this portfolio shall be reached after 4 years – or after year 5 of the programme.

In the case of Costa Rica it is though expected, that only a minor part of 19% of that facility (or US\$ 2.8 million) will actually be destined to Small-scale farmer EbA finance, as commercial farming in the more developed agricultural sector plays a more important role. This represents roughly 17% of the present portfolio.

87. Dominican Republic

In Dominican Republic, at June 2020, a total agricultural loan portfolio of US\$ 38 million to 22,151 clients has been reported – representing the second lowest share of a total microfinance portfolio of US\$498 million with 8%. Almost all loans reported are being managed following individual lending methodologies, with only 1% following group lending methodologies. Community bank lending is not being provided.

The existing loan portfolio of US\$ 38 million to 22,151 clients represents 2.5 times the targeted EbA portfolio, which in case of the Dominican Republic is expected to be destined to Small-scale EbA finance by 39% - or with US\$ 5.8 million roughly a 15% of the current outstanding portfolio.

88. El Salvador

In El Salvador, at June 2020, a total agricultural loan portfolio of US\$ 46 million to 60,507 clients has been reported – representing the third lowest share of a total microfinance portfolio of US\$325 million with 12%. Individual lending methodologies are used for 96% of the portfolio, with 2% following group lending and community bank lending methodologies each for smaller loan amounts – probably in rural areas.

The existing loan portfolio of US\$ 46 million to 60,507 clients represents 3 times the targeted EbA portfolio, which in case of El Salvador is expected to be destined to Small-scale EbA finance by 54% - or with US\$ 8.1 million roughly a 18% of the current outstanding portfolio.

89. Guatemala

In Guatemala, at June 2020, a total agricultural loan portfolio of US\$ 40 million to 21,214 clients has been reported – representing the third highest share of a total microfinance portfolio of US\$312 million with 15%. Individual lending methodologies are used for 96% of the portfolio, with 7% following group lending and community bank lending methodologies each for smaller loan amounts – probably in rural areas.

The existing loan portfolio of US\$ 40 million to 21,214 clients represents 2.7 times the targeted EbA portfolio, which in case of Guatemala is expected to be destined to Small-scale EbA finance by 52% - or with US\$ 7.9 million roughly a 20% of the current outstanding portfolio.

90. Honduras

In Honduras, at June 2020, a total agricultural loan portfolio of US\$ 108 million to 75,176 clients has been reported – representing the third highest share of a total microfinance portfolio of US\$467 million with 23%. Individual lending methodologies are used for 96% of the portfolio, with 3% and 1% following group lending and community bank lending methodologies respectively for smaller loan amounts – probably in rural areas.

The existing loan portfolio of US\$ 108 million to 75,176 clients represents 7.1 times the targeted EbA portfolio, which in case of Honduras is expected to be destined to Small-scale EbA finance by 47% - or with US\$ 7.1 million roughly a 7% of the current outstanding portfolio.

91. Nicaragua

In Nicaragua, at June 2020, a total agricultural loan portfolio of US\$ 47 million to 21,851 clients has been reported – representing the second highest share of a total microfinance portfolio of US\$191 million with 25%. Individual lending methodologies are used for 96% of the portfolio, with 0.5% and 6% following group lending and community bank lending methodologies respectively for smaller loan amounts – probably in rural areas.

The existing loan portfolio of US\$ 47 million to 21,851 clients represents 3.1 times the targeted EbA portfolio, which in case of Nicaragua is expected to be destined to Small-scale EbA finance by 68% - or with US\$ 10.2 million roughly a 22% of the current outstanding portfolio.

92. Panama

In Panama, at June 2020, a total agricultural loan portfolio of US\$ 27 million to 10,397 clients has been reported – representing the second lowest share of a total microfinance portfolio of US\$305 million with 9%. All loans reported are being managed following individual lending methodologies (as opposed to group lending or community bank lending).

The existing loan portfolio of US\$ 27 million to 10,397 clients represents 1.8 times the targeted EbA portfolio, which in case of Panama is expected to be destined to Small-scale EbA finance by 86% - or with US\$ 12.9 million roughly a 48% of the current outstanding portfolio.

93. Comparing these numbers to the results of the financial model, the deployment of the lending facility at the national level could be summarized and compared to the numbers above:

- In average each country could absorb (assuming equal distribution for purposes of discussion) around US\$ 14.7M in funding from the lending facility (US\$ 60M from GCF and US\$ 42.8M from CABEL)
- By rotating this capital at the PFI level, a total disbursement per country over the lifetime of the lending facility of US\$ 51.5M via close to 9,800 loans could be reached.
- This translates to an average disbursement of 980 loans per country, per year, or in average 82 loans per month (due to dynamics of building up the total facility, as well as winding her down, the peak is expected with roughly 1,780 loans in year 3, resulting in 148 loans per month).
- Per country an expected 4 Partner Financial Institutions will gain access to the lending facility, disbursing around 20 loans in average (in peak: 37) per month.

94. Assuming this equal distribution, with exception of Panama, the programme countries will take up concessional EbA microfinance in the range of 7 – 22% of their current agricultural microfinance portfolios. Disbursed over a period of 4 years, and accompanied with increased awareness raising and capacity building activities, it can be expected that the markets in the region present a solid base for its absorption. Further analysis of demand at the country and regional level is available in Annex 30.

95. Further details on the financial mechanisms to be established under this output are provided in Annex 23, including: i) the EbA finance rationale; ii) the governance overview; iii) the funding structure, size and risk management strategies; iv) the structure and approach for technical assistance; v) grant facility operation; vi) lending facility operations; vii) guarantee facility structure and manual; viii) the facilities' closure; and ix) the overall exit strategy targeting additional leveraged finance as well as the proof of the underlying hypothesis (Summary: "EbA finance as de-risking strategy, hence enabling greater investment into EbA by formerly excluded most vulnerable populations"). A detailed description of the technologies that can be financed through these mechanisms is provided in Annex 22. Programme activities that will be implemented under this output are listed below, with technical details provided in Annex 2: Feasibility Study.

- Activity 4.1. Set-up and establish the financial structure for the lending and guarantee facilities.
- Activity 4.2. Financing mechanisms of the EbA lending facility and guarantee facility.
- Activity 4.3. Technical assistance (TA) facility to strengthen technical capacity of accredited and non-regulated financial institutions to access and channel funds for small- and large-scale EbA investments.

96. The table below summarizes the detailed execution arrangements for the components of the programme

COMPONENT 1. Mainstreaming of EbA, water-efficient technologies and natural resource-based businesses into selected catchments			
Output	Activity	Executing Entity	Implementation partner
Output 1 Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures	Activity 1.1. Develop site-specific intervention plans for the 7 target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities and other stakeholders.	FAO	Direct implementation Government authorities (National/municipal) Specialized technical assistance services providers and/or national or international delivery partners
	Activity 1.2. Provide technical assistance to municipal authorities, farmers and rural communities for the implementation of EbA practices as well as water- and resource-efficient technologies		
	Activity 1.3. Provide technical assistance to farmers and rural communities for the development of natural resource-based businesses and alternative climate-resilient livelihoods.		
Output 2 Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.	Activity 2.1. Implement landscape EbA interventions within rural communities across the seven target catchments.	FAO	Direct implementation Government authorities (National/municipal) Specialized technical assistance services providers and/or national or international delivery partners
	Activity 2.2. Implement demonstration water- and resource-efficient technologies within rural communities across the seven target catchments.		
	Activity 2.3. Establish the grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing and start operations.	CABEI	Selected partner through competitive process
Output 3 Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies	Activity 3.1. Establish regional knowledge hub for the dissemination of information on EbA in the Dry Corridor and Arid Zones.	CCAD	UNEP 3.1.2
	Activity 3.2. Raise awareness of financial mechanisms for the implementation of CCA interventions		UNEP: 3.3.2, 3.3.3.;
	Activity 3.3. Enhance capacity of national and local-level policy-makers, in partnership with knowledge network of universities, research centers and private sector, to integrate climate change adaptation and the valuation of natural capital into local policies.		
COMPONENT 2. Financing and implementation of EbA as well as water- and resource-efficient technologies across selected catchments			
Output 4 Financial products and services to finance EbA investments are offered by Partner Financial Institutions (PFI), including PFI access to EbA on-lending funds and support mechanisms.	Activity 4.1. Set-up and establish the financial structure for the lending and guarantee facilities.	CABEI	
	Activity 4.2. Financing mechanism of the EbA lending facility and guarantee facility	CABEI	Selected IFIs/PFIs
	Activity 4.3. Technical assistance (TA) facility to strengthen technical capacity of accredited and non-regulated financial institutions to access and channel funds for small- and large-scale EbA investments.	FAO	UNEP to provide technical support to strengthen technical capacity of IFIs and PFIs financial institutions to access and channel funds for small- and large-scale EbA investments.

97. Intervention strategy.

The programme will be implemented following a sequence that allows for continuous feedback of results and observed impacts of the interventions into the enabling activities to strengthen capacity building, mainstreaming of EbA into

relevant policies and into the regional communication platform. The described sequence below is divided into 5 main phases, with a summarized description of main activities implemented in each of them.

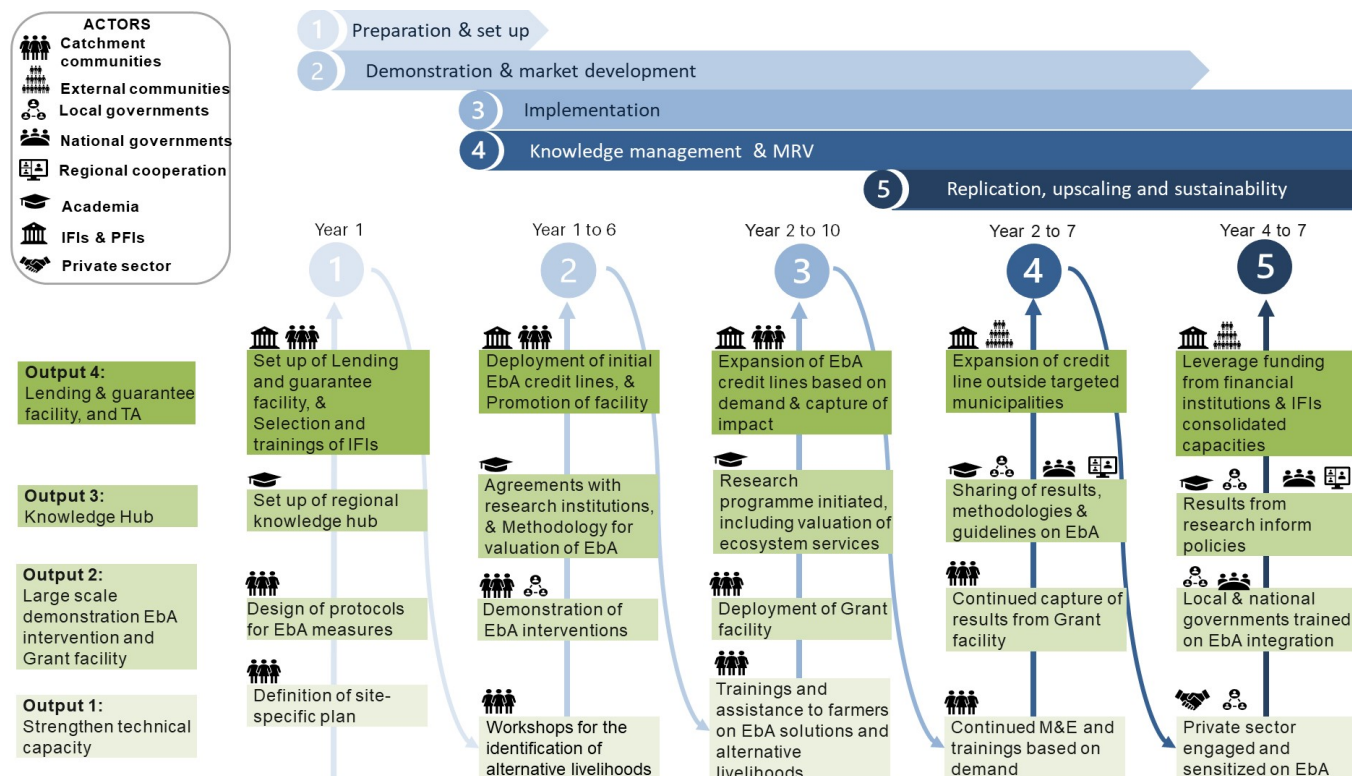


Figure. Graphic representation of implementation strategy

98. 1/Set up and preparedness

During this first stage of the programme, the site-specific implementation plans based on vulnerability and hydrological models will be strengthened and finalized in consultation with communities and key actors including detailed climate risk assessments. These plans have been drafted for Costa Rica and Honduras (annex 31) and will be strengthened and validated with communities during this initial phase of the programme and will be developed for all other 5 countries during the first year following the process described in Annex 31.

99. The specific EbA interventions funded by the grant component will be based on the development of these site-specific plans that will include both a scientific analysis of specific climate projections through the hydrological models, as well as sound stakeholder consultations that will allow to identify the most appropriate interventions. The validation of these indicative interventions will therefore be based on the following criteria:

- Relevance to site-specific climate risks
- Positive impact potential over climate risks
- Familiarity and acceptance by stakeholders
- Positive impact potential over most vulnerable sectors of the community
- Positive impact potential on gender balance
- Technical feasibility of the intervention
- Compliance with social and environmental safeguards
- Potential co-benefits of the intervention
- Minimal level of risks and constraints for implementation
- Potential for self-sustainability
- Potential for sustained monitoring capacity

This stage is critical for the engagement of actors and the Monitoring and Evaluation committees will be established within the participating communities. Technical protocols will be designed for the implementation of the large-scale EbA interventions that will serve as capacity building, demonstration and promotion sites.

Stage 3/ Demonstration and demand generation	Related activities	Year 2		Year 3				Year 4			
		Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16

The elaboration of this site-specific intervention plan will follow a four-step approach in the seven participating countries, as described in the schematic representation below. All participating countries have completed the initial steps, and the remaining steps will be concluded during the first year of programme implementation (activity 1.1).

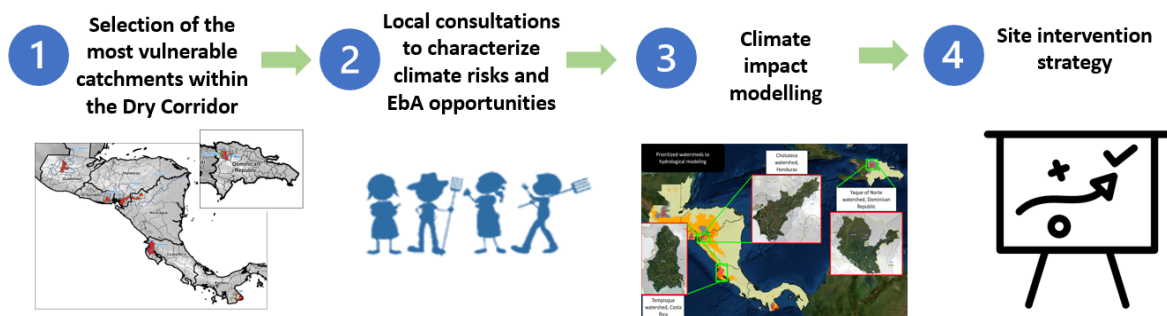
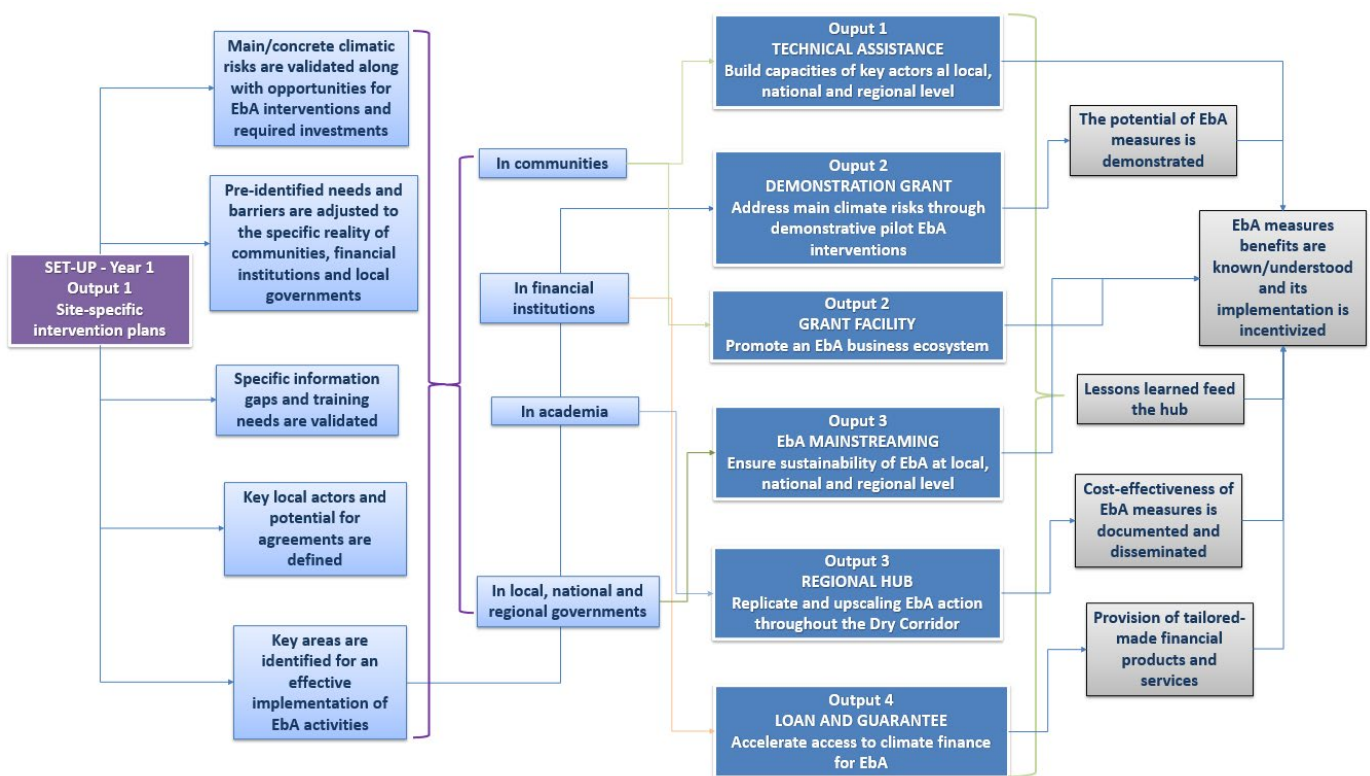


Figure. Steps for the elaboration of site-specific intervention plan

The following figure shows the linkages of the programme components to the site specific EbA plan that will be developed during the first phase of the proposal:



108. Draft sample EbA site-specific plans have been developed for Honduras and Costa Rica and will be validated with key actors in the initial stages of implementation. The other 5 countries will follow the described process below for the completion of their specific plans. This process and the sample plans for Costa Rica and Honduras are further described in Annex 31.

109. *Step 1. Selection of the most vulnerable catchments within the Dry Corridor (already done for the 7 countries)*

Step 1.1 National identification of up to four prioritized catchment areas within the Dry Corridor: The selection of up to four catchments areas of interest per country at the Dry Corridor and the arid zones of the Dominican Republic was done through workshops held with government representatives.

Step 1.2 Regional climate change vulnerability assessment: For each of the prioritized catchment areas, vulnerability assessments were developed. Variables used for exposure, sensitivity and adaptive capacity included access to credit and irrigation, agricultural employment, among others.

Step 1.3 Workshops and consultations for final catchment selection: Governments selected 5-6 bordering municipalities within one of the prioritized catchment areas, based on their vulnerability to climate change and other criteria.

110. *Step 2 Local consultations to characterize climate risks and EbA opportunities (already done for the 7 countries. To be reinforced during the first year of programme implementation)*

Step 2.1 Assessment on territory and identification of key actors: A characterization of the territory and key actors was done. Variables assessed included: demographics, indigenous communities, main regulatory and legal framework, livelihoods, and financial mechanisms available.

Step 2.2 Analysis of main ecosystems, climate risks and perceived climate impacts: The consultation methodology included a land use classification by participants, in which they analyzed main ecosystem's benefits/use, climate risks, perceived impacts, among other, in a sex-disaggregated manner.

Step 2.3 Participatory identification of adaptation options: Consultation participants proposed adaptation measures based on the potential impact on the identified climate risks, ecosystems, livelihoods, and specific needs.

Step 2.4 Perceived barriers and mitigation strategies during consultations: Participants identified main barriers they perceived could jeopardize the implementation of the proposed adaptation measures and the programme in general.

Step 2.5 Alignment with demand analysis from microfinance institutions: Financial institutions in the area were interviewed about their existing portfolios and demand. Climate risks identified during the consultation were compared and aligned with those identified by these institutions.

111. *Step 3 Climate impact modelling (done for Costa Rica, Honduras and Dominican Republic. To be developed for the remaining countries during the first year of programme implementation and validated for the three initial countries)*

Step 3.1 Selection of a sub-basin for the modelling purpose: The Soil Water Assessment Tool (SWAT) is used for the hydrological modelling of climate change potential impacts and impacts of EbA interventions. Water catchment areas are characterized through land use cover, digital elevation model and soil characterization. Based on topography, the basin is divided into sub-basins, and further divided into Hydrologic Response Units (HRU) based on unique combinations of slope, soil, and land cover. SWAT input data are precipitation, temperature, digital model elevation (DEM), soil cover, land use, and drainage network.

Step 3.2 Identification of the main climate risks: The scenarios considered for modelling climate change impacts include a baseline, RCP 4.5 2050, RCP 4.5 2100, RCP 8.5 2050, and RCP 8.5 2100. The hydrological variables to estimate climate change impacts include soil water, percolation, water yield and erosion.

Step 3.3 Impact modelling for main EbA interventions: A modelling scenario is generated to analyse the potential impact of EbA interventions individually or in combination under the RCP 4.5 climate change scenario. Selected interventions included those prioritized through the local consultations and a matrix was generated for the evaluation of impacts for different levels of intervention intensity.

Step 3.4 Conclusions of the climate impact modelling: Results from the hydrological model were discussed through the analysis of changes in the hydrological variables for the Climate change RCP 4.5 scenarios with and without EbA interventions.

112. *Step 4 Site intervention strategy (Pre-drafted for Honduras and Costa Rica. To be developed in consultation with key actors for all remaining countries during the first year of programme implementation)*

The site intervention strategy will include the following components:

Definition of the problem statement: Based on the results from the community consultations and the climate impact modelling, the problem statement can be defined identifying the main climate risks and barriers for adaptation to climate change.

Intervention objective: The specific objective of the programme is defined with a clear identification of the expected results in terms of EbA interventions, enabling conditions and objective population.

Intervention structure: The intervention logic describes the components that will shape the programme, their contribution and their interconnection towards the main objective.

Grant-financed EbA interventions to address prioritised climate risks in the Tempisque basin (Output 1.2): A number of EbA interventions will be financed through a grant mechanism as demonstration pilots for the target territory. This section describes what EbA interventions have been prioritized for the specific climate impacts.

Grant facility established to promote an EbA business ecosystem (Output 1.3): The establishment of a Grant facility will then finance a series of EbA intervention, supporting business and distribution models through a competitive bottom-up approach. A number of criteria is raised to identify potential projects and beneficiaries at the community level, and this facility is also expected to leverage demand for EbA loans.

Loan and guarantee facility to accelerate access to finance for other relevant EbA interventions for the Tempisque basin (Output 2.2): An EbA Lending Facility will be set at below-market conditions to enable financing for small- and large-scale EbA investments at farm-, enterprise- and household-level. A number of criteria for the selection of PFIs and of possible beneficiaries and EbA interventions to be financed is also predefined.

Building capacities of key actors (Output 1 and 3): To ensure sustainability and as a complementary activity of the Grant and Lending facility, trainings will be organized at the local and national level, based on identified needs with targeted stakeholders.

Mainstreaming EbA into key policy and planning frameworks (output 1 and 3): Technical assistance will be provided to strengthen the capacity of policy makers and decision makers and identify entry points of the integration of EbA in policy and regulatory frameworks at the local, national and regional level.

Monitoring and evaluation framework: The implementation strategy of an M&E framework and a proposed set of indicators are included in the intervention plan to support the communities in the established of an M&E framework during the first year of the programme.

113. * *Validation of Site-specific EbA plans*

A final step for validation of the site plans has been added for the pre-defined site plans in Costa Rica and Honduras where the initial year of implementation will be focused on strengthening the hydrological model and validating and confirming the specific sites for the grant interventions and the refinement of the eligibility criteria for the grant and loan component in consultations with communities, experts and the financial institutions.

B.4. Implementation arrangements

114. The proposed programme will be implemented over a period of seven years, from 2021–2028. The Central American Bank for Economic Integration (CABEI) will be the Accredited Entity (AE) for the programme, and will be responsible for overseeing the implementation, financial management, evaluation, reporting and closure of the programme. GCF and CABEI will enter into a single FAA in the form of a trust agreement.

Accredited Entity

115. As the AE, CABEI will oversee programme implementation, using its experience having successfully implemented similar programme activities involving financial intermediation across the region. Over the past 56 years, CABEI's support to the region has resulted in project approvals amounting to more than US\$30.9 billion and disbursements of more than US\$26.2 billion. CABEI will be responsible for financial management and will be accountable for the use of GCF resources under the programme. It will maintain programme accounts, facilitate staff recruitment and procurement processes and will monitor resource mobilisation of baseline as well as co-finance. Financial transactions will be subject to annual audits undertaken by internationally-certified auditors. In addition, as the AE, CABEI will: i) ensure that the programme is executed in accordance with GCF standards; ii) supervise, oversee and manage the implementation of programme interventions; iii) report on programme progress; and iv) ensure that programme activities are well coordinated and aligned with countries' national priorities. To ensure adequate oversight of the programme, CABEI will use its robust fiduciary, technical oversight and quality assurance systems such as its Evaluation Office, its Project Formulation Department (FOPROY) and its Financing for the Majorities Unit (FINAM) to closely support the Executing Entity (EE) and the Regional Project Management Unit (RPMU). Programme implementation by CABEI will also take place in close coordination with a Regional Project Steering Committee (RPSC).

Executing Entity

116. The Food and agriculture organization of the United Nations in collaboration with the Central American Commission for Environment and Development (CCAD) will serve as one of the Executing Entities for the programme through a Subsidiary agreement signed with CABEL. FAO will oversee the execution of the programme at the regional level, hosting the RPMU. At the national level, FAO will guide the day-to-day operations of the programme through National Project Execution Units (PEUs) based in each target catchment, as well as through national executing partners. FAO, through the PEUs, will also provide the technical execution and manage relations with stakeholders and institutions in the respective participating countries. CCAD will also sign a Subsidiary agreement with CABEL for the execution of output 3 ensuring the regional aggregation and dissemination of information to all countries in the region to strengthen the scientific information of the decision making processes at the regional and national levels.
117. The CCAD is the environmental division of the Central American Integration Secretariat (SICA) and has independent legal personality exercised by the Pro Tempore Presidency; the highest authority in the CCAD is the Council of Ministers, which is composed of representatives of the environmental authorities of each Member State. For administrative purposes, all contracts and operational agreements signed by CCAD are run through the Financial Administrative Directorate (DAF) of the General Secretariat of SICA (SG-SICA). SG-SICA provides accounting, purchasing and contracting services, and CCAD follows SG-SICA's regulations on procurement and HR management. The CCAD's mandate in the region includes: i) strengthening of national authorities and regulations; ii) harmonisation of policies and legislation; iii) distribution of information; iv) determining priority action areas; and v) promoting a participatory, democratic and decentralised environmental management. The CCAD is well-placed to execute this output of the programme as a result of its strong institutional presence and its direct links to each participating country. A detailed capacity assessment undertaken for the CCAD has confirmed their capacity to act as the EE for this output of the programme and can be found in annex 29. The CCAD is the coordinating body for Central America in terms of environmental management and climate change strategy and they have extensive experience executing projects with a wide range of donors, including the GEF, the European Union, the World Bank and the Interamerican Development. They also have the capacity to engage with governments at the highest level as they are the convening organisation for the environmental ministries in the region, including all countries participating in the programme.
118. The EEs will be accountable to CABEL as the AE for the effective execution of programme interventions. FAO, through the RPMU, will be responsible for ensuring technical and operational cohesiveness at the regional level and will report to the RPSC on the progress of the programme at both national and regional levels. All operating policies and procedures will follow CABEL's standards of operation. The EEs will work in close coordination with the RPSC and RPMU (described below).

Track record of Executing Entities

119. The FAO is an international organisation with the goals of eradicating food insecurity and malnutrition; eliminating poverty; and promoting sustainable management and utilisation of natural resources. The FAO supports countries in both mitigating and adapting to climate change through a range of programmes and projects based on research and best practices, as part of the 2030 Agenda and the Sustainable Development Goals. With national offices in seven countries of the regional programme and SICA region, the FAO maintains a close relationship with the Ministries of Agriculture and the Ministries of the Environment for implementation related to the Programming Frameworks set with countries. Technical assistance is provided to Executive Secretaries of both councils of ministries for regional implementation arrangements. Since October 2016, the FAO has been accredited by the GCF as an International Organisation for the category of medium projects (up to US\$250 million). It currently has a portfolio of US\$793.4 million at a global level, of which approximately US\$110.1 million is being executed by the FAO's Office for Mesoamerica. In addition, the FAO has made important efforts with countries to facilitate access to environmental financing with the Global Environment Facility (GEF). Projects are implemented both nationally and locally, with experience with local governments, farmers' and women's organisations, and using a multicultural approach.
120. The FAO promotes climate-resilient agriculture as a way to increase productivity, build the resilience of food systems and, wherever possible, reduce GHG emissions. Regionally-relevant examples of this work include:
- GCF FP145 REsilient LIVelihoods (RELIVE) of vulnerable smallholder farmers in the Mayan landscapes and the Dry Corridor of Guatemala, aimed to bolster the resilience of the most vulnerable farmers and indigenous groups (US\$66.7 million);
 - GCF FP089 Upscaling climate resilience measures in the dry corridor agroecosystems of El Salvador (RECLIMA). This programme seeks to restore and reforest degraded ecosystems to protect water sources and

stimulate aquifer recharge through adaptive agroecosystem management, simultaneously improving the resilience of small-scale farmers to climate change impacts (US\$127.7 million); and

- the FAO project 'Climate Smart Agroforestry Systems for the Dry Corridor of Central America' which supports farmers in Guatemala and Honduras to increase the uptake of climate resilient agroforestry systems through the farmer field school approach.

121. Other experiences the FAO has working in the region include:

- the 'Resilient landscape management project' (US\$4.3 million) and 'Strengthening the resilience of multiple-use protected areas to deliver multiple global environmental benefits' (US\$5.8 million) in Nicaragua;
- the 'Promoting climate-smart livestock management' project in the Dominican Republic (US\$1.5 million);
- the 'Protecting biodiversity and recovering degraded ecosystems' project (US\$1.8 million) in Honduras;
- the 'Rural Development with local governments' project (US\$1.5 million) and the 'Adaptation of vulnerability communities to climate change' project (US\$5 million) in Guatemala; and
- the 'Enable the concerted management of FUENTE AL MAR in the Paz river basin' project (US\$1.6 million) and the 'Systematization of agricultural good practices for adaptation and climate change' project (US\$400 million) in Guatemala and El Salvador.

The FAO also supports the implementation of the Forest and Farm Facility, which strengthens and empowers Forest and Farm Producers Organizations (FFPOs), including women, youth and indigenous peoples as primary change agents for climate-resilient landscapes and improved livelihoods (Guatemala, Nicaragua and Panama in Phase I).

122. The CCAD was established with the mission of developing a regional regime for environmental cooperation and integration that contributes to improving the quality of life of the populations of its Member States. The CCAD was appointed by the Tulum Declaration and ratified in Tulum +8 as the leader of regional efforts for the conservation and management of the Mesoamerican Region (MAR) as a shared transboundary ecoregion. The CCAD hosts the Regional Environmental Observatory (REO), a regional digital repository of environmentally-relevant information. Its actions have responded to the Environmental Plans of the Central American Region (PARCA).

123. The CCAD served as the executing agency for the 'Integrated Ridge-to-Reef Management of the Mesoamerican Reef' project in Belize, Guatemala, Honduras and Mexico (US\$9 million). It also served as an executing partner for:

- the 'Increasing Capacity for the Reduction of Flood and Drought Disaster Risks and Promotion of Resilience in Central America' project carried out in Belize, Guatemala, El Salvador, Honduras, Nicaragua, Costa Rica and Panama (US\$2 million); and
- the 'Regional Program for the Reduction of Environmental Depletion and Vulnerability' (PREVDA) in Belize, Guatemala, El Salvador, Honduras, Nicaragua, Costa Rica, Panama and the Dominican Republic (US\$20 million).

The CCAD also provided: i) technical support to the GEF project 'Strengthening of environmental integration for the establishment of the Environmental Plan for the Central American Region' (PARCA II) (US\$4 million) to strengthen the capacities of indigenous communities to protect and manage their natural and cultural resources; and ii) financial cooperation for the 'Consolidation of the Corazon Transboundary Biosphere Reserve' (CTBR) project (US\$12 million) for strengthening of the national system of protected areas in Nicaragua and Honduras.

Technical Advisor

124. UN Environment will play a technical advisory role for the programme through their participation in technical activities where they have a comparative advantage mainly in output 4 contributing with the experience gained through the microfinance for EbA initiatives implemented throughout the region, providing trainings to participating financial institutions, supporting them with the development of MRV systems and guiding the design of technical trainings. To this end, UNEP will be engaged by FAO through a UN to UN agreement for the implementation of programmatic activities agreed upon and in line with the content of the funding proposal. As EE, FAO will be accountable for the use of funds to the AE, and UNEP will be accountable to FAO.

125. UNEP will also contribute technically to the programme through some of the technical activities under output 3, by providing technical advice to CCAD for the development of technical knowledge products on financial mechanisms for EbA, methodologies for ecosystem services valuation and technical guidelines for EbA implementation, as well as by supporting the integration of EbA into relevant local and national policies and regional strategies. As EE for the output 3, CCAD will sign a Project Cooperation Agreement with UNEP, where UNEP will be the service provider for a set of defined goals and activities in accordance to the approved programme's proposal. UNEP will be accountable to CCAD for the use of funds and implementation of agreed upon activities.

Regional Project Steering Committee

126. The RPSC will comprise representatives from *inter alia*: i) FAO and CCAD as the EEs; ii) CABEL as the AE and EE; iii) United Nations Environment Programme (UNEP) – in a technical advisory role; iv) national-level CCAD liaisons; and v) the Regional Project Manager. The RPSC will primarily be responsible for providing technical, oversight and advisory support at the regional level. This will include: i) overseeing programme implementation; ii) review and validation annual regional workplans and programme reports; iii) review and validation of the programme's fulfilment of environmental and financial objectives; and iv) approval of changes to the programme's targets, activities or timelines. In addition, the RPSC will be informed about the final selection of Partner Financial Institutions (PFIs) that will be done by CABEL following their internal regulations and criteria. The RPSC will also make a recommendation on the final selection of recipients of grants through the Grant facility (these selection processes are described further below). The RPSC will be centrally placed to ensure consistent programme progress across all seven countries. Biannual RPSC meetings will be held⁷⁷ to take management-related and technical decisions, discuss the programme's main performance indicators and provide strategic guidance. The RPSC will also be responsible for overseeing the RPMU. In accordance with the AMA the Executing Entities will retain discretion over the use of proceeds and Programme implementation. EEs will have the final decision or a veto right over any recommendation or action taken by the RPSC.

Regional Project Management Unit

127. The RPMU will be based at FAO's regional headquarters in Panama and will be responsible for day-to-day programme administration and execution, and the regional technical coordination of programme activities. It will ensure that programme execution proceeds through clear work plans, terms of reference and carefully designed administrative arrangements that meet CABEL and GCF requirements. A full-time staff member will be recruited to fulfil the role of Project Manager (PM), who will head the RPMU and will be responsible for the daily on-the-ground implementation and management of the programme. The PM will be responsible for ensuring the programme achieves the targets set out in the Results Framework (Section E) to the required standards of CABEL and the GCF and within the specified time and budget allocations. To achieve these targets, the PM will: i) report directly to the RPSC on programme management-related matters; ii) manage the programme in accordance with the specified workplans and allocated budget; iii) ensure that all programme interventions are implemented according to CABEL and GCF guidelines; iii) work closely with PEUs and national institutions and authorities to ensure that the programme is managed effectively and that the needs of all beneficiary groups are considered; and iv) oversee the efficient and effective information and knowledge transfer to relevant programme partners. The PM will be required to fulfil these roles for the duration of the programme.

128. In addition to the PM, the RPMU will be comprised of the following individuals related to the regional management of the programme:

- Financial and Procurement Officer;
- Administrative Officer; and
- Accountant.

129. Regional technical specialists on the RPMU will include a: i) Chief Technical Adviser; ii) Forestry Expert; and iii) Water, EbA and Resource Efficiency Expert. There will also be a Regional Environmental and Social Safeguards Officer, a Regional Gender Officer, a Regional Monitoring and Evaluation Specialist; and a Regional Communications Specialist represented on the RPMU. A Regional Knowledge Hub Coordinator will be based in the CCAD Headquarters in El Salvador to ensure full integration of the programme results into the Regional Environmental Observatory.

National Advisory Committees

130. An NAC will be established in each of the seven participating countries. Each NAC will be comprised of a: i) NAC National Coordinator; ii) national climate change office/department representative; iii) Ministry of Environment representative; iv) Ministry of Agriculture representative; v) Ministry of Planning representative; and vi) representative from Local Government. The RPMU will strongly advise governments to nominate local representatives for the slots allocated for ministries in the NAC. The responsibilities of the NACs will include national programme guidance, planning and oversight, as well as supervision of national financial execution. The NACs will work closely with CABEL and the RPMU to coordinate activities under Output 4 of the programme. They will also participate in fostering national access to the three financial facilities established by the

⁷⁷ with *ad hoc* meetings held whenever necessary

programme. NACs will receive project proposals to the EbA Grant Facility during bi-annual project selection rounds and will coordinate with national ministries to ensure alignment with national policies, adaptation plans and, where applicable, respective regulations and laws. They will also, in coordination with the National Project Execution Units (PEUs), prioritise a selected number of projects in order to act as a first filter, before passing such proposals to the RPMU.

National Project Execution Units

131. National Project Execution Units (PEUs) will be established in the target catchments of each of the participating countries. Each unit will be comprised of: i) a National Project Execution Unit Coordinator (technical lead); ii) two EbA, water and resource technical advisors; iii) a Financial Officer to provide guidance on the three financial facilities; iv) an Administrative Officer; v) a Monitoring and Evaluation (M&E) Specialist; vi) a Communications Specialist; and vii) two researchers (one junior and one senior) to build country-specific evidence bases of the adaptation activities for the regional knowledge hub. PEUs will be responsible for daily execution of the programme in the respective target catchments, as well as communication and capacity-building activities with participating local governments, financial institutions, civil society organisations, farmers and communities. They will provide monitoring, reporting and evaluation services, as well as technical assistance and execution services. The EbA, water and resource experts will provide technical assistance to local governments, farmers and communities on implementing EbA, water- and resource-efficient interventions under the programme. The financial advisors will provide technical assistance to local financial institutions to support EbA, to learn how to use the financial mechanisms established under the programme, and for the valuation of natural assets. The PEUs will also be responsible for the identification of eligible projects for grant financing through the Grant facility. They will engage with local stakeholders at community, association, cooperative and municipal levels to formulate finance applications to be submitted to the NACs, which will subsequently make national selections and pass their recommendations to the RPMU. The M&E Specialists will define and implement the monitoring plan for the programme, and will collect the data required at national levels to track and measure the targets and indicators defined in the programme's results framework

Intermediary Financial Institutions.

132. CABEL network IFIs will be the main counterparts to channel GCF funds and CABEL Co-financing from the Lending Facility towards end beneficiaries. The IFIs will receive the funding for the Lending Facility from CABEL via two funding streams: via on-lending (full recourse) for co-financing from CABEL, and via limited recourse loan for GCF funds: both funding stream will be blended at IFI level and then on-lend to Partner Financial Institutions (PFIS) that will further on-lend to end beneficiaries. The IFIs are also the key partner for further leveraging of EbA finance via the intended Guarantee facility (see paragraphs 53 and 54).

Partner Financial Institutions

133. PFIs in each country will be identified through a process following CABEL's and Intermediary Financial Institutions internal rules and regulations, and through eligibility criteria that CABEL will approve via a no-objection. As its sole Executing Entity, CABEL will have final decision on the selection of each IFI, PFIs and end borrower to be financed under the Lending Facility. The final selection will be communicated to the RPSC. If there were a case in which there is no CABEL-accredited institution within the target catchment area, accredited institutions will be provided with technical assistance to offer an institutional loan to another non-accredited financial institution that can then directly lend to the end beneficiary. The PFIs will work closely with the PEUs to ensure interventions financed directly by the programme, and using the mechanisms established by the programme, are well-aligned with programme objectives. The functioning of the financial facilities and all institutions involved is further described below. Using the internal structures of CABEL, PFI's will be monitored through audits, analyses of relevance, sustainability, and cost-effectiveness of all credit lines, thereby ensuring they are framed within the specific objectives of the programme.

PFIs will be selected on the basis of the following criteria (to refined and confirmed by CABEL during inception of the programme). The criteria will apply throughout the period in which the PFIs participate in the programme.

- PFIs must be duly licensed in the respective countries and have at least three years of operation.
- The PFIs must be and remain in compliance with applicable laws and regulations issued by the local authorities as certified by independent external auditors on an annual basis and by quarterly returns provided by PFIs to the local regulatory authorities
- In case the initial eligibility of the PFIs falls to such a date that their year-end audits have been already completed and do not cover this requirement, then the PFIs would be required to submit a management letter in an acceptable format confirming their compliance with applicable laws and regulations issued by the authorities.

- PFI's board of directors and managers must be considered "fit and proper" by IFIs. It must have qualified and experienced management, adequate organization and institutional capacity for its specific risk profile.
- The PFI must have well defined policies and written procedures for management of all types of financial risks (i.e. credit, operational, liquidity and market risks).
- Operational self-sufficiency: The operational self-sufficiency of the PFI must be over 100% for the preceding financial year.
- The PFI must have adequate liquidity (based on the acceptable prudential ratios dictated by the national financial regulatory body of each country).
- The PFI must have positive profitability and acceptable risk profile (based on the acceptable prudential ratios dictated by the national financial regulatory body of each country).
- The PFI must classify its assets and off-balance-sheet credit risk exposures (at least four times per year) and make adequate provisions.
- The PFI must have adequate portfolio quality, with PAR $\geq$ 90 of at most 5% of total gross loans.
- The PFI must have adequate internal audits and controls for its specific risk profile.
- The PFI must have adequate management information systems, allowing it to separately and individually track all loans made with the EbA facility funds.
- The PFI must agree to engage in financing commercial farms, households, agroindustry and other enterprises.
- The PFI must agree to undergo intensive mandatory technical assistance

Where PFIs are not complying with the above defined minimum requirements, these can adhere to the TA facility to establish respective institutional capacities.

134. The objective of financial facilities is to enable financing — through both loans and grants — of proven EbA solutions and resource- and water-efficient technologies for vulnerable communities in the Dry Corridor and Arid Zones. Eligibility criteria for different activities of the three financial facilities are defined in the respective sections.

135. Further information regarding the operation of the financial facilities is presented in Annex 23: Operational Guidelines.

136. The figures below illustrate the above-described institutional arrangements for programme implementation (Figure 13) and the flow of funds (Figure 14) in diagram format.

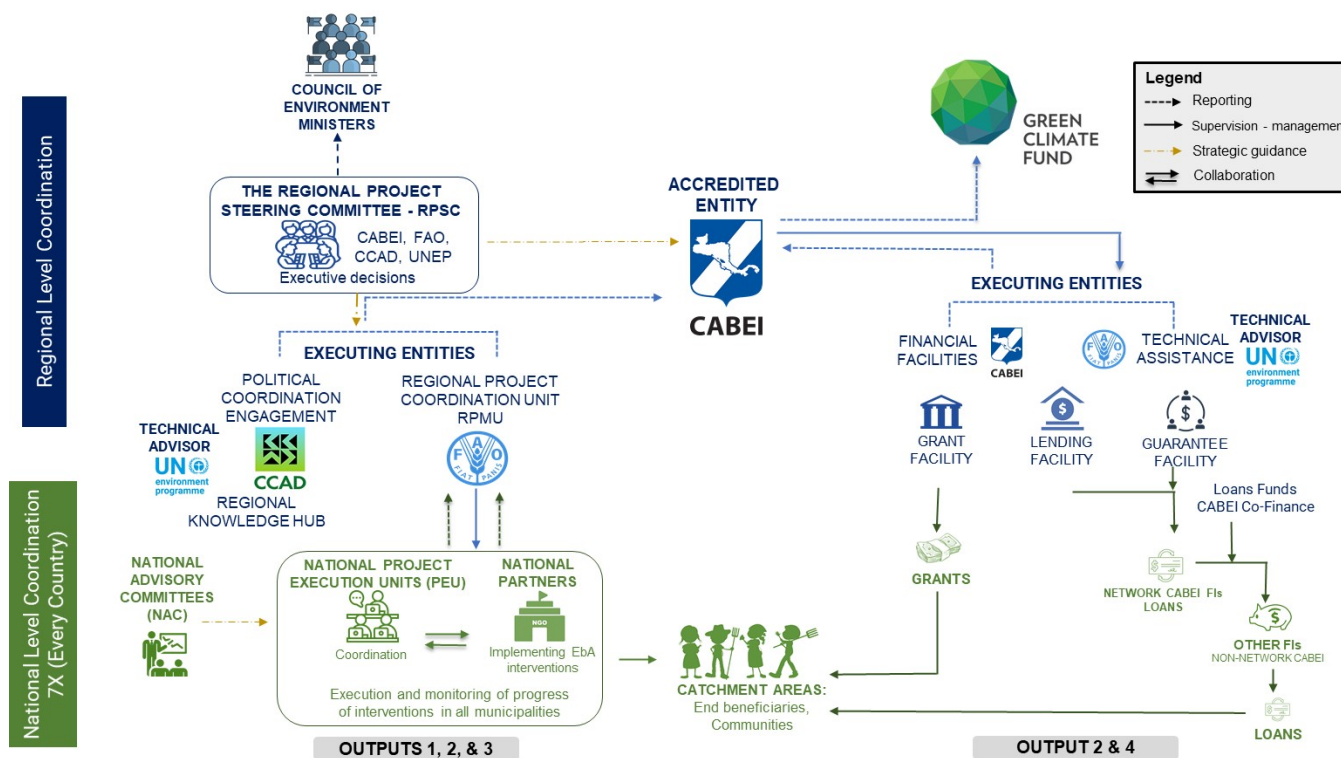


Figure 13. Programme management structure for the proposed programme.

As observed:

- CABEL will act as trustee for GCF funds for the overall Programme
- For the Lending facility CABEL will 1) channel GCF funds and provided limited recourse loans to the selected IFIs, each 1 per country and 2) on-lend its co-financing funds to these same IFIs. the IFIs will each select up to 4 PFIs to on-lend the blended funds (GCF and CABEL co-financing funds) in a two-staged approach: i) a first eligibility check based on a simplified format; and ii) a final selection and lending volume decision based on a fully-fledged due diligence process, including an initial gap analysis of PFI capacity for EbA lending. End beneficiaries can apply for EbA loans directly with the respective PFIs.
- The RPSC will be informed periodically about programme progress also through the selected PFIs in line with CABEL's regulations, as well as for EbA, water- and resource-efficient projects that can be financed under the financial facilities. The RPSC will also be provided with information on monitoring, reporting and verification (MRV) requirements to be complied with by PFIs and other programme entities in line with CABEL's internal regulations.
- National PEUs will work with local stakeholders to identify eligible projects to be financed by grants through the grant facility. They will compile programme proposals and filter out those that do not fulfil minimum requirements; the remainder will be passed to the NACs.
- NACs will undertake selection of eligible projects in each country according to national priorities and will pass their selection to the RPMU.
- The RPMU will undertake a selection of all projects proposed by each NAC and will pass their selection to the RPSC.
- The RPSC will make a preliminary selection of eligible projects at the regional level and will issue the list of grant recipients and final approval will be taken by CABEL.
- For loan applications, beneficiaries can apply for loans directly through PFIs.

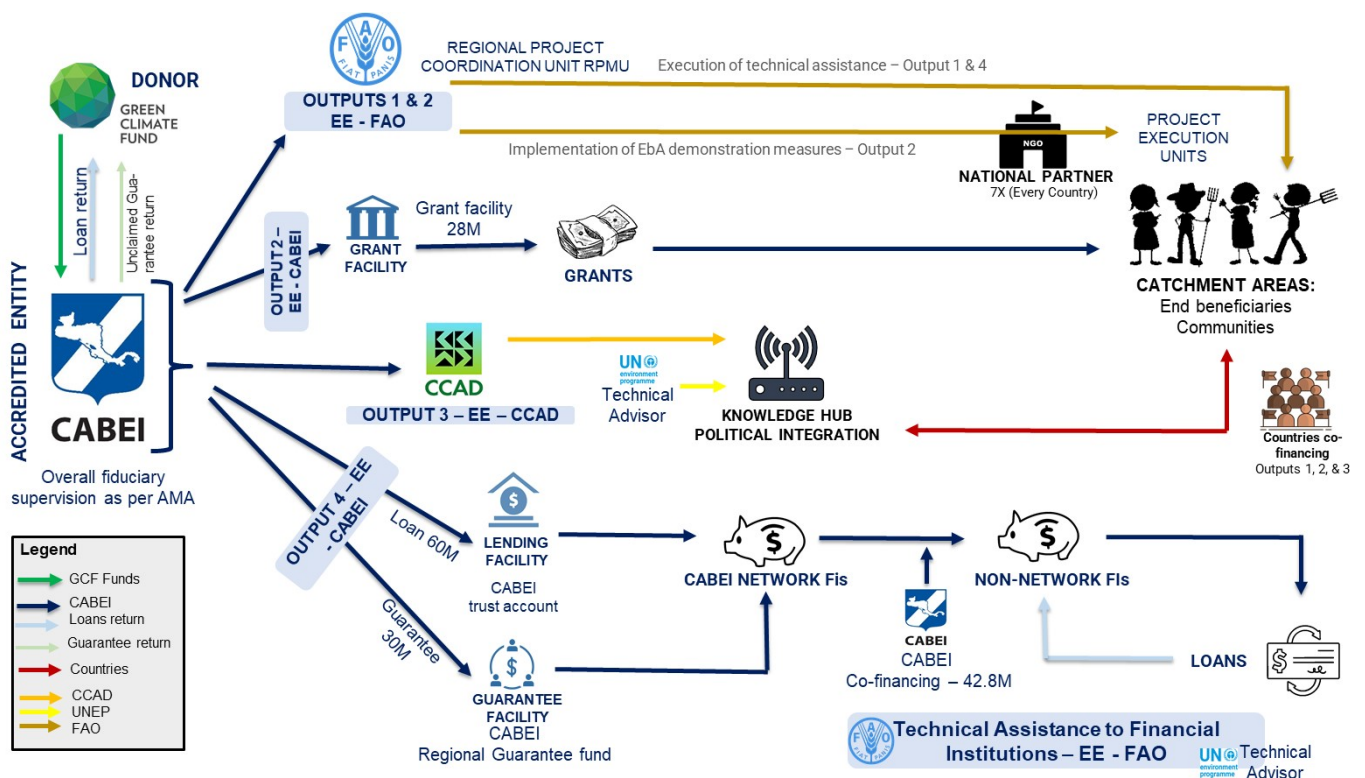


Figure 14. Flow of funds for the proposed programme

B.5. Justification for GCF funding request

137. The seven participating countries are extremely vulnerable to climate change impacts and have experienced severe agricultural and economic losses because of progressively more frequent and intense

droughts and floods in recent years (Table 1). In 2014, 2.5 million people were affected by food insecurity across Honduras, Guatemala and El Salvador as result of the loss of 75% of maize and bean crops⁷⁸, the impact of which was exacerbated by further droughts in 2015. In 2016, 1.6 million people across the Dry Corridor region were food insecure and 3.5 million required humanitarian assistance as a result of the loss of 50–90% of crops⁷⁹. In 2018, a heatwave across Central America caused crop losses across 282,000 hectares, affecting a total of 2.2 million people⁸⁰. A summary of the extreme climate events and associated impacts experienced by the seven participating countries over the last two decades is provided in Table 1 below. Droughts and other extreme climate events linked with these decreasing agricultural yields over the last decade have been more intense and have occurred more frequently than in previous decades. They are also likely to increase in frequency and intensity under future climate conditions.

Table 1. Total number of people affected and damages in US\$ millions for natural disasters occurring between 2000 and 2019⁸¹.

Country	Disaster	Number	Total people affected	Total damage (US\$ million)
Costa Rica	Drought	2		
	Flood	19	478,209	106
	Storm	4	117,574	205
El Salvador	Drought	4	1,486,610	186
	Flood	11	314,431	1,000
	Storm	11	198,145	1,315
Guatemala	Drought	5	5,680,081	158
	Extreme temperature	4	12,634	
	Flood	22	1,034,660	61.9
	Storm	12	987,775	1,639
Honduras	Drought	8	1,517,335	
	Flood	21	940,227	129
	Storm	11	234,636	217
Nicaragua	Drought	4	948,000	
	Flood	16	417,456	0.05
	Storm	14	324,912	3
Panama	Drought	3		272
	Flood	25	143,672	20.8
	Storm	2	24,000	
Dominican Republic	Flood	17	2,984,330	97
	Storm	23	336,429	58

138. The COVID-19 pandemic is imposing additional negative impacts on healthcare systems and economies globally, further reducing the capacity of already-vulnerable Dry Corridor and Arid Zone populations to respond to extreme climate events. With existing limited technical and financial capacities, these populations will require even greater relief interventions as their COVID-19 recovery efforts increase existing debt, while climate threats continue to build.

139. To overcome observed and predicted climate threats in the Dry Corridor and Arid Zones it is necessary to strengthen the technical and institutional capacity for EbA and to introduce novel financial mechanisms to fund adaptation. Currently, governments of the seven participating countries have insufficient resources, knowledge and technical capacity to promote innovative natural resource-based adaptation interventions. The resources required to effect a paradigm shift in the region's approach to natural resource management exceed those that can be supplied through public sector financing. The economic situations of the participating countries in the Latin America and the Caribbean (LAC) region are outlined in Table 2 below.

Table 2. Economic indicators for Guatemala, Honduras, El Salvador, Nicaragua, Costa Rica, Panama and the Dominican Republic. Respective values are shaded green–red to indicate best–poorest economic performance among these countries.

⁷⁸ United Nations Office for the Coordination of Humanitarian Affairs (OCHA), Regional Office for Latin America and the Caribbean, 2014. Drought in Central America Situation Report No. 01 (as of December 10, 2014).

⁷⁹ Food and Agriculture Organisation of the United Nations (FAO), 2016. Dry Corridor, Central America, Situation Report — June 2016.

⁸⁰ FAO, 2018. Central America, Drought causes crop losses in “Dry Corridor” in Central America. GIEWS update 27 August 2018.

⁸¹ Data sourced from EM-DAT: The Emergency Events Database - Université catholique de Louvain (UCL) - CRED, D. Guha-Sapir - www.emdat.be, Brussels, Belgium. Accessed 15 October 2019

Country	GDP per capita ⁸² in US\$ (global rank) ⁸³	GDP growth rate mean (%) (2013–2017) ⁸⁴	Public debt to GDP (ratio) ⁸⁵	Poverty (% population living under US\$ 3.20 per day) ⁸⁶	Human Development Index (global rank) ^{87,88}	Inequality index (Gini) ^{89,90}
Guatemala	8,173 (119)	3.6	30.1	24.2	0.65 (127)	53.0
Honduras	5,499 (134)	3.6	51.9	30.0	0.62 (133)	50.0
El Salvador	8,934 (112)	2.3	68.1	10.3	0.67 (121)	40.0
Nicaragua	5,823 (130)	4.8	46.0	12.8	0.66 (124)	47.1
Costa Rica	17,149 (76)	3.4	66.2	3.8	0.79 (63)	48.7
Panama	24,262 (58)	5.6	38.8	7.0	0.79 (66)	50.4
Dominican Republic	16,956 (78)	6.1	47.1	6.2	0.74 (94)	45.3

140. Although countries in the region are classified from lower-middle to high income based on GDP per capita, their national budgets are constrained, with poverty eradication and sustainable development having been prioritised for funding over CCA. For example, less than 3% of the national budget has been committed for mitigation, adaptation and risk management related to climate change in Guatemala, El Salvador, Nicaragua and Honduras in recent years.

141. Across the LAC region, there is a high dependency on non-reimbursable international support to meet CCA goals. Where the availability of international finance has been limited, countries have had to increase international debt or claim loans in order to respond to climate change impacts. There is a considerable challenge to be met in that the finance available for CCA is far below what is required to meet adaptation needs. For example, a UNDP IFF study⁹¹ in 2011 estimated an increase in the annual cost of adaptation in Costa Rica for the biodiversity sector alone from US\$60 million in 2015 to US\$76 million by 2030⁹². Similarly, in the Dominican Republic, annual adaptation costs for the tourism sector are predicted to increase from US\$16 million in 2015 to US\$57 million by 2030. Aggregate estimates for the LAC region place the cost of adaptation to a 2.5°C rise in temperature at 1.5–5.0% of the region's GDP. The total annual cost for adaptation across all sectors in the LAC region is predicted to increase under both wet and dry climate scenarios up to 2050 (Figure 12). Safeguarding water supplies, implementing flood protection measures and effective management of coastal zones will comprise the bulk of this cost⁹³.

⁸² purchasing power parity

⁸³ IMF, 2017. World Economic Outlook Database. Available at: <http://www.imf.org/external/ns/cs.aspx?id=28>. 2019.

⁸⁴ World Bank, 2019. GDP growth (annual %). Available at: <https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG>.

⁸⁵ CIA Factbook. 2017 estimations. Available at: <https://www.cia.gov/library/publications/the-world-factbook/rankorder/2186rank.html> source.

⁸⁶ World Bank, 2019. World Development Indicators 2017.

⁸⁷ The Human Development Index (HDI) is a composite statistic of life expectancy, education, and income per capita indicators.

⁸⁸ UNDP, 2018. Human Development Report 2018, pp. 22–25. Available at:

http://hdr.undp.org/sites/default/files/2018_human_development_statistical_update.pdf.

⁸⁹ The Gini coefficient is a measure of inequality of income or wealth. Higher numbers indicate greater inequality.

⁹⁰ Nicaragua and Guatemala: estimations from CIA Factbook. 2014. Available at: <https://www.cia.gov/library/publications/the-world-factbook/rankorder/2172rank.html>.

Data for other countries from World Bank, 2016. Available at: <https://data.worldbank.org/indicator/SI.POV.GINI>

⁹¹ UNDP 2011. Adaptation to climate change in Costa Rica: An assessment of necessary investment and financial flows. Project: Capacity Development for Policy Makers to Address Climate Change. Available online at: <http://www.undpcc.org/en/financial-analysis/results>.

⁹² focusing on conservation of terrestrial and marine ecosystems, conservation of inland aquatic ecosystems, prevention of forest fires, and awareness raising

⁹³ Sir Iliff, William. 2010. The costs to developing countries or adapting to climate change: new methods and estimates - the global report of the economics of adaptation to climate change study (English).

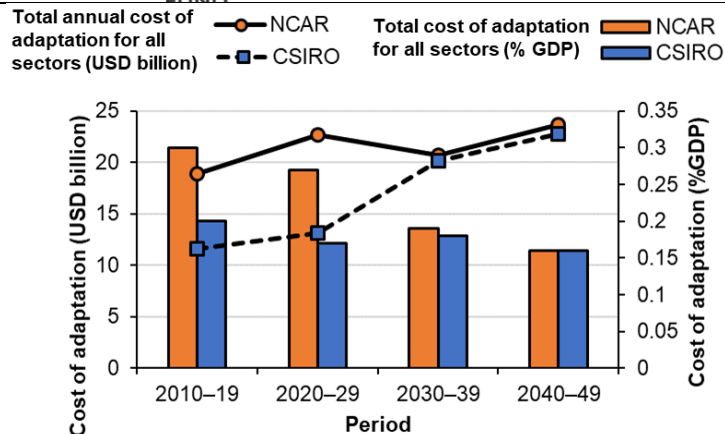


Figure 15. Total annual cost of adaptation ⁹⁴ across all sectors in the Latin America and the Caribbean (LAC) region under a wet (NCAR)⁹⁵ and dry (CSIRO)⁹⁶ scenario. Points indicate total annual costs in US\$ billion based on 2005 values. Bars indicate total annual costs as a percentage of LAC regional GDP.

142. Given this constrained fiscal space, there are limited opportunities to implement climate change adaptation interventions in the region without additional assistance. The seven target countries have already extended their largest possible contributions to finance adaptation and further support is needed to fully address the climate problem. While scope has been identified for partial loan financing at the regional level, the extent of public debt in the individual countries, as well as their sovereign credit ratings (most graded as speculative with significant risk), constrain their borrowing capacities with regards to global recommended borrowing limits. As a result, the countries are unable to take on more sovereign debt to finance adaptation directly, instead relying on alternative financial mechanisms. In addition, and at a local scale, the most vulnerable farmers are often considered to be too high a risk for loans because of their small farm sizes and limited collateral which is unfavourably viewed by banks using conservative practices. This limits these individuals' access to financial resources needed for climate change adaptation.

143. To overcome these constraints, the proposed programme is requesting loan finance to establish a credit line for EbA in the region, with additional guarantees to de-risk investments and support uptake of EbA financing by small holder and commercial farmers. In addition to the loan components, GCF grant financing is requested to establish and administer the credit line and to provide technical assistance for the implementation of EbA. Grant finance is also required to establish a trust fund by CABEL that will support bottom-up implementation of EbA in the most vulnerable communities as well as to support direct implementation of climate change adaptation interventions in the most vulnerable communities that are not in a position to access loan finance.

B.6. Exit strategy and sustainability

144. The sustainability and replicability of the proposed programme interventions will be ensured through six elements in the programme design, namely: i) operations and maintenance; ii) building long-term community ownership; iii) integrating EbA into catchment management and planning; iv) incentivising private sector investments into EbA; v) developing financial instruments for EbA; and vi) monitoring and evaluation. The details of each of these elements are described below.

145. **Post-programme operations and maintenance.** Relevant operations and maintenance (O&M) costs for programme interventions during and post programme implementation are detailed in Annex 21: Operations Manual. Programme stakeholders are committed to undertaking O&M not only during the programme, but also after programme completion. For example, the regional knowledge hub that will be established under the programme will be maintained through the integration and linking of its content with the existing Regional Environmental Observatory (OAR) managed by the CCAD. Because the knowledge hub will be web-based and linked to the OAR, it will not require physical maintenance post programme closure. However, the technical

⁹⁴ Sir Iliff, William. 2010. The costs to developing countries of adapting to climate change: new methods and estimates - the global report of the economics of adaptation to climate change study (English). Washington, D.C.: World Bank Group.
<http://documents.worldbank.org/curated/en/360351468127796135/The-costs-to-developing-countries-or-adapting-to-climate-change-new-methods-and-estimates-the-global-report-of-the-economics-of-adaptation-to-climate-change-study>

⁹⁵ National Centre for Atmospheric Research (NCAR), wettest scenario

⁹⁶ Commonwealth Scientific and Industrial Research Organization (CSIRO), driest scenario

specialists engaged to maintain the OAR will ensure the sustainability of the hub and the updating of its information beyond the programme implementation period. In addition, the knowledge hub will benefit from other centralised information generation projects at the OAR, such as GIZ and NASA programmes, ensuring ongoing consolidated outputs. Other aspects of the programme requiring O&M include financial products and services, EbA interventions, and water- and resource-efficient technologies. These are described in detail in the O&M plan (Annex 21).

146. **Building long-term community ownership.** By engaging closely with target communities — particularly environmental and women's organisations — throughout the programme design and implementation process, the programme will instil a sense of ownership in these communities, which will contribute to gender equality and the long-term sustainability of programme interventions after programme closure. Sub-activities 1.2.6 and 3.1.4 specifically target local cooperatives and women's organisations, respectively, by: i) establishing a training of trainers' programme for the implementation of EbA and water-efficient technologies; and ii) strengthening the technical capacity of community organisations, including women's groups, in revising and disseminating information products to support the adoption of EbA and other climate-resilient practices. Community involvement in programme interventions is also particularly important in Sub-activity 1.1.1 — establishing a community-level monitoring and evaluation committee in each municipality — as well as in Sub-activity 1.3.3 — establishing livelihood training projects in each municipality for farmers and rural communities on locally-appropriate natural resource-based businesses and climate-resilient livelihoods. Communities will further be closely involved in the co-development of all site-specific intervention plans to integrate EbA measures into the seven target catchments (Activity 1.1).
147. **Integrating EbA into catchment management and planning.** Training, monitoring, evaluation and the curation of lessons learned and best practices from programme activities will provide national- and local-level decision-makers with relevant knowledge and information to adopt climate-resilient policies and incentives in the long-term. Output 3 is specifically designed to ensure the information generated through programme activities is disseminated across the region to increase the capacity of institutions to implement EbA interventions, building a strong knowledge-sharing platform. Activity 3.3 will particularly enhance capacity of local-level policy makers to integrate climate change adaptation and the valuation of natural capital into local policies, which will enhance the sustainability of proposed interventions beyond the programme's lifespan.
148. **Integrating EbA into local and national policies and regional strategies.** The programme will support the integration of EbA into local policies focusing mainly on land use management plans, water management plans and water management organizations mainly through technical assistance and trainings under components 1 and 3. The programme will also provide inputs directly to the NAP processes currently under development for most countries in the region and focusing on the sectoral policies for agriculture, water security and ecosystems and forestry which have been prioritized by countries in their NAP processes. At the regional level, the results from the programme gathered in the Regional Observatory will serve as a direct input for the update and strengthening of the regional strategies for Environment, Climate Change, Water Resources and Agriculture all under the SICA mandate.
149. **Incentivising private sector investments into EbA.** The programme will leverage private sector investments into EbA through capacity-building and awareness-raising interventions targeting the private sector. Activities under Output 3 will build the private sector's awareness of the potential benefits of EbA, while Output 1 and 4 will provide financial products and technical assistance for businesses to implement EbA. For example, training commercial farmers on sustainable EbA practices (Sub-activity 1.2.2) will demonstrate the benefits of EbA for ecosystem-based businesses and facilitate private sector investments in these practices as well as in restoration/conservation of ecosystems on which these businesses rely in the long-term. The financial instruments established under Output 4 will further support these farmers to access finance for such investments. By creating a framework for the valuation of ecosystem services, the programme will further facilitate the quantification of long-term economic benefits of large-scale EbA interventions and contribute to the long-term incentive of investing into EbA in the region.
150. **Developing financial instruments for EbA.** The provision of financial mechanisms under Output 4 of the programme for the implementation of EbA and water-efficient technologies, and the comprehensive monitoring of the benefits of these systems, will ultimately result in end beneficiaries who are implementing EbA and other adaptation measures displaying a lower risk profile than individuals who are not. The programme

will showcase commercial EbA finance as an attractive alternative, especially for impact investors, and will catalyse the influx of private commercial finance into this market post programme closure. As a result, the operations of the financial facilities established under the programme will be strengthened and expanded across the Dry Corridor. Indeed, the programme's target areas represent only ~6% of the region's potential sensitised demand for EbA finance; there is therefore great potential for further expansion. Further details regarding the financial operations post programme closure are presented in Annex 12: Operational Guidelines.

151. **Monitoring and evaluation.** Lessons learned from programme activities will be collated by an M&E Specialist in the Regional Project Management Unit and made available to development planners through a regional knowledge hub. In addition, the programme will establish a community-level monitoring and evaluation committee in each municipality and build the capacity of these committees to monitor the biophysical, social and economic conditions in their local catchment. This capacity building will be complemented by assessments of the level of climate vulnerability, risks and opportunities (Sub-activity 1.1.1). Lessons learned from the M&E will be made available to community-level monitoring and evaluation committees to improve the sustainability of the interventions while the committees' local knowledge will, in turn, inform the M&E. In addition, monitoring, reporting and verification (MRV) systems established under Activity 4.3 will enable PFIs to monitor and assess the implementation of EbA and resource- and water-efficient investment projects, as well as to monitor and document their respective economic and ecosystem benefits, thereby encouraging and facilitating further uptake across the region.

FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency			
		174.3		million USD (\$)			
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	US\$60 million	<u>10 years</u>	3	<u>0.75 %</u>		
(ii)	Subordinated loans	0			Enter %		
(iii)	Equity	0			Enter % equity return		
(iv)	Guarantees	US\$30 million	<u>10 years</u>				
(v)	Reimbursable grants	0					
(vi)	Grants	US\$84.3 million					
(vii)	Results-based payments	0					
(b) Co-financing information		Total amount		Currency			
		94.1		million USD (\$)			
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
CABEI		Senior Loans	42.8	million USD (\$)	<u>05</u> <u>2</u>	<u>3.50%</u> <u>fixed</u> ¹⁹⁷	<u>senior</u>
Guatemala		<u>In kind</u>	9.3	million USD (\$)	Enter years Enter years	Enter%	Options
Honduras		<u>In kind</u>	7	million USD (\$)	Enter years Enter years	Enter%	Options
El Salvador		<u>In kind</u>	7.1	million USD (\$)	Enter years Enter years	Enter%	Options
Nicaragua		<u>In kind</u>	10.3	million USD (\$)	Enter years Enter years	Enter%	Options

⁹⁷ Till end of 2021 LIBOR will be the reference rate. Based on the quarterly review by CABEI's Assets and Liabilities Committee (ALCO) this will be adjusted towards the end of 2021. Potentially it will then change to SOFR.

Costa Rica	<u>In kind</u>	8.4	million USD (\$)	Enter years Enter years	Enter%	Options
Panama	<u>In kind</u>	4	million USD (\$)	Enter years Enter years	Enter%	Options
Dominican Republic	<u>In kind</u>	5.2	million USD (\$)			
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	268.4			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	The programme will also leverage in kind and cash local contribution to match with GCF grant contribution. The specific amount will be calculated once the criteria set on grant facility are defined.					

C.2. Financing by component

Output	Activity	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1. Mainstreaming of EbA, water- and resource-efficient technologies and natural resource-based businesses into selected catchments							
Output 1. Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures.	1.1	\$ 9.1 M	\$ 6.1 M	Grants	\$ 3 M	In-kind	Countries' co-financing
	1.2	\$ 2.7 M	\$ 1.4 M	Grants	\$ 1.3 M	In-kind	Countries' co-financing
	1.3	\$ 3.3 M	\$ 2.6 M	Grants	\$ 0.8 M	In-kind	Countries' co-financing
Output 2. Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.	2.1	\$ 59.6 M	\$ 23.4 M	Grants	\$ 36.2 M	In-kind	Countries' co-financing
	2.2	\$ 6.7 M	\$ 2.8 M	Grants	\$ 3.9 M	In-kind	Countries' co-financing
	2.3	\$ 30.3 M	\$ 30.3 M	Grants	Enter amount		
Output 3. Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies.	3.1	\$4.5 M	\$ 3.6 M	Grants	\$ 0.9 M	In-kind	Countries' co-financing
	3.2	\$2.4 M	\$ 0.7 M	Grants	\$ 1.7 M	In-kind	Countries' co-financing
	3.3	\$ 3.2 M	\$ 2 M	Grants	\$ 1.2 M	In-kind	Countries' co-financing
Component 2. Financing and implementation for EbA as well as water- and resource-efficient technologies across selected catchments							
Output 4. Financial products and services to finance EbA investments are offered by Partner	4.1	\$ 1.1 M	\$ 1.1 M	Grants			
	4.2 (sub-activity 4.2.1)	\$ 102.8 M	\$ 60 M	Senior loans	\$ 42.8M	Senior Loans	CABEI

Financial Institutions (PFI), including PFI access to EbA on-lending funds and support mechanisms.	4.2 (sub-activity 4.2.2.)	\$ 30 M	\$ 30 M	Guarantees			
	4.3	\$ 5.4 M	\$ 5.4 M	Grants			
M&E Costs (Million USD)		\$ 0.4 M	\$ 0.4 M	Grants			
Project Management Costs (Million USD)		\$ 6.2 M	\$ 3.8 M	Grants	\$ 2.4 M	In-kind	Countries' co-financing
Contingency		\$ 0.7 M	\$ 0.7 M	Grants			
Indicative total GCF grant cost (Million USD)		\$ 84.3M					
Indicative total cost (Million USD)		\$ 268.4 M	\$ 174.3	\$ 94.1 M			

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☒

152. Capacity building will be provided through several programme activities. Activity 4.3 will provide technical assistance (TA) to strengthen technical capacity of CABEL network as well as non-regulated financial institutions (partner financial institutions) to access and channel funds for small- and large-scale EbA investments. Under this activity, selected partner financial institutions will receive capacity building services for the adequate promotion and managing of EbA finance to end beneficiaries, such as data management for the measurement, monitoring, reporting and verification (MRV) of climate risks and adaptation strategy impacts or effects, the implementation of EbA-oriented awareness raising and marketing campaigns with end beneficiaries and the development of financial and non-financial products and services geared towards EbA investments. Where applicable, the programme will support the integration of external data management solutions to enhance EbA-oriented MRV systems with existing IT infrastructure such as core banking systems. Activity 1.2 will train commercial farmers on sustainable EbA practices, including silvopasture, agroforestry and sustainable land management (SLM) (US\$73,500 under sub-activity 1.2.2) as well as local communities on EbA and water-efficient technologies, including visits to demonstration sites to facilitate farmer to farmer knowledge exchange (US\$147,000 under sub-activity 1.2.7). Local communities will also be trained on locally appropriate natural resource-based businesses and climate-resilient livelihoods (Sub-activity 1.3.3; US\$276,500).

153. Policy- and decision-makers at the municipal level will be trained on climate change impacts, including the impacts of gender inequality, and the role of EbA in mitigating climate change impacts to support political and technical decision-making for climate resilience in the Dry Corridor and Arid Zones (Sub-activity 3.1.1; US\$147,000). This will be supported by building the capacity of these institutions to use the protocols and criteria for the adoption of EbA and implementation of economic incentives for SLM (Sub-activity 3.3.3; US\$275,250). Finally, the technical capacity of community organisations, including women's groups, and local champions will be strengthened for developing, revising and disseminating information products to support the adoption of EbA and other resilient practices (Sub-activity 3.1.4: US\$147,000).

154. Technology transfer will be conducted through Activity 3.1, where a regional knowledge management hub will be established (US\$247,500). Output 2 will also include on-the-ground implementation of EbA activities as well as water- and resource efficient technologies (US\$ 26,106,263).

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

155. The proposed programme will contribute to the achievement of GCF's fund-level adaptation objective of increased climate-resilient sustainable development by facilitating wide-scale investment in EbA and water-efficient technologies across the Dry Corridor of Central America and Arid Zones of the Dominican Republic. In achieving this objective, the programme will contribute to several indicative assessment factors in the GCF Performance Measurement Framework to ensure that GCF funds achieve the desired impact in the region. This will include contributions to fund-level impacts A1.0 — increased resilience and enhanced livelihoods of the

most vulnerable people, communities and regions and A4.0 — increased resilience of ecosystems and ecosystem services.

156. In addition to the fund-level impacts, the programme will contribute to three fund-level outcomes. The specific contribution to each outcome is presented below. Additional details of how the programme interventions interconnect to achieve these fund-level impacts and outcomes are presented in Section B.2. — Theory of Change.
157. *A5.0 — Strengthened institutional and regulatory systems for climate-responsive planning and development.* The programme will build the institutional capacity of local governments in the seven countries to mainstream EbA and other climate-resilient practices into local development plans and policies. This will include facilitating the establishment and operationalisation of financial incentives to promote investment in climate change adaptation.
158. *A7.0 — Strengthened adaptive capacity and reduced exposure to climate risks.* The adoption of EbA and water-efficient technologies across the target sites will reduce the impacts of climate change on vulnerable communities, particularly regarding water scarcity. For example, the combination of improved land-use practices and the adoption of natural resource-based businesses and alternative climate-resilient livelihoods — with technical and financial support provided through the programme — will improve the adaptive capacity of smallholder farming communities under future climate conditions, reducing the impacts of shifting rainfall patterns and more regular droughts. The hydrological modelling undertaken for the basins in Honduras, Costa Rica and the Dominican Republic allow for an estimation of the impact potential of this programme obtained from the implementation of combined EbA interventions at the catchment level. The results of this analysis are detailed in Annex 2, but in general terms, the analysis concludes that a combination of interventions results in substantial improvements for all key hydrological variables in the catchment areas. Soil humidity increases from large losses under the climate change base scenario to positive values under EbA combinations; this results in benefits for water availability both for agriculture and forests that draw water from the top soil layers. The changes in water production (surface water) and percolation (water recharge) show even greater benefits from the implementation of this combination of EbA measures. These values are key for ensuring water availability throughout the year from a scenario of significant water deficit under climate change conditions. Lastly, the implementation of this combination of EbA measures also shows very positive impacts on soil erosion rates.
159. *A8.0 — Strengthened awareness of climate threats and risk-reduction processes.* The proposed programme includes a strong focus on knowledge and awareness as key concepts for building the resilience of vulnerable communities. In particular, the programme will build an evidence base for the effectiveness of EbA and disseminate this information to local decision-makers, private sector investors (including commercial farmers) and local communities.

Adaptation Impact

160. The proposed programme's contribution to climate-resilient sustainable development will directly benefit ~1,084,441 people (30% women) across seven catchment areas — one in each of the participating countries through the proposed interventions. Approximately 134,694 of these people will also directly benefit from the provision of financial mechanisms under Output 4 (lending and guarantee facilities). The programme will also indirectly benefit 1,364,044 people across the region by facilitating increased access to finance for natural resource-based businesses, mainstreaming EbA and other adaptation practices into national policies and building the capacity of governments to implement EbA. The financial mechanisms established under this programme will make ~US\$189.2 million available for investment in EbA and water-efficient technologies in the region, introducing the potential for the restoration of ~142,500 hectares of degraded agricultural land through the implementation of sustainable land use practices. In addition to the technical support and establishment of financial mechanisms, the programme will actively: i) restore ~7,800 hectares of forested areas; ii) protect ~4,100 hectares of forest; iii) establish 36 tree nurseries; iv) establish ~1,900 hectares of agroforestry systems; v) establish ~100 km of firebreaks, living barriers and superficial drainage solutions, respectively; and vi) develop ~5,300 hectares of sustainable fuelwood sources. Finally, at least 100 community buildings will be equipped with rainwater harvesting systems, 156 community-level rainwater reservoirs will be installed, 43 half-orange kilns will be made communally available, and 38 community-level solar-powered water pumping stations will be installed.

D.2. Paradigm shift potential

161. The proposed programme will catalyse a paradigm shift in the regional approach to catchment and water demand management in the Dry Corridor of Central America and the Arid Zones of the Dominican Republic. This shift will see governments promoting large-scale EbA, and the private sector (including commercial farmers and entrepreneurs in rural communities) having access to the financial resources and technical skills required to implement impactful adaptation interventions. The transformative effects of the programme will be achieved through several innovative approaches.
162. The proposed EbA approach is inherently scalable, not only within the targeted catchments, but also across the entire Dry Corridor and Arid Zones. The programme has been specifically designed to create an enabling environment that capitalises on the scalability of EbA, focusing on three core areas of support: financial; policy; and technical. The details of these three areas of support are provided below.

Creation of an enabling environment for investment in EbA and water-efficient technologies

163. The proposed programme will diversify financial systems in the region to create an enabling environment for investment in EbA and water-efficient technologies. This will be based on the establishment of several financial instruments with varying entry-points to make climate finance more accessible to vulnerable communities. In particular, the programme will establish: i) a trust fund by CABEL to support bottom-up promotion of local EbA activities through non-reimbursable financing; ii) an EbA lending facility to channel funds for small- and large-scale EbA investments at farm-, enterprise- and household-level; and iii) a guarantee facility to further de-risk lending to the target beneficiary groups for PFIs. The enabling environment for investment will be further enhanced by building the technical and institutional capacity of PFIs and nonregulated financial institutions to finance and re-finance EbA and water-efficient technologies through all three of the proposed financial instruments.

Contribution to the regulatory framework and policies

164. The enhanced local evidence-base for EbA that will be created through the knowledge hub and the monitoring and evaluation of programme activities will support national and local-level decision-makers to implement policies and incentives for EbA, in particular through improved valuation of natural capital. Decision-makers within local government institutions will be trained to assess and update existing regulatory frameworks and policies to facilitate the mainstreaming of EbA — including making provision for incentives programmes targeting the adoption of EbA practices. Furthermore, training and technical capacity-building activities are included in all programme components. By providing technical support to local governments for the adoption and implementation of climate-resilient land management practices, the programme will help mainstream climate adaptation into local-level development planning. This will include the development of catchment-level land use plans, which will guide the adoption of EbA and sustainable land management practices beyond the scope of the programme's on-the-ground interventions.

Technical support for the enhancement of regional knowledge and learning

165. The proposed programme includes several strategies for knowledge-sharing and learning. This support will include the establishment of a regional knowledge hub through which tailored knowledge products will be created and disseminated. This, combined with the strengthened policy frameworks described above, will catalyse a shift in how decision-makers approach land-use and development planning. In addition, the social, economic and environmental impacts of programme interventions, including the cost-effectiveness of EbA, will be measured through ongoing monitoring and evaluation. These findings will be disseminated to policymakers and decision-makers to catalyse further uptake of EbA and water- and resource-efficient technologies, as well as the enabling financing mechanisms and economic incentives. The programme will also build awareness of the benefits of EbA and the associated financing mechanisms with private investors, as well as local and national governments.

Upscaling of adaptation

166. The programme will catalyse the autonomous upscaling of EbA interventions and water-efficient technologies by combining policy instruments (e.g. economic incentives such as payments for environmental services) with accessible financial mechanisms (e.g. commercial loans, microfinance and guarantee funds). These mechanisms will transform the financial landscape in the region to enable vulnerable groups to access climate finance. Furthermore, the programme will promote the development of climate-resilient, natural

resource-based businesses and alternative livelihoods at the community level. These businesses and alternative livelihoods will be directly linked to EbA activities and the water-efficient technologies promoted by the programme ensuring long-term sustainability. Through comprehensive monitoring, reporting and verification of the benefits realised from the investment in EbA and other adaptation measures, the programme will showcase EbA finance as an attractive alternative for investors, resulting in the expansion of these operations outside the programme's target areas. Indeed, the target programme areas represent only ~6% of the potential demand for EbA finance in the whole of the Dry Corridor and Arid Zones. In addition, the regional replication potential of the programme's interventions is supported by the fact that similar interventions have been identified as appropriate adaptation solutions in countries across the region. For example, an analysis of the Latin America and Caribbean Region⁹⁸ determined that, while it is difficult to prescribe specific national actions based on regional analysis, a general strategy for adaptation, agricultural development and poverty reduction should be based on: i) sustainable management of resources, including dissemination of proven technologies for farmers; ii) improved access to resources, including policies that offer incentives for sustainable land management; and iii) increased competitiveness of smaller farms, including lowering barriers to entry for small businesses. Similarly, a number of initiatives across the region have implemented interventions similar to those proposed in this programme, such as: i) restoring and reforesting degraded ecosystems to protect water resources and stimulate aquifer recharge⁹⁹; ii) introducing innovative agroforestry practices, and improving irrigation and resource technologies¹⁰⁰; iii) establishing a risk-sharing facility to support climate-smart agriculture investments¹⁰¹; and iv) strengthening capacity in financial institutions to develop EbA loans and other services¹⁰². The combined impact of the programme's interventions will be to catalyse EbA beyond the programme's target catchments — incorporating elements of sustainability, scalability, replicability, knowledge-sharing, the creation of an enabling environment for private sector investors, and policy improvements in the programme design.

D.3. Sustainable development

167. The proposed programme will contribute to the achievement of nine of the 17 UN Sustainable Development Goals (SDGs). The main contributions of the programme will be to: SDG 6 – Clean water and sanitation; SDG 11 – Sustainable cities and communities; SDG 13 – Climate action; and SDG 15 – Life on land. In addition, the programme will contribute to: SDG 1 – Poverty reduction; SDG 7 – Affordable and clean energy; SDG 8 – Decent work and economic growth; SDG 9 – Industry, innovation and infrastructure; and SDG 10 – Reduced inequalities. The expected positive environmental, economic, social and gender impacts of the programme are outlined below, as well as the contribution of programme activities to the region's recovery from the COVID-19 pandemic.

Environmental co-benefits

168. The conservation and restoration of ~119,200 hectares of degraded and climate-vulnerable ecosystems — through reforestation, forest management, agroforestry systems and sustainable fuelwood practices — as well as the use of water- and resource-efficient technologies will result in several environmental co-benefits. These co-benefits are listed below.

- EbA interventions, including the conservation and/or restoration of forests and agroforestry systems, will act as natural buffers, controlling and mitigating the impact of floods, droughts, extreme temperatures and wildfires on ecosystems in selected sites.
- Conserved and restored ecosystems will provide habitats for a range of terrestrial species, thereby improving biodiversity on programme sites.
- EbA interventions under the programme will enhance soil fertility through accumulation of organic matter in revegetated areas, as well as enhancing carbon storage capacity and reducing soil erosion at programme sites. Conserved and revegetated areas will further enhance water infiltration and soil moisture retention, improving

⁹⁸ Dixon, J., Gulliver, A. and Gibbon, D. 2001. *Sistemas de Producción Agropecuaria y Pobreza: Cómo Mejorar los Medios de Subsistencia de los Pequeños Agricultores en un Mundo Cambiante*. Hall, M. (editor). FAO and the World Bank. Available online at: <http://www.fao.org/3/a-y1860s.pdf>

⁹⁹ Project: Upscaling climate resilience measures in the Dry Corridor agroecosystems of El Salvador (RECLIMA) (Food and Agriculture Organisation of the United Nations (FAO); ~USD 128 million; ongoing)

¹⁰⁰ Project: Agricultural and energy risk management in El Salvador: an internal strategy to cope with drought and food insecurity (Asociación Coordinadora Indígena y Campesina de Agroforestería Comunitaria Centroamericana (ACICAFOC), World Bank; USD 1.8 million; completed)

¹⁰¹ Low-Emission Climate Resilient Agriculture Risk Sharing Facility for MSMEs (GCF, IDB; USD 158 million; ongoing)

¹⁰² Microfinance for Ecosystem-Based Adaptation (MEbA) (UNEP, Ministry of Agriculture and Rural Development — Colombia, Ministry of Environment (MINAM) — Peru, Ministry of Environment and Sustainable Development (MADS) — Colombia; ~USD 5 million; completed)

water quality and flow regimes in selected sites. The use of water-efficient technologies under the programme will also contribute to improved water quality and availability by decreasing consumption.

Social co-benefits

169. The proposed programme is expected to yield substantial social co-benefits, mostly contributing to social development by improving water and food security of vulnerable communities in the region. These co-benefits are listed below.

- Through the adoption of water-efficient technologies such as rainwater harvesting systems, households will have increased access to clean water for domestic use as well as for food production. This will also improve the targeted communities' health by reducing the risk of water-borne diseases.
- Increased productivity of agricultural systems as a result of improved water supply will lead to enhanced food security and nutritional health. The targeted communities' food security will also be improved in terms of quantity, quality and diversity of food products through the adoption of agroforestry methods.
- The use of agroforestry species according to their suitability under future climate conditions and the promotion of sustainable fuelwood practices will enhance the sustainability of local livelihoods, contributing to cultural preservation in the region.
- The programme will reduce inequality and poverty through increased financial inclusion and improved economic opportunities — especially for women. This will be achieved through interventions such as loans and risk transfer facilities for EbA and water-efficient technologies, as well as with the promotion of natural resource-based businesses and climate-resilient alternative livelihoods. Through its participatory and gender-sensitive approach, the programme will specifically empower women and contribute to social cohesion.

Economic co-benefits

170. Improved access to water and the promotion of EbA will have several economic co-benefits for vulnerable communities in the Central American Dry Corridor and the Dominican Republic's Arid Zones. These are listed below.

- The implementation of water-efficient technologies — such as drip irrigation and rainwater harvesting — in rural communities will improve agricultural practices, increasing agricultural productivity and associated income.
- Short-term work opportunities will be created for the implementation of ecosystem restoration and protection activities under Output 2. Long-term employment opportunities will also be generated by the need for ongoing operations and maintenance of ecological infrastructure established by the programme.
- The programme will provide opportunities to develop sustainable natural resource-based businesses and alternative livelihoods in ecotourism and in the collection of non-timber forest products, taking climate change scenarios and market demand for these products and services into account.
- Increased access to credit will further reduce financial barriers to implementing adaptation measures for the targeted rural communities.
- The implementation of restoration activities and establishment of agroforestry systems under the programme will yield a net present value (NPV) of ~US\$15.8 million worth of benefits which would accrue to rural communities over a 50-year period¹⁰³.

Gender-sensitive development impact

171. Women in Central America tend to hold traditional, unpaid gender roles associated with domestic responsibilities like motherhood, child- and elder-care as well as household management. These roles generally also include water management and collection. During times of water scarcity or following natural disasters, the burden of finding, collecting and storing water increases, with women bearing most of this additional responsibility. The programme has been designed to ensure that interventions are implemented in a gender-sensitive manner. The in-depth understanding of water-related challenges for women in the region (described further in Section G.2) makes inclusion of gender considerations in programme design and implementation critical for the long-term sustainability of programme interventions. By building on this understanding of water-related challenges among women using gender-sensitive learning methods for adaptation, the proposed programme will not only empower women within their communities and promote gender equality, but will also ensure the effectiveness of programme interventions.

Contribution to recovery post COVID-19

¹⁰³ based on an average regionwide NPV of ~US\$1,140/ha under reasonably assumed conditions. Source: World Resources Institute. 2016. The Economic Case for Landscape Restoration in Latin America. Available online at: https://files.wri.org/s3fs-public/The_Economic_Case_for_Landscape_Restoration_in_Latin_America.pdf

172. The COVID-19 pandemic is having severe negative impacts on healthcares systems and economies globally, including further reducing the capacity of already-vulnerable Dry Corridor and Arid Zone populations to respond to extreme climate events. With existing limited technical and financial capacities, these populations will require even greater relief interventions as their COVID-19 recovery efforts increase existing debt, while climate threats continue to build. There are a number of solutions in priority sectors¹⁰⁴ that can be integrated into countries' recovery plans, depending on their respective socio-economic and political circumstances. One of these high impact opportunity areas is to make ecosystems, food and rural livelihoods resilient. The fragmentation and loss of natural habitats facilitates the spread of infectious diseases, and the conservation of ecosystems to improve their connectivity can dramatically reduce this risk. In addition, parallel threats posed to ecosystem health and food security by climate change require cost-effective national interventions, while securing the availability of food will directly impact the population's ability to cope with future disease outbreaks. EbA is vital in this recovery process as EbA measures are cost-effective and facilitate the sustainable provision of ecosystem services that support the economic development of local populations while reducing loss from natural disasters, improving people's ability to cope with climate change and disease. The promotion and facilitation of EbA by this programme will allow participating countries to undergo economic development that is not only climate resilient, but also provides benefits that improve the ability to cope with disease, lessening the potential impact of future outbreaks. The programme will also reduce these countries' economic burdens of post-COVID-19 recovery by providing external funding that will aid in the recovery of critical sectors, including food production and water security, as well as tourism — the latter being related to landscape restoration.

D.4. Needs of recipient

Vulnerability

173. The Dry Corridor region of Central America and the Arid Zones of the Dominican Republic are extremely vulnerable to the impacts of climate change (see Sections B.1 and B.5)¹⁰⁵. A large proportion of people in these densely populated regions live in poverty, relying on subsistence agriculture and plantation forestry for their livelihoods. Rising temperatures, shifting rainfall patterns and the increased intensity of droughts are predicted to have several impacts on these vulnerable rural populations (see Section B.1), including: i) decreased harvests of important crops; ii) shifting suitability of land area for important crops; iii) increased frequency of wildfires; iv) increased frequency of pest outbreaks; and v) increased flooding after extreme rainfall events. These impacts will result in, *inter alia*, increased poverty, food insecurity and migration from rural areas¹⁰⁶. Women in rural communities will be disproportionately affected as they are responsible for, *inter alia*, water collection, food security and unpaid family care. Through the proposed programme, the vulnerability of rural communities in the Dry Corridor will be reduced. This will be achieved through improved catchment and water management, as well as increasing rural communities' access to financial resources to fund adaptation interventions. The programme will engage women's organisations to ensure that women have access to training, financial mechanisms and knowledge projects that will increase their resilience to climate change.

¹⁰⁴ Identified in 'Post-COVID-19 Recovery: how to articulate integrated responses to the health, economic and climate crises in Latin America and the Caribbean' developed by UNEP (2020)

¹⁰⁵ Honduras, Nicaragua, Guatemala, the Dominican Republic and El Salvador are among the fifteen most vulnerable countries in the world to climate events. Kreft et al., 2016. Global Climate Risk Index 2017. Who suffers most from extreme weather events? Weather-related loss events in 2015 and 1996 to 2015. Bonn: Germanwatch e.V. Available online at: <https://germanwatch.org/sites/germanwatch.org/files/publication/16411.pdf>

¹⁰⁶ Hannah et al., 2017. Regional modeling of climate change impacts on smallholder agriculture and ecosystems in Central America. Climatic Change 141: 29-45.

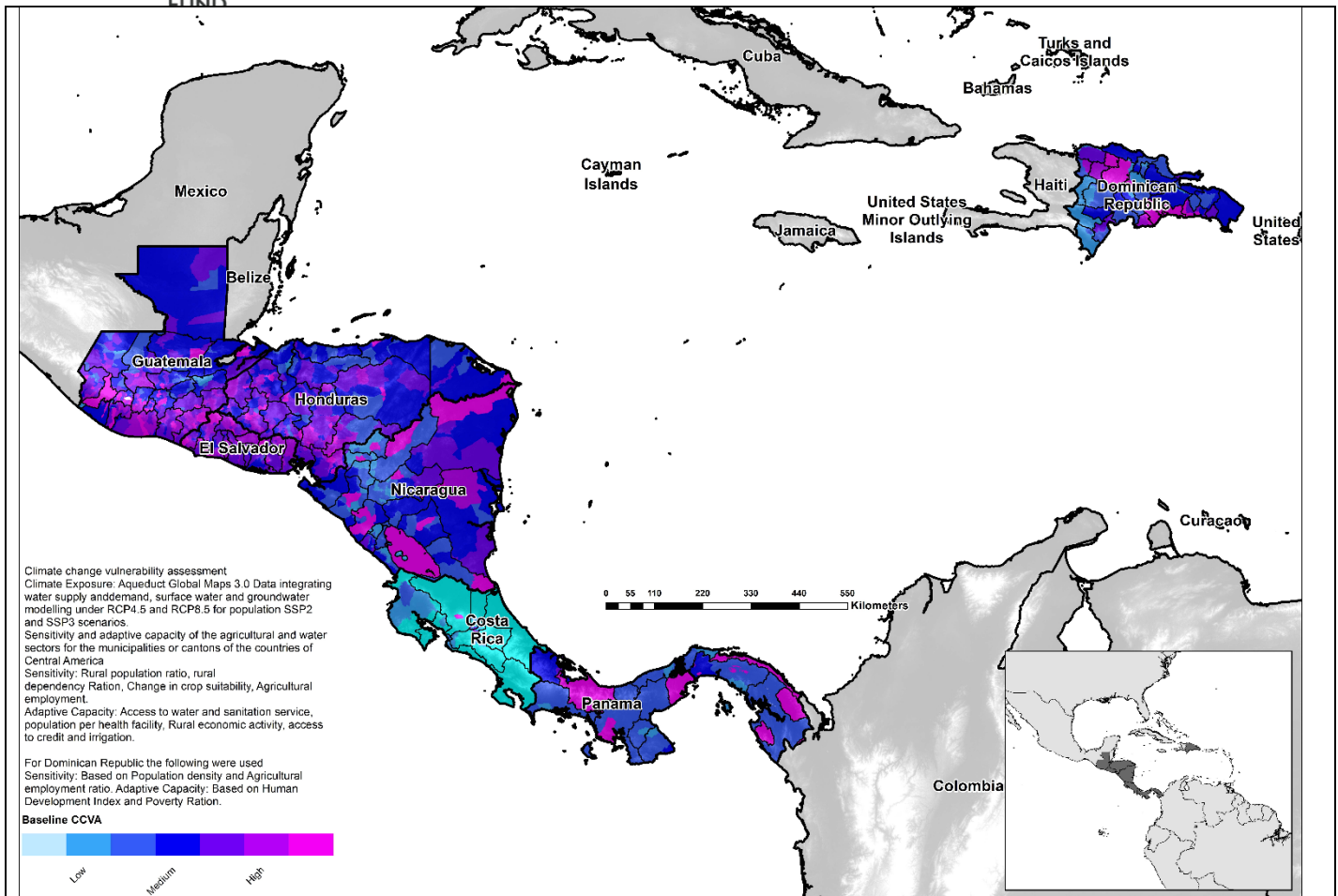


Figure 16. Map showing relative climate change vulnerability of the seven countries.

Economic and social development

174. In the last decade, the Latin America and Caribbean (LAC) region — under which the seven beneficiary countries fall — has achieved an average annual economic growth rate of 3.2%. This growth rate contributed to an overall reduction in poverty and expansion of the middle class¹⁰⁷. Currently, the seven beneficiary countries are classified as low middle to high income, with medium to high Human Development Index (HDI) scores (Table 2, Section B.5). Nonetheless, ~40% of households in the LAC region remain vulnerable and economically insecure, with a notable wealth gap existing between urban and rural populations. This economic insecurity is compounded by climate change impacts, including extreme rainfall and droughts. Economically insecure households — including vulnerable rural communities — are often slow to recover effectively from climate impacts¹⁰⁸. Rural communities have little or no access to adaptation finance, which limits their adaptive capacity. By improving access to financial resources for adaptation interventions, the adaptive capacity of rural communities in the Dry Corridor will be increased. With increased financial and technical capacity, communities will be able to withstand the negative impacts of severe droughts.

Absence of alternative sources of financing

175. The national budgets of Dry Corridor countries are limited, and poverty eradication and sustainable development have been prioritised for funding over CCA. There is a high dependency on non-reimbursable international support to meet CCA goals across the LAC region where the finance available for CCA is far below what is required to meet adaptation needs. The added burden of the COVID-19 pandemic on the economies of these countries will leave even fewer funds available for CCA in the participating countries. In the absence of donor support, the countries have limited financial resources to: i) implement large-scale EbA interventions at the landscape level; ii) implement small-scale water- and resource-efficient technologies at community- or

¹⁰⁷ Báez et al., 2017. Shaking up economic progress: Aggregate shocks in the Latin America and the Caribbean. International Bank for Reconstruction and Development/The World Bank. Available online at: <https://openknowledge.worldbank.org/bitstream/handle/10986/28892/121664-WP-Final-Report-Aggregate-Shocks-and-Economic-Progress-English-web.pdf?sequence=1&isAllowed=y>

¹⁰⁸ Báez et al., 2017.

household-level; iii) establish effective financial mechanisms that develop climate-resilient, natural resource-based livelihoods; and iv) strengthen policies, institutions and knowledge-sharing for the implementation of EbA and economic incentives for EbA. The proposed programme will improve access to financial resources for private sector and community-level adaptation interventions, as well as mainstream climate change adaption into government planning and policy-making. This mainstreaming will enable the uptake of large-scale EbA and other adaptation measures across Dry Corridor countries.

Need for strengthening institutions and implementation capacity

176. At the local level, vulnerable rural communities do not have the financial resources, knowledge base, or technical capacity to: i) develop, implement, and maintain EbA and other adaptation measures; ii) co-ordinate the frequent, cross-sectoral, multi-stakeholder engagement that is necessary for EbA and other adaptation interventions; and iii) capitalise on EbA interventions by generating new private income streams from the ecosystem goods and services produced by restored climate-resilient ecosystems. The proposed programme will provide access to technical and financial resources, knowledge products and government support structures that will enable rural communities to adapt effectively to climate change in the Dry Corridor.

Country-specific allocation of GCF funds

177. Owing to the different socio-economic, development and environmental contexts of the different participating countries, their respective needs for climate finance differ. With regards to the financial mechanisms being established under Output 4 of the programme, the participating countries will likely have different existing capacities for the uptake of EbA finance, meaning that some will be able to absorb higher levels of finance than others. There will also be different levels of demand for EbA finance in each country. The Regional Project Steering Committee will take into account all of these factors to agree on concentration limits to be finally decided upon by CABEL and the financial institutions for investing GCF Proceeds in a single country under the lending and guarantee facilities. The initial disbursement of the loan will be equally distributed among the 7 countries and this will serve as a basis for decisions on allocations of subsequent loan disbursements. With regards to GCF grant finance (Component 1), on-the-ground interventions to be implemented in the target municipalities in each country were selected based on extensive stakeholder engagement, availability of area for implementation and appropriateness of interventions. A breakdown of grant proceeds distributed per country is provided in Annex 17.

D.5. Country ownership (max. 500 words, approximately 1 page)

Alignment with regional and national policies and strategies

178. Climate change adaptation is recognised as a priority in the Central American region and has been integrated into several regional and national policies and strategies. Specifically, the programme is closely aligned with several policies and strategies of the Central American Integration System (SICA)¹⁰⁹, including those listed below.

- The Regional Environmental Strategy Framework (2015–2020)¹¹⁰ provides a framework and objectives for the regional integration of actions on, *inter alia*: i) climate change and risk management; ii) biodiversity; iii) water resource management; and iv) finance for environmental management.
- The Regional Climate Change Strategy (2010)¹¹¹ sets out projections for the impacts of climate change and variability on the region and provides a framework for regional responses to climate change.
- The Regional Strategy for the Conservation and Sustainable Use of Biodiversity in Mesoamerica (2003)¹¹² articulates priorities for the regional integration of conservation and biodiversity management efforts.
- The Regional Strategic Program for Forest Ecosystems Management (2014)¹¹³ aims to support the sustainable management and protection of forest ecosystems by promoting regional cooperation, through decentralised governance and through recognising the multifunctional nature of these forests.

¹⁰⁹ The Central American integration System (SICA) is a regional institution that facilitates diplomatic and economic integration between Central American states. The members of the organisation are Belize, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama and the Dominican Republic. https://www.sica.int/sica/sica_breve_en.aspx

¹¹⁰ CCAD, 2014. Estrategia Regional Ambiental Marco 2015 – 2020 (ERAM). Available at: <http://www.sica.int/download/?94463>

¹¹¹ CCAD, 2010. Estrategia Regional de Cambio Climático (ERCC). Available at: <http://www.sica.int/download/?96568>

¹¹² CCAD, 2003. Estrategia Regional para la Conservación y Uso Sostenible de la Biodiversidad en Mesoamérica. Available at: <http://www.sica.int/download/?9032>

¹¹³ CCAD & CAC, 2014. Programa Estratégico Regional para el Manejo de los Ecosistemas Forestales (Perfor): región de Centroamérica y República Dominicana. Available at: <http://www.sica.int/download/?96569>

- The Regional Agro-environmental and Health Strategy of Central America (2009–2024)¹¹⁴ aims to promote sustainable, intersectoral management of agricultural ecosystems, with a focus on sustainable land management, climate change, biodiversity, agro-environmental businesses and healthy lifestyles.
- The Food and Nutrition Security Policy for Central America and the Dominican Republic (2012–2032) promotes regional action to develop sustainable, healthy and culturally-appropriate food and nutrition systems.
- The Regional Strategy for Sustainable and Climate-adapted Agriculture for the SICA region (2017)¹¹⁵ sets out priorities for the agricultural sector in terms of climate change adaptation, sustainable livelihoods and the development of sustainable, low-carbon agricultural landscapes.
- The Central American Strategy for Rural Development (2010–2030)¹¹⁶ seeks to build the capacity of rural communities, government institutions and CSOs to promote innovative and inclusive rural development mechanisms.
- The Strategy and Plan for Integrated Water Resource Management (IWRM) in Central America (2010)¹¹⁷ promotes regional cooperation for IWRM to improve water security and access to water and promote the efficient and sustainable management of water resources.

179. In addition to this regional alignment, the programme contributes to achieving national priorities and aligns with numerous policies in each of the seven target countries (summarised in Table 3). Further details of the programme's alignment with the legal and policy framework of each country are given in Section 8 of Annex 2: Feasibility Study.

Table 3. Summary of national policies and strategies relating to climate change adaptation in each of the seven target countries.

Country	Policy, strategy or plan
Guatemala	National Climate Change Policy (2009)
	Law on Climate Change (2013)
	Framework Law to Regulate Vulnerability Reduction, Compulsory Adaptation to the Effects of Climate Change and Mitigation of Greenhouse Gases (2013)
	Nationally Determined Contribution (2017)
	National Action Plan for Climate Change Adaptation and Mitigation (nd)
Honduras	Climate Change Act (2013)
	Intended Nationally Determined Contribution (2015)
	National Climate Change Strategy (nd)
	Law and Action Plan for Adaptation to Climate Change (nd)
El Salvador	National Climate Change Strategy (2013)
	Intended Nationally Determined Contribution (2015)
	National Climate Change Plan (2015)
	National Plan for Climate Change Adaptation (nd)
Nicaragua	Nationally Determined Contribution (2018)
	National Environmental Climate Change Strategy (nd)
	National Policy for Mitigation and Adaptation to Climate Change and Creation of the National Climate Change Response System (nd)
	Plan for Adaptation to Climate Variability and Change in the Agricultural, Forestry and Fishing Sectors (nd)
Costa Rica	National Strategy on Climate Change and Plan of Action (2009)
	Framework Law on Climate Change (2013)
	Intended Nationally Determined Contribution (2015)
Panama	National Climate Change Policy (2007)
	National Climate Change Strategy (nd)
	Nationally Determined Contribution (2016)
Dominican Republic	National Strategy to Strengthen Human Resources and Skills for Advancing Green Development with Low Emissions and Climate Resilience (2012)
	National Strategy for Adaptation to Climate Change in the Agricultural Sector (2014–2020)
	Intended Nationally Determined Contribution (2015)
	Climate Change Policy (2016)

¹¹⁴ COMISCA, CCAD & CAC. 2008. Estrategia Regional Agroambiental y de Salud de Centroamérica 2009-2024. Available at:

<http://www.sica.int/download/?23679>

¹¹⁵ CAC, 2017. Estrategia Agricultura Sostenible Adaptada al Clima para la región del SICA (2018-2030). Available at: http://apps.iica.int/observatorio-girsa/Content/Archivos/Publicaciones/Archivos/10112017_EstrategiaASAC-CAC.pdf

¹¹⁶ CAC, 2010. Estrategia Centroamericana de Desarrollo Rural Territorial 2010 - 2030 (ECADERT). CAC, SICA, San José. Available at: <http://www.sica.int/download/?80566>

¹¹⁷ CCAD, 2010. Estrategia y Plan para la Gestión Integrada de Recursos Hídricos en Centroamérica (ECGIRH). Available at: <http://www.sica.int/download/?79057>

Capacity of the Accredited and Executing Entities to deliver

180. The Central American Bank for Economic Integration (CABEI) will be the Accredited Entity (AE) for this programme. CABEI has supported economic and social development in the region since its establishment in 1960, including channelling more than US\$600 million to support MSMEs in the agricultural sector. They have worked extensively in all the target countries and are acting as the AE for the GCF project 'Productive Investment Initiative for Adaptation to Climate Change' (CAMBio II)¹¹⁸ which was approved in 2018. CABEI also served as the Executing Agency on the original CAMBio¹¹⁹ project funded by GEF, which aimed to assist MSMEs in contributing to sustainable development and the protection of biodiversity through their products and services.
181. Through a network of intermediate financial and microfinance institutions, CABEI has achieved a comprehensive regional footprint, including having a significant presence in rural areas that will be targeted by this programme. In addition, through their large portfolio of projects and programmes, CABEI has developed strong relationships with local and national decision-makers. The extensive experience and proven track record of the AE will enable them to support and equip the Executing Entity to effectively execute the programme and contribute to its successful implementation.
182. The programme will be co-executed by three entities: CABEI, FAO and CCAD. FAO has been working for more than ten years to expand climate investments for the agricultural sectors, building its partnership with the Global Environment Facility (GEF) and the Green Climate Fund (GCF) with a view to fostering climate-resilient development trajectories. As an accredited entity to the GCF, FAO has the capacity and expertise to mobilize, through its funding proposals, large flows of climate finance for climate change mitigation and adaptation initiatives and to act as an implementation partner for grants granted under the GCF Readiness Support Programme. As FAO's GCF portfolio is growing exponentially, the Organization is prepared to assist countries whose ambitions for climate action are growing. FAO acts as an implementing entity in relation to GCF grants for small and medium-sized projects (up to USD 250 million) with low and medium levels of environmental and social risk and assists Member States in the development and implementation of country-led projects. The Central American Commission for Environment and Development (CCAD). This entity has implemented a number of initiatives similar to the proposed programme in the region, using funding from the GEF, the European Union, USAID, as well as through bilateral agreements with individual countries. In addition, it has acted as the main regional coordination entity for the development and implementation of the Regional Environmental Strategy for Central America. The Council of Ministers, which acts as the highest authority within the CCAD, will ensure that the that programme implementation proceeds ensuring alignment with the national priorities and visions of each of the participating countries.

Stakeholder Engagement

183. The development of this programme was initiated by the Council of Ministers of the CCAD, which includes members from each of the seven target countries. In addition to engaging with the CCAD, regional and national stakeholder consultations have been conducted throughout programme development. Initial consultations took place in June–July 2016 in six countries¹²⁰. Through these national workshops, priorities and focus areas for the programme were identified. The draft Concept Note was circulated, and further consultations were conducted in October–November 2017 with representatives from each country. A regional workshop was also held in October 2017 to obtain input on the draft Concept Note from representatives of governments and NGOs in the region. These inputs contributed to refining the programme concept and incorporating lessons learned from previous initiatives in the region.
184. Further consultations took place at the International Central American Dry Corridor Symposium in October 2018 and in each of the target countries from December 2018 to March 2019. These consultations

¹¹⁸ GCF. 2018. Funding Proposal: Productive Investment Initiative for Adaptation to Climate Change. Available at: <https://www.greenclimate.fund/projects/fp097>

¹¹⁹ Central American Markets for Biodiversity (CAMBio): Mainstreaming biodiversity conservation and sustainable use within Micro, Small and Medium-sized Enterprise Development and Financing. Implemented from 2006–2017. Further information available at: <https://www.thegef.org/project/central-american-markets-biodiversity-cambio-mainstreaming-biodiversity-conservation-and>

¹²⁰ Nicaragua was incorporated into the programme after its inception and was therefore not included in the initial consultations.

contributed to the finalisation of the Concept Note and the development of this Funding Proposal. Final national consultations were conducted in October 2019 during the concluding phase of programme development. These consultations were conducted in communities in each of the seven target catchments to get community inputs on their needs and locally-appropriate interventions. The outputs of these consultations are presented in Annex 7.

185. As for financial sector stakeholders the CABEL team counts with information of 4 financial institutions in Costa Rica, Dominican Republic, El Salvador and Panama, 3 of which participate in the current UN MEbA project and are actively involved in distributing targeted EbA Finance in the programme areas. Further consultations are underway with up to 50 financial institutions out of a total of 134 in the programme areas via interviews and online surveys with field personnel. Focus is on observed end beneficiaries' EbA-aligned agricultural practices in programme areas, the panorama of offered financial and non-financial products and services potentially adjusted for EbA finance as well as appetite for participation in the programme.

D.6. Efficiency and effectiveness

Cost-effectiveness and efficiency of programme interventions

186. The programme targets interventions on multi-stakeholder levels in order to prepare for EbA mainstreaming and upscaling by combining forces between public and private actors in selected areas. While US\$84.3 million in grants are geared towards concrete and tangible benefits for an estimated total of 2,228,147 direct and indirect beneficiaries (outputs 1-3) via the strengthening of stakeholders' capacities to better understand, develop and implement EbA solutions, a further US\$102.8 million in debt finance and US\$ 30 million as guarantee facility leveraging further US\$ 129,974,200 will build upon such created capacities, with an estimated number of 134,694 direct beneficiaries and 83,544 indirect beneficiaries (total of 218,238 for lending and guarantee facilities). Smallholder farming units, rural community projects and commercial farms are expected to access loans to promote EbA in their regions¹²¹. The senior loans that make up the debt finance will be supported by a guarantee fund, which will de-risk the loans, creating incentives for financial institutions to engage in EbA finance, for which there exist no or only limited offerings as of today. While the grant solutions provide the basis for the establishment and expansion of a body of knowledge in target regions, the debt solutions will be managed fully self-sustained on a commercial basis, targeting potential further investment into the mechanisms created. Hence, the programme foresees – considering the overall costs considered throughout all components, as well as direct (financial) and indirect (economic) benefits resulting from their implementation – an FIRR of 10.1% with a NPV of US\$ 353,851,655 and an EIRR of 17.5% with an NPV of US\$ 2,005,009,949 (both NPV discounted at an average of issued public bond rates as alternative investment).

Application of best practices

187. This programme draws on best practices from ground-breaking projects in the region and further afield. One of these is the Microfinance for Ecosystem-Based Adaptation (MEbA) project which informed the selection of appropriate EbA solutions for the proposed programme. Phase one of the project (MEbA I) was implemented in Colombia and Peru, while activities were expanded to Ecuador, Costa Rica, El Salvador, Dominican Republic, Senegal and Benin during phase two (MEbA II). EbA solutions drawn from MEbA will be supported by implementing locally appropriate water- and resource-efficient technologies, such as rainwater harvesting systems and half-orange kilns, for adapting to the increasing impacts of climate change, which were selected through consultations with local communities. Furthermore, effective methods for increasing commercial and smallholder farmers' access to financial resources for adaptation were drawn from the MEbA project as well as the Productive Investment Initiative for Adaptation to Climate Change (CAMBio II) and projects. These methods were applied to the design of activities under Component 2, in particular, to ensure the sustainable use of financial resources for adaptation over the long term. Lessons learned from numerous baseline investments into climate change adaptation in the seven participating countries in the Central American Dry Corridor and Arid Zones of the Dominican Republic were also used to guide the design of all programme interventions. Such lessons include appropriate mechanisms for ensuring that all programme activities are implemented in a participatory, gender-sensitive and sustainable way. Details on all best practices and lessons learned relevant to the programme are presented in Section 10 of Annex 2: Feasibility Study.

188. The primary international best practices and lessons learned identified for the programme include:

¹²¹ Assuming 5 members per household

- increasing commercial and smallholder farmers' capacity and access to finance for implementing EbA solutions and promoting the uptake of water- and resource-efficient technologies to respond to the impacts of climate change;
- effectively managing the available financial and human resources to ensure the efficient and cost-effective implementation of programme interventions;
- building an effective evidence base drawing on relevant past and ongoing initiatives to inform the implementation of a suite of complementary EbA strategies, promote the uptake of new financial mechanisms and resources for adaptation and support capacity-building interventions;
- undertaking regular monitoring and evaluation of adaptation interventions during the implementation phase to ensure that the most effective solutions are being implemented and that an adaptive management approach is being applied that considers changing climatic conditions; and
- engaging and collaborating extensively with all relevant programme stakeholders within each of the seven participating countries will encourage buy-in from national- and local-level decision-makers therefore contributing to the sustainability of proposed adaptation interventions over the long term.

LOGICAL FRAMEWORK

E.1. Paradigm shift objectives

- ☐ Shift to low-emission sustainable development pathways
- ☒ Increased climate resilient sustainable development

E.2. Core indicator targets

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	Click here to enter text. t CO ₂ eq
	Lifetime	Click here to enter text. t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	<u>N/A</u> USD
	(b) Requested GCF amount	<u>N/A</u> USD
	(c) Expected lifetime emission reductions	<u>N/A</u>
	(d) Estimated cost per t CO ₂ eq (d = a / c)	<u>N/A</u> Choose an item.
	(e) Estimated GCF cost per t CO ₂ eq removed (e = b / c)	<u>N/A</u> Choose an item.
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	<u>N/A</u> USD
	(g) Public source co-financed	<u>N/A</u> USD
	(h) Private source finance leveraged	<u>N/A</u> USD
	(i) Total Leverage ratio (i = f / b)	<u>N/A</u>
	(j) Public source co-financing ratio (j = g / b)	<u>N/A</u>
	(k) Private source leverage ratio (k = h / b)	<u>N/A</u>
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	1,084,441 TOTAL 325,335 females (30%) 759,106 males
	Indirect	1,364,044 TOTAL 545,621 females (40%) 818,423 males
	<i>By country Data provided in Annex 17</i>	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	
	Indirect	
<i>For a multi-country proposal, leave blank and provide the data per country in annex 17.</i>		

E.3. Fund-level impacts						
Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	A1.2 Number of males and females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.)	PFI Financial and Programmatic reports (related to the lending facilities); and Site surveys (related to the lending facilities)	EbA focused finance is limited to isolated examples in El Salvador and Guatemala. Stronger baseline to be determined at inception phase	Commercial: 700 females 1,300 males Subsistence: 7,000 females 13,000 males	Commercial: 3,500 females 6,500 males Subsistence: 59,500 females 110,500 males	Farmers receiving EbA focussed finance will use it for nature-based business and climate-resilient livelihood options
A4.0 Improved resilience of ecosystems and ecosystem services	A4.1 Coverage/scale of ecosystems protected and strengthened in response to climate variability and change	Implementation reports; and Site surveys and remote sensing data.	0 ha of new forest protection zones 0 ha of new natural forest 0 ha of restored forest 0 ha of restored pine forest 0 km of living fence agroforestry systems 0 ha of new agroforestry systems for natural shade	2,000 ha of forest protection zones across all seven target catchments. 900 ha of natural forest in major recharge areas and riparian zones in all seven target catchments. 2,500 ha of restored forested areas across all seven target catchments. 400 ha of pine forests restored in Guatemala, Honduras and Nicaragua. 600 km of living fence agroforestry systems across all seven target catchments. 300 ha of agroforestry systems for natural shade in coffee plantations in Guatemala, Honduras, Nicaragua and Costa Rica.	4,160 ha of forest protection zones across all seven target catchments. 1,820 ha of natural forest in major recharge areas and riparian zones in all seven target catchments. 5,170 ha of restored forested areas across all seven target catchments. 820 ha of pine forests restored in Guatemala, Honduras and Nicaragua. 1,293 km of living fence agroforestry systems across all seven target catchments. 612 ha of agroforestry systems for natural shade in coffee plantations in Guatemala, Honduras, Nicaragua and Costa Rica.	Sustainable land management practices will improve the provision of ecosystem services, including provision of water and agricultural productivity

			0 km of diversified living fence silvopasture	600 km of diversified living fence silvopasture systems across all seven target catchments.	1,280 km of diversified living fence silvopasture systems across all seven target catchments.	
			0 ha of new silvopasture systems	600 ha of silvopasture systems using individual trees across all seven target catchments.	1,280 ha of silvopasture systems using individual trees across all seven target catchments.	

E.4. Fund-level outcomes

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	<i>A5.1 Institutional and regulatory systems that improve incentives for climate resilience and their effective implementation</i>	Municipal policy documents and action plans. Key informant interviews Policy uptake scorecard ¹²² Interviews with PMU	No new incentives developed for investment in EbA	15 municipalities with at least Level 1 policy uptake	36 municipalities with at least Level 2 policy uptake	Developing accounts profiles of natural capital, developing guidelines for economic incentives and training local governments will lead to incentive programmes being integrated into policies.
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	<i>A7.1 Use by vulnerable households, communities, businesses and public-sector services of Fund-supported tools instruments, strategies and activities to respond to climate change and variability</i>	Household surveys Digital management reports on EbA capacity index verified by PFIs	0 investments ¹²³ made into EbA interventions with an expected increase in productivity /hectare (in relation to a control population) of at least 5%, of which: • 0 are commercial operators.	10,000 investments made into EbA interventions with an expected increase in productivity / hectare (in relation to a control population) of at least 7.5%, of which: • 1,000 are commercial	36,482 investments made into EbA interventions with an expected increase in productivity / hectare (in relation to a control population) of at least 10%, of which: • 2,094 are commercial	Financial mechanisms developed by the programme are suitable for vulnerable communities to access finance for EbA and water/resource-efficient technologies.

¹²² This policy uptake scorecard will have four levels and will measure the extent of use of the incentives for EbA in relevant local government policies and plans.

Level 0: Incentives not integrated meaningfully into municipal policy.

Level 1: Incentives narrative woven through the draft municipal policy – including for the payment of ecosystem services (PES) and the establishment of water funds – but not yet operational.

Level 2: Action plans have been developed for implementation of incentives for EbA – including PES and water funds – with communities aware of the availability of incentives, but no projects operating under the incentive programme yet.

Level 3: Incentive programmes fully operational, with budgets allocated to implement the incentives and at least one EbA project under implementation under the incentive programme.

¹²³ Baseline to be determined through activity 1.1 “Develop site-specific intervention plans for the 7 target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities and other stakeholders”.

			• 0 are subsistence farmers.	• 9,000 are subsistence farmers.	• 34,388 are subsistence farmers.	
A8.0 Strengthened awareness of climate threats and risk-reduction processes	A8.1 Number of males and females made aware of climate threats and related appropriate responses	Household surveys Training reports Participant surveys following training sessions	Limited awareness among vulnerable communities of locally appropriate adaptation interventions	104,000 males 56,000 females	227,500 males 122,500 females	Training of trainers programmes are effective in disseminating information on adaptation options to local community members All community members who attend a training workshop or access information on the knowledge hub are considered to be aware of climate risks and related appropriate responses

E.5. Project/programme performance indicators

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Component 1: Mainstreaming of EbA, water- and resource-efficient technologies and natural resource-based businesses into selected catchments						
Output 1. Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures.	Technical capacity of local government institutions to plan for and implement EbA and other adaptation measures, as measured by the capacity scorecard.	Capacity scorecard ¹²⁴ focused on independent stakeholders	Baseline to be determined at inception phase	15 local institutions with at least a capacity score of 3 out of 8	36 local institutions with at least a capacity score of 7 out of 8	Local monitoring systems provide sufficient detail to guide effective planning.
	Number of males and females implementing EbA, water/resource-efficient technologies, and/or climate-resilient livelihood options as a result of improved knowledge on how to do so.	Household surveys Monitoring reports from local-level M&E committees Participant exit surveys linked to training workshops	Baseline to be determined at inception phase	104,000 males 56,000 females	227,500 males 122,500 females	The training of trainers programme, supported by training workshops and farmer-to-farmer knowledge exchanges are sufficient to enable community members to adopt climate-resilient practices. At least half of the direct beneficiaries from the lending facility will have benefitted from some kind of training, knowledge sharing activity or participated in demonstration activities
Output 2. Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.	Tons of fuelwood produced through plantations to reduce impacts of fuelwood collection on natural forests	Site surveys Remote sensing data Implementation reports	Baseline to be determined through activity 1.1 in year 1 ¹²⁵	13,000 tons of fuelwood produced	212,200 tons of fuelwood produced	Introducing SLM practices will enhance the provision of ecosystem services.
	Rainwater harvesting capacity	Site surveys Implementation reports	Baseline to be determined through activity 1.1 in year 1	1,250 m ³ rainwater harvesting capacity	2,725 m ³ rainwater harvesting capacity	Communities will continue to use the introduced technologies and will not revert to old practices
	Rainwater storage capacity	Site surveys Implementation reports	Baseline to be determined through activity 1.1 in year 1	35,000 m ³ rainwater storage capacity	78,000 m ³ rainwater storage capacity	Communities will continue to use the introduced technologies and will not revert to old practices

¹²⁴ The indicator scale is based on four-step criteria of capacity assessment for each stakeholders group. Each question is scored from 0–2, with: 0 = not at all, 1 = partially; and 2 = to a large extent/completely. An overall score is calculated, with a maximum score of 8 given four criteria.

i) Are the stakeholders aware of the current and expected impacts of climate change and the availability of climate-resilient practices available to adapt to these impacts?
 ii) Do local institutions have access to the EbA and water/resource-efficient technology protocols?
 iii) Do local institutions have the capacity to monitor and evaluate the state of ecosystems and the effectiveness of EbA interventions at the landscape level?
 iv) Do local institutions have the capacity to train farmers on the adoption of EbA and water/resource-efficient technologies?

These four criteria will be elaborated, reviewed and validated at inception phase of the programme.

¹²⁵ Baseline to be determined through activity 1.1 "Develop site-specific intervention plans for the 7 target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities and other stakeholders".

	Use of resource- efficient technologies	Site surveys Implementation reports	Baseline to be determined through activity 1.1 in year 1	20 half-orange kilns in use	43 half-orange kilns in use	Communities will continue to use the introduced technologies and will not revert to old practices
<i>Output 3. Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies.</i>	Level of integration of climate change adaptation in municipal-level political and technical decision-making for climate resilience.	Policy uptake scorecard ¹²⁶	Limited integration of CCA in municipal-level political and technical decision-making for climate resilience	36 municipalities with at least Level 1 policy uptake	36 municipalities with at least Level 2 policy uptake	Lessons learned from programme activities are sufficient to support the mainstreaming of CCA into local and national policies.
Component 2: Financing and implementation of EbA as well as water- and resource-efficient technologies across selected catchments						
<i>Output 4. Financial products and services to finance EbA investments are offered by Partner Financial Institutions (PFI), including PFI access to EbA on-lending funds and support mechanisms.</i>	Cumulative investments (US\$) into EbA through the lending facility and expected average interest rates	Management reports of financial or support mechanisms Digital management reports Audit reports (general + programme-specific)	Approx. US\$10M (from previous disbursements in the region, e.g. MEbA). An average interest rate of 22% is assumed for non-EbA loans ¹²⁷ .	Baseline plus US\$149.643 M in investments, financed via EbA loans, with average interest rates expected at relative 15% below average rates for non-EbA control populations.	Baseline plus US\$381.880 M in investments, financed via EbA loans, with average interest rates expected at relative 10% below average rates for non-EbA control populations.	PFI's need specific EbA financing and support (e.g. risk transfer) mechanisms to fully take advantage of last-mile solutions. End beneficiaries will adhere to specific EbA finance products and non-financial services to invest into EbA capacity once available. The concessionality will diminish over time, as function of more leveraged capital crowding in – which will have only the guarantee facility benefits – but not GCF fund blending.
	EbA investments financed by leveraged finance only and covered by the EbA guarantee facility.	Management reports of financial or support mechanisms Digital management reports Audit reports (general + programme-specific)	0 investments financed by loans covered by CABEL cofinancing, leveraged finance and covered by the EbA guarantee facility.	0 investments financed by loans covered by CABEL cofinancing, leveraged finance and covered by the EbA guarantee facility.	At least 80,000 investments financed for a total amount of at least US\$250,00, stemming leveraged and covered finance by the guarantee facility.	PFI's need specific EbA financing and support (e.g. risk transfer) mechanisms to fully take advantage of last-mile solutions End beneficiaries will adhere to specific EbA finance products and non-financial services to invest into EbA capacity once available The guarantee facility will leverage finance at interest

¹²⁶ This policy uptake scorecard will have four levels and will measure the extent of CCA mainstreaming in relevant local government policies and plans. Level 0: CCA not integrated meaningfully into municipal policy; Level 1: Evidence-base in place to guide policy adjustments in local municipalities; Level 2: CCA narrative woven through the draft municipal policy; Level 3: Policies updated to fully account for climate change adaptation.

¹²⁷ No data is available for baseline values of average interest rate for EbA loans, but will be collected in year 1.

						rates more aligned with but still below market interest rates for non-ebA financing.
	Institutional capacity of financial institutions to channel funds for small- and large-scale EbA investments	Institutional capacity scorecard ¹²⁸	Baseline to be determined at inception phase	12 institutions with at least Level 3 (out of 5) capacity	28 institutions with at least Level 4 (out of 5) capacity	PFI's need specific EbA financing and support (e.g. risk transfer) mechanisms to fully take advantage of last-mile solutions

E.6. Activities

Activity	Description	Sub-activities	Deliverables
Activity 1.1. Develop site-specific intervention plans for the 7 target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities, and other stakeholders.	A community-based approach will be adopted to identify site-specific interventions in each of the seven municipalities. A community-level monitoring and evaluation committee will be established and capacitated in each municipality to monitor biophysical, social and economic conditions in their local catchment areas as well as to assess the level of climate vulnerability, risks and opportunities. Baseline assessments conducted by these committees – with support from regional M&E specialists – will underpin the refinement of site-specific intervention plans, focussing on the selection of appropriate land for the implementation of interventions identified through stakeholder consultations during the programme preparation phase. The outcomes of these assessments will be fed into the Regional Knowledge Hub (Activity 3.1). The suite of identified EbA interventions includes the conservation and restoration of natural forest, as well as the introduction of agroforestry, silvopastoral and agrosilvopastoral systems, along with sustainable fuelwood plantations. Specific consideration will be given for connectivity of	1.1.1 Establish a community-level monitoring and evaluation committee in each municipality and build the capacity of these committees to monitor the biophysical, social and economic conditions in their local catchment areas as well as to assess the level of climate vulnerability, risks and opportunities. 1.1.2 Hold community engagement workshops in each municipality to develop a framework for site-specific intervention plans in each of the seven targeted catchments. 1.1.3 Draft site-specific intervention plans for each of the seven targeted catchments. 1.1.4 Hold a stakeholder workshop to validate and get stakeholder buy-in for site-specific intervention plans in each of the seven targeted catchments.	36 monitoring and evaluation committees established 7 site-specific intervention plans

¹²⁸ Within the MEbA II project an institutional capacity scorecard has been developed that assesses the capacity of a given institution in five main dimensions and expresses findings over 100 points in total, implemented with 13 institutions in Latin America and Sub-Saharan Africa:

- EbA solutions: does the institution have a basic understanding of concrete EbA and efficient water management solutions and is it able to distinguish and verify the application of these solutions by target end beneficiaries (survey-based)?
- Information management: is internal information management structured in a way that enables the assessment of loan applications based on EbA and water/resource-efficient technology protocols (e.g. via MEbA II EbA capacity and EbA verification indicators) and spur ongoing learning to enable autonomous and effective EbA by target end beneficiaries?
- Risk management: is the institution in a position to integrate climate and ecosystem risk management (identification, measurement, monitoring and mitigation of these risks) into daily operations, including incorporating business continuity and contingency plans?
- Products & services: is the portfolio of financial and non-financial products sufficiently sophisticated to integrate EbA and water-efficient technology-focused products and services relevant to PFI's areas of operations?
- Marketing: is the marketing management incorporating all aspects needed to raise awareness of target end beneficiaries regarding the importance and potential impact of climate change, ecosystem services and water scarcity

Questions are scored between 1 and 10 points with different weights applied; 20% percentiles define 5 segments of development status of an institution. [we do not have it as that – we use it to steer TA and close gaps] Each question is scored from 0–2, with: 0 = not at all, 1 = partially; and 2 = to a large extent/completely. An overall score is calculated, with a maximum score of 8. These four criteria will be elaborated, reviewed and validated at inception phase of the programme.

	targeted ecosystems to maximise ecosystem services, as well as for land-tenure and the needs of local communities for the land to ensure long-term sustainability. Finally, sites will be identified for the implementation of water- and resource-efficient technologies within rural communities across the seven target catchments. Details of proposed interventions for each site are presented in Section 14 of Annex 2: Feasibility Study.		
Activity 1.2. Provide technical assistance to municipal authorities, farmers and rural communities for the implementation of EbA practices as well as water- and resource-efficient technologies	Effective implementation of EbA practices and water-/resource-efficient technologies requires training for both the recipients and the local-level authorities that provide support services to the recipients. Under the proposed programme, protocols will be developed for a range of sustainable practices, including : i) conservation and restoration of forested areas; ii) establishment existing agroforestry systems; iii) development of sustainable fuelwood source; iv) water-efficient technologies; and v) resource-efficient technologies that reduce fuelwood demand. To facilitate/promote the adoption of these protocols, cooperation agreements will be developed with commercial farmers and training will also be provided to build their technical capacity to implement sustainable practices. In addition to the direct training provided to commercial farmers, a training of trainers programme will be established in each country to facilitate long-term knowledge transfer to vulnerable communities. To increase the reach of the training programmes, trainers will be selected from a range of organisations, including local departments of environment, CBOs, water committees, women's organisations and local cooperatives within each target country. Formal training workshops will also be arranged within each municipality, which will include site visits to demonstration sites established under Output 4.	1.2.1 Develop cooperation agreements with commercial farmers for forest restoration and sustainable land management. 1.2.2 Train commercial farmers on sustainable EbA practices, including silvopasture, agroforestry and SLM. 1.2.3 Develop protocols for the implementation of sustainable EbA practices, including the: i) conservation and restoration of forested areas; ii) establishment existing agroforestry systems; and iii) development of sustainable fuelwood source. 1.2.4 Develop protocols for the adoption of water-efficient technologies by households (e.g. rainwater harvesting systems), by communities (e.g. water intakes) and by smallholder and commercial farmers (e.g. (drip irrigation, solar water pumping) 1.2.5 Develop protocols for the adoption of resource-efficient technologies that reduce fuelwood demand, (e.g. efficient charcoal kilns, fuelwood drying, efficient biomass stoves). 1.2.6 Establish a training of trainers programme for representatives of local departments of environment, CBOs, water committees, women's organisations and local cooperatives within each target country for EbA and water-efficient technologies based on the protocols developed under sub-activities 1.2.3–1.2.5. 1.2.7 Organise training workshops linked to community meetings within each target municipality to train local communities on EbA and water-efficient technologies, including visits to demonstration sites to facilitate farmer to farmer knowledge exchange.	• Cooperation agreements with commercial farmers • 3 EbA implementation protocols • 1 protocol for the adoptions of water-efficient technologies • 1 protocol for the adoptions of resource-efficient technologies • 7 training of trainers programmes • 36 training workshops

Activity 1.3. Provide technical assistance to farmers and rural communities for the development of natural resource-based businesses and alternative climate-resilient livelihoods.	A community-based approach will be taken to improve the climate-resilience of vulnerable, natural resource-based livelihoods in the target municipalities. This will involve the establishment of technical training programmes to farmers and rural communities on locally appropriate natural resource-based business opportunities and climate-resilient practices that will improve the sustainability of natural resource-based livelihoods. This training will be underpinned by a detailed livelihood assessment that will identify climate risks impacting the livelihoods of local communities, as well as potential alternative practices that will reduce this vulnerability. The results of this assessment will be used to develop livelihood action plans for each municipality identifying at least three natural resource-based businesses and/or climate-resilient livelihoods options per municipality. Targeted training programmes workshop will be designed for each livelihood option, taking into consideration the local context and inputs from community consultations to customise the training for each municipality. These programmes will then be rolled out through community training workshops, with one workshop per livelihood being held per target municipality.	1.3.1 Conduct a livelihood assessment within each of the seven target catchments, focusing on climate risks and alternative options. 1.3.2 Conduct community engagement workshops to develop livelihood action plans for each target community based on the assessments conducted in 1.3.1. 1.3.3 Establish livelihood training programmes in each municipality for farmers and rural communities on locally appropriate natural resource-based businesses and climate-resilient livelihoods as identified in the livelihood action plans.	• 7 livelihood action plans • 108 training workshops
Activity 2.1. Implement landscape EbA interventions within rural communities across the seven target catchments.	On-the-ground EbA interventions will be implemented on public land across each of the target municipalities. Interventions are listed below. • Conservation of existing forest ecosystems to maintain the provision of ecosystem services. • Active restoration of partially degraded forest ecosystems, including assisted natural regeneration and enrichment planting. • Establishment of agroforestry, silvopastoral and agrosilvopastoral systems. • Establishment of fire breaks. • Establishment of mixed-use plantations using native species with multiple ecosystem services, including non-timber forest products. • Establishment of sustainable fuelwood plantations to reduce pressure on natural forests. • Soil conservation measures. These interventions will serve as demonstration sites to promote the uptake of EbA practices through the financial mechanisms established through Output 1. Details on specific interventions and targets for each municipality are provided in Annex 17b.	2.1.1 Establish tree nurseries focused on native species. 2.1.2 Establish forest protection zones. 2.1.3 Protect and restore natural forest in major recharge areas and riparian zones. 2.1.4 Restore forested areas across seven catchments. 2.1.5 Restore pine forests in Guatemala, Honduras and Nicaragua. 2.1.6 Establish agroforestry systems using diversified living fence arrangements in basic grains crops. 2.1.7 Establish agroforestry systems for natural shade in coffee plantations. 2.1.8 Establish silvopasture systems using diversified living fence arrangements. 2.1.9 Establish silvopasture systems using individual trees. 2.1.10 Establish sustainable fuelwood and timber plantations. 2.1.11 Establish firebreaks for forests and plantations. 2.1.12 Construct living barriers for soil conservation.	• 36 tree nurseries • 4,160 ha of forest protection zones. • 1,820 ha of restored and protected natural forest • 5,170 ha of restored forest across seven catchments • 820 ha of restored pine forest • 1,293 km of agroforestry systems established using diversified living fence arrangements • 612 ha of agroforestry systems established in coffee plantations • 1,280 km of silvopasture treelines • 1,280 ha of silvopasture systems using individual trees • 5,305 ha of sustainable fuelwood plantations • 129 km of firebreaks • 118 km of living barriers • 118 km of superficial drainage solutions

		2.1.13 Construct superficial drainage for soil conservation. 2.1.14 Technical design and guidance for the implementation of Forestry Measures.	
Activity 2.2. Implement demonstration water- and resource-efficient technologies within rural communities across the seven target catchments.	Water- and resource-efficient technologies will be installed in each target municipality. Rainwater harvesting interventions will target community buildings such as schools and clinics where they will benefit the most vulnerable people. By targeting communities, these interventions will also serve as demonstration sites of the technologies to promote uptake of interventions through the financial mechanisms established in Output 1. Similarly, resource-efficient technologies will be made communally available to maximise beneficiaries and promote upscales within the target communities and surrounding communities across the region. Upscaling will be promoted through the training and demonstration site visits implemented through Activity 2.3. Details on specific interventions and targets for each municipality are provided in Annex 17b.	2.2.1 Install rainwater harvesting systems on public or community buildings using rooftop capture and plastic or geomembrane deposits for storage (25 m³) to supply water for 8-12 families. 2.2.2 Install community-level rainwater reservoirs (500 m³). 2.2.3 Install half-orange kilns for efficient charcoal production to be administered by existing local cooperatives. 2.2.4 Install community-level water pumping systems by solar panels. 2.2.5 Technical design and guidance for the implementation of Water & Resource Measures.	• 109 rainwater harvesting systems on community buildings • 156 community-level rainwater reservoirs • 43 half-orange kilns for efficient charcoal production • 38 community-level water pumping systems by solar panels
Activity 2.3. Establish the grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing and start operations	The grant facility is intended to identify and promote site-specific and relevant EbA and water-efficient technologies. A bottom-up approach to identification of eligible projects will be established. Eligible projects can be community-level investments, the establishment of water funds, or the support of selected solution providers (to be defined), among others. The grant facility will channel investment to selected projects via PFIs in each of the 7 catchment areas. Operational expenses will be borne by the fund's overall activities.	2.3.1 Define eligibility criteria from Steering Committee guidelines and align to national and municipal adaptation policies and initiatives, define final operability of the grant facility, including the definition of a competitive selection model. 2.3.2 Raise awareness on grant availability for EbA investment in conjunction with community engagement workshops under 1.1.1. 2.3.3 Monitor success of selected projects, including definition and documentation of underlying economics (knowledge gained to be disseminated to inform scale-up finance) 2.3.4 Communicate results via specific events and media initiatives. 2.3.5 Secure initial set-up and running cost of grant facility	• 1 Grant facility operational manual developed and approved • 28 awareness raising and promotion events executed • Marketing and communication materials • Audit reports
Activity 3.1. Establish regional knowledge hub for the dissemination of information on EbA in the Dry Corridor and Arid Zones.	A regional knowledge hub will be established to facilitate the sharing of best practices and lessons learned for EbA across the seven target countries. The platform will be hosted by the CCAD, building on the existing Environmental Observatory portal. The knowledge hub will serve two main purposes. First, it will act as a repository for climate change adaptation information, including	3.1.1 Establish a knowledge management hub linked to the Environmental Observatory to disseminate best practices and lessons learned from the programme, to support decision-making related to EbA implementation. The hub will incorporate specific modules for engagement	• 1 knowledge management hub • Knowledge products for: ○ sustainable landscape management standards and procedures; ○ opportunities for access to financing and women's economic empowerment; ○ sustainable natural resource-based business opportunities; and

	all protocols developed by the programme as well as the best practices and lessons learned collected through the programme's monitoring and evaluation processes. Second, it will provide users with specific knowledge products that will guide policy- and decision-makers to effectively implement sustainable landscape management practices. This will include guidance on how to access opportunities for accessing finance provided through the programme, with specific focus on women's economic empowerment. The knowledge products will also include technical assistance for creating business opportunities as well as technical guidelines for the assessment of ecosystem services. Training will also be provided to policy- and decision-makers at the municipal level on how to use the knowledge hub and associated tools. Finally, community organisations – including women's groups – and traditional leaders will be trained to act as champions in developing, revising and disseminating information products to support the adoption of EbA and other resilient practices. This approach will facilitate the sharing of ideas and sustainable practices between communities from the grass root level, using existing structures and traditional methods which will not only extend the reach of the dissemination campaign, but also the uptake by local communities.	with universities, research centers and private sector in the region for EbA Measures. 3.1.2 Develop knowledge products to be disseminated through the knowledge hub, including sustainable landscape management standards and procedures, opportunities for access to financing and women's economic empowerment as well as technical assistance, business opportunities, technical guidelines for the assessment of ecosystem services and their contribution to human well-being. 3.1.3 Train policy- and decision-makers at the municipal level on climate change impacts, including the impacts of gender inequality, and the role of EbA in mitigating climate change impacts to support political and technical decision-making for climate resilience in the Dry Corridor and Arid Zones. 3.1.4 Strengthen the technical capacity of community organisations, including women's groups, and local champions in developing, revising and disseminating information products to support the adoption of EbA and other resilient practices.	○ technical guidelines for the assessment of ecosystem services. ● 36 training workshops 36 community champions
Activity 3.2. Raise awareness of financial mechanisms for the implementation of CCA interventions.	Output 1 is establishing several financial mechanisms that will be accessible to communities to support the adoption of sustainable land management practices. To ensure the effective uptake of these mechanisms, training will be provided at three levels. First, public officers and/or agencies will be identified each municipality to serve as community liaisons. These officers/agencies will be trained on how to access the various financial mechanisms and how to disseminate this knowledge to their communities. Second, medium—large scale private sector organisations and farmers will be trained on how to access loans through the EbA credit line. This training will be closely linked to the technical training provided under Outcome 2. And third, awareness will be raised among the most vulnerable communities on the EbA Credit Line and Trust Fund, and how to access them. This will include targeting community organisations, including women's groups, to	3.2.1 Train public officers/agencies at the municipal level to facilitate the uptake of financial mechanisms in their communities. 3.2.2 Train medium—large scale private organisations/ farmers to access loans through the EbA credit line. 3.2.3 Raise awareness among vulnerable communities and their organisations, including women's groups, about available financial mechanisms, including the EbA Credit Line and Trust Fund.	● 36 municipal-level public officers/agencies trained ● 36 training workshops targeting medium—large scale private sector organisations and commercial farmers ● 36 awareness campaigns, involving at least one community organisation per municipality.

	maximise the reach of the awareness campaign.		
Activity 3.3. Enhance capacity of national and local-level policy-makers, in partnership with knowledge network of universities, research centers and private sector, to integrate climate change adaptation and the valuation of natural capital into local policies.	To facilitate the mainstreaming of climate change adaptation – sustainable land management practices in particular – it is necessary to build a strong evidence base for the efficiency and effectiveness of sustainable practices. To achieve this, information will be collated from lessons learned through the programme interventions as well as from other comparable ecological zones in other countries, supported by the development of methodologies for assessing the value of ecosystem services and natural capital. These methodologies will provide empirical evidence for the positive economic benefits of sustainable land management practices, which will support the mainstreaming of such practices into local development plans. Furthermore, an effective methodology for the valuation of ecosystem services will facilitate the establishment of economic incentives for SLM, including water funds and payment for environmental service schemes. Local governments will then be trained to make the policy changes to implement economic incentives for investment in SLM practices, with country-specific guidelines being developed to support this training.	3.3.1 Build an evidence base, using both in-country research and findings from comparable ecological zones in other countries, to clearly demonstrate the value and effectiveness of proposed adaptation activities to municipal-level policy-makers. 3.3.2 Develop or adjust a methodology to value ecosystems services to develop accounts profiles of natural capital so that it is integrated into development plans in the Dry Corridor. 3.3.3 Develop guidelines for local governments outlining protocols and criteria for implementation of economic incentives for SLM (e.g. water funds, payments for environmental services). 3.3.4 Train local governments to make policy changes, including the use of protocols and criteria for the adoption of EbA and implementation of economic incentives for SLM (e.g. water funds, payments for environmental services). 3.3.5 Organise workshops targeting municipalities within the Dry Corridor and Arid Zones to disseminate the evidence resulting from local government experiences to promote the integration of climate resilience in broader policies and actions across the region.	• Refined methodology for valuing ecosystem services • 7 country-specific guidelines for the implementation of economic incentives • 36 training workshops
Activity 4.1. Set-up and establish the financial structure for the lending and guarantee facilities	Lending and guarantee facilities, as well as their respective investment and coverage criteria, are set up following relevant regulations and legal frameworks.	4.1.1 Set-up relevant fiscal, legal and regulatory compliance structures for the Lending facility and Guarantee facility's operation. 4.1.2 Create legal agreements, including regulatory, legal and fiscal approvals for the lending and guarantee facilities. 4.1.3 Develop Steering Committee statutes and operational guidelines for the two facilities including eligibility criteria and processes for the blended EbA lending facility. 4.1.4 Define investment criteria from Steering Committee guidelines and align with national laws and regulations, responsible banking principles, national and municipal adaptation policies and initiatives, define final operability of the	• 1 lending facility established and operational • 1 guarantee facility established and operational • 21 technical studies carried out in 7 countries • 1 EbA lending operational manual developed and approved, including all contractual framework

		blended EbA lending facility. 4.1.5 Establish the guarantee facility to support EbA finance via the lending facility - define guarantee coverage criteria. Based on Regional Project Steering Committee guidelines and aligned with national laws and regulations, responsible banking principles, national and municipal adaptation policies and initiatives, the final operability of the EbA Guarantee Facility will be defined.	
Activity 4.2. Financing mechanism of the EbA lending and guarantee facility	The EbA lending facility will provide an EbA credit line to PFIs that have been selected based on outreach and financial performance, to on-lend to end beneficiaries for investment into EbA or water-efficient technologies. Such investment can be targeted towards community-level investment, rural MSMEs including smallholder farmers or other institutions that can foster EbA for end beneficiaries across different sectors.	4.2.1 Deployment of the EbA lending facility 4.2.2 Deployment of the guarantee facility	• 28 partner financial institutions onboarded including initial capacity evaluation • A minimum of 90,000 EbA loans disbursed
Activity 4.3. Technical assistance (TA) facility to strengthen technical capacity of accredited and non-regulated financial institutions to access and channel funds for small- and large-scale EbA investments	The financial facilities are accompanied by training and technical assistance activities to communities in the areas of the programme. Successful implementation will catalyse the uptake of EbA and EWT solutions to be financed by specific financial mechanisms. The EbA credit line will be accompanied by a TA facility that will develop capacity within PFIs for the adequate promotion and management of EbA finance to end beneficiaries, such as data management for the measurement, monitoring, reporting and verification (MRV) of climate risks and adaptation strategy impacts or effects, the implementation of EbA-oriented awareness raising and marketing campaigns with end beneficiaries and the development of financial and non-financial products and services geared towards EbA investments. This will enhance the capacity of PFIs to monitor and assess the implementation of EbA and EWT investment projects as well as to monitor and document the respective economic and ecosystem benefits. Where possible Information and Communication Technologies will be introduced to provide data analytics and satellite data insights into the EbA lending portfolio and climate impacts on the portfolio.	4.3.1 Develop or adjust training curriculums and technical assistance methodology targeted to fill gaps identified during initial due diligence based on ToT or online course methodologies and targeting certification financial institution personnel 4.3.2 Initiate and execute tailor-made technical assistance programme for PFIs based on gaps identified during initial due diligence 4.3.3 Periodic monitoring of TA and training success via reassessment based on scorecard used during initial due diligence 4.3.4 Adjust TA methodology and training curriculum according to periodic reassessment 4.3.5 Promote the blended EbA lending facility with potential network institutions and non-network institutions in each of the 7 countries	• Standardised technical assistance and training package, including ToT or online course and certification • At least 28 partner financial institutions received technical assistance and increased their EbA capacity

E.7. Monitoring, reporting and evaluation arrangements

189. The Project Manager (PM) will be responsible for the overall monitoring of programme progress against the outcome-based indicators during implementation, as well reporting programme progress to the Regional Project Steering Committee (RPSC). For on-the-ground monitoring of programme interventions, a Regional Monitoring and Evaluation (M&E) Specialist, supported by seven National M&E Specialists (one per beneficiary country), will be employed to develop practical guidelines for — and operationalise — a performance monitoring framework to track the programme's progress towards achieving its targets. The Regional M&E Specialist will primarily be responsible for: i) establishing a performance monitoring framework to define biannual targets for the programme to meet the mid-term and final targets defined in the Logical Framework (Section E.3–5 of this Funding Proposal); ii) measuring the indicators at least 1–2 times per year to evaluate the progress of the programme in meeting the targets and the application of gender-disaggregated indicators; iii) reporting to the PM and RPSC on the performance of the programme according to programme indicators; and iv) participating in the production of reports. Given the scale of the programme, National M&E Specialists will be procured in each beneficiary country to support the Regional M&E Specialist in monitoring the performance of programme interventions at a national level. Specifically, the National M&E Specialists will be responsible for: i) following guidelines developed by the Regional M&E specialist to conduct performance monitoring for the programme, aimed at meeting the targets defined in the programme document by the end of the implementation phase; ii) measuring the national indicators at least 1–2 times per year to evaluate the progress of the programme in meeting the targets and indicators; iii) reporting to the NAC on the performance of the programme according to programme indicators; iv) ensuring that all activities under the programme comply with the Programme's ESMF process and procedures; v) supporting environmental screening of sub activities and help to follow the activity specific Environmental Management Plans (EMPs); and vi) participating in the production of reports to the RSSPC and RPMU. Both the Regional and National M&E Specialists will be supported by a Gender Officer (GO), who will be employed on a part-time basis to oversee and monitor the application of gender-disaggregated indicators.
190. The Regional M&E Specialist and GO will perform regular visits — at least once a year to each country — randomly selecting municipalities to monitor on-the-ground implementation of programme interventions. They will use these visits to cross-check the information provided by PFIs and beneficiaries in programme monitoring reports. During each field visit, the Regional M&E Specialist and GO will collect qualitative data on the outcomes and difficulties experienced during implementation through focus group discussions with different beneficiaries, as well as through in-depth interviews with key stakeholders. Using this information, the Regional M&E Specialist, supported by the National M&E Specialists, will compile six-monthly programme reports to be shared with the RPMU and RPSC. These reports will outline programme progress in each country as well as regionally, and will allow feedback and advisory support from the RPMU and RPSC in the case of potential programme delays or other challenges that may arise during implementation. Annual programme progress reports and annual gender reports will also be produced to be sent to the GCF Secretariat with information regarding programme progress measured against expected outputs.
191. CABEI, as the Accredited Entity (AE) will perform supervision missions from year two of programme implementation onwards, selecting three countries each year to visit and assess the status of programme implementation. This process will allow CABEI to maintain oversight throughout the programme's lifetime. In addition, CABEI's Credit Supervision Department — which oversees the destination and use of funds disbursed to PFIs — will monitor the programme's compliance with policies and procedures, as well as the alignment between implementation and programme objectives.
192. As per the Accreditation Master Agreement (AMA), an independent consultant will be contracted by CABEI to conduct a Mid-Term Evaluation (MTE) and the Terminal Evaluation (TE) of the programme. Both evaluations will comply with GCF guidelines and international standards and will make use of qualitative as well as quantitative data analyses, including interviews with relevant stakeholders, data provided by the Regional and National M&E Specialists and GO, participative workshops and focus groups with beneficiaries to promote participative learning. The MTE will provide an assessment of programme performance against its targets at the programme's mid-point. This will be a formative exercise and will include analysing whether the programme is on track, what problems and challenges the programme is encountering, and what corrective actions are required, so that the programme can achieve its intended outcomes by programme completion in the most efficient and sustainable way. The RPSC and RPMU will be informed of all MTE findings and will develop a

management response to the evaluation recommendations along with an implementation plan. It will be the responsibility of the PM to monitor whether the agreed recommendations are being implemented during the remainder of the programme's operational life.

193. The TE will take place at the end of programme implementation and will be an independent assessment of programme performance against standard evaluation criteria (e.g. strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability). This assessment will be done based on documented evidence, stakeholder interviews and, in most cases, a field mission. In addition to an assessment of expected results and impacts, unexpected outcomes will be evaluated, including unexpected co-benefits, best practices and lessons learned. The draft TE report will be sent to programme stakeholders for comment and subsequently disclosed publicly.

194. The costs for results monitoring and performance evaluation will be covered by the salaries of the Regional and National M&E Specialists, which are presented in Annex 4: Detailed budget and Annex 11

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

195. There are no United Nations Security Council (UNSC) sanction regimes currently under implementation in any of the seven countries involved in the programme. In addition, no individual, entity or other group listed on the UNSC sanctions list, including the consolidated list, will be involved in any manner with the programme or its activities, either as a counterparty, implementer or beneficiary.

196. The identified risks to the proposed programme's implementation and sustainability, associated impacts and mitigation measures are described in detail below, with summaries provided in the following two tables. Risks were classified according to a risk matrix (Table 4), with a summary presented in Table 5 which shows the risk level based on the probability and impact of the risk factors. The mitigation potential is based on the mitigation actions detailed below and is combined with the risk level to determine overall risk after mitigation.

Table 4. Risk matrix to determine risk categorisations¹²⁹.

		Impact			
		Low	Medium	High	
Probability	Low				Very Low Risk
	Medium				Low Risk
	High				Medium Risk
					Risk
					Very Low Risk
					Low Risk
					Medium Risk
					High Risk
					Extreme Risk

Table 5. Programme risk factors and mitigated risks.

Risk Factor #	Probability	Impact	Risk	Mitigation Potential	Mitigated Risk
1	Medium	Medium	Medium Risk	High	Very Low Risk
2	Medium	Low	Low Risk	Medium	Very Low Risk
3	Medium	Low	Low Risk	High	Very Low Risk
4	Medium	Medium	Medium Risk	High	Low Risk
5	Medium	High	High Risk	High	Medium Risk
6	Low	Medium	Low Risk	Low	Low Risk
7	Low	Medium	Low Risk	Medium	Very Low Risk
8	Medium	Low	Low Risk	Low	Very Low Risk
9	Medium	Medium	Medium Risk	Medium	Low Risk
10	Low	High	Medium Risk	High	Very Low Risk
11	Low	Medium	Low Risk	Medium	Very Low Risk

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Behaviour change does not occur and the implementation of proposed programme activities (including operations and maintenance of those established under Output 2) cease after conclusion of the programme, because rural communities are unwilling to invest in EbA interventions. Ecosystem degradation and unsustainable practices continue.		
Mitigation Measure(s)		
The programme will facilitate long-term change in land-use practices and uptake of EbA by making financial instruments available to support investment into EbA. In addition, the programme will engage directly with communities		

¹²⁹ United Nations Office for Disaster Risk Reduction. 2017. National Disaster Risk Assessment, Governance System, Methodologies, and Use of Results.

as well as with local and national governments to create an enabling environment for EbA interventions, including through:

- incorporating comprehensive engagement with communities for planning, implementation and monitoring of EbA initiatives (for example Activities 4.1 and 1.1);
- providing training for communities, commercial farmers and decision-makers to develop capacity for supporting and implementing EbA and to build a common understanding of the benefits of these approaches (for example Activity 1.2);
- developing protocols and guidelines for the adoption of EbA and water/resource-efficient technologies that will guide ongoing training(Activity 1.2);
- developing an evidence base for the effectiveness of EbA in the local context including assessing the economic benefits of these interventions (Activity 1.3); and
- mainstreaming climate adaptation and EbA into local- and national-level policies, to promote ongoing implementation of EbA interventions beyond the programme lifetime (Activity 3.3).

A detailed Exit Strategy has been developed to ensure the proposed programme's long-term sustainability and to decrease the likelihood that this risk will occur (Section B.6). In addition, an operations and maintenance (O&M) manual is provided in Annex 21 which further details how interventions will be maintained post-programme implementation.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Medium	Low
Description		
High turnover of staff in implementing institutions, frequent changes in government bodies and limited institutional memory results in a disruption and/or delays in the programme implementation and may jeopardise the sustainability of the programme.		
Mitigation Measure(s)		
Decisions, best practices and lessons learned will be documented throughout the programme to support institutional memory that will sustain programme activities. This memory will also be strengthened through the regional knowledge hub that will be established under Activity 3.1. Protocols for EbA will be developed in both English and the main local languages to guide new staff who become involved in EbA implementation during and after the programme. Furthermore, through the programme, PFIs and communities will be trained and engaged in the implementation as well as the monitoring and evaluation of EbA as well as water- and resource-efficient interventions, thereby strengthening the capacity of local communities and institutions to plan and implement EbA. Where possible, the RPSC will make use of established regional and national structures to capitalise on well-established practices and systems that are familiar to government staff.		

Selected Risk Factor 3

Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Overlapping or misalignment between the proposed programme activities and those being implemented by other departments or organisations in the programme area.		
Mitigation Measure(s)		
The proposed programme has been designed with extensive stakeholder engagement and desktop research to ensure that there is alignment with any past and ongoing initiatives in the region. Regular engagements will be conducted with relevant individuals/organisations to prevent duplication of effort, support collaboration and ensure that current and future projects are aligned with the proposed programme's interventions. Coordination mechanisms — including the regional knowledge hub which will facilitate the sharing of lessons learned — will be established at the regional and national levels to ensure synergy among efforts of government and other national and regional stakeholders.		

Selected Risk Factor 4

Category	Probability	Impact
Technical and operational	Medium	Medium

Description		
Inadequate engagement with local-level stakeholders and programme partners. This may result in limited integration of local knowledge into programme activities.		
Mitigation Measure(s)		
The programme design has involved extensive engagement with local-level stakeholders over a period of three years to ensure support for programme activities and extensive engagements with programme partners and stakeholders will continue during the programme's implementation (detail provided in Annex 7: Stakeholder engagement plan). Implementation activities will build on the local knowledge and experience of PFIs (Output 4) and other local organisations. In addition, communities will be engaged in the planning of EbA interventions and their monitoring and evaluation (for example Activity 1.1) to ensure alignment between programme activities and the needs of local stakeholders.		
Selected Risk Factor 5		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
Limited technical capacity among PFIs, government authorities and local communities to channel funds for and implement EbA interventions.		
Mitigation Measure(s)		
Training and technical assistance will be provided for relevant PFIs, government authorities and local communities that will be involved in programme activities. In particular, a technical assistance facility will be established to facilitate financial institutions' abilities to access funds for EBA interventions. In addition, technical capacity of government authorities and local communities will be strengthened to improve their abilities to: i) implement EbA protocols in a scientifically rigorous manner; and ii) monitor and evaluate the success of these interventions.		
Selected Risk Factor 6		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Natural disasters undermine the: i) implementation of EbA interventions and result in economic loss and/or damage to the interventions; and ii) restoration and sustainable management of forests, wetlands and agro-systems.		
Mitigation Measure(s)		
The programme's EbA and water/resource-efficient interventions will target areas that have been identified as being vulnerable to climate change impacts. For example, EbA will be implemented in sites that are prone to drought as a measure to reduce the risk or physical impact of this type of climate-related natural disaster. Water/resource-efficient solutions will also increase the resilience of these communities to droughts. Furthermore, the programme will integrate information on local climate change risks and impacts into training for PFIs and communities to ensure that forecasts are taken into consideration during the planning and implementation of EbA interventions. The RPMU will also oversee the incorporation of forecasts into programme planning and implementation.		
Selected Risk Factor 7		
Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Social tensions may arise as existing power relations within the community might disrupt equal opportunities to use and benefit from water- and resource-efficient technologies.		
Mitigation Measure(s)		
The programme's activities have been designed to be transparent and will maintain a broadly consultative approach throughout the implementation period. The beneficiaries of the water- and resource-efficient technologies have been identified through local stakeholder consultations. In addition, the activities have been designed to ensure that all individuals are afforded access to the technologies, which will be installed in public facilities — for example rainwater		

harvesting in schools — rather than in private households. Mechanisms for resolving conflicts and addressing disputes and complaints will be established to ensure that the programme's activities are transparent and that the programme management team can be held accountable by beneficiary communities.

Selected Risk Factor 8

Category	Probability	Impact
Governance	Medium	Low

Description

Changes in political leadership, which may result in delays in implementation.

Mitigation Measure(s)

Regular engagement with non-political actors — such as government agency officials, local authorities, staff and local community members — will be prioritised during programme implementation. Additionally, formulation of local ordinances, mainstreaming of programme interventions in local planning and budgeting systems will ensure that political issues will have negligible impact on the sustainability of programme interventions. Finally, support from regional bodies such as the CCAD will promote continued political support.

Selected Risk Factor 9

Category	Probability	Impact
Governance	Medium	Medium

Description

Insufficient coordination, collaboration and information-sharing between countries and implementing partners at a regional level and between relevant government departments, research institutions and private sector entities within the countries. This can result in the limited participation of relevant institutions and agencies, reducing the timely delivery and effectiveness of the programme.

Mitigation Measure(s)

Detailed implementation arrangements have been designed to ensure effective coordination of programme activities between regional stakeholders. These arrangements are outlined in Section B.4. The relevant institutions were involved in the design of the programme activities and have agreed on all roles and responsibilities. Coordination between them will be supported by regular meetings/engagements between implementing partners to facilitate information sharing.

In addition, programme countries, relevant government departments, research institutions and the private sector will be encouraged to share information through a number of technologies and systems that will be established or strengthened by the programme. Technologies and systems will be selected based on the most appropriate approaches to sharing information in both the relevant regional and local contexts. Examples include:

- a regional knowledge hub to facilitate sharing of best practices and lessons learned across countries and sectors as well as to provide policy- and decision-makers with relevant information regarding EbA;
- cooperation agreements with stakeholders to promote an understanding of roles and responsibilities within the programme and facilitate effective participation; and
- training workshops for local authorities, community organisations and partner financial institutions, among others, to ensure a shared understanding of EbA and water/resource-efficient strategies.

Selected Risk Factor 10

Category	Probability	Impact
Other	Low	High

Description

Inadequate incorporation of gender, and environmental and social safeguard (ESS) considerations into the implementation of the proposed programme activities as well as its management structures and systems.

Mitigation Measure(s)

The programme has been assessed according to the GCF's environmental and social protection standards and associated risks and mitigation measures have been identified. In addition, a Gender Action Plan has been developed and will be implemented to ensure that gender equality is incorporated into all interventions. An Environmental and

Social Management Plan (ESMP) has also been designed based on a detailed screening process. The ESMP will provide a framework for ensuring that programme activities do not have a negative impact on the environment or local communities.

Selected Risk Factor 11

Category	Probability	Impact
Credit	Low	Medium

Description

Deteriorating economic situation of end beneficiaries and in consequence payment capacity as well as lack of management capacity at the level of PFIs and IFIs can lead to increased credit risk of funds provided by GCF and CABEL. This increased credit risk not only can affect the funds provided by GCF and CABEL, but also put the overall success of the programme at risk. Hence the potential impact is considered 'medium, while the probability is considered as 'low' as the financing mechanisms – three financial facilities – build upon proven and established structures and the programme as a whole is targeting the increase of end beneficiaries' resilience, lowering their credit risk.

Mitigation Measure(s)

The mitigation of credit risk within the programme relies upon mitigation measures on three different levels:

1. Eligibility of IFIs, PFIs and end beneficiaries: CABEL will define strict eligibility criteria for all actors involved down-stream of its provision (co-finance) and management (GCF funds) of funds, hence the selection of counterparts will be used to limit the credit risk
 - a. IFIs: only institutions forming part of CABEL's already accredited network of IFI will gain access to the programme; these institutions regularly lead their respective markets in terms of credit risk management capacity (as expressed per their ratings)
 - b. PFIs: CABEL's minimum eligibility criteria will ensure that only the most solid institutions will gain access to the programme; further, by assigning part of the PFI's credit risk to IFIs (via the on-lending stream of funds), these IFIs will ensure to strengthen the selection of low-risk PFIs
2. Promotion of EbA finance: targeting the investment into EbA solutions that increase resilience and hence lower credit risk of end beneficiaries as main objective of the programme ties the activities of the programme directly to concrete measures to mitigate credit risk
3. Introduction of solid MRV system: via the introduction of proven digital tools, will enable a centralized monitoring of credit risk tendencies in resulting portfolios and sub-portfolios, with the aim to introduce early warning routines as well prioritize contingency plan activities, following methodologies UNEP's MEbA project

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment

197. The environmental and social risk associated with the programme was evaluated in accordance with GCF's Environmental and Social Safeguards (ESS). According to the assessment, the programme is considered to be Category I-2 or Medium Risk. This categorisation is supported by an Environmental and Social Assessment and associated Management Framework (Annex 6). The assessment and management plan comply with GCF requirements by: i) defining the context and baseline of each target area in which the interventions will be implemented; ii) identifying all negative and positive impacts associated with each of the proposed interventions; and iii) providing a management plan to describe how any adverse impacts identified during risk screening will be accounted for or mitigated throughout the programme implementation period. The primary environmental and social risks that may be associated with the programme — as identified by the assessment — are presented below.

198. Overall, the environmental and social risks assessed to be presented under this programme are minor and can be mitigated and/or effectively managed with the implementation of an appropriate strategy. The programme is expected to result in net-positive environmental and social benefits in the long term, which include: i) supporting economic security and environmental benefits by facilitating access to micro-finance and grant facilities for the implementation of small-scale EbA interventions; and ii) increasing water security through improved catchment management and the provision of water-efficient technologies and systems. However, several risks have been identified that, if not adequately managed, will reduce the resilience of certain communities or result in adverse environmental, social or economic impacts. These potential impacts are detailed below.

Environmental impacts

199. Several potential environmental impacts have been identified as being associated with the implementation of the proposed activities. These impacts are, however, avoidable and can be mitigated effectively through appropriate management measures. The main environmental impacts include:

- i) a possible increase in the use of pesticides and fertilisers in the EbA options, which may create both environmental and social risks by polluting natural resources;
- ii) an increase in the emission of short- and long-lived climate pollutants, by the execution of protocols and EbA measures, especially if the implementation of agroforestry systems includes the use of chemically sourced nitrogen fertilisers;
- iii) methodological approaches to assessing ecosystem services may generate biases at the time of assessment, affecting the perceived importance of specific ecosystems, therefore impacting on biodiversity and habitats; and
- iv) unintended promotion of deforestation for fuelwood to be used in the furnaces and stoves being marketed under the programme.

200. The environmental impacts listed above can be effectively managed by, for example: i) the dissemination of specific guidelines on the use of fertilisers; ii) awareness-raising on strategies for the sustainable harvesting of fuelwood; and iii) clear methodological approaches to assessing ecosystems. For further information on specific management and mitigation strategies relating to each activity, please refer to Annex 6: Environmental and Social Management Framework (ESMF).

Social impacts

201. The potential for adverse social impacts under the programme has been assessed as more likely than environmental impacts. These adverse impacts largely pertain to the potential for economic and cultural displacement, adverse economic impacts resulting from debt accrued through the use of financing mechanisms and restrictions on land use. The main potential social impacts include:

- i) non-recognition or alteration of Indigenous People's livelihood strategies, cultural practices, religious beliefs or governance structures;
- ii) economic displacement and an increase in economic insecurity in the event that credit conditions are not suitable for small producers, or that interest rates are not commensurate with the return rate of EbA interventions;
- iii) increased restrictions on land-use for small scale farmers, Indigenous Peoples or other marginalised groups;
- iv) unequal opportunities for indigenous peoples and other marginalised groups to benefit from the programme interventions, particularly if the opportunities are not promoted in an accessible, culturally appropriate and inclusive manner; and
- v) limited inclusion of all stakeholder priorities given the large geographic extent, diverse stakeholder groups and range of sociocultural contexts.

202. These impacts are all manageable or avoidable and can be effectively reduced and/or mitigated through the implementation of appropriate management strategies. For further information please refer to Annex 6: ESMF.

Indigenous Peoples

203. Potential adverse impacts on Indigenous Peoples are a critical concern for the GCF and although no unavoidable impacts on Indigenous Peoples have been identified in the risk screening, certain interventions may result in unintended impacts. However, it is important to note that the programme has been designed to be open and inclusive and it will consider the languages, dialects and cultural practices of all beneficiary groups throughout implementation to ensure equal access and fair distribution of programme benefits. In addition, because adaptation measures have been designed with an ecosystem-based approach that inherently respects Indigenous Peoples' cultural relationship with their territory, it is assumed that Indigenous Peoples will not be affected. CABI — as the AE — has experience in facilitating the inclusion of Indigenous Peoples in GCF-financed projects and will maintain oversight.

Grievance Redress Mechanism

204. The established CABI Grievance Redress Mechanism (GRM) will be implemented for this programme. The mechanism is approved by the GCF and has been used for a prior GCF project. The GRM will be used at all stages of the programme and will provide an accessible, rapid, fair and effective response to concerned stakeholders, especially any vulnerable groups who may lack access to formal legal systems or support. For further information please refer to Annex 6: ESMF.

G.2. Gender assessment and action plan

205. During programme preparation, a gender assessment was conducted to understand the differential vulnerability to, and impact of, climate change on women and men in the Dry Corridor and Arid Zones of the seven participating countries. From the findings of this assessment, and an analysis of the proposed programme solution, a gender action plan (GAP) was developed. The process to produce these components included a desktop review of available literature, as well as stakeholder and beneficiary consultations held in Guatemala, Honduras, El Salvador, Nicaragua, Costa Rica, Panama and Dominican Republic. The assessment does note that gender-disaggregated data is limited, particularly regarding indicators and information on land tenure, participation in decision-making, livelihood activities, and access to credit and finance.
206. Traditional gender roles and stereotypes in the programme area, as well as the resulting asymmetrical power differences they produce, create a culture of social exclusion that leaves women vulnerable to the impacts of climate change, with reduced capacity or opportunity to adapt. The patriarchal and sexist roles, norms and stereotypes create inequalities in social relations and cultural institutions, wherein men maintain 'productive', income-generating roles and women perform mostly 'reproductive', unpaid duties. Notably, this manifests in women's: i) reduced knowledge of information systems (such as early warning or forecasts); i) low participation in training; iii), limited access to finance and credit; iv) inequitable management and control of natural resources and land; v) greater exposure to violence; vi) limited participation in decision-making institutions; and vii) exclusion from accessing benefits and public resources for development. In addition to undermining their agency, capabilities and safety, this system limits the realisation of women and girls' rights, which is particularly evident in rural and indigenous contexts where gender inequalities are most pronounced and baseline vulnerability (for all genders) is highest.
207. The profile of migration from the region has changed in the past decade, with women increasingly becoming active agents in the migration process, rather than passive actors relative to the husbands or male family members' decisions. This is notable because migrant women face additional risks to men because of the same gender conditions that shape their vulnerability at origin. This can manifest in isolation, lack of social networks, precarious employment conditions, unsafe living environments, and exposure to violence, exploitation and even enslavement.
208. Although potentially more relevant to mitigation activities, the assessment also notes that female environmental and rights activists in the region are particularly targeted by extractive industries (mining, hydroelectric, monocultures and/or oil) for violence, stigmatisation and criminalisation. These conditions, wherein women experience discriminatory, violent and exclusionary practices, are unfavourable to their continued participation and representation in activist spaces, which may result in unfavourable biases and even further inequality.
209. Regarding climate change, consultations conducted for the assessment described the effects of droughts and floods on women, which include: i) multiplication of the 'reproductive' day, wherein women's burden of work increases; ii) negative impacts on agriculture and fishing for smallholder families, particularly for female-headed households; iii) increased migration and displacement; iv) decreased forest cover reducing access to resources like water, food, fuels, and non-timber forest products (including herbs and medicinal plants) which has additional implications for household health; v) reduced opportunities for women in income-generating work such as agriculture or handcrafts (as a result of reduced resources); vi) reduced access to clean drinking water, which has further opportunity costs for women and girls' exposure and time spent collecting water; vii) negative impacts on women's largely-marginalised physical and mental health due to stress, hunger and malnutrition, heatstroke and kidney disease, and 'household' diseases such as Zika and mites.
210. The assessment describes how the conditions and vulnerabilities described above will influence programme implementation, particularly noting how social exclusion may preclude women from participating meaningfully in decision-making processes and accessing the benefits of the programme, such as training and economic empowerment. To mitigate against biasing against women as programme beneficiaries during implementation and subsequently, the gender action plan outlines several interventions in response. These interventions will help ensure the programme does not unintentionally contribute to perpetuating or worsening gender equality and gender-based climate vulnerability in the Dry Corridor and Arid Zones. The GAP seeks to

ensure women are empowered and enabled to meaningfully contribute to, and benefit from, programme activities and that stakeholders' actions and programme processes make considerations for the needs, interests and vulnerabilities of women. The GAP includes clear targets, gender-sensitive design features and quantifiable performance indicators to promote women's participation and benefits from the programme.

211. The actions of the GAP include:

- developing women-oriented financial products and services to finance EbA investments;
- capacity-building and awareness-raising amongst stakeholders on the importance and implementation of gender mainstreaming/gender equality;
- strengthening the technical capacities of women's organisations at the local level; and
- ensuring the active participation and empowerment of women producers and women's organisations in programme processes.

212. Implementation arrangements of the programme also include a Gender Officer under the RPMU and a gender expert in each of the PEUs.

213. The full gender assessment and programme-level gender action plan are included as Annex 8.

G.3. Financial management and procurement

214. CABEL, through its current structure, including its Representation Offices in each country of the Programme, the Office of Compliance, the Office of Environmental and Social Safeguards, the Office of Supervision of Credit Operations, and the Internal Audit Office, will oversee, monitor and follow up the implementation of the Dry Corridor Programme. It is through this structure and the legal obligations derived from the contracts between CABEL and the IFIs, and the contracts between CABEL & FAO and CABEL & CCAD that CABEL can guarantee that the programme will be implemented under CABEL's policies and procedures. In particular:

- The Private Sector Division, through CABEL's Representation Office in the countries will be responsible for the coordination of actions and activities regarding the follow up, monitoring and oversight of the programme; the management and process of the disbursements needed by the programme executing entities, the IFIs and the PMU to carry out the programme's planned activities.
- The Credit Division, through its Office of Credit Operations Supervision will be responsible for leading the oversight process of the physical and financial execution of the programme.
- The Office of Environmental and Social Sustainability will be responsible for designing, completing and validating the I-BCIE questionnaire, as well as generating the calculation and Final Evaluation Report ex ante, which is an integral part of the intervention's credit opinion, which is presented with the operation to the CABEL approval authorities, as well as accompanying the supervision process with the Office of Credit Operations Supervision.
- The CABEL Independent Evaluation Office is responsible for the validation of the I-BCIE questionnaire, the development of the mid-term and ex post calculation and Final Evaluation Report.
- CABEL's Procurement Unit, in coordination with CABEL's Office in the countries, will oversee the procurement activities required by the programme, to ensure that the implementation is in line with CABEL's Policy for Procurement of Goods, Works, Services and Consulting with Bank resources and its application rules. The purpose of these regulations is to establish the guidelines and procedures that the executing entities must comply with for procurement in the execution of operations.
- The CABEL's Integrity and Compliance Office will be responsible for analyzing and proposing a course of action of the reports/allegations related to prohibited practices to an Integrity Committee, in accordance with CABEL's internal regulations and the applicable terms to the different contractual relationships with counterparties.
- The Finance Division through its Partnerships and International Cooperation Department is responsible for acting as the GCF focal point for the follow up of the programme execution and coordinating the

development and delivery of all reports agreed with GCF in order to guarantee a proper monitoring and programme execution.

215. Regarding the Fiduciary Compliance, CABEL is an International Multilateral Development Bank, who has policies and procedures in place regarding antifraud/anticorruption, AML/CFT and sanctions in accordance with CABEL's nature and are currently in force and in accordance with current international standards.
216. Policy and Procedural Manual is applied and observed in all active and passive operations, and in those involving procurement of goods and services in which the Bank interacts or is the counterpart of another natural or juridical person. CABEL's Money Laundering Prevention Policy, Manual and procedures are currently in force and in accordance with current international standards. This Policies and Manuals will be applied in order for the programme and its activities to conform to any UNSC sanctions programs that may be applicable, also to prevent that no entities or individuals involved in the programme as counterparties or implementers are listed on any UNSC sanctions lists.
217. CABEL shall neither accept nor maintain a commercial relationship or operation if it suspects that funds of a client or potential client may originate from criminal activities.
218. CABEL has a Risk Rating System, which is comprised of comprehensive and periodic reviews of all private and non-sovereign sector operations.
219. CABEL uses Sentinel (World Check) for the screening of all counterparties. We use Google in order to review our direct counterparties and PEP. Additionally, CABEL has implemented a swift sanctions screening to monitor all incoming and outgoing CABEL transactions and has acquired banker's almanac.
220. On the other hand, CABEL has implemented several mechanisms (web, email, voice mail, among others) in which reports and allegations of irregularities, unethical situations, the commission of prohibited practices or regulatory breaches in the use of its funds or resources administered by CABEL can be submitted. Reports about environmental and/or social issues or damages related to CABEL's projects can be reported as well. Any natural or juridical person, including CABEL's staff members may submit a report. Anonymously reports are permitted.
221. The Integrity and Compliance Office is in charge of analysing and proposing a course of action of the reports/allegations related to prohibited practices to an Integrity Committee, in accordance with CABEL's internal regulation and the applicable terms to the different contractual relationships with counterparties.
222. As demonstrated by its status as an Accredited Entity to the GCF, CABEL has extensive expertise in working with international funds and has a good track record in implementing projects using international financial management practices. For the proposed programme, CABEL will be responsible for the fiduciary aspects and will be accountable for all financial and investment activities.
223. The procurement of goods and services (including consultants' services) for activities financed by the GCF will be carried out following CABEL's standard practices and procedures — including its procurement and consultants' guidelines — which are aligned to those of the GCF, as well as to all national laws. Further details on the procurement of goods and services, as well as CABEL's applicable procurement procedures are provided in the Procurement Plan in Annex 10. Procurement executed by EE's will abide to each EE's regulations, that can be found in annex 29. The EEs will be accountable to the sound management of funds to the AE while applying its internal control of procedures, guaranteeing the adequate internal control for operability of administrative areas.
224. CABEL will provide to the GCF:

- Annual progress reports on the status of programme implementation, including the disbursements made during the relevant period — or more frequent progress reports if requested by the GCF; and
- an annual audited financial statement of the specific account prepared by an independent auditor or evaluation body.

225. CABEL will observe the highest ethical standards during the procurement and execution of this programme. CABEL will ensure: i) substantive quality of programme implementation; ii) effective use of both international and national resources allocated to it; and iii) proper coordination among all programme stakeholders, in particular national, sub-national and local partners. Details of the programme management structure of the programme and the respective roles that will be played by CABEL are provided in Section B.4. Implementation arrangements.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☒ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Description of EbA, water- and resource-efficient technologies eligible for finance
- ☒ Annex 23 Operational guidelines for financial mechanisms
- ☒ Annex 24 Forestry and agricultural sectors baseline
- ☒ Annex 25 Food and water security baseline
- ☒ Annex 26 Description of selected municipalities

☒	Annex 27	Past and ongoing projects in the region
☒	Annex 28	ToR for main programme posts
☒	Annex 29	EE Capacity assessment
☒	Annex 30	EbA finance demand and capacity assessment
☒	Annex 31	EbA site specific plans for Costa Rica and Honduras

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



Despacho de la Ministra
Ministerio de Ambiente y Energía

To: The Green Climate Fund ("GCF")

San José, Costa Rica, September 9th, 2021

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic

Dear Madame, Sir,

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in Costa Rica as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Minister of Environment and Energy, Andrea Meza Murillo, the NDA of Costa Rica.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Costa Rica has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of Costa Rica;
- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.



**Despacho de la Ministra
Ministerio de Ambiente y Energía**

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

ANDREA
MEZA
MURILLO
(FIRMA)

Firmado digitalmente por ANDREA
MEZA MURILLO (FIRMA)
Fecha: 2021.09.09 19:25:12 -06'00'

Andrea Meza Murillo
Minister of Environment and Energy
Costa Rica

MARN-DCC-UCI-057-2021

San Salvador, El Salvador
September 8, 2021

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic

Dear Madam, Sir,

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in El Salvador as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Ministry of Environment and Natural Resources, Fernando Andrés López Larreynaga, the NDA of El Salvador.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

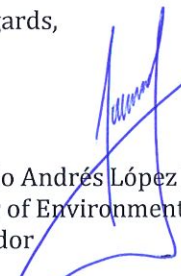
- (a) The government of El Salvador has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of El Salvador;
- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Fernando Andrés López Larreynaga
Minister of Environment and Natural Resources
El Salvador





Ministro

MINISTERIO DE AMBIENTE Y RECURSOS NATURALES
GUATEMALA, C.A.

To: The Green Climate Fund ("GCF")

Guatemala, september 9, 2021
Oficio MI-1162-2021/AELM-gpvg

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic

Dear Madam, Sir,

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in Guatemala as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Minister of Environment and Natural Resources, Mario Roberto Rojas Espino, the NDA of Guatemala.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Guatemala has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of Guatemala;



Ministro

MINISTERIO DE AMBIENTE Y RECURSOS NATURALES
GUATEMALA, C.A.

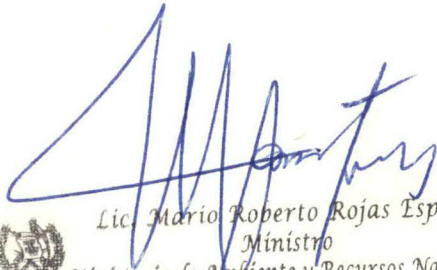
- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Lic. Mario Roberto Rojas Espino
Ministro
Ministerio de Ambiente y Recursos Naturales





GOBIERNO DE LA
REPÚBLICA DE HONDURAS



SECRETARÍA
DE RECURSOS NATURALES
Y AMBIENTE

Tegucigalpa, Honduras
September 6th, 2021

OFICIO No. UCEMR-DMA-0301-2021

Mr. Yannick Glemarec
Executive Director
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic

Dear Mr Yannick Glemarec,

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in Honduras as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Secretariat of Natural Resources and Environment; The National Designated Authority of the Republic of Honduras.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Honduras has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of Honduras;



OFICIO No.UCEMR-DMA-0301-2021

2/2

- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

LIC. LILIAM RIVERA

Secretariat of Natural Resources and Environment
Honduras



LRH/COOPEX
Cc: Archivo



Dirección: Despacho de Recursos Naturales y Ambiente 100 ms. al Sur del Estadio Nacional,
Tegucigalpa M.D.C., Honduras, C.A.
Teléfono: 2232-9200 Página Web: <http://www.miambiente.gob.hn>

Gobierno de Reconciliación y Unidad Nacional
Unida Nicaragua Triunfa



Managua, Nicaragua, September 8, 2021
MHCP-VM-E-0890-08-2021

Mr.

Yannick Glemarec

Executive Director

Green Climate Fund

G-Tower, 24-4 Songdo-don, Yeonsugu Incheon City

Republic of Korea

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic.

Dear Mr:

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in Nicaragua as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Ministry of Finance and Public Credit, Mr José Adrián Chavarría Montenegro, the NDA of Nicaragua.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The government of Nicaragua has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.



Ministerio de Hacienda y Crédito Público
Avenida Bolívar, Frente a la Asamblea Nacional
Teléfonos 22227061, 22227718, 22281255
Fax. 22222509 www.hacienda.gob.ni

Gobierno de Reconciliación y Unidad Nacional
Unida Nicaragua Triunfa



Managua, Nicaragua, September 8, 2021
MHCP-VM-E-0890-08-2021

Pág. 2

- b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of Nicaragua.
- c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


José Adrián Chavarría Montenegro

Vice Minister General

Ministry of Finance and Public Credit Nicaragua



Cc.:
Archive

Ministerio de Hacienda y Crédito Público
Avenida Bolívar, Frente a la Asamblea Nacional
Teléfonos 22227061, 22227718, 22281255
Fax. 22222509 www.hacienda.gob.ni

Panamá, 6 de septiembre de 2021
DM-1655-2021

Señor
YANNICK GLEMAREC
Director Ejecutivo
Fondo Verde para el Clima (FVC)
En su despacho

Re: Propuesta de financiamiento para el GCF del Banco Centroamericano de Integración Económica (BCIE) sobre Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de República Dominicana.

Estimado Señor. Glemarec,

Nos referimos al programa titulado Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de la República Dominicana en Panamá como se incluye en la propuesta de financiamiento que nos presentó el Banco Centroamericano de Integración Económica (BCIE) el 27 de noviembre de 2019.

El abajo firmante es el representante debidamente autorizado del Ministerio del Ambiente, Milciades Concepción, la Autoridad Nacional Designada / Punto Focal de Panamá.

De conformidad con la decisión B.08 / 10 del GCF, cuyo contenido reconocemos haber revisado, por la presente comunicamos nuestra no objeción al programa Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas del República Dominicana como se incluye en la propuesta de financiamiento.

Al comunicar nuestra no objeción, se da a entender que:

(a) El gobierno de Panamá no tiene ninguna objeción al programa Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de la República Dominicana como se incluye en la propuesta de financiamiento;

(b) El programa Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de la República Dominicana, tal como se incluye en la propuesta de financiamiento, se ajusta a las prioridades, estrategias y planes nacionales de Panamá;

(c) De acuerdo con las salvaguardas ambientales y sociales del GCF, el programa Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de la República Dominicana, tal como se incluye en la propuesta de financiamiento, se ajusta a las leyes nacionales pertinentes, y regulaciones.

También confirmamos que nuestro proceso nacional para determinar la no objeción al programa Adaptación basada en ecosistemas para aumentar la resiliencia climática en el Corredor Seco Centroamericano y las Zonas Áridas de la República Dominicana, como se incluye en la propuesta de financiamiento, se ha seguido debidamente.

También confirmamos que nuestra no objeción se aplica a todos los proyectos o actividades que se implementarán dentro del alcance del programa.

Reconocemos que esta carta se pondrá a disposición del público en el sitio web del GCF.

Atentamente,


MILCIADES CONCEPCIÓN
Ministro de Ambiente
Autoridad Nacional Designada / Punto Focal de Panamá.

Albrook, Calle Broberg, Edificio 804
República de Panamá
Tel.: (507) 500-0855

Panama, September 6 , 2021
DM-1655-2021

Mr.
YANNICK GLEMAREC
Executive Director
Green Climate Fund

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic.

Dear Mr. Glemarec:

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in Panama, as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Ministry of Environment, Milciades Concepción, the NDA / Focal Point of Panama.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Panama has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of Panama;
- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

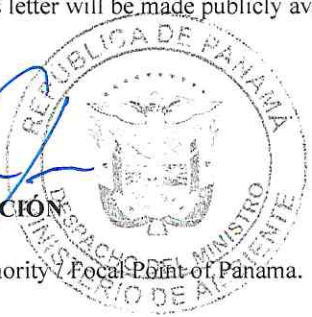
We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


MILCIADES CONCEPCIÓN
Minister of Environment
National Designated Authority / Focal Point of Panama.



Albrook, Calle Broberg, Edificio 804
República de Panamá
Tel.: (507) 500-0855

VCI-0139-2021

Santo Domingo, Dominican Republic
September 6, 2021

To: The Green Climate Fund (“GCF”)

Re: Funding proposal for the GCF by Central American Bank for Economic Integration (CABEI) regarding Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic

Dear Madam, Sir,

We refer to the programme titled Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic in the Dominican Republic as included in the funding proposal submitted by Central American Bank for Economic Integration (CABEI) to us on November 27th, 2019.

The undersigned is the duly authorized representative of the Ministry of International Cooperation, Milagros De Camps German, the NDA of the Dominican Republic.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Dominican Republic has no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal;
- (b) The programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Dominican Republic;



- (c) In accordance with the GCF's environmental and social safeguards, the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Milagros De Camps German
Deputy Minister of International Cooperation
Ministry of Environment and Natural Resources
Dominican Republic



Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Central American Bank for Economic Integration (CABEI)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Costa Rica, Dominican Republic, El Salvador, Guatemala, Honduras, Nicaragua, and Panama.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, May 6, 2021
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the national language of all participating countries.
Link to disclosure	English: https://www.bcie.org/en/operations/in-process-operations/dry-corridor-program Spanish: https://www.bcie.org/operaciones-y-adquisiciones/programa-del-corredor-seco-centroamericano
Other link(s)	N/A

Remarks	An ESMS consistent with the requirements of a Category I-2 programme is contained in the “Environmental and Social Management Framework”, developed according to CABEL’s policies and procedures on Environmental and Social Safeguards (ESS).
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, May 28, 2021
Place	Office and website of the NDAs of Costa Rica, Dominican Republic, El Salvador, Guatemala, Honduras, Nicaragua, and Panama. Costa Rica: https://minae.go.cr/noticias-minae/documentos/153-marco-de-gestion-ambiental-y-social Dominican Republic: https://ambiente.gob.do/transparencia/informes-de-seguimientos-a-los-programas-y-proyectos/#1017-wpfd-marco-de-gestion-ambiental-y-social-mgas El Salvador: https://marn.gob.sv/programas/adaptacion-basada-en-el-ecosistema-para-aumentar-la-resiliencia-climatica-en-el-corredor-seco-centroamericano-y-las-zonas-aridas-de-la-republica-dominicana/ Guatemala: https://www.marn.gob.gt/paginas/Programa del Corredor Seco Centroamericano Honduras: http://www.miambiente.gob.hn/static/documentos/MARCODEGESTIONAMBIENTALYSOCIALMGAS.pdf Nicaragua: http://www.hacienda.gob.ni/documentos/documentos-mhcp/documentos-despacho-del-ministro/publicaciones/Annex%206-Marco%20de%20Gestio301n%20Ambiental%20y%20Social-MGAS.pdf/view Panama: https://dcc.miambiente.gob.pa/adaptacion-y-resiliencia/

Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBC
Date of GCF's Board meeting	Monday, October 4, 2021

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP174

Proposal name:	Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic
Accredited entity:	Central American Bank for Economic Integration (CABEI)
Country/(ies):	Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama and Dominican Republic
Programme/programme size:	Large

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme design, featuring three different financing facilities (loan facility, grant facility and guarantee facility), will encourage participation of the private sector in ecosystem-based adaptation (EbA) type of investments and diversify financial systems in the Central American region to create an enabling environment for investment in EbA and water/energy-efficient technologies at scale.	The implementation arrangements include several different entities and therefore result in complex contractual obligations and higher project management costs.
The technical assistance facility is tailored to increase technical capacities of local governments, farmers and non-regulated financial institutions to monitor and assess the results of the implementation of EbA and water/energy-efficient investments as well as document the economic and ecosystem benefits of these investments.	GCF and CABEI will be lending to different entities and therefore GCF will be taking higher risks. Checks and balances will be further defined in the term sheet.
Target catchments were selected following a rigorous vulnerability assessment that included hydrological modelling of the three watersheds and extensive consultations with key stakeholders in each country. The comprehensive feasibility study, detailed methodology of catchment development; two of the seven catchment analyses developed based on the methodology as well as further documentation submitted with the second re-submission following non-endorsement by the independent Technical Advisory Panel provided sufficient evidence on the strong climate rationale of this project.	The programme is heavy on grant financing for capacity-building and technical assistance, and the accredited entity (AE) administration costs are higher.

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVIII, titled “List of proposed conditions and recommendations”, respectively.

II. Summary of the Secretariat’s assessment

2.1 Programme background

3. **Background/history of the programme to inform assessment of the funding proposal:** The term “Dry Corridor” emerged in the last three decades to describe the increasing frequency and intensity of droughts in the region. With more than one million families in the Central American Dry Corridor and the arid zones of the Dominican Republic reliant on agriculture for their livelihoods, the impacts of increasingly severe droughts over the past five years have resulted in the need for considerable increase in climate finance in the region. In 2014 alone, intense drought resulted in an estimated USD 465 million worth of agricultural losses across the region, with more than 2.5 million people experiencing food insecurity. From 2015 to 2018, food insecurity has increased annually in the region, with indigenous populations, women and rural communities the most likely to suffer from chronic hunger and malnutrition. By the last quarter of 2018, 2.2 million people had experienced crop losses that year, with 1.4 million urgently needing food assistance. These impacts are expected to continue to worsen under future climate scenarios.

4. EbA measures, such as forest conservation, reforestation, planting of agroforestry systems, improved irrigation systems and instalment of vegetative barriers, result in substantial improvements for all key hydrological variables in watershed areas where such agriculture activities take place. Furthermore, EbA measures result in soil humidity improvements and benefits for water availability for both agriculture and forests. The changes in water production (surface water) and percolation (water recharge) show even greater benefits from the implementation of a combination of EbA measures. The Dry Corridor and arid zones experience significant water shortages, and EbA is seen as key to ensuring water availability throughout the year. Lastly, the implementation of this combination of EbA measures also shows very positive impacts on soil erosion rates.

5. The programme will address the following barriers: i) access to credit for adaptation interventions among vulnerable populations; ii) access to financial resources for the implementation of EbA, water- and energy-efficient technologies among the private sector; iii) knowledge and understanding of climate change impacts as well as appropriate adaptation measures among decision-makers, communities and commercial farmers; iv) limited integration of EbA and other adaptation measures into sectoral policies; v) technical capacity within governments and local communities to implement EbA and other adaptation measures; and vi) development of economic incentives for investment in sustainable natural resource management and EbA.

6. Without comprehensive adaptation interventions, agricultural losses will continue to increase, along with ecosystem loss, water shortages, poverty and food insecurity. These impacts will ultimately increase the exposure and sensitivity of communities in the Dry Corridor and arid zones who are already extremely vulnerable to climate change.

7. **Climate objective:** The Central American Dry Corridor and the arid zones of the Dominican Republic are among the world’s most vulnerable tropical regions to the impacts of climate change, with the Dominican Republic, El Salvador, Guatemala, Honduras and Nicaragua recognized as being among the 16 most vulnerable countries in the world to extreme climate events for the period 1998–2017. Currently, the Dry Corridor encompasses ~64 per cent of the municipalities in Central America and is expected to extend to ~85 per cent by 2050. Climate

projections in the region indicate that, by the end of the century, temperatures will increase by 3–3.5 °C under Representative Concentration Pathway 4.5 and by 6–7 °C under Representative Concentration Pathway 8.5. At the same time, mean annual rainfall is programmed to decrease by 11 per cent on average. These changes will be compounded by further changes in the El Niño-La Niña cycle, which will result in prolonged droughts as well as more frequent and extreme rainfall events across the region. If average global temperatures reach 1.5 °C above pre-industrial levels, extreme El Niño events are likely to happen twice as often, approximately every 10 years as opposed to every 20 years, and have severe consequences for agriculture and food security in the region.

8. The hydrological modelling undertaken for the programme areas has demonstrated that a combination of adaptation interventions, such as forest conservation, reforestation, transition to agroforestry systems, introduction of irrigation systems and vegetative barriers, would result in substantial improvements for all key hydrological variables in the selected catchment areas. In response to the above-described climate risks, GCF financing will be used to create a paradigm shift in the way EbA investments are deployed at a regional level in the Dry Corridor and arid zones. This shift will see governments of the seven participating countries—Costa Rica, the Dominican Republic, El Salvador, Guatemala, Honduras, Nicaragua and Panama—promoting large-scale EbA initiatives. In addition, smallholder farmers and commercial farmers, as well as entrepreneurs in rural communities, will have increased access to the financial resources and technical skills required to implement EbA interventions.

9. Together, these interventions will catalyse the autonomous upscaling of EbA and water- and energy-efficient technologies by combining technical capacity-building, policy instruments and accessible financial mechanisms. This approach will transform the financial landscape in the region to enable vulnerable groups to access climate finance and implement climate-resilient measures to strengthen their adaptive capacity. The proposed programme will directly benefit ~522,200 people across the target catchments through on-the-ground interventions, with ~286,000 of these people also able to access loans for the implementation of adaptation measures through financial mechanisms established under the programme.

10. **Financing information and environmental and social safeguards (ESS) category:** A combination of GCF loan and grant finance, a guarantee, and a senior loan provided as co-finance will be used to establish three financial facilities at the regional level, namely: i) a grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing (USD 28 million); ii) an EbA lending facility for small- and large-scale EbA investments at farm-, enterprise- and household-level (USD 102 million); and iii) a guarantee facility that will provide an additional layer of security to financial institutions in order to promote EbA finance (USD 30 million). The facilities will be managed by CABEL with a blended financing structure, where funds in the form of senior loans are channelled via CABEL and intermediary financial institutions (IFIs) to partner financial institutions (PFIs) towards end beneficiaries. The loan facility is structured for GCF to assume the credit risk of the PFIs using a limited recourse loan. By taking this additional risk compared to CABEL which is lending to larger IFIs, GCF will lower the interest rate to the PFIs and ultimately the end borrowers, improving access to affordable finance for smallholders in the target catchments. The guarantee facility will encourage the IFIs and PFIs to mobilize more of their own resources, while also allowing them to lend to more vulnerable borrowers who would not otherwise be able to meet their collateral requirements. Total GCF financing is USD 84.2 million in the form of grants and USD 60 million in the form of loans, with co-financing from countries of USD 51.2 million in the form of in-kind and USD 42.8million in the form of senior loan, thus bringing the co-financing ratio to 1:0.51. The programme environmental and social safeguards category is IS-2. The entity has submitted relevant documentation (an environmental and social management framework) as part of the funding proposal package.

2.2 Component-by-component analysis

11. The programme will target seven catchments for programme interventions, one in each of the participating countries (except for the Dominican Republic, which will target two catchment areas). The programme's regional approach will create an enabling environment for scalable and sustainable catchment and water demand management, facilitating large-scale investment in and implementation of EbA and water-efficient technologies. The total grant request of US\$84.3 million will be used in the following ways: i) US\$10.0 million will be used to strengthen the technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures through the activities comprised in output 1; ii) US\$ 28.8 million will be used to implement large-scale adaptation interventions — including EbA, water- and resource-efficient technologies — in rural communities across the region throughout output 2; iii) US\$28 million will form the grant facility through which vulnerable communities can access on-grants to foster the creation of a business ecosystem around EbA available technologies as per activity 1.2.; iv) US\$6.3 million will be used to generate and share information on climate change adaptation and finance across the region, and mainstream it into local and national policies within the output 3; v) US\$6.4 million will be used to set up the financial mechanisms and strengthen the technical capacity of CABEL accredited and non-regulated financial institutions to access and channel funds through them throughout output 4; vi) US\$3.8 million will be used for overall management and monitoring of programme activities (Independent evaluations and PMC costs); and vii) US\$0.7 million is requested for contingency. The EbA lending facility (EbA credit line) will be established and operationalised with the GCF senior loan provision of US\$60 million and CABEL US\$42.8 million co-finance together being offered to PFIs as blended finance mechanism at below-market conditions for small- and large-scale EbA investments at the farm-, enterprise- and household-level.

Component 1: Mainstreaming of EbA, water-efficient technologies and natural resource-based businesses into selected catchments (total cost: USD 121 million; GCF cost: USD 72.6 million, or 60 per cent)

12. The component includes the mainstreaming of EbA and water- and energy-efficient technologies through further strengthening of technical capacities of local governments, farmers and rural communities. A community-based approach will be adopted to identify site-specific interventions in each of the 36 municipalities targeted by the programme during its first year. This will be supported by the establishment of community-level monitoring and evaluation committees, which will promote country ownership and sustainability of programme interventions. The technical capacity of these committees will be strengthened to enable the monitoring of biophysical, social and economic conditions in their local catchment areas, as well as to assess the level of climate vulnerability, risks and opportunities.

13. Technical assistance will also be provided to farmers and rural communities in the 36 municipalities for the implementation of on-the-ground EbA practices, including silvopasture, agroforestry and sustainable land management¹. It will support the establishment of natural resource-based businesses (NRBs), which will support the implementation of tree nurseries and the manufacturing, distribution or repair of water-efficient technologies as well as the non-destructive use of ecosystem goods and services provided through EbA, such as businesses developed around the ecotourism industry or non-timber forest products.

14. The technical assistance under this component will also support the integration of EbA in policy and regulatory frameworks at the local level focusing mainly on water management regulations and water management community structures as well as local land-use planning

¹ Technical assistance will not be limited to farmers accessing the financial mechanisms from output 1; it will be available to farmers in the target municipalities that are open to adopting improved practices but lack the technical knowledge to do so.

instruments. Country-specific implementation protocols will be developed for: i) the implementation of sustainable EbA practices; ii) the adoption of water-efficient technologies; and iii) the adoption of energy-efficient technologies that reduce fuel wood demand.

15. A USD 28 million grant facility will be established to support site-specific EbA and water-/energy-efficient technologies for the most vulnerable communities, which are unable to access loan finance. Based on approved eligibility criteria, a selected set of interventions will be financed through this facility. On-the-ground EbA-oriented grant support will benefit community-based institutions including *mancomunidades*,² municipal associations, farmers' cooperatives, solution providers and/or civil society organizations. Eligible programmes will include community-level investments, the establishment of water funds, seed funding for payment for ecosystem services and support for the adoption of selected EbA practices and/or water- and energy-efficient technologies.

16. Under component 1, a regional knowledge hub will also be established. The knowledge hub will act as a repository for EbA information, including, but not limited to, all protocols developed by the programme as well as the best practices and lessons learned collected through the programme's monitoring and evaluation processes and previous and ongoing complementary programmes. This will include guidance on how to access opportunities for accessing finance provided through the programme, with specific focus on women's economic empowerment. Researchers will be hired in every country to assist with the synthesis of findings from catchments, programme advances, impacts and experiences learned for dissemination across the region.

17. Lastly, this component will support the preparation of a methodology for the valuation of ecosystem services and development of accounting profiles for natural capital for integration into development plans in the Dry Corridor and arid zones. The methodology will draw on best practices and lessons learned from ongoing initiatives, including the: i) "incentivo forestal" in Guatemala; ii) payment for environmental services in Costa Rica and Panama; iii) Water Fund of the city of Quito in Ecuador; and iv) water funds established in cities in Costa Rica, the Dominican Republic and Guatemala. Ultimately, the valuation of ecosystem services will be used to promote the establishment of economic incentives for EbA investments such as water funds and payments for environmental services.

18. The final activity under this component includes large scale EbA landscape interventions financing through grant support. Financing will be provided to the most vulnerable communities in the 36 municipalities³ that lack the capacity to access financing and equipment to implement adaptation interventions and cannot receive loans through the loan or grant facilities. Such landscape interventions include: i) establishing one tree nursery per municipality focusing on native species and fuelwood trees (36 total); ii) establishing 4,160 ha of forest protection zones across the seven catchment areas; iii) protecting 1,820 ha of natural forest in major recharge areas and riparian zones, including assisted regeneration; iv) restoring 5,170 ha of natural forest areas across the seven catchments; v) restoring 820 ha of native pine forests in Guatemala, Honduras and Nicaragua; vi) establishing 1,293 km of agroforestry systems using diversified living fence arrangements in basic grain crops; vii) establishing 612 ha of agroforestry systems for natural shade in coffee plantations; viii) establishing 1,280 km of silvopasture systems using diversified living fence arrangements; ix) establishing 1,280 ha of silvopasture systems across the 36 target municipalities; x) establishing 5,305 ha of sustainable fuelwood and timber plantations across the 36 target municipalities; xi) establishing 129 km of firebreaks for forests and plantations; xii) constructing 118 km of living barriers for soil conservation around agricultural plots; and xiii) constructing 118 km of superficial drainage for

³ A description of the target municipalities and catchments is provided in section 12 of annex 2: feasibility study.

soil conservation xiv) and establishing 109 rainwater harvesting systems, 156 community-level rainwater reservoirs, etc.

Component 2: Financing and implementation for EbA as well as water- and energy-efficient technologies (total cost: USD 139 million; GCF cost: USD 96.4 million, or 70 per cent)

19. Under this component the proposed programme will establish and operationalize financial mechanisms to facilitate investment in EbA practices, as well as the adoption of water- and energy-efficient technologies. This component will focus on providing grants and blended finance for EbA investments as well as water- and energy-efficient technologies. Such mechanisms will promote bottom-up identification of adaptation interventions and at the same time de-risk financial mechanisms available for financing adaptation.

20. Under this component, the EbA lending facility will be established and operationalized with the GCF senior loan provision of USD 60 million and the CABEL USD 42.8 million co-finance offered to PFIs as a blended finance mechanism at below-market conditions for small- and large-scale EbA investments at the farm-, enterprise- and household-level. By providing a blended financing mechanism composed of concessional finance from GCF in combination with a market rate senior debt as co-financing from CABEL, incentives are created for financial institutions on different levels to engage in EbA finance. IFIs in CABEL's established network will manage the EbA lending facility via the establishment of trusts for direct on-lending⁴ to local PFIs such as cooperatives, small banks, NGOs and rural community banks, among others. PFIs in turn will on-lend to end beneficiaries. Investments will be targeted towards community-level investment, rural micro, small and medium-sized enterprises or other institutions that can foster EbA, as well as water- and energy-efficient technologies, for end beneficiaries across different sectors.

21. A guarantee facility will be linked to the EbA lending facility to secure IFIs' on-lending of CABEL's co-financing funds as well as mobilize additional lending from IFIs' own resources and additional public and private investors' resources by mitigating against credit risks associated with commercial EbA finance. CABEL will manage the guarantee facility as a window in its recently established Guarantee Fund. The guarantee facility will cover end beneficiary loans which lack real guarantee means; where end beneficiaries are unable to provide full coverage, for example via real estate or movable assets, the guarantee facility will cover the missing participation and cover up to 75 per cent of the resulting loss in case of non-payment. Hence, it enables more end beneficiaries to access EbA finance as well as increases security of collateral provided from PFIs to IFIs in the form of generated EbA lending portfolio.

22. Lastly, to facilitate access to and operation of the facilities, these will be accompanied by a technical assistance facility that will develop capacity within municipalities and PFIs to monitor and assess the implementation of EbA and water/resource-efficient investment projects. Technical assistance will be a mandatory condition for PFIs to access the EbA lending facility. This will ensure that capacities, especially for adequate monitoring, reporting and verification systems, are in place to manage the EbA finance. Such technical assistance will focus on developing capacity within municipalities and PFIs to monitor and assess the implementation of EbA and water/energy-efficient investment programmes as well as to monitor and document the respective economic and ecosystem benefits. Where possible, information and communication technologies will be introduced to provide data analytics and satellite data insights into the EbA lending portfolio and climate impacts on the portfolio.

⁴ "On-lending" in this context refers to the process whereby selected PFIs will receive money via two mechanisms: from CABEL co-financing through a senior loan as well as from IFI trust structures at below-market rates. They will then lend this money with its associated rates (on-lend) to farms, enterprises and households for implementation of EbA and other adaptation measures.

Component 3: Programme management (total cost: USD 6.1 million; GCF cost: USD 3.8 million, or 61% per cent).

23. Programme managements costs equal to USD 6.1 million and will be used for the supervision and monitoring of the implementation of programme activities.

III. Assessment of performance against investment criteria

24. The programme will combine investments in water supply and irrigation, agroforestry, silvopastoral and forestry systems, and farmer and institutional capacity-building for climate resilience.

3.1 Impact potential

Scale: Medium to high

25. The proposed programme's contribution to climate-resilient sustainable development will directly benefit 1,084,441 people (50 per cent women) across seven catchment areas. While USD 84.3 million in grants are geared towards concrete and tangible benefits for an estimated total of 949,747 beneficiaries via the strengthening of stakeholders' capacities to better understand, develop and implement EbA solutions, a further USD 102.8 million in debt finance will build upon such created capacities, with an estimated number of 766,634 direct beneficiaries and 475,556 indirect beneficiaries (total of 1,084,441). The programme will also indirectly benefit 1,368,044 people across the region by facilitating increased access to finance for NRBs, mainstreaming EbA and other adaptation practices into national policies and building the capacity of governments to implement EbA.

26. The programme will reduce climate change vulnerability for farmers and producers' organizations by improving ecosystem services, primarily water regulation. Impacts of climate change have been described in the proposal based on the existing information of regional climate projections and downscaled through hydrological models at the basin level for three catchments in Costa Rica, Honduras and the Dominican Republic. The results of the hydrological modelling developed show a consistent trend in the impacts of climate change in substantially decrease of precipitation that leads to reduced water yield, reduced water infiltration, groundwater storage and increased erosion risk. Models used for the simulation of various climate change impacts in the target locations and suggested adaptation interventions provide for positive impacts over these variables when applied over climate change scenarios showing the capacity of these interventions to dramatically reduce the potential impacts of climate change.

27. Furthermore, the programme includes the development of site-specific intervention plans through activity 1.1. (sub-activity 1.1.3) for each of the seven catchment areas as a first step in the implementation of the programme. Two draft sample site-specific intervention plans were developed for Costa Rica and Honduras, and a methodology to develop site-specific adaptation plans was also clearly elaborated in the funding proposal. Those site plans will be validated during the initial stages of implementation in consultation with key actors in both countries.

28. The allocation of grants will be based on the impact and activities identified in the site specific adaptation plans, and will follow a clear set of principles set out in the funding proposal, which will ensure that activities to be funded contribute to a cohesive catchment approach targeting actions and reflect the principles that reduce pressure on ecosystems and the services they provide, as well as protecting, restoring or using biodiversity and ecosystems in a sustainable manner.

29. With the financial support under this programme it is estimated that the following will be the impact on the ecosystems and ecosystem services in the region: i) ~142,500 ha of

degraded agricultural land restored through the implementation of sustainable land use practices; ii) ~7,800 ha of forested areas restored; iii) ~4,100 ha of forest protected; iv) 36 tree nurseries established; v) ~1,900 ha of agroforestry systems established; vi) ~100 km of firebreaks, living barriers and superficial drainage solutions established; and vii) ~5,300 ha of sustainable fuelwood sources developed. Finally, at least 100 community buildings will be equipped with rainwater harvesting systems, 156 community-level rainwater reservoirs will be installed, 43 half-orange kilns will be made communally available, and 38 community-level solar-powered water pumping stations will be installed.

3.2 Paradigm shift potential

Scale: High

30. The proposed programme will catalyse a paradigm shift in the regional approach to catchment and water demand management in the Dry Corridor of Central America and the arid zones of the Dominican Republic. This shift will see governments promoting large-scale EbA investments as well as increased private sector participation, including commercial farmers and entrepreneurs in rural communities who would now have access to the financial resources and technical skills required to implement impactful adaptation interventions. The transformative effects of the programme will be achieved through several innovative approaches anchored into the programme design.

31. The programme has been specifically designed to create an enabling environment that capitalizes on the scalability of EbA, focusing on three core areas of support: financial, policy and technical. The programme will diversify financial systems in the region to create an enabling environment for investment in EbA and water-efficient technologies. This will be based on the establishment of several financial instruments with varying entry points to make climate finance more accessible to vulnerable communities. The enabling environment for investment will be further enhanced by building the technical and institutional capacity of PFIs and non-regulated financial institutions to finance and re-finance EbA and water-efficient technologies through all three of the proposed financial instruments.

32. The enhanced local evidence base for EbA that will be created through the knowledge hub and the monitoring and evaluation of programme activities will support national and local-level decision-makers to implement policies and incentives for EbA, in particular through improved valuation of natural capital.

33. Lastly, the programme will catalyse the autonomous upscaling of EbA interventions and water-efficient technologies by combining policy instruments (e.g. economic incentives such as payments for environmental services) with accessible financial mechanisms (e.g. commercial loans, trust funds, microfinance and guarantee funds). These mechanisms will transform the financial landscape in the region to enable vulnerable groups to access climate finance. Furthermore, the programme will promote the development of climate-resilient NRBs and alternative livelihoods at the community level.

3.3 Sustainable development potential

Scale: High

34. The programme will have significant sustainable development potential, with environmental, social and economic benefits. Some of the environmental co-benefits include conservation and/or restoration of forests and agroforestry systems that will act as natural buffers, controlling and mitigating the impact of floods, droughts, extreme temperatures and wildfires on ecosystems in the selected sites. Conserved and restored ecosystems will provide habitats for a range of terrestrial species, thereby improving biodiversity in programme sites.

35. Social co-benefits will mostly contribute to social development by improving the water and food security of vulnerable communities in the region. This includes improvement of the

targeted communities' health by reducing the risk of water-borne diseases, increased productivity of agricultural systems, and reduced inequality and poverty through increased financial inclusion of those participating in the proposed financing schemes. The programme will provide opportunities to develop sustainable NRBs and alternative livelihoods in ecotourism and in the collection of non-timber forest products.

3.4 Needs of the recipient

Scale: Medium to high

36. In the last decade, the Latin America and Caribbean region—under which the seven beneficiary countries fall—has achieved an average annual economic growth rate of 3.2 per cent. This economic insecurity is compounded by climate change impacts, including extreme rainfall and droughts. Economically insecure households, including vulnerable rural communities, are often slow to recover effectively from climate impacts. Rural communities have little or no access to adaptation finance, which limits their adaptive capacity. By improving access to financial resources for adaptation interventions, the adaptive capacity of rural communities in the Dry Corridor will be increased. With increased financial and technical capacity, communities will be able to withstand the negative impacts of severe droughts.

37. The national budgets of Dry Corridor countries are limited, and poverty eradication and sustainable development have been prioritized for funding over CCA. There is a high dependency on non-reimbursable international support to meet CCA goals across the Latin America and Caribbean region where the finance available for CCA is far below what is required to meet adaptation needs. The added burden of the COVID-19 pandemic on the economies of these countries will leave even less funds available for CCA in the participating countries. The proposed programme will provide access to technical and financial resources, knowledge products and government support structures that will enable rural communities to adapt effectively to climate change in the Dry Corridor.

3.5 Country ownership

Scale: Medium to high

38. CCA is recognized as a priority in the Central American region and has been integrated into several regional and national policies and strategies, including the Regional Climate Change Strategy (2010), Regional Strategy for the Conservation and Sustainable Use of Biodiversity in Mesoamerica, Regional Strategic Program for Forest Ecosystems Management, and Regional Agro-environmental and Health Strategy of Central America, among others.

39. In addition to this regional alignment, the programme contributes to achieving national priorities and aligns with numerous policies in each of the seven target countries. A summary of how it meets each country's national priorities is provided in table 3 of the funding proposal. Of the seven participating countries, three have submitted draft country programmes to GCF as part of readiness grant deliverables to date, and two of these countries (Guatemala and Honduras) feature the current proposal in their country programme. The funding proposal is also part of the entity work programme of the AE, CABEI.

40. All no-objection letters in support of this proposal have been provided.

3.6 Efficiency and effectiveness

Scale: Medium to high

41. The AE provided a comprehensive demand and supply analysis to justify the loan request as well as the programme's financial models, which foresee direct financial and indirect economic benefits resulting from implementation of the programme's activities. The demand and supply analysis includes details on the absorption capacity at the country level and establishes a comparison with the numbers of expected loans and financial entities through this

programme, which provides sufficient evidence that there is growing interest to engage in green finance by the financial institutions and that the financial infrastructure to serve the expected demand for EbA finance does exist, but could be strengthened. The financial internal rate of return of the programme stands at 10.3 per cent, net present value (NPV) at USD 491,083,311 and an economic internal rate of return at 17.3 per cent, with an NPV of USD 2,255,887,065 (both NPV discounted at an average of issued public bond rates as alternative investment).

42. While the grant solutions provide the basis for the establishment and expansion of a body of knowledge in target regions, the debt solutions will be fully self-sustained on a commercial basis, targeting potential further investment into the mechanisms created. The senior loans that make up the debt finance will be supported by a guarantee fund, which will de-risk the loans, creating incentives for financial institutions to engage in EbA finance, for which there exists no or only limited offerings as of today.

43. The deployment of the loan facility will follow a phased approach along the project implementation period to ensure the cohesive and meaningful impact of the financed activities as well as proper vulnerability assessments and identification of priorities for other vulnerable regions within the Dry Corridor where the intervention could be scaled up. This phased approach will also allow for more time to strengthen capacities of the financial institutions and to promote the financial mechanism with communities and producers. The first phase will be the launching of the facility, the second will be the replicating phase and the third one will be the scale-up of the facility. Each of these phases will have eligibility criteria for beneficiaries and predefined areas of loan coverage that will also be guided by a number of milestones that will need to be met in order to release the amount for the following phase.

44. The programme has a co-financing ratio of 1:0.54. The target countries will make in-kind contributions and CABEI will contribute grants and loans. Although the co-financing ratio is not high, the programme expects to leverage significant funding from IFIs and PFIs primarily from the USD 30 million guarantee provided by GCF that will cover the credit risk of 25–75 per cent of PFI loan portfolios. It should also be noted that GCF is lending to the PFIs, and the PFIs will in turn lend to the end beneficiaries. Therefore, the PFIs will take the end beneficiary credit risk for the full programme loan amount.

45. The programme will reach 1,084,441 direct beneficiaries and 1,364,044 indirect beneficiaries for a total of 2,448,485 beneficiaries. This equates to USD 120/beneficiary for the programme and USD 79/beneficiary for GCF.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

46. **Programme background.** The programme aims to overcome the barriers to the adoption of climate adaptive approaches by strengthening the capacity of vulnerable people, including smallholder and commercial farmers, as well as entrepreneurs in the rural communities of the target areas. The programme aims to contribute to the conservation and restoration of the degraded landscapes by introducing EbA approaches and other adaptation measures. It also aims to unlock financing for the adoption of sustainable resilient practices. The environmental co-benefits of the programme include the reduction of adverse impacts of floods, droughts, extreme temperatures and wildfires on the ecological habitats due to the buffering effect of the EbA interventions that could result in an overall increase in terrestrial biodiversity. Improvements in soil fertility, water infiltration and soil moisture retention are also expected to reduce erosion and enhance water quality and availability. The promotion of natural resource-based activities and alternative livelihoods, including the adoption of water-efficient

technologies, have expected social co-benefits of increased access to clean water, reduced exposure to waterborne diseases and improved food security.

47. **Environmental and social risk category and safeguards instrument.** The programme is classified as Category B or Medium Risk by the AE. The Secretariat confirms the environmental and social risk categorization since the potential risks and impacts are generally avoidable and can be mitigated effectively through the implementation of appropriate management measures. The AE has prepared an Environmental and Social Management Framework (ESMF) which contains the relevant legal and institutional frameworks, general environmental and social baseline in the target countries, and provisions for stakeholder engagement and a grievance redress mechanism (GRM).

48. **Compliance with GCF environmental and social safeguards (ESS) standards.** Subparagraphs (a)–(h) below summarize the programme’s compliance with GCF ESS standards and requirements:

- (a) **ESS 1: Assessment and management of environmental and social risks and impacts.** An ESMF has been prepared for the programme where the potential environmental and social impacts have been identified, assessed and ranked on their significance. The ESMF has also defined feasible mitigating measures that can be used to avoid and manage the environmental and social impacts posed by the programme. The ESMF also contains a detailed plan for the monitoring of critical environmental and social impacts and risks during the implementation of the programme;
- (b) **ESS 2: Labour and working conditions.** The programme will provide employment opportunities for the local populations and indigenous communities with a number of labourers involved in the implementation phase of the programme. The ESMF considered the potential risks and impacts of the activities on working conditions, terms of employment, workers’ organization, non-discrimination, equal opportunities, child labour, and forced labour of direct, contracted and third-party workers. The ESMF provided for the development of a Labor Management Guide and a GRM to address labour and working condition concerns of workers from the local population and of contractors;
- (c) **ESS 3: Resource efficiency and pollution prevention.** Through promotion of the EbA options and water-efficient technologies, the programme encourages sustainable restoration and management of forests, wetlands and agroforestry landscapes including ensuring sustainability of available water. Nevertheless, the programme may result in the use of pesticides and fertilizers that may pollute land and water resources. The programme aims to address these impacts with the dissemination of specific guidelines on the use of pesticides and fertilizers. The ESMF also indicates the development of an integrated pest management and agrochemicals use guidelines for the programme including the establishment of an efficient solid and liquid waste management module for the programme;
- (d) **ESS 4: Community health, safety and security.** The inadequate handling of pesticides and fertilizers may likely expose communities to hazardous materials which can be harmful to their health and safety. However, as the ESMF indicates that an integrated pest management and agrochemicals use guidelines will be developed, the probability of the community being exposed to these risks is low. The ESMF also makes proposals for mitigating public health risks from potential contamination of water in collection systems and the development of an Emergency Response Plan should there be a need to develop one as assessed during programme execution;
- (e) **ESS 5: Land acquisition and involuntary resettlement.** The programme does not envision the acquisition of land in the implementation of its activities. Nevertheless, the programme identifies the possibility for land use restrictions in implementing

conservation and restoration of the forest areas in the target basins. The programme may also lead to modification of current livelihood sources and/or reduction of income as a result of adopting sustainable practices and proposed EbA protocols. To mitigate this, the programme will implement inclusive consultation and decision-making processes such that affected communities will be kept well informed with the view to them being agreeable on the proposed interventions. Further, the programme will develop an economic compensation strategy should the need arise;

- (f) **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The programme activities are expected to enhance biodiversity through restoration, reforestation and adoption of sustainable EbA interventions. Nevertheless, the establishment of new mixed plantations of native species and agroforestry systems including the development of sustainable sources of firewood can potentially negatively affect the ecosystem if improperly managed and implemented. For instance, mixed species plantations and restoration of the agroforestry systems may induce increased use of agrochemicals. Uncontrolled sourcing of firewood for charcoal may also exacerbate deforestation and expansion of agricultural lands instead of increasing reforestation. To mitigate these unwanted outcomes, trainings related to biodiversity conservation, efficient resource management and pollution prevention will be implemented. Technical support on sustainable practices such as on the use of organic fertilizers, vermicomposting, soil conditioning and other agrosilvopastoral management practices will be encouraged. Further, should significant risks and impacts related to biodiversity be identified, the ESMF provides for a Biodiversity Management Plan that can be prepared and implemented;
- (g) **GCF Indigenous Peoples Policy and ESS 7: Indigenous peoples.** The programme will affect indigenous peoples in some of the countries in the Dry Corridor and arid zones where the programme is to be implemented. The AE recognizes that there is a risk of non-recognition or alteration of indigenous people's livelihood strategies, cultural practices, religious beliefs or governance structures, increased restrictions on land use for indigenous peoples and unequal opportunities for indigenous peoples. Where indigenous peoples are potentially affected by the project, the AE has indicated that it will require an Indigenous Peoples Plan (IPP) to be developed that will mitigate potential adverse effects on indigenous peoples and to improve positive impacts. The guidelines contained in the ESMF for these IPPs importantly describe the process of continuous free, prior and informed consent (FPIC) consultations and the participation of indigenous peoples (including both men and women) in the project implementation and operation processes and emphasizes that FPIC will be used in all cases, even where the legislation refers only to "consultation". Also, any IPP must develop a culturally appropriate strategy that clearly describes measures to enable indigenous peoples to take advantage of the opportunities generated by the project and sustainably preserve and manage the use of natural resources on which they depend. The GCF Independent Redress Mechanism and the Secretariat's indigenous peoples' focal point will be available for assistance at all stages, including before a claim has been made, as required by paragraph 70 of the GCF Indigenous Peoples Policy; and
- (h) **ESS 8: Cultural heritage.** Some of the regions where the programme will be implemented may have cultural value and may be implemented in communities that have various cultural practices. Therefore, the programme ensures that this tangible and intangible cultural heritage is adequately assessed and will not be adversely affected. The ESMF provides for continued access to any cultural heritage that may be affected by the programme including continued access to these sites and the use of cultural practices and religious beliefs during implementation.

49. **Implementation arrangements.** CABEL as the AE will be responsible for overseeing the implementation of the programme. A Regional Project Management Unit (RPMU) will be based at one of the executing entity's (EE) regional headquarters in Panama and will be responsible for the daily project administration, execution and regional technical coordination of project activities. Various technical specialists will be based at the RPMU including a Regional Environmental and Social Safeguards Officer. Each country is expected to have a national officer on Gender, Social Engagement and Safeguards, who will have the responsibility of ensuring that the ESMF is put in place in-country.

50. **Stakeholder engagement.** The programme has conducted an extensive community engagement with various stakeholders that enabled them to opine on the various components of the programme and how the programme is going to affect them from all perspectives during programme development. It has also developed a stakeholder engagement plan which serves as the programme's participation strategy and guidance for stakeholder participation and consultation throughout the life of the programme.

51. **Grievance redress mechanism (GRM).** The AE has an established a GRM which provides for avenues where stakeholders and potentially affected communities can provide feedback or grievances, and receive responses, with regard to the implementation of the programme. The ESMF also provides for the establishment of a four-tier activity level GRM to receive complaints and claims, collect, store and distribute information, resolve issues with corresponding follow-up, monitoring and reporting and finally inform and communicate to concerned stakeholders.

4.2 Gender policy

52. The AE has submitted a gender assessment and gender action plan; therefore, it complies with the requirements of the GCF Gender Policy. The gender assessment was completed based on a desk review, primary data collection and stakeholder consultation at various levels including women and other vulnerable groups.

53. Cultural practices, patriarchal values and existing customary and statutory laws on marriage, divorce, inheritance, custody and land tenure have significant implications on access, use and control of resources by women and men. Feminization of poverty, sociocultural and traditional beliefs, inadequate basic social services, low representation of women in politics and violence against women, among others, are existing barriers to achieving gender equality and women's empowerment throughout the Dry Corridor. Social and cultural norms in Dry Corridor countries play a key role in creating greater vulnerability and gaps in the realization of women's and girls' human rights and especially in rural and indigenous contexts. Women and girls especially continue to assume primary responsibility for unpaid family care and domestic work, and climate change is increasing the workload of women and girls as they are responsible mainly for the search for food, water and biomass for energy on a daily basis. About 60 per cent of the population of the Dry Corridor lives in extreme poverty and women are disproportionately represented in the poorest of the poor. The most severe poverty is located in El Salvador, Guatemala, Honduras and Nicaragua and has an amplified impact in rural areas. The feminization of poverty is very significant and is exacerbated by the lack of access to services including health, education, water and transportation. This limits entrepreneurship opportunities for women and is being exacerbated by the impacts of climate change on agriculture, livestock and fisheries. Women face discrimination in access to resources and control of land and property and this limits access to financing, training programs, and membership in farmers' cooperatives and local development organizations and their role in decision-making. The participation of women producers at the local level within producer associations, irrigation organizations, neighbouring boards and water committees is limited and their presence in positions of social and political leadership is still clearly a minority.

54. Women and girls are migrating to escape the extreme poverty derived from drought, loss of crops and livestock and, above all, lack of access to food. Sex disaggregated data is limited, including information on land tenure, participation in decision-making, the level and characteristics of the types of agricultural and livestock activities they carry out, their participation in activities and their access to credits or other financial activities, among other important aspects. There is a worrying increase and sharpening of violence against women, including slavery and sexual torture, throughout the Dry Corridor region amid a culture that tolerates and normalizes violence against women. In general, there are few institutions that offer protection to women and girls in situations of gender-based violence (GBV). Forced pregnancies of adolescent girls and forced and early marriages and unions are also common in many areas within the Dry Corridor. There is a migration crisis stemming from poverty, violence and the impacts of climate change.

55. The AE has provided a gender action plan and therefore complies with the requirement of the Gender Policy of GCF. The gender action plan provides activities that address the challenges faced by women and includes baseline data, indicators, targets, timelines and budgets. The programme will include seven full-time national gender experts and one regional gender expert to ensure gender issues are integrated throughout the project, including gender expertise in decision-making bodies. Activities in the gender action plan are designed to address gender gaps in a number of areas. One focal area is designed to generate enabling conditions for women's access to financial schemes, which facilitates access to credit for women producers and women's cooperatives to help strengthen women's entrepreneurship initiatives at the local level and with CCA and mitigation activities. This is complemented by building the capacity and awareness of public officials in the municipalities involved in the programme on the importance of the gender approach and how to incorporate it into agricultural public policies and take specific actions to strengthen the capacities of municipality technicians on social and environmental resilience and gender issues. Protocols will be developed to address the specific barriers women face in the adoption of efficient water technologies and ensure the inclusion of measures to ensure effective involvement of women in water committees. The programme will also develop actions with municipalities to facilitate the legalization of property cooperatives and women's community organizations and strengthen the technical capacities of women's organizations at the local level. Measures will be taken to ensure the active participation of women in producers' organizations. In general the programme will generate spaces of empowerment for women producers that ensure the inclusion of their priorities into decision-making. The programme will promote quota systems for the representation of women in community and producer organizations related to the programme. To complement these actions, the programme will generate community dialogue spaces, which will include discussions among men about positive masculinities. Finally, the programme will mainstream gender and sex disaggregated data into monitoring systems and decision-making. The programme will be conducting more country and programme-specific needs assessments to collect more information and incorporate these into country and programme-level gender analyses and action plans.

56. Since GBV has been identified as a critical challenge, the programme is strongly encouraged to take measures to minimize the risk of increased GBV as a result of project intervention and develop measures to ensure that women have equal access to the grievance mechanism that will provide all female beneficiaries with a clear and safe process for reporting incidents of GBV. The AE is also encouraged to conduct a full analysis of the enabling environment (including international conventions ratified, legal, policy and institutional support framework, national strategic development plans, etc.) that demonstrates the commitment to the pursuit of gender equality of the countries involved in this programme.

57. In general, the gender action plan includes measures designed to address barriers to gender equality through addressing biased norms, increase women's engagement in decision-making, reduce gender gaps and support progress towards a more gender-equal society.

4.3 Risks *(This section of the assessment has been redacted and full assessment shared with the Board on limited distribution basis)*

4.4 Fiduciary

58. The EEs in the project are FAO, the Central-American Commission for Environment and Development (CCAD) and CABEL, which will also perform the role of AE in the project.

59. As an EE of the project, FAO will play a role in the execution of capacity-building, technical assistance and implementation of the EbA demonstration sites throughout the region for outputs 2 and 4 of the programme. CCAD, on the other hand, will be involved in the political coordination among participant countries through the direct execution of output 3, which is focused on gathering and disseminating key results and analyses derived from the execution of the project with the aim of generating evidence to inform and mainstream adaptation into key national policies and regional strategies. Additionally, the United Nations Environment Programme (UNEP) will provide technical advice to CCAD for the development of technical knowledge products on financial mechanisms for EbA, methodologies for ecosystem services valuation and technical guidelines for EbA implementation.

60. In its role as AE, CABEL will be responsible for overseeing the implementation, financial management, evaluation, reporting and closure of the project. As an EE, CABEL will be responsible for the set-up and management of the three financial facilities of the project under component 1, with FAO as EE for the technical assistance facility supported by UNEP as technical advisor.

61. The programme will have a Regional Project Steering Committee (RPSC) comprising of representatives from FAO, CCAD, CABEL and UNEP and the regional project manager (PM). The RPSC will primarily be responsible for providing technical, oversight and advisory support at the regional level, which will include: i) overseeing project implementation; ii) review and validation of annual regional workplans and project reports; iii) review and validation of the project's fulfilment of environmental and financial objectives; and iv) approval of changes to the project's targets, activities or timelines. In addition, the RPSC will be informed about the final selection of PFIs, which will be done by CABEL following internal regulations and criteria, as well as make a recommendation on the final selection of recipients of grants through the grant facility. In its central role, the RPSC will ensure consistent project progress across all seven countries. The RPSC will also be responsible for overseeing the Regional Project Management Unit (RPMU).

62. The RPMU will be based in FAO's regional headquarters in Panama and will be responsible for the day-to-day project administration and execution and the regional technical coordination of project activities. It will ensure that project execution proceeds through clear workplans, terms of reference and carefully designed administrative arrangements that meet CABEL and GCF requirements. For the duration of the project, a PM will head the RPMU and will be responsible for the daily on-the-ground implementation and management of the project. The PM will be responsible for ensuring the project achieves the targets set out in the Results Framework to the required standards of CABEL and GCF and within the specified time and budget allocations. The PM will: i) report directly to the RPSC on project management-related matters; ii) manage the project in accordance with the specified workplans and allocated budget; iii) ensure that all project interventions are implemented according to CABEL and GCF guidelines; iv) work closely with Project Execution Units (PEUs) and national institutions and authorities to ensure that the project is managed effectively and that the needs of all beneficiary groups are considered; and v) oversee the efficient and effective information and knowledge transfer to relevant project partners.

63. National PEUs will be established in the target catchments of each of the participating countries and will be responsible for the daily execution of the project in the respective target catchments, as well as communication and capacity-building activities with participating local governments, financial institutions, civil society organisations, farmers and communities. They will provide monitoring, reporting and evaluation services as well as technical assistance and execution services. The PEUs will also be responsible for the identification of eligible projects for grant financing through the grant facility. They will engage with local stakeholders at community, association, cooperative and municipal levels to formulate finance applications.
64. CABEI will be responsible for the fiduciary aspects of the project and will be accountable for all financial and investment activities. Through its Credit Supervision Department—which oversees the destination and use of funds disbursed to PFIs—CABEI will monitor the project's compliance with policies and procedures as well as the alignment between implementation and project objectives.
65. CABEI will provide to GCF annual progress reports on the status of project implementation, including the disbursements made during the relevant period and an annual audited financial statement of the specific account prepared by an independent auditor or evaluation body.
66. The PM will be responsible for the overall monitoring of project progress against the outcome-based indicators during implementation as well reporting project progress to the RPSC. As per the accreditation master agreement (AMA), CABEI will conduct the mid-term evaluation and the terminal evaluation of the project.
67. It is recommended that as a condition precedent for FAA effectiveness, CABEI submits country level budget containing the indicative unit costs and quantities of Eligible Expenditures, in accordance with the Accredited Entity's policies and procedures that is in form and substance satisfactory to the Fund.

4.5 Results monitoring and reporting

68. The Theory of Change clearly explains how the results chain will cascade from the goal statement of the project to the proposed activities and properly articulates the relevant interlinkages among logic levels, barriers and risks. The "if, then, because" logic is clearly formulated in the diagram provided in the funding proposal, and the stated results and outcomes are defined in a manner that are adequately supportive of meeting the ultimate project goal.
69. As an adaptation project, the intervention expects to benefit 1,084,441 direct and 1,364,044 indirect beneficiaries, of which women comprise 30 per cent of the direct and 40 per cent of the indirect beneficiaries as per the gender-disaggregated metrics for the GCF core indicator.
70. The logical framework is assessed by the Secretariat to have applied results measurements against the GCF results management framework/performance measurement framework. Fund-level or project-level indicators provided sufficiently for quality at entry considerations. Comments and requests for final minor revisions have been communicated to the AE, and the Secretariat expects that there will not be any difficulties from the AE side to reflect these suggestions for further improvements.
71. The implementation timetable has been provided in a format that would enable progress assessment during the implementation period, with the milestones and deliverables properly integrated to determine implementation performance in concrete terms.

4.6 Legal assessment

72. The Accreditation Master Agreement was signed with the Accredited Entity on 6 September 2017 (the “AMA”), and it became effective on 8 November 2017.
73. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme. It is recommended that, prior to submission of the Funding Proposal to the Board (a) the Accredited Entity has obtained all its internal approvals and (b) the Fund has received a certificate or legal opinion from the Accredited Entity in form and substance satisfactory to the Fund confirming that all final internal approvals by the Accredited Entity have been obtained and that it has the authority and capacity to implement the programme.
74. The proposed programme will be implemented in seven countries, out of which in six countries GCF is not provided with privileges and immunities, while in one country an agreement on privileges and immunities has been concluded but has not entered into force yet. This means that, amongst other things, GCF is not protected against litigation or expropriation in these countries which risks need to be further assessed.
- (a) The Republic of Costa Rica: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in May 2016. The Secretariat met with AILAC delegations during COP 22 in November 2016. In March 2019, the Secretariat has sent a formal reply to the ‘Note from Costa Rica’, in English and Spanish, informing that English is the working language for future negotiations. To date, the negotiations are still ongoing.
- (b) The Dominican Republic: The Secretariat met with the Dominican Republic delegation during COP 21 in December 2015 after which a draft agreement with background note on privileges and immunities was sent in January 2016. The Secretariat sent the latest draft agreement together with background notes on P&Is to the Counsel to the President and the Ambassador to the United States in October 2017. In December 2020, the Secretariat sent another latest draft agreement to the Vice Minister of the Ministry of Environment and proposed having a conference call. However, no formal response has been received so far.
- (c) The Republic of El Salvador: The Secretariat sent a draft agreement together with background notes on P&Is to the NDA in September 2015. The Secretariat met with El Salvador delegation during COP 21 in December 2015, after which the draft agreement with background note was resent to the NDA. In October 2018, the Secretariat met with NDA to provide information about P&I Agreement and the rationale for P&Is and clarified queries regarding the importance of signing P&I Agreements. However, no formal response has been received so far.
- (d) The Republic of Honduras: The Secretariat met with the Honduras delegation during COP 21 in December 2015 and a draft agreement and background note on privileges and immunities were sent in December 2015. The Agreement on Privileges and Immunities was concluded and signed on 17 March 2016, but the Agreement is not in force yet. The Secretariat sent an inquiry regarding the entry in force of the agreement in July 2020. However, no formal response has been received so far.
- (e) The Republic of Guatemala: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the NDA in September 2015. The Secretariat met with a representative of Guatemala during COP 22 in November 2016. Latest draft agreement together with background notes on privileges and immunities were sent to the Foreign Minister in October 2017. However, no formal response has been received so far.
- (f) The Republic of Panama: The Secretariat sent a draft agreement together with background notes on privileges and immunities to the Minister of Presidency in

September 2017. In October 2018, the Secretariat met with the NDA and provided information about the draft agreement and its rationale. The Secretariat has sent a follow up email in November 2020; however, no formal response has been received so far.

- (g) The Republic of Nicaragua: The Secretariat sent a draft agreement on privileges and immunities together with background notes to the NDA in April 2018. The NDA has confirmed receipt of the Agreement but indicated that it would need to pass through the review and approval of the Congress, which is a lengthy process. The NDA requested the Secretariat to provide a non-official Spanish translation of the document in October 2018 to expedite the process; however, no formal exchanges has occurred so far.

75. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities.

76. The proposed programme envisages the implementation of a lending facility to be financed by a portion of the GCF funding and by the Accredited Entity's co-financing, with such resources channelled through intermediary financial institutions (IFIs) in the host countries that will provide downstream loans to smaller partner financial institutions (PFIs), which in turn will provide sub-loans to the final borrowers. In respect to its own co-financing, the Accredited Entity will enter into (full recourse) loan agreements with the IFIs. However, with respect to the GCF funding, a limited recourse loan instrument is proposed between CABI and the IFIs for purposes of conserving and passing down the GCF concessionality to the PFIs and the final sub-borrowers. However, since the Accredited Entity has not used a limited recourse loan in its prior operations, the applicable terms and conditions for such on-lending to the PFIs may be subject to further changes or revisions prior to the execution of the FAA (and thus be subject to the completion of the legal due diligence by the Secretariat).

4.7 List of proposals conditions (including legal)

77. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP174

Proposal name:	Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic
Accredited entity:	Central American Bank for Economic Integration (CABEI)
Country/(ies):	Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama and Dominican Republic
Programme/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1. Earlier versions of this funding proposal were submitted for review at the twenty-eighth and twenty-ninth meetings of the Board (B.28 and B.29), and did not receive the endorsement of the independent Technical Advisory Panel (TAP). In its assessments, the independent TAP raised various concerns and provided specific actionable feedback, including:

- (a) Completion of a catchment-specific climate rationale for each of the seven catchments where the Ecosystem-based Adaptation (EbA) demonstration interventions and grant facility will be applied, sourcing historical observed data for each catchment, and using these data to validate the choice of model that is used to make future projections about key climate parameters;
- (b) Completion of a catchment-specific EbA plan for each of these seven catchments;
- (c) Clear articulation of the adaptation impact of the project as per the GCF investment criteria indicators; and
- (d) Clear articulation of the relationship between the grant portion of the project, to be implemented in the seven target catchments of the participating countries, and the loan and guarantee portion of the project, to be implemented across the entire Dry Corridor of the seven countries.

2. The core concern raised by the independent TAP in its previous assessments was that the funding proposal did not provide sufficient site-specific data to support the series of grant-based EbA interventions proposed. The current funding proposal provides some additional information around the availability of climate data and proposes an ex-ante validation process for interventions, which is intended to address the concerns of the independent TAP. The validation process, if implemented robustly, could ensure that proposed EbA interventions are supported by the scientifically determined adaptation needs of the impacted communities, among other things. This solution is not the independent TAP's preferred outcome as made clear in previous assessments, but it may suffice to meet a minimum standard for endorsement in this case. However, the funding proposal does *not* provide enough information on how this validation process will be implemented and how it can assure GCF that grant funding will only be disbursed for interventions that are supported by the evidence-based needs of the catchments. For that reason, an endorsement here must be conditional on greater clarity regarding the proposed validation process and satisfaction that it is fit-for-purpose.

1.1 Funding proposal overview

3. The term “Dry Corridor” describes arid zones in parts of Central America and the Caribbean (Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama and Dominican Republic). The Central American Bank for Economic Integration (CABEI), the accredited entity (AE), states in the funding proposal that these zones are experiencing increasing frequency and intensity of agricultural droughts and heatwaves, as well as an increase in the frequency and severity of extreme rainfall events and hurricanes in the wet season, causing storm damage and flash floods (in turn causing reductions in aquifer recharge) – because of climate change. The funding proposal reports that more than one million families in the Dry Corridor are reliant on agriculture for their livelihoods and are susceptible to the impacts of increasingly severe droughts. In 2014 alone, intense drought resulted in an estimated USD 465 million worth of agricultural losses across the relevant region, with more than 2.5 million people experiencing food insecurity. From 2015 to 2018 food insecurity has increased annually in the region. By the last quarter of 2018, 2.2 million people had experienced crop losses that year, with 1.4 million urgently needing food assistance. According to the AE, these impacts are expected to continue to worsen under future climate scenarios.

4. The funding proposal aims to use a mix of implementation grants, capacity-building grants, financial guarantees and on-lending to further EbA measures such as forest conservation, reforestation, planting of agroforestry systems, improved irrigation systems and installation of vegetative barriers. These interventions are expected to mitigate the impacts of climate change in the Dry Corridor. The total financing scale of the proposed programme is USD 268.4 with GCF funding expected to make up 65 per cent of this amount (USD 174.3).

5. The funding proposal seeks to address the above-described climate threats by focusing, in part, on seven targeted catchment areas (one in each of the countries in the Dry Corridor) and using a mix of grant and capacity-building approaches to promote EbA interventions in those catchments. Additionally, the funding proposal aims to deploy loans and guarantees (along with capacity grants) across the entire region to finance interventions that would improve water demand management in the Dry Corridor, among other potential adaptation benefits. The goal, as laid out in the funding proposal, is to provide smallholder and commercial farmers, as well as entrepreneurs in rural communities across the seven countries, with increased access to the financial resources and technical skills required to implement adaptation interventions. The specific financial interventions proposed are as follows:

- (a) The total grant request of USD 84.3 million will be used in the following ways according to the AE: (i) USD 6.4 million will be used to set up the financial mechanisms and strengthen the technical capacity of CABEI-accredited and non-regulated financial institutions to access and channel funds through them; (ii) USD 28 million will form the grant facility through which vulnerable communities can access on-grants channelled through financial institutions; (iii) USD 10 million will be used to strengthen the technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures; (iv) USD 6.3 million will be used to generate and share information on climate change adaptation and finance across the region, and mainstream it into local and national policies; (v) USD 28.8 million will be used to implement large-scale demonstration interventions, including EbA, water- and energy-efficient technologies in rural communities across the region. These interventions represent a list of specific interventions previously agreed between the AE and each of the seven countries in the funding proposal; and (vi) USD 3.8 million will be used for overall management and monitoring of programme activities with an additional USD 0.7 million for contingency;
- (b) An EbA lending facility would be established and operationalized with a GCF concessional loan of USD 60 million and approximately USD 43 million in market rate co-finance from CABEI. The financing is intended for small- and large-scale EbA investments at the farm, enterprise and household level. The GCF funds would be

channelled through intermediary finance institutions (IFIs) with at least one located in each of the seven Dry Corridor countries. In turn, the IFIs would on-lend to partner financial institutions (PFIs) in the relevant country who would then directly on-lend to end beneficiaries; and

- (c) A GCF-funded guarantee facility intended to provide an additional layer of security to financial institutions to promote EbA finance (USD 30 million). The guarantee facility would be made available to de-risk CABEL's co-finance as well as that from other providers of non-concessional finance. The main aim is to increase the appetite of IFIs to take on further financing to channel to PFIs by guaranteeing the PFI EbA finance performance.

1.2 Impact potential

Scale: Medium

6. There is evidence that the Central American Dry Corridor and the arid zones of the Dominican Republic are among the world's tropical regions most vulnerable to the impacts of climate change, with Honduras, Nicaragua, Guatemala, the Dominican Republic and El Salvador recognized as being among the 16 most vulnerable countries in the world to extreme climate events for 1998–2017. Currently, the Dry Corridor encompasses approximately 64 per cent of the municipalities in Central America and is expected to extend to around 85 per cent by 2050. Climate projections for the region indicate that, by the end of the century, temperatures will increase by 3–3.5°C under RCP4.5. These changes could be compounded by further changes in the El Niño–La Niña cycle, which could result in prolonged droughts as well as more frequent and intense extreme rainfall events across the region.

7. The independent TAP recognizes that targeted and catchment-specific EbA interventions could reduce hydrological vulnerability associated with increased droughts and periods of aridity. However, the key is for these interventions to be part of an EbA plan that is tailored to the relevant catchment. The impact potential of EbA interventions is dependent on the specific climate challenges of each catchment and whether such interventions are part of a reinforcing suite of actions that, in suitable local combinations, best mitigate those impacts.

8. The original funding proposal submitted in B.28 did not contain actionable catchment-specific EbA plans for the seven catchments where the demonstration interventions and the EbA grant facility would be applied. In the amended funding proposal for B.29, the AE clarified its intention to develop site-specific intervention plans for each of the seven catchment areas as a critical first phase to ensuring that activities to be funded contribute to a cohesive catchment-specific plan and landscape approach. The AE proposed that part of the grant funding request would first be used to develop the catchment-specific EbA plans. In support of this approach, the AE further explained that some work has already commenced in some of the catchments, including using existing information on regional climate projections and downscaling through hydrological models at the basin level for three catchments. Two draft sample site-specific intervention plans were developed – for Costa Rica and Honduras – and these were provided to the independent TAP to illustrate the approach the AE intends to take.

9. In its previous assessments, the independent TAP expressed its preference for first completing the climate rationale at the catchment scale for each country's target catchment, followed by the development of a bespoke EbA catchment plan. The independent TAP appreciated that the amended funding proposal built in a sequencing as part of the programme whereby the catchment plans would be completed first, prior to most interventions being funded. However, the independent TAP remained concerned because of (i) the gaps in the data and approaches proposed to validate interventions, and (ii) the request for grant funding for pre-selected demonstration projects without a persuasive case that these interventions will generate catchment-wide adaptation benefits or that they are critical elements of a catchment-specific EbA plan.

10. The current funding proposal provides some additional information in support of the climate needs of the region and clarifies where the AE believes that a lack of historical climate data has constrained its ability to fill in analytical gaps in support of its climate assertions. The AE stated that it encountered limited availability of site-specific climate information needed to generate more accurate and informative analyses of climate risks and vulnerabilities for resilient development. The independent TAP's clear preference is that the use of climate models must be validated against observed historical data sets from meteorological stations, wherever these are available, and not against computer-generated data sets. However, the independent TAP understands that in some circumstances this preferred approach may be difficult to achieve due to the lack of historical data. The AE asserts that the availability of historical observed meteorological data and site-specific climate change projections in this case are very limited. CABEL observes in the funding proposal that there are only 24 meteorological stations within the vast territory of the Dry Corridor and notes that "only for three of the countries, there were stations within the prioritized pilot sites and just one of them (Honduras) had complete observed rainfall and temperature data for a period of over 30 years with less than 15% of discontinuity."

11. The funding proposal also re-states that the catchment-specific plans will be completed first as part of the programme's initial phase and has now made clear that no grant-funded interventions will be approved prior to the finalization of the relevant catchment plan. The AE further asserts that no intervention will be funded until it has been through a validation process whereby a proposed intervention is confirmed to meet the needs of the specific region in line with the catchment plans. The validation of actual site-specific interventions is to take place during the initial stages of implementation and EbA interventions will be validated from a wide range of identified EbA interventions that are adjusted to the specific climate risks and socio-environmental context of the catchment. According to the AE, "the validation of these indicative pre-selected interventions will be based on the development of the site-specific plans that will include both a scientific analysis of specific climate projections through the hydrological models, as well as sound stakeholder consultations that will allow to identify the most appropriate interventions." The funding proposal also provides a list of specific criteria that will be applied for the selection and validation of these EbA interventions, including:

- (a) Relevance to site-specific climate risks;
- (b) Positive impact potential over climate risks;
- (c) Familiarity and acceptance by stakeholders;
- (d) Positive impact potential over most vulnerable sectors of the community;
- (e) Positive impact potential on gender balance;
- (f) Technical feasibility of the intervention;
- (g) Compliance with social and environmental safeguards;
- (h) Potential co-benefits of the intervention;
 - (i) Minimal level of risks and constraints for implementation;
 - (ii) Potential for self-sustainability; and
 - (iii) Potential for sustained monitoring capacity.

12. The independent TAP notes that the list of validation criteria appears to be a reasonable starting point, but the effectiveness of a validation process will depend on (i) the quality of the catchment plans themselves and (ii) the manner and rigour by which these criteria will be applied. The funding proposal does not provide a clear sense or example of how the ex-ante validation of interventions will occur. The independent TAP could endorse a phased programme whereby complete and data-driven catchment plans are first developed and subsequently EbA

interventions are selected, but only if these are subject to rigorous validation, through a known process, against those plans. It is also critical that the process would prevent funding of proposed interventions that are not supported by the needs of the catchment. The funding proposal does not provide enough information as to the operation of the validation process for the independent TAP to make a determination of adequacy. Developing a detailed process for validation that is fit-for-purpose is thus a necessary first step and is included as a condition for the independent TAP's endorsement of the proposal.

13. With respect to the USD 28.4 million grant request for the “pre-identified EbA demonstration interventions”, the concerns of the independent TAP were acute. Under this component of the B.29 funding proposal, grants would be provided to communities in 36 municipalities across the selected catchments that the AE believes cannot receive funding through the loan or grant facilities.¹

14. Although the activities listed in the B.29 funding proposal may offer localized environmental benefits, they were *not* selected based on a clear nexus to catchment-specific EbA plans, and no site-specific argument was made for how they will specifically reduce the vulnerability of communities to climate change impacts. The current B.30 version of the funding proposal has been amended so that these interventions are now deemed to be solely “indicative” of the type of large-scale demonstration projects that the AE proposes to fund through grants. According to the AE, “the selected interventions remain as indicative and are subject to validation during the inception phase and in consultation with the specific communities.” The independent TAP appreciates that the demonstration interventions listed in the funding proposal are no longer formally considered pre-selected and certain. The independent TAP agrees that any intervention should first be validated against a site-specific catchment plan that, for each catchment, explains its specific climate challenges; how the challenges are affecting ecosystem service provision; how this makes communities vulnerable; how these vulnerabilities to hazards can be mapped spatially; and what interventions are needed to maintain or restore these specific ecosystem services, to reduce vulnerability. The validation of the demonstration interventions should confirm the expected adaptation impact, underpinned by a climate rationale for each catchment, and accompanied by a clear EbA plan for each catchment. Only by having a robust validation process can the independent TAP be certain of the truly “indicative” nature of these interventions.

15. **EbA loan programme** – The funding proposal envisages that GCF funding would also support a lending mechanism for EbA as well as water- and resource-efficient technologies in all seven countries. This would be accomplished by providing a blended financing mechanism composed of concessional finance from GCF in combination with market rate debt as co-financing from CABEL. The blended rate is intended to create incentives for regional and local financial institutions to engage in EbA finance. IFIs in CABEL's established network would manage the EbA lending facility via direct on-lending to local PFIs such as cooperatives, small banks, non-government organizations, rural community banks, among others. PFIs in turn would on-lend to end beneficiaries. The goal is for investments to be targeted across the entire Dry Corridor/Arid Zone regions of the seven countries (not just the seven catchments receiving

¹ Such landscape interventions include (i) establishing one tree nursery per municipality focusing on native species and fuelwood trees (36 municipalities in total); (ii) establishing 4,160 ha of forest protection zones across the seven catchment areas; (iii) protecting 1,820 ha of natural forest in major recharge areas and riparian zones, including assisted regeneration; (iv) restoring 5,170 ha of natural forest areas across seven catchments; (v) restoring 820 ha of native pine forests in Guatemala, Honduras and Nicaragua; (vi) establishing 1,293 km of agroforestry systems using diversified living fence arrangements in basic grain crops; (vii) establishing 612 ha of agroforestry systems for natural shade in coffee plantations; (viii) establishing 1,280 km of silvopasture systems using diversified living fence arrangements; (ix) establishing 1,280 ha of silvopasture systems across the 36 target municipalities; (x) establishing 5,305 ha of sustainable fuelwood and timber plantations across the 36 target municipalities; (xi) establishing 129 km of firebreaks for forests and plantations; (xii) constructing 118 km of living barriers for soil conservation around agricultural plots; and (xiii) constructing 118 km of superficial drainage for soil conservation, establishing 109 rainwater harvesting systems, 156 community-level rainwater reservoirs, and so on.

grant support) towards community-level investment, rural micro, small and medium-sized enterprises or other institutions that can foster EbA, as well as water- and energy-efficient technologies, for end beneficiaries across different sectors. The loan programme is not limited to the catchment areas in the grant portion of the funding proposal and does not rely on the grant-funded interventions for its success. In response to a question from the independent TAP on the link between the catchment-specific grants and the wider loans, the AE noted that the deployment of the loan facility will follow a phased approach starting first in a select group of five municipalities in the identified catchments and expanding from there geographically to cover additional municipalities. According to the AE, this phased approach will also allow for more time to strengthen capacities of the participating financial institutions during the initial stages and to promote the financial mechanism with communities and producers. The additional municipalities for the loan expansion are to be selected based on proper vulnerability assessments and climate impact projections as well as the analysis of demand and the potential for replication with the participating financial institutions to ensure economies of scale in the learning process and the internal processes already developed in the initial municipalities.

16. The phased nature of the loan programme is critical from an efficiency and impact standpoint. To that end, the independent TAP has been informed by the GCF Secretariat that the specific performance thresholds for the loan facility described in the funding proposal will also be reflected as conditions for GCF disbursement to be set in the funded activity agreement (FAA). The FAA will include a disbursement schedule with milestones to be met, hence ensuring that GCF proceeds are disbursed in accordance with pre-agreed terms. The independent TAP concurs with this approach and views this as a condition to endorsement.

17. The loan programme would operate alongside a GCF-funded guarantee facility intended to secure on-lending by IFIs of CABEL's co-financing funds and mobilize additional lending from IFI resources, as well as additional resources from public and private investors, by mitigating against credit risks associated with commercial EbA finance.

18. Technical assistance would be provided for PFIs to access the EbA lending facility. This is meant to ensure that capacities, especially for adequate monitoring, reporting and verification systems, are in place to manage the EbA finance and to determine whether would-be borrowers are eligible for such loans.

19. The independent TAP notes that the loan and guarantee component of the project seeks to re-orient existing financial sectors to funding activities that would lead to climate benefits and ultimately prove to be financially sustainable, contributing to impact potential. This typically requires a mix of initial concessionality, knowledge sharing and improved institutional capacity. This funding proposal aims to do that at scale and across seven countries. However, the independent TAP notes that the investment objectives of the loan facility are broad: (i) implementing integrated catchment management and restoring catchments; (ii) improving hydrological flow and filtration of rainwater into groundwater reserves through forest and ecosystem restoration; and (iii) reducing demand for scarce water resources by implementing water-efficient technologies at the farm and household level. The funding proposal does not include investment eligibility criteria to ensure climate impacts are achieved. According to the AE these criteria will be defined by the Regional Project Steering Committee, and will be used by the project executive units, the national coordination units and regional lending committees when reviewing eligibility. However, the independent TAP considers that the absence of the criteria makes it difficult to assess the overall impacts of EbA interventions. Scattered interventions might not add to the overall objectives of the lending operation in the catchment areas, particularly regarding objectives 1 and 2. The independent TAP will require as a condition to disbursement that the investment eligibility criteria be finalized.

20. In the amended funding proposals presented for B.29 and B.30, the AE has provided improved data on the market size and potential demand for the loan and guarantee products. In the original independent TAP assessment at B.28, the independent TAP questioned market

study proxy data assumptions regarding the numbers of smallholder farming units, rural community projects and commercial farms that were expected to access loans to promote EbA in their regions. A completely new section on the demand analysis was added to the B.29 funding proposal with details provided to facilitate understanding of the absorption capacity at the country level and establishing a comparison with the numbers of expected loans and financial entities through this programme. The new information provides a more convincing demonstration of the potential uptake of the proposed products.

21. Information from the Microfinance Network of Central America and the Caribbean (REDCAMIF) was used to analyse the supply capacity for EbA finance and the feasibility of the proposed lending facility. According to latest statistics from the REDCAMIF MIF Index for June 2020, the microfinance institutions that belong to the network manage a total inclusive agricultural portfolio of USD 324 million across all seven countries for 221,477 clients. This represents more than three times the total size of the targeted lending facility's volume of USD 102.8 million (GCF plus CABI co-financing). The microfinance sector's involvement in inclusive agricultural finance is a useful proxy for its ability to serve EbA finance, which will primarily be targeted at small-scale farmers.

22. Moreover, the amended funding proposal has introduced a clearer process for phasing disbursements against milestones, although the details were yet to be finalized at the time of this assessment, and no draft term sheet had been shared with the independent TAP. The amended funding proposal also reduces the overall size of the loan facility in response to a suggestion by the independent TAP to consider scaling down the loan facility both in volume and possibly in geographic and intervention focus. The adjustment in size helps slightly to mitigate the risk of deployment failure, which could be damaging to the perception of bankability of EbA solutions across the region, including among government partners, IFIs and PFIs. This concern is relevant considering the experience of comparative efforts such as Microfinance for Ecosystem-Based Adaptation (MEbA) and the Productive Investment Initiative for Adaptation to Climate Change (CABio II), where loan disbursement rates have been very slow – partly because of the impact of the COVID-19 pandemic, but also possibly because of capacity constraints in partner institutions. In addition to these reasons, the independent TAP has a real concern that the slow uptake may also relate to limited absorptive capacity by the market. The Secretariat informed the independent TAP that the specific performance thresholds for the loan programme described by the AE in the amended funding proposal will be reflected as conditions for GCF disbursement to be set in the FAA, which will include a disbursement schedule with milestones to be met to ensure that GCF proceeds are disbursed in accordance with pre-agreed terms.

23. It is the view of the independent TAP that, with the phasing safeguards not just included in principle, but as a covenant in the FAA and the relevant governing agreements for the programme, the loan and guarantee component of the funding proposal could be useful in testing and educating the market for EbA demand. This is in fact a core role for GCF finance. Of course, the effectiveness of the phasing approach will hinge on the ability and willingness of the GCF to monitor disbursement rates and adjust loan amounts as necessary to adjust to actual capacity constraints of partner institutions and demand. For this reason, the independent TAP is endorsing the project subject to the inclusion of a covenant determining this phased approach. The impact of the loan and guarantee component could thus be positive, but, as with the grant component, must be underpinned by a clear adaptation rationale, so that the theory of change for how precisely these loans will reduce vulnerability to climate change is clear. In the case of these loans, the sheer diversity of interventions under consideration and the region-wide application makes this more challenging. The independent TAP notes that, unlike the grant components which are focused on the target catchments, the loan programme is intended to eventually allocate funds to the entirety of the Dry Corridor region across the seven countries.

24. The overall rating for the impact potential criterion for the project is Medium, on the assumptions that (i) a covenant on the phased disbursement of the loan funding is included; (ii) greater clarity and certainty around the investment eligibility criteria and validation process are provided; and (iii) a condition that no grant monies would be disbursed for any given intervention until such times as the relevant catchment plan has been approved (supported by site-specific climate rationale) and the intervention has been successfully validated.

1.3 Paradigm shift potential

Scale: Medium

25. Corporate or large agricultural producers in the Dry Corridor region, where they exist, usually apply climate risk management strategies. According to the funding proposal, PFIs across the region already have lending programmes for more sophisticated customers, including funding for interventions that the funding proposal categorizes as EbA. However, this is regularly not the case for small and medium-sized farming units. Increased resilience of such farming units could play a role in mitigating agricultural losses due to climate impacts, along with biodiversity loss, water shortages, poverty and food insecurity.

26. The proposed programme intends to catalyse a paradigm shift in the regional approach to catchment and water demand management, as well as managing climate change-intensified hazards such as drought, heatwaves, flood and hurricanes, in the Dry Corridor of Central America and the arid zones of the Dominican Republic. This shift will see governments promoting large-scale EbA investments as well as increased private sector participation, including by commercial farmers and entrepreneurs in rural communities who would then have access to the financial resources and technical skills required to implement impactful adaptation interventions.

27. It is the view of the independent TAP that a lending programme could offer a building block for future expansion if (i) new and additional demand for EbA finance can be confirmed; (ii) there is a path to financial sustainability for this expanded lending sector; and (iii) the links between the interventions financed through the lending programme and the proposed EbA interventions and the expected climate impacts are clear. The paradigm shift nature of the proposal is further enhanced by building the technical and institutional capacity of PFIs and non-regulated financial institutions to finance and re-finance EbA and water-efficient technologies by the proposed financial instruments.

28. In the view of the independent TAP, the highest potential for paradigm shift comes from the ability to contribute to improving the regulatory framework and policies that could catalyse EbA demand and acceptance. The funding proposal envisages enhancing the local evidence base for EbA through the creation of a regional knowledge hub, through which tailored knowledge products will be created and disseminated. Additionally, the monitoring and evaluation of project activities will support national and local level decision-makers to implement policies and incentives for EbA. Decision-makers within local government institutions will be trained to assess and update existing regulatory frameworks and policies to facilitate the mainstreaming of EbA, including making provision for incentives programmes targeting the adoption of EbA practices. By providing technical support to local governments for the adoption and implementation of climate-resilient land management practices, the project will help to mainstream climate adaptation into local-level development planning.

29. **Comprehensiveness:** The funding proposal intends to create an enabling environment that capitalizes on the scalability of EbA, focusing on three core areas of support: financial, policy and technical. If successful, the programme will diversify financial systems in the region to create an enabling environment for investment in EbA and water-efficient technologies. This will be based on the establishment of several financial instruments with various entry points to make climate finance more accessible to vulnerable communities. The enabling environment for investment will be further enhanced by building the technical and institutional capacity of PFIs

and non-regulated financial institutions to finance and re-finance EbA and water-efficient technologies through all three of the proposed financial instruments.

30. The independent TAP is of the opinion that the overall approach of the programme is comprehensive in marrying capacity-building, technical support and finance. However, in the absence of further preparatory work, the paradigm shift potential would be significantly limited for the grant facility by the large degree of uncertainty remaining about whether the grants and loans will in fact reduce the vulnerability of Dry Corridor farming communities to climate change impacts, given the gaps in the climate rationale and EbA plans for catchments. For this reason, the endorsement of the grant facility by the independent TAP is contingent upon the review and acceptance of the condition around the completion of a validation process, and binding agreement that any intervention should first be validated against a site-specific catchment plan prior to funding. For the loan facility, which is applied beyond the seven catchments, in the wider Dry Corridor/Arid Zone in the seven countries, it is even more difficult to show that the disparate loans covering a wide range of interventions in varying geographies add up to effective climate change adaptation. This is offset, however, by the paradigm shift involved in developing the capacity of the IFIs and PFIs to tailor their loan portfolio to effectively managing climate risk. For this reason, the independent TAP considers the paradigm shift potential to be Medium.

1.4 Sustainable development potential

Scale: High

31. Notwithstanding questions around the climate-specific impacts of the funding proposal mentioned in previous paragraphs, the proposed programme will result in environmental co-benefits. Some of the environmental co-benefits include conservation and/or restoration of forests and agroforestry systems which will act as natural buffers, controlling and mitigating the impact of floods, droughts, extreme temperatures, and wildfires on ecosystems in selected sites. The programme will also increase access to finance for underserved populations, which should support economic development generally and improve the financial sustainability of agricultural livelihoods.

1.4.1. Alignment with SDGs

32. According to information presented in the funding proposal, if adequately planned and implemented, the proposed project will contribute to the achievement of nine of the 17 United Nations Sustainable Development Goals (SDGs). The main contributions of the project will be to SDG 6 – Clean water and sanitation; SDG 11 – Sustainable cities and communities; SDG 13 – Climate action; and SDG 15 – Life on land. In addition, the project will contribute to SDG 1 – Poverty reduction; SDG 7 – Affordable and clean energy; SDG 8 – Decent work and economic growth; SDG 9 – Industry, innovation and infrastructure; and SDG 10 – Reduced inequalities.

1.4.2. Economic co-benefits

33. The programme is intended to increase access to credit which in turn should further reduce financial barriers to implementing adaptation measures for the targeted rural communities.

1.4.3. Social co-benefits

34. The proposed project is expected to yield substantial social co-benefits, mostly contributing to social development by improving the water and food security of vulnerable communities in the region.

1.4.4. Environmental co-benefits

35. The conservation and restoration of some degraded and climate-vulnerable ecosystems – through reforestation, forest management, agroforestry systems and sustainable fuelwood practices – as well as the use of water- and energy-efficient technologies will result in several environmental co-benefits.

1.4.5. Gender-related co-benefits

36. Women in Central America tend to hold traditional, unpaid gender roles associated with domestic responsibilities such as motherhood, child- and elder-care as well as household management. These roles generally also include water management and collection. During times of water scarcity, the burden of finding, collecting and storing water increases, with women bearing most of this additional responsibility. Mitigation of climate impacts should alleviate water-related challenges for women in the region. The funding proposal describes how, by building on this understanding of water-related challenges among women using gender-sensitive learning methods for adaptation, the proposed project will not only empower women within their communities and promote gender equality but will also ensure the effectiveness of the project interventions.

37. Taking all these co-benefits into account, the independent TAP considers the sustainable development potential to be High.

1.5 Needs of the recipient

Scale: High

38. From an overall perspective, the Latin America and the Caribbean region — under which the seven beneficiary countries fall — has achieved an average annual economic growth rate of 3.2 per cent over the last ten years. However, this growth rate varies substantially between the countries and even within countries between urban and rural communities. Rural communities have little or no access to adaptation finance, which limits their adaptive capacity. By improving access to financial resources for adaptation interventions, the adaptive capacity of rural communities in the Dry Corridor will be increased. With increased financial and technical capacity, communities will be able to withstand the negative impacts of severe droughts, flood and severe storms.

39. Owing to the different socioeconomic, development and environmental contexts of the different participating countries, their respective needs for climate finance differ. With regards to the loan and guarantee financial mechanisms being proposed by the AE, the participating countries will likely have different existing capacities for the uptake of EbA finance, meaning that some will be able to utilize higher levels of finance than others. There will also be different levels of demand for EbA finance in each country. To that end, the on-the-ground interventions to be implemented in the target municipalities in each country were selected on the basis of extensive stakeholder engagement, availability of area for implementation and appropriateness of interventions.

40. At the local level, vulnerable rural communities do not have the financial resources, knowledge base or technical capacity to (i) develop, implement and maintain EbA and other adaptation measures; (ii) co-ordinate the frequent, cross-sectoral, multi-stakeholder engagement that is necessary for EbA and other adaptation interventions; and (iii) capitalize on EbA interventions by generating new private income streams from the ecosystem goods and services produced by restored climate-resilient ecosystems.

41. The national budgets of Dry Corridor countries are limited, and poverty eradication and sustainable development have been prioritized for funding over climate change action. There is a high dependency on non-reimbursable international support to meet climate change goals across the region where the finance available is far below what is required to meet adaptation needs. The added burden of the COVID-19 pandemic on the economies of these countries will

leave even fewer funds available for climate action in the participating countries. The proposed programme will provide access to technical and financial resources, knowledge products and government support structures that will enable rural communities to adapt effectively to climate change in the Dry Corridor. For all these reasons, the independent TAP considers the needs of the recipient to be High.

1.6 Country ownership

Scale: High

42. **Relevance to national policies and objectives** – Climate change adaptation is recognized as a priority in the Central American region and has been integrated into several regional and national policies and strategies. In addition to this regional alignment, the project contributes to achieving national priorities and aligns with numerous policies in each of the seven target countries as well detailed in the Feasibility Study provided with the funding proposal.

43. The development of this funding proposal was initiated by the Council of Ministers of the Central American Commission for Environment and Development (CCAD), which includes members from each of the seven target countries. In addition to engaging with CCAD, regional and national stakeholder consultations have been conducted commencing in 2016. Through these national workshops, priorities and focus areas for the project were identified. The draft concept note was circulated, and further consultations were conducted in October–November 2017 with representatives from each country. A regional workshop was also held in October 2017 to obtain input on the draft concept note from representatives of governments and non-governmental organizations in the region. These inputs contributed to refining the project concept and incorporating lessons learned from previous initiatives in the region.

44. The funding proposal also includes no-objection letters from each of the seven relevant countries.

45. Importantly, the funding proposal includes USD 51.3 million in aggregate grant funding from the countries and in-kind contributions. As for co-financing (USD 40.1 million), each country will identify additional investment sources from their national budgets to increase the interventions of this output. At the moment there is limited clarity as to the timing of this co-finance and its application.

46. Based on the above analysis, the independent TAP considers that the country ownership of the project may be high if the project is successfully implemented as proposed (i.e. country co-financing).

47. **Capacity of accredited entities and executing entities to deliver** - The project implementation arrangements involve several entities and layers of actors at the regional, national and local level. CABEI will oversee the project implementation and the Food and Agriculture Organization of the United Nations (FAO) and CCAD will jointly act as executing entities. UN Environment will play a technical advisory role. Moreover, there will be several financial institutions involved.

48. The independent TAP believes that the overall project governance structure is complex and that it is critical that all the relevant actors have clear and common objectives in order to achieve the expected climate change impacts of the project. This involves clear investment criteria (based on climate change rationale) and EbA plans.

49. As reflected above, the independent TAP remains concern around the implementation by CABEI of similar projects, such as the GCF approved project CAMBIO II which, after 3 years since approval, has not started to disburse loans. During a due diligence call with the independent TAP, CABEI explained that only very recently has it been able to deliver some consultations in the countries to start delivering the loans.

50. **Engagement with civil society organizations and other relevant actors** - The project proposal presents some references (in annex 7) of the consultations delivered in the Council of Ministers up to November 2019 and some stakeholder consultations at the country level up until the same year. Much has happened in the region since 2019, and further consultations need to be developed to ensure that the project contemplates the current economic situation of the targeted beneficiaries, their current needs and willingness to take part of the project.

1.7 Efficiency and effectiveness

Scale: Medium

51. Efficiency and effectiveness of the grant portion of this funding proposal would be Low-Medium, if the uncertainties around the expected adaptation impact, given the absence of completed EbA plans for the target catchments, were not addressed. However, the independent TAP is of the view that, considering the additional climate data clarifications provided, the clear regional impact of climate change and the proposed validation process, this programme could rate Medium on the GCF assessment scale, providing the conditions (elaborated below) are fulfilled. Ultimately this will depend on the robustness of the validation process that requires further elaboration and demonstration.

52. Efficiency and effectiveness of the loan and guarantee element of the funding proposal is considered to be medium. The independent TAP remains concerned about the ability of the project partners to disburse loan funding at such a large scale within the project implementation period. The loan and guarantee portion of the project draws from the experiences of the MEbA project which, according to the funding proposal, informed the selection of appropriate EbA solutions for the proposed project. Phase one of the project (MEbA I) was implemented in Colombia and Peru, while activities were expanded to Benin, Costa Rica, the Dominican Republic, Ecuador, El Salvador and Senegal during phase two (MEbA II). EbA solutions drawn from MEbA will include implementing locally appropriate water- and energy-efficient technologies, such as rainwater harvesting systems and half-orange kilns, for adapting to the increasing impacts of climate change. Furthermore, effective methods for increasing commercial and smallholder farmers' access to financial resources for adaptation were drawn from the MEbA project as well as CAMBio II projects, approved by the GCF at B.21.

53. The independent TAP notes that the MEbA programme has only been able to issue USD 30 million in loans since 2014 across five countries. The MEbA loans represent just over 17,000 projects. Moreover, the majority of the MEbA loans are focused on a single intervention (solar dehydrators). The outcomes of the MEbA programme are substantially more modest than those proposed by the AE in the Dry Corridor region.

54. The CAMBio II programme was approved at B.21 (2018) and is substantially smaller (USD 15.5 million GCF contribution) than the funding proposal under consideration. To date, CAMBio II has faced deployment issues and has not been able to generate the expected results. According to the AE, some of this lack of performance is related to the difficult year 2020, in which the Central American region has been affected by various climatic, health and economic crises. The AE informed the independent TAP that it expected the execution of the CAMBio II programme to accelerate in 2021; however, the independent TAP understands that progress so far in 2021 has been below expectations.

55. Notwithstanding the above concerns, the independent TAP notes that the overall efficiency and effectiveness of the funding proposal has been improved by incorporating a phased approach to the deployment of the loan facility. If a phased or tranching approach is appropriately structured through appropriate covenants in the FAA and other governing agreements, this should allow funding disbursements to be better tailored to actual demand over time, to increase use of the guarantee product over time, and allow greater time for vulnerability assessments and identification of priorities for other regions within the Dry Corridor where the intervention could be scaled up. This phased approach will also allow for

more time to strengthen the capacities of the financial institutions and to promote the financial mechanism with communities and producers.

56. The funding proposal projects a co-financing ratio of 1:0.51 and a private source leverage ratio of 1:0.67. The programme's leverage primarily results from the USD 30 million guarantee provided by GCF that is expected to cover the credit risk of 25–75 per cent of PFI loan portfolios. The inclusion and use of a guarantee product is a positive feature of the proposal, as is the loan structure whereby the GCF is lending to the PFIs via a trust structure, and the PFIs will in turn lend to the end beneficiaries. Therefore, the PFIs will take the end beneficiary credit risk for the full programme loan amount even though they are not required to contribute their own resources beyond the GCF and AE resources.

57. Based on the above, the overall rating for efficiency and effectiveness for both the grant and loan portion of the funding proposal is thus considered as Medium, subject and dependent on the fulfilment of the conditions herein.

II. Overall remarks from the independent Technical Advisory Panel

58. The AE states that the Dry Corridor suffers and would continue to suffer from the effects of climate change. The funding proposal seeks to ameliorate these impacts by focusing on specific catchment areas (grant-funded interventions) and by providing a region-wide loan and guarantee facility. At the catchment level, a successful EbA approach must be part of a cohesive catchment-specific plan and landscape approach, underpinned by a clear climate rationale for each catchment. The B.30 funding proposal provides some additional information to demonstrate how climate change may impact the areas of intervention in the future as well as its explanation of the limits of data and climate information availability in the selected zones of the programme, and the scarcity of meteorological stations in these zones.

59. In this amended funding proposal, the AE further proposes a sequencing process by which the programme includes the development of site-specific intervention plans for each of the seven catchment areas as a first step in the implementation of the programme. As set out in the funding proposal, the first year of the project would be focused on developing the site-specific intervention plans, establishing the community monitoring committees and initial activities to ensure stakeholder engagement and identification of priority interventions. The independent TAP's preference, as expressed in relation to the funding proposal presented for B.29, was that this effort should occur before endorsement, so that the proposed interventions (particularly the demonstration interventions) can be clearly shown as part of a cohesive approach that, in aggregate, produces the desired adaptation impacts to justify the GCF investment.

60. The B.30 funding proposal alleviates some of these concerns by making the demonstration projects indicative and by subjecting *all* interventions (including the demonstration projects) to an ex-ante validation process. According to the funding proposal, the selection/validation of the specific EbA interventions will be based on scientific climate risk assessments and sound stakeholder consultations. The funding proposal further provides a list of criteria that would be examined in such a validation process; however, the funding proposal is thin on the mechanics of the process and the standards to be applied, and the endorsement of the independent TAP is conditional on the firming up and detailed specification of the validation process. Considering the broad evidence of climate adaptation needs of the subject region, as described in the funding proposal, and the potential for a robust validation process to ensure that interventions are aligned with the actual climate needs of specific catchments, the independent TAP can endorse this funding proposal subject to the conditions below. The independent TAP notes that even with these conditions, the funding proposal only minimally satisfies the GCF investment criteria and the independent TAP urges the Secretariat to devote sufficient time and resources to ensure the conditions are thoroughly met.

61. The independent TAP maintains its preference for the completion of a catchment-specific EbA plan for each of these seven catchments based on catchment level climate rationale as discussed herein and in previous assessments. However, the independent TAP is endorsing this proposal subject to certain conditions which, if performed adequately, could minimally address its concerns. From the independent TAP's perspective, the conditions listed below are necessary to meet the GCF investment criteria, otherwise the impact potential and efficiency and effectiveness of the grant component of the funding proposal do not suffice for endorsement.

62. Pursuant to the above, the Programme is endorsed for approval with the following conditions to be included in the relevant funded activity agreement (FAA):

Conditions precedent to the execution of the FAA:

- (a) The AE shall submit to the Fund, in a form and substance satisfactory to the GCF Secretariat, an ex-ante EbA Intervention Validation Process that will be utilized to ensure that no EbA intervention under the Programme can receive any grant funding until such time as the proposed intervention has been successfully validated by the AE, and that shall contain the following:
 - (i) A step-by-step description of the EbA Intervention Validation Process that covers each of the validation criteria provided in the funding proposal, including:
 - (1) Sufficient detail on each proposed intervention including site determination, duration, cost, and expected EbA impact among other relevant information so as to permit analysis and evaluation of the proposed intervention in the context of the validation criteria and the relevant catchment plan;
 - (2) For each intervention, a clear articulation of the adaptation impact of the proposed EbA intervention to be grant funded as per the GCF investment criteria indicators;²
 - (3) For each validation criterion, a clearly articulated and objective standard by which each intervention will be measured against and scored against. A scoring system shall also be designed that can help select amongst interventions. The standard and scoring system to be devised must be deeply rooted in the relevant EbA site specific catchment plan to be endorsed by the Secretariat; and
 - (4) The process description shall include information on the data and materials to be collected and presented for each criterion as part of the validation process;
 - (ii) A third-party audit process/external assessment for the demonstration EbA site specific interventions; and
 - (iii) A sample intervention validation document and supporting information as a template.

Covenants for inclusion in the FAA:

- (a) The Accredited Entity shall ensure that the provision of grant funding to any EbA interventions (including demonstration projects) under the Programme is subject to a requirement that proposed interventions be first successfully validated under the EbA Intervention Validation Process prior to funding;

² The criteria state that "Project proposals should describe the expected change in loss of lives, value of physical assets, livelihoods, and/or environmental or social losses due to the impact of extreme climate-related disasters and climate change in the geographical area of the GCF intervention."

- (b) The Accredited Entity shall ensure that GCF funds intended for grant funding of EbA interventions (including demonstration projects) in a select catchment shall only be disbursed to the Executing Entities or Final Recipients (as applicable) after the completion and submission by the Accredited Entity to the Fund of a catchment specific EbA plan (each, an “**EbA Plan**”) for that area, in form and substance acceptable to the GCF Secretariat.
- (c) Prior to approving the EbA plans, the Accredited Entity shall:
 - (i) Deliver such plans to the Fund for its review and no-objection. The EbA plans shall include an independent external assessment of the plan and its underlying climate rationale analysis. This external assessment shall (1) be in writing and utilized by the GCF Secretariat to determine whether to non-objecting to a given plan; (2) be conducted by a person or entity previously approved by the GCF Secretariat and funded by the Accredited Entity; (3) confirm the level of availability of site-specific climate information and whether the plan makes use of such data to generate accurate and informative climate risks and vulnerability analysis for resilient development. The Accredited Entity shall incorporate the GCF Secretariat’s comments, if any, in the finalization of such EbA Plans, and
 - (ii) Deliver the final copy of the EbA Plans within five (5) days of their approval by the Accredited Entity:
- (d) The Accredited Entity shall approve the EbA plans only if they contain the following:
 - (i) The catchment EbA plan includes a catchment specific climate rationale, sourcing available historical observed data for the catchment, and utilizes this data to validate the choice of a model that is used to make future projections about key climate parameters related to drought and other hazards such as flood and storms, as the case may be; and
 - (ii) A validated site-specific catchment plan which shall include spatial mapping of the hazard vulnerability of the catchment, and proposed interventions for the catchment that will maintain or restore ecosystem services that reduce the vulnerability of communities to the expected impacts of specific climate hazards.
- (e) The FAA shall provide for a phased structure of the loan facility as described in the funding proposal, including a disbursement schedule for loan proceeds from GCF to the Accredited Entity subject to milestones, to ensure that the relevant GCF proceeds are disbursed in accordance with pre-agreed terms. In case such milestones are not satisfied in volume, timing or other material condition, GCF shall have the right to suspend or alter the disbursements for the loan facility to the Accredited Entity.

Condition precedent to first disbursement of GCF Proceeds under the FAA:

- (a) The AE shall submit to the Fund of an updated investment eligibility criteria for the loan facility (Activity 4.1), in a form and substance satisfactory to the GCF Secretariat, taking into consideration all recommendations from the independent TAP. In particular, the referred criteria shall be specific and transparent to provide greater substance to the broad investment objectives currently in the funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP174)

Proposal name:	Ecosystem-based Adaptation to increase climate resilience in the Central American Dry Corridor and the Arid Zones of the Dominican Republic
Accredited entity:	Central American Bank for Economic Integration (CABEI)
Country/(ies):	Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama and Dominican Republic
Programme/programme size:	Large

Impact potential

More detailed climate projections to 2050 and 2100 have been included in the revised version that confirm the trends of increased aridity in the Dry Corridor region, strengthening the argument that the selected measures are appropriate to the selected catchments. As a response to iTAP's concerns, CABEI has included a set of criteria for the validation of the selected EbA landscape interventions of the programme to be financed through the grant component.

In terms of observed data, these are very limited in the region, and it will be necessary to agree on alternative methods to evaluate climate impacts and the potential for adaptation of the proposed interventions when observed data are not available

The pre-defined pilot interventions have been selected through a site-specific consultative process (annex 7) to understand site-specific modelled (when possible) and perceived climate risks as well as other relevant aspects that determine the relevance of the selected interventions. The positive impacts of these interventions have also been estimated through the hydrological model showing consistence on positive trends (section 13 of FS) for water availability, groundwater recharge and erosion and are therefore relevant to the needs of the catchments.

Also, in response to iTAP's concerns, CABEI has introduced a phased approach for the loan facility to ensure better alignment with climate risks, actual demand over time and stronger capacities of the key stakeholders.

Paradigm shift potential

As the iTAP points out, the real paradigm shift potential lies on the potential of the enabling environment that capitalizes on the scalability of EbA, focusing on three core areas of support: financial, policy and technical.

For the financial sector, technical capacities will be built within PFIs and non-regulated financial institutions to assess risk, provide advice to local producers and steer investments into more resilient practices. In addition, the programme foresees also the development of capacities of vulnerable communities to better integrate climate risk and risk management options into policy and regulatory mechanisms that will increase this transformation also from the land use, agriculture and water management perspective at different administrative levels.

The need for adaptation funding in the region is far larger than the actual size of this programme and continues to increase rapidly with local communities facing years of severe droughts followed by years of extreme rainfall events. This programme has the potential to create a sustainable funding mechanism unlocking the private sector participation to implement adaptation measures at scale.

Sustainable development potential

iTAP's evaluation under this criterion is appreciated and highlights how the EbA interventions included in the proposal are no-regret options that provide not only climate resilience benefits, but other environmental, social, and economic co-benefits.

Needs of the recipient

iTAP's evaluation under this criterion is appreciated and highlights how the on-the-ground interventions to be implemented in the target catchments in each country were selected based on extensive stakeholder engagement, availability of area for implementation and appropriateness of interventions.

The assessment also highlights how the region is highly dependent on international support to meet climate change goals across the region where the finance available is far below what is required to meet adaptation needs. The added burden of the COVID-19 pandemic and the hurricanes in late 2020 on the economies of these countries will leave even fewer funds available for climate action in the participating countries.

Country ownership

Country ownership is indeed very high for this programme. It was initiated by the Council of Ministers of the Central American Commission for Environment and Development (CCAD) and there is a key opportunity at the moment to align and inform the development of the National Adaptation Plans that are currently under development for at least 5 of the 7 participating countries.

The strong consortium that will be leading the execution of this programme will ensure an integrated approach between political, financial, and technical aspects that will favour an enabling environment for the scalability of EbA.

Efficiency and effectiveness

CABEI considers that this criterion could be assessed at least as Medium High, since the value for money to create impact is high compared to other similar projects. As a matter of fact, this programme is proposing a comprehensive approach addressing all key elements in an integrated manner: capacity, policy and include a strong consortium of leading political and technical institutions and a sustainable financial mechanism unlocking the private sector participation to implement adaptation measures at scale.

Overall remarks from the independent Technical Advisory Panel:

CABEI believes this proposal is well aligned with the recently communicated investment priorities from the GCF and its core role in adaptation finance testing and educating the market for EbA demand. This programme provides a unique opportunity to build more sustainable and actual demand driven funding mechanisms to scale adaptation investments in the region.

Responding to iTAP's recommendations in its previous assessments, CABEI made adjustments and improvements to the proposal, including more detailed information on the climate rationale for all programme sites, developing the first site specific intervention plans for 2 countries in the region (Costa Rica and Honduras) and strengthening the demand analysis for the credit and guarantee components of the program. In addition, phased structures for the grant and the loan facilities have been introduced to ensure a stronger alignment with the climate rationale and market adjustments of the interventions.

CABEI takes note of the established conditions from iTAP to the endorsement of the proposal and will coordinate the necessary arrangements with the GCF Secretariat in order to fulfil these conditions.

Gender Analysis/Assessment

Part I: Gender Analysis/Assessment:

1) CONTEXT OF THE DRY CORRIDOR

The social, political, economic and environmental conditions of the countries occupying the dry corridor, especially the countries of Central America, remain a trigger for the extreme poverty experienced by a part of their population, especially the rural and indigenous population. Economic factors include the economic model that deepens poverty, violence and social exclusion, as a result of lack of opportunities and employment; among the political factors we find an institutionality historically weakened by economic and political elites who defend sectarian interests, justify and sustain acts of corruption and impunity; Social factors mainly include common and organized violence, domestic violence, among others ¹These conditions set up a scenario that systematically pushes many people and increasingly women to have to migrate forcefully.

Most violence in these countries is generated by organized crime and especially gang groups and leaves "visible impacts" such as injuries and assaults on physical health, disabilities, neglect, dispute areas deterioration of public infrastructure, limitations in services in hospitals, health centers and schools, presence of massive security forces, threats, forced displacement, killings, use of women and girls as *an object* of extortion and others, but also "less visible consequences" such as family disintegration, extreme devaluation of women and girls, loss of physical and economic capital, breakdown of social fabric, mistrust among the population, the loss of public spaces and the loss of freedom of mobilization in the territory that slowly and devastatingly contributes to the loss of leadership and community organization, to the deterioration of the quality of life ².

According to the Central American Dry Corridor Characterization Study³, in Central American countries, the percentage of rural population varies from 38% in El Salvador to 52% in Honduras and Guatemala. Of the total rural families, the percentage of families producing basic grains ranges from 54% in El Salvador and Honduras to 67% in Guatemala, with an average of 62%. The rest of the rural population (38%) engaged in salaried work and other agricultural and non-agricultural activities.

In Guatemala 61% of the producers of basic grains are belonging to the indigenous peoples of the highlands. Producers of basic grains, especially maize and beans, from Guatemala to Nicaragua, produce on an average, small area: 1.3 ha in Guatemala and El Salvador, 2.4 ha in Honduras and 2.8 ha in Nicaragua. The local population grows for food purposes for family consumption and only trade when there are surpluses or no urgent non-food needs

Understanding that climate change affects men and women farmers differently is a critical factor in the development of new practical and sustainable climate-adapted technologies that help producers cope with the impacts of change Climate.

Better understanding of the vulnerabilities of men and women to climate change and how they differ and what actions should be taken into account to reduce these vulnerabilities is key. To do this, it is

¹ Childhood Without Borders. Magazine No. 1. Apr/18. Regional Observatory on Child Migration/OREMI. Year 2, No. 2, Jul-Dec/17. Pp3

² Current situation of humanitarian access, March2018. ACAPS

³ According to the Central American Dry Corridor Characterization Study. Food and Agriculture Organization of the United Nations (FAO)

necessary to understand the differentiated uses that give to forests and trees among others, to support the development of sustainable practices adapted to the climate that are sensitive to the gender. In this sense, the perspective of gender equality is critical when it comes to decision-making, policy-making and adaptation and mitigation strategies. Women are not only victims of the phenomenon, they are agents of change that bring knowledge and solutions. While women have important abilities as agents of transformation, they also have specific vulnerabilities that tend to be exacerbated in the presence of extreme weather events because they typically have fewer opportunities access and control of resources. On the one hand, climate change exacerbate gender inequalities, but gender inequalities also contribute to greater negative impacts from climate change. In this way understanding the differentiated risks and impacts of climate change in men and women help to ensure a greater impact on the programme's response.

A large majority of women live in conditions of social exclusion that are expressed in limitations to mobilize outside the domestic space, Less access to information in the event of a threat, Less access to early warning systems and forecasts of climate variability, as well as difficulties in participating in training processes.

Women should be included, not because they are more *vulnerable*, but because they have different perspectives and experiences with which they can contribute; they are nonetheless under-represented in decision-making on relevant aspects of change climate change: reduction of greenhouse gas emissions, as well as adaptation and mitigation strategies.

2) HELD BELIEFS, PERCEPTIONS, AND STEREOTYPES RELATED TO GENDER IN THE PROGRAM FOOTPRINT COUNTRIES OF INTERVENTION

2.1 Dominant masculinities

Traditional gender roles and stereotypes attributed to men and women, as well as asymmetrical power differences between men and women are some of the factors contributing to the definition of a sexist social fabric in the areas of intervention of the dry corridor. In this context, gender inequality is rooted in a social construction of masculinity, whereas boys learn to exercise power over girls and women. Which means not listening to the voices of women and girls, subordinating their desires and decisions to men, considering the bodies of women and girls as an object of man who has no rights.

Social and cultural norms are sexist in dry corridor countries. Gender roles, norms and stereotypes are marked by strong inequalities in social relations and patriarchal cultural constructs. Which set expectations about women's femininity, reproductive role and sexuality and play a key role in creating greater vulnerability to violations and gaps in the realization of women's and girls' human rights and especially in rural and indigenous contexts.

Children are socialized based on a hegemonic model of masculinity that defines them as strong, dominant, providers and that they must adhere to a rigid code of honor that forces them to compete or use violence to verify that they are "*male*." They are also socialized with the expectation that they will be well informed and powerful in terms of productivity activity. Being a "male" in these countries often means demonstrating your virility (through multiple heterosexual relationships), discussing your conquests with your male peers, and verifying your fertility by having children. In addition, in most communities children are raised so as not to worry about their own health and not to seek help when they feel stressed. As a result they believe they cannot express doubts or concerns about their bodies, sexuality or reproductive health, or seek help. Nor are they prepared to take responsibility for the exercise of their sexuality so they do not commit to the care of themselves and their partner for the

prevention of STIs, unwanted pregnancies, etc. These conditions have differential consequences for them and for them.

There is a growing recognition in dry corridor countries, which the hegemonic model of masculinity puts women at risk of being victims of gender-based violence and sexual and reproductive health problems, such as the infection of Daddy virus human hill - which is one of the agents of cervical cancer - Meanwhile, men suffer higher mortality rates from external causes (accidents, homicide, suicide); higher rates of alcohol and drug abuse; violence and exclusion when they do not meet the expected male model; worrying STI rates and very low opportunities for participation in the upbringing of their sons and daughters.

2.2 The situation of rural and peasant women

In the social and political context where the dry corridor communities identified for intervention are located. There are certain setbacks of human rights, they form systems of discrimination and privileges that feed, the intersection between gender inequality and other forms of exclusion, such as class, ethnicity, race, age, the place of residency and disabilities. In this sense, the multiple discriminations faced by women in terms of the benefits of state initiatives, access to decision-making spaces and participation in policy-making processes is a constant . ⁴And it is very important that this situation of inequality that women live, is not detached from the analysis of the impact on life of populations by climate change

a) Feminization of Poverty

The Dry Corridor is the most densely populated region in Central America, with a population of approximately 10.5 million people. About 60% of the population of the Dry Corridor lives in extreme poverty. The most severe poverty is located in Nicaragua, Honduras, El Salvador and Guatemala and has a special impact in rural areas. The effects of extreme events such as drought and flooding particularly affect women who in general are in the situation of greater poverty. It is caused by the loss of crops, productive activities, problems with livestock and fishing.

b) Gender-based violence

The dry corridor region faces a crisis of violence and systematic human rights violations. Large national territories controlled by organized crime or criminal gangs, economic policies that deepen inequality and favor private interests, meddling in public affairs of religious hierarchies, militarization, stripping of territories and natural goods to native communities and peoples, as well as policies and laws that restrict and criminalize social protest or that violate fundamental rights, are some of the expressions of this crisis that is presented to varying degrees and nuances according to the country.

Impunity is present throughout the region to alarming degrees. At the world's level Nicaragua, Honduras and El Salvador rank sixth, seventh and eighth respectively in the impunity index⁵. Impunity affects women in particular, especially those living in remote rural areas. The InterAmerican commission of Human Rights (ICHR) has recognized that in several countries there is a pattern of systematic impunity in prosecution and actions around cases of violence against women.

In this context, there is a worrying increase and sharpening of violence against women. Increasingly cruel forms of violence, including slavery and sexual torture, are present throughout the dry corridor region amid a culture that tolerates and normalizes violence against women. There is also an increase

⁴ Central American Commission on Environment and Development (CCAD),

⁵ Center for Studies on Impunity and Justice (CESIJ) of the University of the Americas Puebla. <http://www.udlap.mx/cesij/resultadosigi2015.aspx>

in femicide/femicide rates aggravated by cruelty in the mode of death, the planning of the death and its execution in the presence of sons and daughters.

Violence against women is an extreme form of gender inequality and affects one in three women in the Americas, who have been victims at some point in their lives of violence by their partner or stranger. It takes many forms, but sexual, physical and emotional violence by a male partner are the most prevalent forms⁶. Violence against women has lasting consequences that profoundly affect their physical and mental health, the health and psychosocial development of their children, the well-being of families and communities, and the budget and economic development of nations⁷ and can have profound and lasting consequences for survivors' health, such as physical injury, unwanted pregnancy, abortion, sexually transmitted infections (including HIV/AIDS infection) and various negative outcomes in mental health matters.

This context of violence in which women live must be taken into account within the framework of the areas of intervention, as it also affects women in the selected communities. In the areas of intervention, the house or the home that are considered as *private* space where women and girls live, is not the safest place for them. The sphere of *private* space is built around the belief that *from doors to the inside* what happens is a family-only theme. Even when they may be in violation of women's right to a life free from violence. Under this paradigm, multiple forms of violence towards women and girls are justified and undefeated in rural communities. In turn, physical violence is the most visible expression of the exercise of power by the husband or partner with the woman. It highlights the asymmetry in power relations between men and women, which perpetuates the subordination of women to men in the private space.

In the areas of intervention of the program, the family nuclei are structured around the male gender at all decision levels. From the domination and power of the husband that clearly affects the relationship with women and the role that they have socially assigned to suppress, dependence and accept physical violence. Psychological violence has a profound impact on the assessment of themselves and self-esteem and their mental health. Which has a direct impact on the physical and psychological health of your sons and daughters. Rural women and women belonging to indigenous peoples may be at greater risk of violence than other women.

c) Access to services

Living in rural areas of the dry corridor and being a peasant woman increases socioeconomic disadvantages due to lack of access to services including health, education, water, health and transportation conditions.

The lack of services for reproductive health care, childbirth care, as well as access to contraceptive methods is a challenge for rural women. The limited access to maternal and child health services, the absence of doctors and health providers causes lack of care for women and early childhood. Lack of services prevents pregnant women from checking their pregnancies. During pregnancy they do not have access to care centers, since they are concentrated in the urban areas of the main cities and have no resources to pay for transportation. In turn, the roads to move are in very poor condition and low transport. Often when they go to the controls is at the end of pregnancy or when they are in labor and have no more alternatives. Many decide with no more options to give birth in their own homes and some give birth on the roads, when they are on their way to being treated.

⁶ Multi-country study on health of women and domestic violence against woman: WHO; 2005. http://www.who.int/gender/violence/who_multicountry_study/en/.

⁷ Global and regional estimates of violence against women: Prevalence and effects of conjugal violence and non-conjugal sexual violence in health OMS; 2013. 51 p.

d) Femicide

The problem of femicides in dry corridor countries is particularly complex, as it is often linked to other challenges pointed out in the backlines; violence in the couple or family members; organized crime; public insecurity; trafficking in people and girls; migration of women and girls in unsafe conditions; and discrimination, exclusion and structural racism against rural women and girls and indigenous women. These challenges are more likely to be present in countries with persistent inequality and discrimination against women and girls who replicate violent and non-cohesive societies

Of the 25 most critical countries in the world, 14 are in Latin America and the Caribbean. According to the gender observatory of Latin America and the Caribbean. In absolute terms, if you compare the rate per 100,000 women, the phenomenon reaches an extent in El Salvador that is not parallel in another country in the region: 10.2 femicides per 100,000 women. In 2016, Honduras registered 5.8 femicides per 100,000 women. In Guatemala, the Dominican Republic also saw high rates in 2017, equal to or greater than 2 cases per 100,000 women.

The path to the end of gender-based violence is arduous. According to the agency UN Women, 30% of Latin American women suffer from this phenomenon including women from dry corridor countries. This figure, coupled with the number of victims of femicide daily, achieves the criteria that the World Health Organization establishes as an epidemic.

e) Violence against women environmental defenders in the dry corridor territories, derived from the presence of extractive companies projects

In some territories they have been denounced by the organizations of rural women, that some extractive projects of transnational companies have promoted a context of violence, stigmatization and criminalization against community leaderships especially towards women who promote the rights of rural women. Reference is made that human rights violations are committed and that this is accompanied in some cases by the persecution of women and young defenders of the environment

There are large tracts of territory that have been granted to commercial companies for mining, hydroelectric, monocultures and/or oil. These large projects that are causing an impact on the ecosystems of the area also coexist with the smaller territories for family production.

According to the report of the Mesoamerican Initiative of Women Human Rights Defenders⁸, in El Salvador, Guatemala, Honduras, Mexico and Nicaragua, it shows that in 2015 and 2016 there were a total of 2,197 assaults on human rights defenders in the region, an increase of 30% compared to the previous period, which reported a total of 1,688 assaults. In Guatemala, Mexico and Honduras, most of the aggressions were concentrated, which doubled from 2015 to 2016 in these three countries. Following the trend of previous reports, land, territory and natural property defenders continue to be the most assaulted. All of this has a considerable impact on women producers in communities who protect their territories and work to mitigate the effects of climate change for years.

The situation of helplessness in which some communities and their leaders are placed pose scenarios of impunity that contribute to strengthening the factors of discrimination and violence against

⁸ ***Violence against women human rights defenders in Mesoamerica 2012-2014 repport.***
Mesoamerican Initiative for Women Human Rights Defenders

women, and thus to fostering unsafe environments, hostile and exclusionary, unfavorable to the defense of rights and their participation as agents of change

f) Feminization of migration

In this context of climate change, the migration of adolescent women and girls has ceased to be an invisible phenomenon, where the woman perceived herself as a passive actor *accompanying* the husband or *waiting for* the husband to emigrate. The current situation shows the active role that women producers play in the migration process. They have a role from decision-making to migrating, decision-making family networks support the destination countries. As well as in the shipment and administration of remittances from the destination country.

The trigger to start the journey is the crisis stemming from the serious effects of climate change and women and girls undertake migrant projects *voluntarily* or forcefully, to escape the extreme poverty derived from drought, loss of crops, livestock and above all lack of access to food. In the country of origin, irrigation also exists when women should look for alternatives to agricultural work. As precarious jobs that also puts them at risk of labour exploitation, for example in maquilas or even sexual exploitation.

Women who migrate due to climate change also live a great vulnerability in their status as migrants. And during the journey they go through situations of isolation, lack of social and affective networks, precariousness in their economic, employment living conditions, their ignorance about their rights in the country of arrival. In this context of crisis, the migration project has also led to the increase of new forms of slavery through the recruitment of women through trafficking and trafficking networks for commercial exploitation purposes. Most of the time, they are tricked through advertisements in local newspapers social media or even on the streets.

A 2017 study published by⁹ the World Food Program analyzed data on migrants from El Salvador, Guatemala, and Honduras that were rejected by Mexican immigration authorities when they tried to reach the United States. The study found that 50 percent of them had been working in the agricultural sector before leaving. Contrary to the common perception that migration in these countries is driven mainly by violence and insecurity, 65% cited unemployment or economic hardship due to the loss of agriculture, and 19 % said low incomes and poor working conditions were reasons to leave.

The document also notes that 47% of the families interviewed suffered from food insecurity. Such levels of food insecurity had not previously been recorded in the region in assessments carried out over the past three years in the Dry Corridor.

About 72% of the families interviewed¹⁰ from dry corridor countries said they were resorting to "emergency" measures to deal with their situation, such as selling their land, animals and tools to buy food. Meanwhile, 78% of family members who remain in their countries receive a consignment on a monthly basis, and in fact 42% of the families surveyed report that remittances were their only source of fixed income, as they did not have access to the agricultural work due to climate change. More than half of the money sent by migrants is used by families to buy food, followed by agricultural investments – land and animal purchases – and investment in small businesses

⁹ “Food Security and Emigration: Why People Flee and The Impact This Has on Families Who Remain in El Salvador, Guatemala, and Honduras”

¹⁰ Op. Cit

g) the situation of girls in rural areas of intervention

the analysis of the situation of women in dry corridor countries, it is very important to contemplate women throughout their life cycle. Girls and adolescents also have a very important role in the intervention areas of the proposal. Their living conditions are often the most disadvantaged and suffer multiple forms of violation of their rights as girls. Among the most visible problems in rural communities that are closely linked to the extreme poverty that families experience due to the effects of climate change are: Forced pregnancies of adolescent girls and forced and early marriages and unions. These problems are often invisible and normalized by the communities themselves and have a decisive impact on the lives of the girls in the communities. Since they cause a disruption to your life project and have an impact on the local development of the community.

Girls and adolescents work the land with their families and have a very important role in caring for families. However, your opinion and participation in decisions is often devalued.

- **Forced pregnancy**

The list of the 50 countries with the highest fertility rates in adolescents in the world includes almost all countries in the region; Nicaragua, Dominican Republic, Guatemala, Honduras, countries with particularly high rates (Rico and Trucco, 2014). Pregnancy and motherhood are spoken of as an inevitable consequence of the lives of girls and adolescents in the situation of poverty and precariousness to which they are exposed in communities. By implying that girls and adolescents have no choice but about their lives. And in turn, it would be a natural condition closely linked to becoming a woman.

Pregnancies among teenage girls in communities are often unplanned and the couple is rarely found to be of the same age. They report with a large majority of 15-year-old pregnancies considered to be the result of sexual violence and coercion. The age of onset of sexual intercourse was dropping and is often not with peers of the same age. The age of the first pregnancy is evident in community areas in girls under the age of 14.

- **Forced and early child marriage and unions**

In Latin America and the Caribbean in 2017, 23% of women aged 20 to 24 married at age 18 and 5% married 15.¹¹ The countries in the region with the highest prevalence of women aged 20 to 24 who were married or entered a union before the age of 18 are the Dominican Republic with 36%, Nicaragua with 35%, Honduras with 34%, Guatemala with 30% and El Salvador with a 26%.¹² There are significant age differences between girls who marry in childhood and their partners, but here too, data is scarce; a 2005 UNFPA report estimated that this difference would be between 6 and 7 years, although the figure may have changed significantly since then.¹³

The poverty generated by the effects of climate change puts producing families in a complex situation to feed their daughters. Delivering a girl in marriage or union with a man who is often older, is thought to be a way out of the lack of food for the upbringing of girls. However, the consequences on girls' lives are devastating as it disrupts their life project.

¹¹ Unicef. 2017. *State of the World's Children 2017*. New York: UNICEF.

¹² Op cit.

¹³ UNFPA, 2017

2. 3 Roles And Responsibilities

Gender analysis in statistics shows that the information available is still limited; Women are still not represented in statistics that collect central information on land tenure, participation in decision-making, the level and characteristics of the types of agricultural and livestock activities they carry out, their participation in activities and their access to credits or other financial activities, among other important aspects.

The social dimension in which people, their homes, families, organization, from the different positions they occupy and opportunities that men and women have in society. It is also one of the areas that is poorly documented in agricultural censuses and surveys.

In general, smallholder families grow basic grains for their livelihoods in all zones. In addition, they have a yard for poultry and one or more pigs (depending on the amount of food available); and for the cultivation of fruits, vegetables and herbs. Other forms of land use depend on circumstances and opportunities. First, it depends on the size of the property.

The average age of central American basic grains producers is 49 years, 14 years older than the average age of the economically active rural population which is 35 years; the difference may indicate the lack of generational takeover in the field. The presence of other opportunities, cultural changes, ambitions, capacities to meet basic needs, etc. mean that there is a loss of interest among young people in continuing the traditions of their parents.

Gender roles are very marked:

The woman is assigned a reproductive and family care role. She raises children and cares for the sick and the elderly. Their tasks often focus on the private space of the home. They are responsible for preparing basic food for consumption at home and are the ones who manage the smaller and family plots and domestic use. They have their central work in growing food and rearing poultry and livestock. Because of this, they are responsible for food production for the family, resulting in significant workload and stress of ensuring that food is not lacking. In general, it alternates these activities with informal work in precarious conditions, with low wages and with seasonal value. In addition, a large part of women and even girls participate in unpaid family-worked economic activities that do not have a decision-making power over the profits collected. Women in rural areas of dry corridor countries spend many hours of the day working and do not have free time.

Caring for minor species care is one of the works that is basically designated to women, as they are the ones who care about seeing how they can have poultry and if possible some pigs, since these species provide food to the family.

Women in their role as caregivers manage to meet the needs of their families through extensive knowledge of natural goods. For the treatment of diseases they are highly dependent on their ancestral knowledge as they resort to the use of medicinal plants and herbs to treat the symptoms of the sick, although the shortage of plants has affected the care for the sick and adds work and investment of money for women in their role as caregivers. To this we add that most plant collectors, seed preservatives are women however their medicinal knowledge is often ignored and underestimated.

Traditionally they remain in their properties during planting times, however, in the context they live by the impact on land of climate change, they seek other economic options outside of agricultural production to generate resources.

This situation has a very big impact on women who must constantly 'solve' in order to meet the needs of the family nucleus and often the extended family. This creates an additional strain for women who in a situation of so much instability this does not allow them to plan.

Men have a productive role and is the one that occupies its time in the crops aimed at the market. they are characterized by being directly involved in production, providing physical work and productive management, with the male who makes decisions about the production and marketing of income crops, re-presenting to the family, taking the decisions and participating economic organizations

Because of the effects of climate change they sometimes have to move away from their place of origin in order to find alternative jobs. There are isolated livelihoods where the farm's income is low and labour is cheap; obviously these are areas of high vulnerability. This phenomenon reaches a maximum when it becomes permanent migration abroad of certain family members to support their family nucleus with the sending of remittances.

2.4 Control and Access To Resources

Most small and medium-sized producers grow for self-consumption and local markets, which suffer from low productivity, as their plots are usually located in mountainous areas with little access to irrigation water, road infrastructure to take out its production, credit, modern production technology, improved seeds, tools and basic supplies. abnormal weather conditions contribute to crop loss in addition to the diseases that attack the plantations have caused severe havoc.

Underemployment of wealthy farm owners for part of the year in most livelihoods. It has provided labour to well-off agricultural producers, rural enterprises, coffee companies, agribusiness, domestic employment, hotels and, more recently, maquilas. It also causes migration to urban centers, other countries in the region, the United States (USA) and Spain.

There are factors that make it difficult for women to engage in paid work. From a livelihood approach, these factors are related to women often having neither access nor control over natural resources: land and water.

Water is one of the goods of which women are in the largest percentage in charge of providing in the home in which they invest hours to get the resource, however they are not the ones who have control of births or ecosystems , which visibly affects them

Not having land ownership is very difficult for women to apply for and obtain credits or even save the resources generated. It also has major challenges in accessing social resources such as their participation in the community structures that take definitions. This makes it difficult to engage politically in community decision-making.

Other resources such as energy, technology and transport also limit their options for production and therefore their opportunities for access to paid work. Men often have control and access to resources, as well as decision-making within the family, even if resources are provided by women.

2.5 Participation

The participation of women producers is reduced at the local level within producer associations, irrigation organization, neighboring boards, water committees. And also their presence in positions of

social and political leadership is still clearly a minority. In community spaces they have a weak participation also because their voices are not heard or taken into account

2.6 Women And Girls Vulnerability And Adaptive Capacity To Climate Change

Holistic analysis is required to talk about climate change, in a complex context such as the dry corridor region with territories with high environmental and climate vulnerability. In turn, in addition to the adverse environmental conditions, as mentioned lines ago, they also present a political, social context and with a strong inequality of gender. That make populations and ecosystems more or less vulnerable to the impacts of climate change

Among the most recent and important impacts of extreme events, the El Niño phenomenon that began in 2015 has been one of the most harmful and its impact to this day is still evident. As a result, some 3.5 million people need humanitarian aid with 1.6 million in moderate or severe food insecurity in the countries most affected: El Salvador, Guatemala and Honduras¹⁴

Several climate change-related phenomena are becoming stronger according to local consultations with women. Droughts have increased causing a decrease in crop yields. This has fostered the presence of pests and diseases, and also contributes to the spread of devastating fires. In turn, access to water has been decreased and its quality has deteriorated. On the other hand, extreme rains, hail waves and cyclones have also contributed to the damage of crops.

The few studies that exist, complaints from women's environmental organizations and conversations with women during local consultations show that environmental devastation and the effects of climate change have a major impact on women and girls. The effects of climate change live with more challenges, especially those associated with access to water and biodiversity and care of livestock.

Identifying and taking into account the differentiated impacts of climate change on women is key, taking into account the central role that women play in the processes of adaptation and mitigation of this phenomenon. Among the effects highlighted during consultations of the different dry corridor countries are the following:

- **Multiplication of the reproductive day**, with health concerns, generating greater poverty. Women have more work because they need to mobilize to address food shortages. The workload is increased by the need to travel large routes for water and wood harvesting.

On the other hand, the consultations indicated that women's work is also increased due to flooding, which forces women to ensure that livestock survive and take them to safer places. In addition, reconstruction work on lost infrastructure, such as fences or pens, also adds additional work to women.

- **Negative impacts on agriculture and fishing, for smallholder families especially for women.** Family farming as the main livelihood for women is at risk.

Most of the causes of environmental vulnerability mentioned in local consultations and mentioned by women are interconnected. Indiscriminate deforestation causes soil erosion

¹⁴ <http://www.fao.org/news/story/es/item/422220/icode/>

and degradation, increased intermittent flows, flooding, landslides, river sedimentation, loss of waterways, loss of water sources, drying of rivers during part of the year, deterioration of roads and roads.

Soil degradation has led to loss of fertility and loss of crop yield. In addition, they highlight the inappropriate use of fertilizers and chemical pesticides favors health problems of the population, as well as pests, decreases the use of production system's own resources, increases costs and creates more poverty among producing families.

This results in financial loss for the family. They lower family income from corn and bean sales. Women clams and fish fishermen of livestock production also expressed concern about the absence of income.

Another element of concern is that genetic diversity has been lost because people buy certified seed in some cases even transgenic and have lost their heirloom variety seeds.

All of these are mentioned as threats that are affecting families, and losses in this sector are forcing rural populations to migrate to find work for the family's livelihood or payment of credits or expenses made by the crops that have been lost.

This has a social impact due to the disintegration of families and often leaving children in the care of others in the family or neighbors, which can create various risks for them. When women are not the ones migrating, they take care of the family and try to cover the financial expenses of the house and the credits obtained, through the sale of food, jobs in local shops or domestic services.

- **Migration and displacement by the risk of natural disasters.** The World Disaster ¹⁵ Report shows that displacement as a result of natural risks or conflict substantially increases the risk of physical abuse towards women and girls. Women's safety is at risk in addition to other deprivations they suffer when displaced. The most frequent threats mentioned in the areas of the dry corridor are: irregularity of rainfall, drought, floods, landslides, environmental degradation, low crop yields, low prices of products on the market, high costs of inputs and supplies. transport, access roads in poor condition or lack thereof.
- **Decreased forest cover.** Women say that with climate change forest cover has been lost significantly and with it, there has been decreased flows from water sources. The forest has been lost either by the hand of man or by the attack of the weevil. There is a concern on the part of women in some countries about illegal logging and marketing

Women use forests to obtain food, fuels (wood or coal for cooking), raw material as inputs of products they can sell, herbs, medicinal plants. As forest resources decline, they will be devoid of subsistence supplies.

¹⁵ World Disaster Report 2018 © International Federation of Red Cross and Red Crescent Societies, 2018

For the treatment of diseases they are highly dependent on their ancestral knowledge as they resort to the use of medicinal plants and herbs to treat the symptoms of the sick, in recent years they have observed that some of these species are harder to find, so loss or limited access to diversity of medicinal plants and herbs is another impact of climate change, which affects the care of sick people.

- **Limited access to formal work** through food marketing, many of these informal employment opportunities depend directly or indirectly on natural assets, for example: water.

For women who make handicrafts as an alternative resource, through the realization of products such as baskets, dumplings or other products that need raw material that has disappeared due to drought, the effects are negative and make it difficult to access extra resources for the family economy.

In some countries, women are not recognized as farmers in their identity document, which makes it difficult for them to access agricultural aid or grants from governments.

- **Limited access to water.** The availability and access to water has serious difficulties, with an impact on health. Lack of access to safe drinking water has also been accompanied by increased health problems for families, parasites and other diseases of their own by a lack of water. A situation in the region is the "chocolate rivers" that will flow into the gulf, which is reflected in pollution on the coast, and increased sedimentation.

The water supply is primarily carried out by women and girls. the child's phenomenon has caused the water lines to decrease or dry completely, so they have to move to other aspects that are removed from the community, generating long lines of women waiting to fill their containers with water to take to their homes. This can lead to conflicts between communities as before the drought they have had to share the water sources, which has generated disputes and some local consultations fear even wars. The investment of time for water supply can last all day. The reversal of this time indicates is because they have to be aware of the moment when the water will fall and fill the containers that are necessary for daily tasks. The absence or reduction of girls' schooling for their obligations in the search for water impacting their potential as a factor of change and reaffirming gender roles. The risk of gender-based violence increases, the absence of water puts women and girls at risk, as they may be victims of sexual violence on the routes to water sources.

- **Impact on women's health.** The impact of women's health is significant and often unvisibilized as their role as caregivers they have little time for self-care. There is an increase in kidney disease due to water contamination. Also mentioned in local consultations are the increase in what they call household diseases that are related to skin diseases arising from mosquito bites, such as zika, or mites

Chronic tiredness also causes overload that significantly affects their health, making an impact on their mental health, with stress and anguish.

Lack of food is causing hunger and malnutrition in women and children. Diet change have been forced because a lack of dairy and protein. Uncertainty of crops yields and livestock deaths, as well as high costs contribute to malnutrition with serious health impacts.

One phenomenon that is occurring in recent years is what people refer to as "sunstroke" which is the feeling of higher temperature in the body, due to the increase in temperature.

This is thought to cause dehydration and hypertension and skin diseases, so it is recommended that there be no sun exposure during midday.

2. 7 Needs and Priorities

The challenges of mobilizing women in local consultations demonstrate the barriers rural women experience when participating in consultation spaces for interventions in the areas where they live. Despite this, some women were able to move some key issues that are evident to them such as needs and priorities to be taken into account by women

- Generating spaces for economic empowerment and strengthening of women's technical capacities to implement Ecosystem-based Adaptation. It is constant that lack of technical capacity in some areas, and land tenure and lack of territorial order make conservation difficult.
- Provide support mechanism for women living violence.
- Increasing the negotiating power of women at home and in the community space. --Strengthening the role of women as decision-makers.
- Full recognition of women's rights, legal and social status.
- Generate productive economic development opportunities for women producers. Support in facilitating access to credit. Legal accompanying for access to land titling
- Financial mechanisms: Necessary for the implementation of the programme and ensure sustainability. Currently, financial mechanisms benefit agriculture before than afforestation. There are no loans to plant trees, but for activities that promote deforestation.
- Incorporation of Ecosystem-based Adaptation and knowledge management: Very important. But governance issues (land ownership, agricultural incentives/deforestation) must be resolved for the programme to continue over time (sustainability).
- Water boards/committees define leadership roles for women: community-level bodies responsible for ensuring the basins and water projects that supply each community. It is mentioned that for women the protection of the forest as a key factor for the generation of water for the territory.
- Funds invested in municipalities considering the needs of women and that really contribute to improving resilience.

Adaptation measurement	Needs and priorities
Restoration of forest cover/water resource.	• Accessibility to the water resource. • Generation of employment opportunities. • Family wellbeing (clean air and water) • Water retention and distribution: Time savings for women, when there is more water
Sustainable production of wood and firewood	• Improvement in the exploitation of the natural forest. • Protection and conservation of natural habitats • Opportunity to improve income. • Increased job opportunity. • Improves family quality of life.
Rainwater collection	• Generating jobs during dry season women through entrepreneurship

Efficient irrigation technology.	• Food safety. • Availability of resources for other investments. • Time savings, physical effort and economic savings.
Implementation of agrosilvopastoral systems.	• Increased availability of time to dedicate to the family. • Reduction of migration.
Implementation of agroforestry systems.	• Its products improve food safety (e.g. fruits from shaded systems). Some are sold. • Healthy environment (shade and lower temperature) during the agricultural labor.
Ancestral seed banks and improved seed for adaptation (Drought resistant)	• Reduced production costs, accessibility to buy other type of foods for the family, to cover other family needs. • Improves food, nutrition and women's health
Environmental Education. Formation of values.	• Raise the level of education. • Increased environmental commitment and awareness taking into account the knowledge of women producers
Implementation of home adaptation resources (efficient stoves, ovens)	• Decreased workload. • Economic savings.

3) CONCLUSIONS

The high poverty caused by the negative effects of climate change in which women and girls live in the context of dry corridors has a profound impact on the living conditions of women and their families. The feminization of poverty is very significant and is exacerbated by the lack of access to family entrepreneurship initiatives as an alternative to the problems that climate change has generated in agriculture, livestock and fisheries.

Women face discrimination in access to resources and control of land and property. Owning the land is important to women for the economic value that this represents, but also to be able to use it as a means of production and generated additional income¹⁶. It is a way of being able to access training programs, linking to women's farmers' cooperatives or local development organizations. When the woman does not own the land she also causes the different tasks she performs in the territory to be invisibilized and makes it more difficult to participate in community development spaces. As for their participation in decision-making, it is generally limited and when it is done in a shared way with men, it happens that women seek the legitimization of their decisions in the male figures of the family.

Women and girls especially continue to assume primary responsibility for unpaid family care and domestic work. In this scenario, the serious situation caused by the impact of climate change has increased the workload of women and girls, as they are responsible mainly for the search for food and water on a daily basis. The burden falls disproportionately on girls and women, perpetuating gender inequality and violence against them.

The root of the violence that is rooted in the imbalances of power and structural inequality between men and women is manifested as a social mechanism of people's social relations. And it is

¹⁶ Measurement of women's contribution in agricultural activities in the Dominican Republic. National Bureau of Statistics 2018

underpinned by entrenched social and cultural norms that normalize gender-based violence in communities.

Gender violence cannot be detached from the impacts associated with climate change, as it impacts women's and girls' lives and puts them in situations of extreme vulnerability to their active contribution to remedy climate change. The different forms of violence against women and girls expressed in numerous ways are a widespread and deep-rooted practice that is perpetuated by the poverty generated by climate change, which in turn perpetuates inequality between men and Women. The lack of available data does not make visible the extent of the problem of gender-based violence in the selected areas of dry corridor intervention, nor the significant impact on women's and girls' health. Especially in the mental health, invisibilized, which is causing in all of them, very serious impacts that affect the self-care and care of their families.

The start of the migration project is carried out by the extreme poverty caused by the effects of climate change. During the journey, women and girls experience situations of violence, such as new forms of slavery through the recruitment of women through trafficking and trafficking networks for commercial exploitation purposes. They are often tricked through advertisements in local newspapers on social media or even on the streets.

Other forms of violence caused by poverty experienced by families that are increasing by the effects of climate change are those experienced by girls and adolescents. Girls are undervalued and face discrimination on the basis of their age, gender and ethnicity. When their families do not have resources or food for their survival, they often no longer take responsibility for girls by allowing them to marry or enter unions with much older men who can take care of them. Early child forced marriages and unions are evidenced as a standardized and invisibilized problem. This violation of girls' human rights has also increased due to the current situation of extreme vulnerability faced by girls due to poverty resulting from climate change. And that it is often linked to sexual exploitation and trafficking in girls in the context of crisis. Early child forced marriages and unions brings together all forms of violence that can be exerted against a girl and adolescent, as it is closely linked to high levels of domestic violence in the private space of the home and has devastating effects on the life project girls, their access to school and their health. It is one of the causes of forced motherhood, putting their lives at risk from sexual violence and the complications that pregnancy and childbirth cause in the lives of girls

Lack of access to quality and warm health services, as well as their difficult access because of poor road conditions, have a very serious impact on women in rural communities. Care for pregnant women and girls and care for childbirth is precarious. And especially those in the farthest communities such as indigenous women. In addition, in general there are no institutions that offer protection to women and girls in situations of gender-based violence, they are in an extremely precarious situation and without budget for human resource recruitment. Incoordination between the different bodies creates an additional barrier for women to decide to report and seek legal support.

There is a significant challenge in strengthening health and protection services, in how to respond adequately to cases of gender-based violence and provide good quality confidential medical and psychosocial responses to survivors in accordance with national protocols and in accordance with the basic principles of gender-based violence and the women's and girls' approach.

The constant stress of lack of food is causing serious problems in the health of women and their children. Women's mental health problems, stress, anxiety, fear and depression have been exacerbated by a lack of access to food for the children and their children. This represents an

additional situation of violence towards women, who bear the burden of ensuring the survival of their families in a context of climate change

There are areas in the Dry Corridor, where agriculture is not feasible, due to the prevailing climate, which is extremely dry and soil conditions are not adequate, so diversifying production and complementing traditional sources of through women's entrepreneurship is important. Building spaces for women's economic empowerment can contribute to greater success in poverty reduction.

A key element for the strengthening of women's capacities is the empowerment and participation in decision spaces in order to have their demands heard and incorporated, which leads women to make decisions as well as to the economic empowerment.

Literacy is also necessary, as some women producers often struggle to read, some had been in second grade. Lack of schooling impacts the degree of participation women can have in decision-making processes, about their livelihoods and ecosystems. This puts women at a disadvantage because, in the case of having access to information and documents, they do not have the necessary tool to make this information affordable.

To promote women's participation in decision-making spaces, individuals surveyed mentioned that it is necessary to strengthen capacities with a percentage of selection

It is necessary to point out that women form social networks within their communities and are aware of the structures and capacities within their territories. Therefore, it is important to include women in the training and decision-making processes related to the management of natural assets. By incorporating the gender approach into the adaptation process, it will not only benefit women, but also men in the long term.

The sustainable provision of financial services to low-income households has become an important tool for combating poverty and especially women who, in these areas, are mostly heads of families. Savings, leasing, insurance, remittances, as well as business loans, housing, health and education allow low-income families to increase their household incomes, capitalize on, plan their future, send more children to school for longer, and reduce their vulnerability to the crises that make up part of their daily lives. Access to financial services also significantly improves women's safety and self-esteem.

Historically in the Central American region, these benefits have been largely made possible by the work of micro-financial institutions, which have opted to serve that population that was not credited for formal banking. As a result, governments, development agencies, international cooperation agencies and other microfinance practitioners have been actively involved in creating or supporting mechanisms that encourage microfinance development in different countries around the world, with a special emphasis on gender.

4) RECOMMENDATIONS

Output 1. Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures.

- **Capacity building and awareness-raising of public officials in the municipalities involved in the programme on the importance of the gender approach and how to incorporate it into agricultural public policies:**
 - Use, management and equitable control of natural resources.
 - Empowering women from disaster risks.
 - Equitable participation in decision-making processes.
 - Equitable distribution of benefits and public resources for development.
 - Equitable access to training and training processes.
 - Access to information systems (early warning systems, systems to detect critical places regarding climate change and risk management).
 - Recognition of the needs and interests of men and women of different ethnic groups and of different ages.
 - Recognition of differential impacts and the various vulnerabilities that men and women may suffer according to their age and ethnicity.
 - Promotion of affirmative actions or temporary measures for women, in processes that promote sustainable development.
- **Develop protocols that guarantee the specific needs of women, for the adoption of technologies efficient in the use of water by households. Ensuring measures for the involvement of community women in water committees.**
- **Take specific actions to strengthen the capacities of municipality technicians on social, environmental resilience and gender.** To contribute to the development of public policies for climate adaptation and mitigation. These policies are to be based on the voice and proposals of women, who are the majority of the population not in the construction processes, much less are represented in the proposals for climate change policies and plans.
- **Develop actions with municipalities to facilitate the legalization of property cooperatives and community women's organizations**

Output 2: Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.

- **Ensure the active participation of women producers and women's organizations in the programme's decision-making processes, as well as in water committees.** Women's participation in the decision-making spaces of the programmes activities is a priority. One of the main gender gaps is precisely in decision-making processes. Women have historically been excluded from these spaces by denying not only their needs, interests and knowledge, but also being invisibilized as producers (workers) and unaware of their contributions and their ability to influence economic and social development, local

Women's organizations at the community or municipality level that advocate for gender equality and women's rights will be important to involve them because they will be able to account for the different demands and perspectives of rural women. Including representatives of local rural women's organizations in climate change decision-making processes contributes to the sustainability of the programme.

Participation must be equitable, this means that the same number of men and women are part of the workshops and focus groups and that their participation is equally active and meaningful. It is

essential that training and training initiatives for the active participation of women in decision-making and consultation spaces for policy-making are essential.

A key element for the strengthening of women's capacities is the empowerment and participation in decision spaces in order to have their demands heard and incorporated, which leads women to make decisions as well as to the economic empowerment.

- **Generate spaces of empowerment for women producers that ensure the inclusion of topics that they identify. Some examples:** Training women in new technologies and promoting the exchange of successful experiences in this regard. institutional strengthening of women's community organizations, through a wide range of trainings: from courses focused on various agricultural issues, to gender awareness workshops and local leadership promotion.

A community-based child care space will need to be generated during sessions that mobilize women. In order to give women the opportunity to participate more actively in training spaces and thus have a more active and productive role in rural economies.

- **Promote quota systems for the representation of women in Community and producer organizations related to the programme.** It is a key element in the framework of the strategy for women's participation. It will increase women's bargaining power and raise its voice to remind the community and its representatives that women's development is an effective strategy to combat poverty and improve the livelihoods of rural populations. As well as for women to play an active role in actions to combat climate change

Women's participation in all social and political processes is a key element in consolidating a quality democracy. Therefore, to strengthen their capacities, that they know their rights and how they can claim them.

- **Generate community dialogue spaces, which include sessions of discussions between men about positive masculinities:** In the context of rural areas, traditional social norms reinforce traditional gender roles, limiting access to important productive resources for the family. To advance gender roles and strengthen capacities on the benefits of gender equality for local development. It is important that gender equality is not perceived as a threat but as an opportunity for the community and to work hand in hand in curbing climate change.

- **Implementation monitoring strategies.** It is important to generate strategies and indicators to monitor efforts to reduce gender gaps in each rural area or in each particular agricultural condition. It is recommended that women's organizations linked to the agricultural and business sector be developed.

- **Promote an awareness plan on environmental education that** involves women producers Who say in consultations that a lot of environmental education is needed at all levels to solve the problems that are being experienced. Promoting the conservation of the environment, flora and fauna. The population needs to be aware.

- **Generate actions that allow women to self-recognize their fundamental role in agriculture and adapt to variability and climate change.** This is because in the rural area there are women who do not yet have the same rights as men

Output 3: Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies.

- **Strengthen the technical capacities of women's organizations at the local level. This represents an element of women's empowerment, facilitating access to resources, both financial and training and support of human resources, certification advice and/or legal persons.**
Organizing itself has an impact on their self-esteem and empowerment and generates more opportunities for all of them. Rural women's participation in cooperatives and associations also increases their participation in local development spaces.

Output 4. Financial products and services to finance sustainable investments are offered by Partner Financial Institutions (PFI), including PFI access to on-lending funds and support mechanisms.

- **Generate enabling conditions for women's access to financial schemes, which facilitates access to credit for women producers and women's cooperatives. To help strengthen women's entrepreneurship initiatives at the local level and with climate change adaptation and mitigation activities.**

Women's economic empowerment is potentially significant at the community level, and beyond, because it increases their visibility as agents of economic growth and as policy decision-makers.

Women's cooperative groups can form the basis of collective action to address gender inequalities in a community, including issues such as gender-based violence, access to resources, and decision-making in the local environment. They can be an entry point to promote greater political and social mobilization of women around gender issues.

In the context of the Dry Corridor, there are some fundamental limiting factors to increase access from female farmers to financing schemes derived from traditional gender structures and assigned roles. In this component, activities will focus on addressing some of those limiting factors to increase the access to financing schemes for female farmers. CABI has developed and is currently implementing a programme (Programa Regional de Financiamiento Empresarial para Mujeres - FEM) that can be used as a reference for the development of methodologies, tools, guidelines and processes that foster women's access to the established financial mechanisms.

5) METHODOLOGICAL CLARIFICATION

The gender assessment was developed through a desk review and a consultative process. The desk evaluation was led by a regional gender specialist with the support of seven local consultants and implied the review of the existing available data for the selected watersheds, and a regional literature assessment of common issues. A stakeholders' mapping was conducted to include the main actors and identify key women's organizations. The national consultants elaborated an overview of the selected municipalities with disaggregated data for men and women for several indicators, including the education level, life expectancy, school dropout rate, statistics of the labour situation, etc. (See Annex 26).

Furthermore, an initial socialization activity was conducted with potential end beneficiaries in October 2019, through participative workshops in each of the targeted watersheds: the details of these consultations can be found in Annex 7. A stakeholder engagement plan was defined with the gender

expert, with a participatory strategy that was designed with gender sensible indicators and activities. The detailed list can be found in Annex 7, this being the summary table of the participants numbers (available on page 39):

Country	Date	Municipalities	Attendees		
			Women	Men	Total
Honduras	14/10/2019	Choluteca, Marcovia, Duyure, Morolica, Apacilagua y Orocuina	8	25	33
Nicaragua	14/10/2019	Somoto	13	16	29
El Salvador	16/10/2019	San Miguel	9	21	30
Panama	18/10/2019	Pedasí	12	18	30
Costa Rica	18/10/2019	Nicoya, Guanacaste	11	24	35
Guatemala	22/10/2019	Santa Cruz del Quiché	1	28	29
Dominican Republic	22/10/2019	San Ignacio de Sabaneta, Prov. Santiago Rodríguez	8	14	22

The information gathered during the workshops fed the initial gender assessment, keeping in mind this would be complete during the inception phase of the programme – indeed, if the Guatemala example is taken where only one woman participated, her testimony isn't representative, although this singular presence can be considered as an indicator in itself.

This process didn't cover all necessary information, but it also helped in revealing information gaps that must be filled during the inception phase of the programme. This analysis contributed to the definition of recommendations (section 4 above) for the programme development and the elaboration of the Gender Action Plan (above).

Part II: Gender Action Plan: Template (Project/Programme Level)

1) Introduction

This Gender Action Plan (GAP) forms the basis for operationalizing the results and recommendations of the gender analysis section 1. It contains specific gender elements to be considered in the programme design and during the implementation of programme measures and activities. Moreover, it helps to monitor implementation of these measures and activities. Hence, the GAP ensures an effective gender mainstreaming and integration of a consistent gender-perspective in the programme in order to maximize climate and development co-benefits. The aim is to promote opportunities, drivers of change and positive gender dynamics as well as to manage and mitigate potential adverse risks over the duration of the programme. The GAP ensures that the programme is compliant with GCF's gender policy (GCF/B.09).

In striving for Ecosystem-Based Adaptation, and strengthening resilience at the community level, the regional program will take concrete measures to break down barriers to women's participation and decision-making at all levels. Transforming gender relations includes building on existing collaborative gender norms and practices at the household and community levels and implementing complementary strategies to improve them.

The program will contribute to gender equality through promoting exchanges among women and men, in both mixed-gender and separate moments for reflection. These findings will be used to decrease gender forms of discrimination in terms of everyday practices, as well as those that may exist in legal and normative frameworks or other structures.

Promoting non-violent, equality-promoting masculinities at the household, community and government levels will contribute to breaking down one of the most important barriers to gender and other inequalities. Effective partnerships between women and men based on more equal gender relations is likely to result in higher productivity, increased sources of and amount of household income, as well as other gains.

2) Impact Statement

The programme's objective is to increase climate-resilient sustainable development by facilitating wide-scale investment in EbA and water-efficient technologies across the Dry Corridor of Central America and Arid Zones of the Dominican Republic.

The programme aims to Increase and strengthen the resilience of vulnerable communities, especially women and girls, in the Dry Corridor of Central America and Arid Zones of the Dominican Republic to the negative impact of climate change. Through strengthening the capacities of the population in adaptation strategies based on Ecosystems (AbE), female heads of household; less time and work required by women for household chores; Increase in time savings, women economic empowerment

3) Outcome Statement:

Climate-resilient sustainable development by building the adaptive capacity of vulnerable communities across the Dry Corridor of Central America and the Arid Zones of the Dominican Republic increased

The programme has been specifically designed to create an enabling environment that capitalises on the scalability of EbA, focusing on four core areas of support: financial; policy; and technical. The details of these areas of support are provided below.

Component 1: Mainstreaming of EbA, water- and resource-efficient technologies and natural resource-based businesses into selected catchments

- Contribution to the regulatory framework and policies
- Technical support for the enhancement of regional knowledge and learning
- Upscaling of adaptation

Component 2: Financing and implementation for EbA as well as water- and resource-efficient technologies across selected catchments

- Creation of an enabling environment for investment in EbA and water-efficient technologies

4) Considered outputs and activities

The program outputs and activities that will contribute to meeting these goals include: equitable numbers of women as direct and indirect beneficiaries; capacity building for women and men, both programme personnel as well as producers and organization/community members; technical accompaniment through extension workers; public awareness; and monitoring and reporting on formal commitments, among others.

Through these strategies, women and men can contribute equitably to making sound economic and political decisions on an equal basis, in order to achieve gender equality, women's empowerment and strengthened adaptation to climate change and sustainability.

5) Methodology for the elaboration of indicators and targets

The GAP includes a set of indicators and targets for each activity of the programme, as detailed below. These indicators are additional to the ones included in the Logical Framework. The targets were defined based on the experience of UNEP, FAO and CCAD in the region¹⁷. Main assumptions used for their elaboration are:

- 1) The proportion of female-headed farms is below 30% in all Latin-American countries where data is available:
 - a. Between 20-30%: Panama, Nicaragua
 - b. Between 10-20%: Dominican Republic, Costa Rica, El Salvador
- 2) The access of female farmers to technical assistance can be between 5 and 40% lower than male farmers. This access is below 20% in the Central American countries where data exists.

¹⁷ Please refer to sources::

- FAO 2014. Recomendaciones de políticas de agricultura familiar (pages 104, 110-113)
- FAO 2018. Las políticas públicas aseguran la conquista de derechos para las mujeres.

- 3) Between 5 and 13% of rural women have access to productive credit. Up to 50% fewer women than men have access to agricultural credit.

Based on this, only a number of indicators establish equality between men and women, while most of them are based on a 30% of women participation. Regardless, all investment criteria elaborated, communication material and guidelines will include gender considerations.

6) Enabling Women's Participation in Program Activities

A number of measures will be established by the regional program to enable women's participation in activities. Some of them will consist of:

- sessions held at times and locations that are safe and convenient given women's diverse activities;
- childcare provided during capacity building sessions;
- holding events in local languages or providing translation/interpretation when needed; and
- Training and other materials mainstream gender by, in part, reflecting women's needs, interests, priorities and potentials.

Regarding capacity building, technical assistance and extension services:

- All programme personnel, including RPMU, PEU members, PFIS, and government staff will receive training in the following topics: 1) mainstreaming gender equality in relevant technical areas, etc.; 2) a specialized component on gender equality that will address violence against women, care economy, non-violent and equality-promoting masculinities, etc.
- The public communication and education strategies will emphasize gender equality. The strategies will be developed using participatory methods.

Among the different products planned during the programme implementation, the RPMU and PEUs will prioritize the elaboration of a Gender Equality Component or a disaggregated GAP of several of them, in accordance to the gender equality approach used in the programme. This concerns mainly, but is not limited to, the following:

- The financial mechanisms and their guidelines,
- the technical facility assistance,
- the site-specific intervention plans for the 7 target catchments,
- the protocols for the implementation of EbA practices as well as water- and energy-efficient technologies,
- the regional knowledge hub for the dissemination of information on EbA.

Finally, given the sensibility of GBV issues, the programme's grievance mechanism will engage with existing UN inter-agency mechanisms in the countries with standard operating procedures to ensure safety, dignity and confidentiality of potential users, as well as a referral pathway to a UN GBV focal point for each country. GBV protection will be mainstreamed in the programme implementation through the following actions to be taken by the programme's teams:

- i. the PEUs will be in regular contact with local organizations, and the mapping of actors elaborated during the initial baseline phase of the programme will include which local entities can provide relevant information on gender action, including GBV.
- ii. Regular dialogue with local authorities will also be ensured, as well as the PEU's understanding of local grievance mechanisms for GBV already in place and that might be complementary to the multiagency mechanism mentioned above.

- iii. Consultative processes will ensure respect of local customs and norms, and enabling actions are included in the GAP to promote positive integration of women's participation.

Persons from the affected population will have several avenues available to seek protection. Thus, dedicated hotlines, email addresses, instant messaging systems or applications, and letters and other written communication, to the extent possible, will substitute channels that require face-to-face contact and can jeopardize confidentiality.

7) GAP's Budget

The programme's GAP will be executed in all components in the design, implementation, planning, monitoring and evaluation phases. The budget for the GAP includes a Regional Gender Officer and National Adviser in gender, social participation and safeguards in each country for the 7 years of implementation. Gender enabling actions are also included in activities' budget, as can be found in the Budget notes' references included below and in Annex 4.

An estimate budget is included in the GAP, based on the following methodology:

- 100% of regional gender expert's budget,
- 60% of national advisors on gender, social participation and safeguards experts' budget,
- 5% of regional manager and national coordinators' budget,
- 10% of all relevant activities that will develop a focus on gender, as specified in the GAP.

This totals 2.6M, equivalent to 3.1% of total GCF Grant budget for this programme. This estimate will require further consideration during implementation and is provided for information purposes only.

Activities	Enabling gender action	Indicators	Baseline	Target	Timeline	Budget Notes and Estimate in USD	Responsibilities
Output1. <i>Strengthened technical capacity of local government, farmers and rural communities to implement EbA and other adaptation measures.</i>							
Activity 1.1. Develop site-specific intervention plans for the 7 target catchments to integrate EbA measures through a participatory process with municipal authorities, local communities and other stakeholders.	The focus of the gender action plan under this activity will be to ensure that differentiated opportunities for climate action and needs are identified and reflected in the site-specific action plans for the 36 participating municipalities and to ensure equal participation in the capacity building activities. Specific activities will include: a) Gender equality needs assessment incorporated in the site-specific action plans b) Ensure women membership and participation in the community-level monitoring and evaluation committee in each catchment c) Ensure women's participation in the community consultations for the validation of the implementation plans d) Include the participation of women farmers in the selection	a) Number of site-specific action plans that include results from gender needs assessment b) % of women in the community-level monitoring and evaluation committee in each municipality c) number of trainings on gender equality at community level d) number of stakeholder workshops for site action plans that include topics on gender equality e) % of women participating in the selection process of the sites for implementation of efficient technologies in water and energy within rural communities in the seven target basins.	a) 0 b) 0 c) 0 d) 0 e) 0	a) 100% of site-specific action plans include results from gender needs assessment b) At least 30% of the community-level monitoring and evaluation committee in each municipality are women c) At least one annual training on gender equality at community level d) At least 7 community consultations for site action plan validation that include topics on gender equality e) At least 30% of women participating in the selection process of the sites for implementation of efficient technologies in water and energy within rural communities in the seven target basins.	By year 2	A1 A7 A11 A17 A5 B25 A44 Estimated at: \$226,998	Accredited Entity/Executing Agency Gender advisor

	process of sites for the implementation of efficient technologies in water and energy within rural communities in the seven target basins.						
Activity 1.2. Provide technical assistance to municipal authorities, farmers and rural communities for the implementation of EbA practices as well as water- and resource-efficient technologies	The focus of the gender action plan under this activity will be to ensure that differentiated opportunities for action and needs are identified and reflected in the implementation protocols for the EbA interventions and to ensure equal participation in the capacity building activities and training programmes as well as to ensure that cooperation agreements are signed with female commercial farmers Specific activities will include: a) Guarantee representation of commercial women farmers in the development of cooperation agreements with commercial farmers b) Include gender equality in the capacity building activities to strengthen the capacities of commercial farmers in sustainable EbA practices that guarantee c) Guarantee the needs of women in the protocols for the adoption of	a) Number of cooperation agreements with women commercial farmers b) Number of trainings to commercial farmers that include gender equality c) Number of protocols developed that guarantee gender equality and women specific needs d) number of consultations on the specific needs of women in relation to water and energy management e) Number of men and women as lead trainers. f) Number of men and women participants in training of trainers' sessions. g) number of training programs for trainers per year includes gender equality and positive masculinities	a) 0 b) 0 c) 0 d) 0 e) 0 f) 0	a) at least 30% of cooperation agreements with women commercial farmers b) at least 7 annual gender equality trainings to commercial farmers c) 100% of developed protocols include gender considerations d) at least 30% of women as lead trainers e) equal participation in training of trainers' sessions. f) At least half of the training programs for trainers per year includes gender equality and positive masculinities	<i>A, b, d by year 3</i> <i>C by year 5</i> <i>e, f, g by year 5</i>	A19 A21 A24 A28 A31 A34 A5 B25 A44 Estimated at: \$268,910	Accredited Entity/Executing Agency Gender advisor

	water efficient technologies, energy efficiency on firewood, d) -In the framework of the training for trainers, include a module on gender equality and positive masculinities e) Ensure participation of women as trainers in the ToT programme f) Include women-run farms in the visits to demonstration sites during the training workshops in each municipality.						
Activity 1.3. Provide technical assistance to farmers and rural communities for the development of natural resource-based businesses and alternative climate-resilient livelihoods.	The focus of the gender action plan under this activity will be to ensure that differentiated opportunities for action and needs are identified and reflected in the livelihood assessment within each of the seven target catchments, focusing on differentiated climate risks and alternative options for resource-based business and climate resilient livelihoods ensuring women's participation Specific activities will include: a) Develop assessments that integrate gender differentiated opportunities for action and needs on climate risks and alternative options for resource-based business and climate resilient livelihoods	a) Number of assessments that include gender differentiated opportunities for action and needs on differentiated climate risks and alternative options for resource-based business and climate resilient livelihoods b) Number of women participating in livelihood assessment in each municipality c) Number of training programs that include gender differentiated economic empowerment, business plans based on natural resources integrating the results from the previous assessment d) % of women organizations with increased technical capacities for EbA business opportunities	a) 0 b) 0 c) to be determined at inception phase d) 0	a) 100% of livelihood assessments include gender differentiated opportunities for action and needs on climate risks and alternative options for resource-based business and climate resilient livelihoods b) Equal number of women and men participating in livelihood training programs in each municipality c) At least 30% of women demands for technical assistance to create business opportunities and technical guidelines d) At least 30% of women organizations demonstrate increased technical capacities for EbA business opportunities	By 1-3 year	A38 A43 A5 B25 A44 Estimated at: \$207,748	Accredited Entity/Executing Agency Gender advisor

	b) Establish livelihood training programs in each municipality for communities that include gender differentiated economic empowerment; business plans based on natural resources integrating the results from the previous assessment c) Strengthen the technical capacities on EBA of women's organizations at the local level. d) Strengthen the capacities of municipality technicians on social, environmental resilience and gender equality	e) number of trainings of municipality technicians on social, environmental resilience and gender equality		e) At least 7 trainings of municipality technicians on social, environmental resilience and gender equality			
Output 2 . Demonstration adaptation interventions implemented in rural communities across seven target catchments in the Dry Corridor and Arid Zones.							
Activity 2.1. Implement landscape EbA interventions within rural communities across the seven target catchments.	The focus of the gender action plan under this activity will be to ensure equal participation and needs consideration when implementing EbA activities in the rural communities of the target areas. Specific activities will include: a) At the landscape and farm levels, ensure that women specific needs are considered in the decision-making processes for establishing the exact areas of intervention. b) At the farm level, ensure that the beneficiaries of the implementation of EbA measures include farms owned or managed by women.	a) % of women producers, representatives or organizations that participate in the decision-making process for selecting exact areas of intervention at the landscape and farm levels. b) % of farms benefited by the EbA pilot measures at the farm level that are owned or managed by women. c) % of women that are hired for the on-the-ground implementation of EbA pilot activities. d) Number of monitoring & evaluation plans that are drafted considering the needs expressed by women and that measure the differentiated impacts for both women and men.	a) 0 b) 0 c) 0 d) 0	a) At least 30% of the participants in the decision-making processes for selecting exact areas of intervention at the landscape and farm levels are women. b) At least 10% of the farms benefited by the EbA pilot measures at the farm level are owned or managed by women. c) At least 30% of the work force hired for the on-the-ground implementation of EbA pilot activities are women. d) Every monitoring & evaluation plan that is developed is drafted considering the needs	By 1-7 year	B33 A5 B25 A44 Estimated at: \$221,574	Accredited Entity/Executing Agency Gender advisor

	c) Guarantee that women are hired for the on-the-ground implementation of EbA pilot measures. d) Develop a gender-sensitive monitoring & evaluation plan for the EbA pilot activities.			expressed by women and measures the differentiated impacts for both women and men.			
Activity 2.2. Implement demonstration on water- and resource-efficient technologies within rural communities across the seven target catchments.	The focus of the gender action plan under this activity will be to ensure equal participation and needs consideration when implementing water and energy efficient technologies in the rural communities of the target areas. Specific activities will include: a) Guarantee the consultation of women to understand their specific needs related to access to water and energy efficient technologies/activities. b) At the community level, ensure women participation in the decision-making processes about the implementation of water and energy efficient technologies/activities. c) Promote opportunities for women to influence decision-making about the implementation of water and energy efficient technologies/activities. d) Ensure that the results from consultation are considered when deciding the exact location for installing water and energy efficient technologies/activities.	a) Number of consultations that collect information on the specific needs of women related to water and energy-efficient technologies/activities. b) % of participants in decision-making processes about the implementation of water and energy efficient technologies who are women. c) % of women participants in decision-making processes who have key positions at water committees and/or water-related organizations. d) % of variables in decision criteria for determining exact location of interventions that are related to an improved access to water for women. e) % of community members trained and that actively work on the management, use and maintenance of water and energy efficient technologies who are women.	a) 0 b) 0 c) 0 d) 0 e) 0	a) At least one consultation in each of the 5-6 municipalities in the targeted catchments where the specific needs of women related to water and energy-efficient technologies/activities are examined. b) At least 30% of the participants in decision-making processes about the implementation of water and energy efficient technologies are women producers, representatives or organizations. c) At least 35% of the women participants (equivalent to 10% of all participants) in decision-making processes have key positions at water committees and/or water-related organizations. d) At least 40% of the variables in decision making criteria for determining the exact location of interventions are related to an improved access to water for women (e.g.: shortened distances to	By 1-7 year	B38 A5 B25 A44 Estimated at: \$216,931	Accredited Entity/Executing Agency Gender advisor

	e) Promote women technical training and active work on management, use and maintenance of water and energy efficient technologies.			wells, time gained for productive work, rest or care, etc.). e) At least 35% of the community members trained and that actively work on the management, use and maintenance of water and energy efficient technologies are women.			
Activity 2.3 Establish the grant facility to support bottom-up selection and promotion of local EbA activities through non-reimbursable financing and start operations	The objective under this activity will be to promote equal access to the grant facility for male and female farmers as well as organizations. Specific activities will include: a) Integration of gender equality aspects into the eligibility criteria for the grant facility in line with national regulations and CABEL's gender policy b) Development of campaign materials specifically directed to promote women's requests to the grant facility c) Integration of gender equality indicators in the monitoring, reporting and verification mechanism developed under sub-activity 1.2.3 d) Integrate gender equality messages in the communication of results through specific events and media initiatives	a) # of gender considerations incorporated in eligibility criteria for the grant facility, in line with national regulations and CABEL's gender policy b) Number of campaign materials specifically directed to promote women's requests to the grant facility c) Number of gender equality indicators integrated in monitoring, reporting and verification mechanism d) Number of specific events and media initiatives that integrate gender equality messages in the communication of results of the grant facility	a) 0 b) 0	a) At least 2 gender considerations incorporated in the set of eligibility criteria for the grant facility for each country, in line with national regulations and CABEL's gender policy b) at least 1 annual campaign focused on gender equality in entrepreneurship initiatives c) At least 2 gender indicators included in the framework for monitoring, reporting and verification mechanisms developed in each country d) At least 2 annual messages on gender equality in entrepreneurship initiatives socialized by the media	By year 1 eligibility criteria and monitoring framework Annually for awareness campaigns and media messages	B44 B50 A5 B25 A44 Estimated at: \$199,350	Accredited Entity/Executing Agency Gender experts

Output 3:
Information on climate change adaptation and its financing disseminated across the region and mainstreamed into local and national policies.

Activity 3.1. Establish regional knowledge hub for the dissemination of information on EbA in the Dry Corridor and Arid Zones.	The objective under this activity will be to promote equal access to the knowledge hub for both men and women and to ensure the products' elaborated by the knowledge hub include gender-equality issues to generate evidence on the differentiated vulnerability to climate change and the opportunities for more efficient climate action including the gender perspective. Specific activities will include: a) Ensure knowledge products include a gender-specific focus b) Integration of gender-neutral language in the knowledge hub platform and all knowledge products elaborated. c) Ensure the exchange of best practices and lessons learned include experiences on gender issues' integration in the seven target countries. d) Include a module on climate change impact on gender inequality in annual training for policy- and decision-makers under sub-activity 3.1.3 e) Ensure women's group participation and representation in the community champions'	a) % of knowledge products elaborated include gender-specific issues b) Number of research products that deals with gender and ecosystem-based adaptation c) number of knowledge products elaborated following gender-neutral language's recommendations¹⁸ d) number of exchanges of best practices and lessons about gender equality and EbA published in the regional hub e) Number of training modules on climate change impact on gender inequality f) Number of women's group participation in training g) number of female community champions.	a) 0 b) to be determined at inception phase c) 0 d) 0 e) 0 f) 0	a) at least 50% of knowledge products include gender-specific issues b) 1 piece of research deals with gender and EbA c) at least 80% of knowledge products are produced following gender-neutral language's recommendations d) at least 7 exchanges of best practices and lessons about gender equality and EbA are published e) At least 1 module per training includes climate change impact on gender inequality f) At least 60% of women's groups participate in trainings g) At least 1 community champion per country is female.	a, b and c between years 1 to 7 d and e between years 2 and 6	C1 C3 C7 C8 A5 B25 A44 Estimated at: \$392,737	Accredited Entity/Executing Agency Gender advisor
--	--	--	--	---	---	--	---

¹⁸ Using guidelines such as UN guidebook in Spanish (also available in English): <https://www.un.org/es/gender-inclusive-language/guidelines.shtml>

	developed under sub-activity 3.1.4.						
Activity 3.2. Raise awareness of financial mechanisms for the implementation of CCA interventions	The objective under this activity is to ensure sensitization of potential clients of the financial mechanisms for EbA interventions for both men and women. Specific activities will include a) Ensure women's participation in training on loan access through the EbA credit line. b) Map out the different needs and preferred information channels of both women and men for the financial mechanism during trainings under sub-activities 3.2.1 and 3.2.2 c) Awareness-raising campaigns under sub-activity 3.2.3 include gender equality messages in the communication of results through specific events and media initiatives, using results from assessment realized in (b)	a) Number of trainings carried out with gender considerations (based on schedules, transportations and other needs) b) Number of training curriculums that include key messages to promote women's access to the financial mechanisms c) Number of training reports that include disaggregated information on men and women's need regarding financial mechanism d) Number of local campaigns that target women to access financial mechanisms e) Number of local campaigns increase awareness on gender equality	a) 0 b) 0 c) 0 d) 0 e) 0	a) at least 70% of trainings are carried out with gender considerations b) 100% of training materials include key messages for women's access to loans through the EbA credit line. c) 70% of trainings attended by women include disaggregated information on men and women's needs for financial access. d) At least 7 local campaigns (one per country) target women specifically regarding access to financial mechanisms e) at least 7 of local campaigns integrate gender equality messages	(a), (b) and (c) by years 2 to 7 year (d) and (e) on years 1, 2, 4 and 6	C12 C14 C16 A5 B25 A44 Estimated at: \$227,708	Accredited Entity/Executing Agency Gender advisor
Activity 3.3. Enhance capacity of local-level policy-makers to integrate climate change	The objective under this activity is to contribute to the amplification of gender-affirmative actions in the proposed response models to climate change. a) Mapping good practices led by women's organizations and environmental defenders,	a) # of women-led initiatives identified related to climate change adaptation and ecosystems' valuation. b) # of gender-sensitive indicators included in the	a) 0 b) 0	a) At least 7 women-led initiatives in LAC identified, at least 3 of them are from the 7 target countries b) At least 3 gender-sensitive indicators included in the methodology under sub-activity 3.3.2	(a), (b) and (c) between years 1 to 7 (b) between years 2 and 3	C22 C26 C28 A5 B25 A44	Accredited Entity/Executing Agency Gender advisor

adaptation and the valuation of natural capital into local policies	including in other regions or countries. b) Include gender-sensitive indicators in the development of the methodology under sub-activity 3.3.2. c) Integrate gender-affirmative actions in the guidelines for local governments outlining protocols and criteria for implementation of economic incentives for SLM under sub-activity 3.3.3. d) Include module on gender equality and climate change in all the training of local governments that are carried out, to promote changes in public policies and protocol uses e) Ensure participation of female public officials in the municipalities in the capacity building programme on the integration of climate resilience in broader public policies	methodology to value ecosystem-services. c) % of guidelines including recommendations on gender-affirmative actions d) # of modules on gender issues included in training and capacity building activities e) % of female public officials in the municipalities involved in the capacity building programme on the integration of climate resilience in broader public policies		c) At least 60% of the guidelines produced include recommendation on gender-affirmative actions. d) At least 1 module per training and capacity building activities is focused on gender issues e) at least 50% of female public officials in the municipalities participate in the capacity building activities	(c) <i>between years 2 and 4</i>	Estimated at: \$217,548	
---	--	--	--	---	----------------------------------	-------------------------	--

Output 4
Financial products and services to finance sustainable investments are offered by Partner Financial Institutions (PFI), including PFI access to on-lending funds and support mechanisms.

Activity 4.1. Set-up and establish the financial structure for the lending	The focus of the gender action plan under this activity will be on the definition of investment criteria and operational guidelines that facilitate	a) Number of guidelines for technical, environmental, financial requirements of sustainable investments and financial eligibility requirements for application	a) 0 b) 0 c) 0	a) At least 2 sets of guidelines developed for IFIs, PFIs for the lending and guarantee facilities that integrate gender equality considerations for the	By year 1 Annually Revised and updated if necessary	D3 D6 D9 A5 B25 A44	Accredited Entity/Exe cuting Agency Gender experts
--	---	---	-------------------------------------	---	--	---	---

and guarantee facilities.	and promote gender equality in access to finance instruments. Specific activities will include: a) Inclusion of gender equality considerations into the operational guidelines for the lending and guarantee facilities b) Integration of gender equality aspects into the investment criteria for the lending facility in line with national regulations and CABEL's gender policy and FEM programme c) Integration of gender equality indicators in the monitoring, reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5	to loan funds and support mechanisms guarantee gender focus b) Gender equality aspects integrated in investment criteria approved by Steering Committee and CABEL c) Number of gender equality indicators in the monitoring, reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5		operationalization of the financial schemes b) At least 2 gender equality aspects integrated in the investment criteria approved by Steering Committee and CABEL in year 1 of the programme c) At least 2 gender equality indicators are integrated in the monitoring, reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5 including recommendations from the RPMU and PEU gender experts		Estimated at: \$99,174	
Activity 4.2. Financing mechanism of the EbA lending facility and guarantee facility.	The objective under this activity will be to promote equal access to the lending and guarantee facilities for male and female farmers as well as organizations. Specific activities will include: a) Inclusion of gender equality considerations into the operational guidelines for the lending and guarantee facilities b) Integration of gender equality aspects into the investment criteria for the lending facility in line with national regulations and CABEL's gender policy and FEM programme	a) Number of guidelines for technical, environmental, financial requirements of sustainable investments and financial eligibility requirements for application to loan funds and support mechanisms guarantee gender focus b) # of gender equality aspects integrated in investment criteria approved by Steering Committee and CABEL c) Number of gender equality indicators in the monitoring,	a) 0 b) 0 c) 0 d) To be determined at inception phase e) To be determined at inception phase	a) At least 2 sets of guidelines developed for IFIs, PFIs for the lending and guarantee facilities that integrate gender equality considerations for the operationalization of the financial schemes b) At least 2 gender equality aspects integrated in investment criteria approved by Steering Committee and CABEL in year 1 of the programme	A, b, c at the end of year 1 D, e by year 7	D3 D6 D9 A5 B25 A44 Estimated at: \$99,174	Accredited Entity/Executing Agency Gender advisor

	c) Integration of gender equality indicators in the monitoring, reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5 d) Ensure equal access for male and female farmers to financing schemes to support water efficiency measures as well as to information on how to apply for them	reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5 d) increase in number of women producer's organizations applicants who successfully apply to on-lending funds and support mechanisms in comparison to other lending funds from CABEL e) increase in % of women producer's organizations that access similar amount of funding than men organizations in comparison to other lending funds from CABEL		c) At least 2 gender equality indicators are integrated in the monitoring, reporting and verification mechanism developed under sub-activities 1.1.4 and 1.1.5 including recommendations from the RPMU and PEU gender experts d) At least 20% increase of women producer's organizations applicants who successfully apply to on-lending funds and support mechanisms in comparison to other lending funds from CABEL. e) At least 20% increases of women producer's organizations that access similar amount of funding than men organizations in comparison to other lending funds from CABEL			
Activity 4.3 Technical assistance (TA) facility to strengthen technical capacity of CABEL network and non-regulated financial institutions to access and channel	The objective under this activity will be to promote equal access to the lending and guarantee facilities for male and female farmers as well as organizations through the integration of gender considerations into the training materials for financial entities at different levels. Specific activities will include: a) Develop and adjust training curriculums and technical assistance methodology targeted to fill gaps identified within gender equality approach,	a) % of women trained in the ToT programme b) number of financial institutions that received technical assistance and increased their AbE capacity include women in their management team c) Number of key messages to promote women's access to the financial mechanisms to potential network institutions and non-network institutions in each of the 7 countries	a) 0 b) 0 c) 0	a) At least 30% of women trained in the ToT programme b) At least 10 of the 28 associated financial institutions that received technical assistance and increased their AbE capacity include women in their management team c) At least one annual campaign with gender focus for awareness on financial mechanisms	By end of programme	D13 D14 D25 A5 B25 A44 Estimated at: \$218,648	Accredited Entity/Executing Agency Gender advisor

funds for small- and large-scale EbA investments.	b) Promote the blended EbA lending facility with potential network institutions and non-network institutions in each of the 7 countries with key messages to promote women's access to the financial mechanisms						
---	---	--	--	--	--	--	--
